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‘*Carnal Knowledge* is a new materialist collection, not only because it brings together new materialist work, but also because it consistently and constantly reflects on what is new about new materialisms (it does not apply conceptual tools such as affect, but traverses bodies of thought).’

– Iris van der Tuin, *Women’s Studies International Forum*

Carnal Knowledge
Towards a
'New Materialism'
through the Arts

Edited by **Estelle Barrett**
and **Barbara Bolt**

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FOREWORD

This edited collection of essays began as a conversation between a group of Australian and Finnish scholars who had been working in the emerging field of new materialism. What distinguished their work from other investigations in this field was their commitment to understanding and articulating a new materialism through the creative arts.

Under the sway of postmodernism, writing about the arts concerned itself with meaning, situating art within the broader discursive formations. Theorized through this interpretative frame, the place, role and power of materiality in art was subsumed under the rubric of discourse or ignored. *Carnal Knowledge: Toward a "New Materialism" Through the Arts* is part of a renewed energy and interest in the relationship between the arts and materiality. This has been described as the "material turn". It raises the question: What is this new materialism? This anthology addresses this question and offers perspectives on "the new materialism" from a number of established and emerging writers who are working in this exciting new field. The key questions guiding this project are:

- What is/was the "old materialism"?
- How does classical materialism differ from Marxist materialism?
- How does the new materialism help to shift the focus from representationism and the hegemony of language/signification in our understandings of discursive practices, material processes and the production of knowledge?
- What role can the arts and current understanding of artistic practice play in answering some of these questions and illuminating how matter is crucial to understandings of cultural production and social/historical change?

We believe that addressing such questions requires a re-thinking of the relationship between materiality and signification and perhaps even a redefinition of these key concepts. Thinkers and writers from Lucretius and Epicurius to Nietzsche, Heidegger, Deleuze, Kristeva, Kirby, Lyotard, Grosz and Haraway (to name a few) have in various ways, needed at these categories. Their work often returns us to the notion that the "new materialism" has some resonances with the "old materialism" resting as it does on assumptions that are ultimately metascientific, though never

metaphysical. Such assumptions go beyond empirical science, though not beyond physical reality, beyond matter and the agency of the non-human.

Julia Kristeva (1980) suggests that through the arts, we may discover a mode of production that devises an alternative, original, mobile and transformative knowledge. Creative practice “does not name, but enunciates the very place of the materialist dialectic that no human science has yet approached”. This anthology draws together a range of scholars whose research spans dance, music, film, fashion, design, photography, literature, painting and stereo-immersive VR. It will demonstrate how art allows us to map the complex relations between nature and culture, between the body and language and knowledge.

Estelle Barrett and Barbara Bolt

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INTRODUCTION

Toward A “NEW MATERIALISM” THROUGH THE ARTS

Barbara Bolt

At the center of Raphael's 1508–1511 fresco, *The School of Athens*, two central figures are locked in animated conversation as they stride toward us. Both figures grasp bound books in their left hands and gesture with their right. On the left, the elderly scholar Plato, treatise *Timaeus* in hand, gestures to the heavens, the world of Ideal Forms. His gesture motions to the immaterial, a world that can be apprehended through speculation with the eye of the mind (reason). On his right, Plato's student Aristotle confidently points outwards toward us and the material world around. Aristotle's world is the practical world, one that can be seen with the eyes (empiricism).

Placed at the center of the painting, Plato and Aristotle represent the two dominant strands of Western philosophical thought that have framed Western conceptions of the world. Structurally, the use of one point perspective in the image implicates us as viewers in the narrative of the picture. According to the logic of perspective, the viewer takes up a position that is identical with the position originally occupied by the painter, the maker of the picture. Implicit in the image's structure, is a humanist metaphysics that places man-as-subject (*subiectum*) at the center of all relations. “Man” is the one who orders the world and produces a picture of the world. In this relation of representation, the human is the active subject whilst the world is passive matter to be acted upon. There is, however a tangential movement or *clinamen* operating in this picture of the Greek philosophical world. Off to the side, at the far left hand corner of the picture plane, stands a figure with his back turned to Plato and Aristotle. Oblivious to everything else going on, he leans against a column totally absorbed in his own matters. This figure is Epicurus.

In keeping with his position in the painting, Epicurus' thought offered a radical departure from the dominant Greek debates of the time. Whilst better known to us for his hedonistic ethics, Epicurus' physics diverged from the world of Ideal Forms and also from ideas concerning the appearance of things. Instead, it focused on the matter of things. Epicurus drew from the atomistic materialist theories of Democritus,



1. Raphael (originally Raffaello Sanzio, 1483–1520), *The School of Athens*, 1508–1511, fresco, Rome, Vatican, Stanza della Segnatura

to offer an interpretation of being that was concerned with matter and the processes of matter. Whilst Democritus' systematic materialist theory held that reality consists only of "atoms and void", Epicurus ascribed a dynamic freedom to atoms through what he identified as the unpredictable swerve of atoms (Stack 1998). For Epicurus, the swerve sets things in motion and initiates new patterns and movements (Sedley 1998, 2005). It is through these material movements and productive processes of matter that the world is made. Lucretius named this dynamic the *clinamen*.

Through the figure of the *clinamen*, Epicurus offers the possibility of active or agential matter. This thread runs from the ancient thought of Democritus thought, through the scientifically based materialist theories of the eighteenth and nineteenth century to what is now being called the "material turn" or the new materialism. What draws all these theories under the one umbrella "materialism" is an understanding or theory of the world asserting that: all entities and processes, including human beings, "are composed of – or are reducible to – matter, material forces or physical processes" (Stack 1998).

How are we to understand the resurgence of materialist theories at the beginning of twenty-first century? With its acknowledgement of agential matter, neo-materialism questions the anthropocentric narrative that has underpinned our view of humans-in-the-world since the enlightenment, a view that posits humans as makers of the world and the world as a resource for human endeavors. The new materialist

discourse derives its urgency from the ethical, ecological and political imperatives that loom as a consequence of this view of the world. In his review essay on new materialism, “Things-Beyond-Objects”, Andrew Poe suggests that “What should cause theorists” alarm are how divisions between the human and the world – between subject and object – have had the anathema result of allowing for the destruction of the actual (which is to say ecological) world. The subject, it seems, has come to care too much for itself’ (Poe 2011: 153). Put simply, the idea that the world is a passive resource for use by active humans is no longer sustainable. The matter of the world can no longer be a mere resource for human endeavor. What is at stake here, is the very ground on which humanism has been built and sustained.

The emergence of neo-materialism “now” may be understood as the result of the butterfly effect – a confluence of currents across the disciplines that have validated a rethinking of the relationship between humans and non-humans. Whilst humanist thought placed the human subject firmly at the center of the social and physical world, discoveries in science, particularly around quantum physics and nanotechnology, and the emergence of new human-technological relationships have decentered the subject. These movements, in concert with social, political and philosophical theories that question the privilege given to humans in the human/non-human binary, underpin discourses of new materialism. At the core of the material turn, is a concern with agential matter. As Iris van der Tuin notes, scholars from diverse (inter)disciplinary fields, continents and generations are developing theoretical tools for “dealing with agential matter rather than (gendered) passive matter” (van der Tuin 2011: 271).

The “I” of new materialism is no longer the sovereign human subject, but is conceived of as a material-semiotic actor, an articulation that, according to Donna Haraway (1991) encompasses the human and the non-human, the social and physical, and the material and immaterial. Susan Hekman, drawing on Andrew Pickering’s notion of the mangle, proposes that “the ‘I’ is a mangle composed of multiple elements” (Hekman 2010: 100). This reconfiguration of human/non-human relationships has emerged around a *vital materiality* where it is not just humans who do things (Bennett 2010). In these new configurations the material and the discursive mingle and mangle.

In the humanities and social sciences, particularly in feminist theory, the new materialism involved a reaction against the cultural turn that defined cultural and political theory in the late twentieth century. Exponents of the new materialism argue that where social constructivist theories thrive, matter becomes mute. New materialism aims to return to matter, the vivacity denied by social constructivist theories that posit all social processes and, indeed reality itself, as socially and ideologically constituted. Hence, in their introduction to *Materialist Feminisms* (2008), Stacy Alaimo and Susan Hekman claim that the new feminist materialism aims “to bring the material, specifically the materiality of the human body and the natural world, into the forefront of feminist theory and practice” (Alaimo and Hekman 2008: 1).

What relevance might this have for the creative arts and for this anthology? In the 1980s and 1990s the influence of cultural theory spread beyond the boundaries of social and political theories and came to infect the arts and leach “matter” out of art.

However, this influence needs to be understood within the context of the history of modernism, its assumptions and the fact that artistic relationship *embodies* a materialist dialectic.

Whilst each artistic form has a unique relation to the matter of things, the humanist view of relationship of matter to form was codified in modernism's notion "truth to materials". In 1911, the modernist visual art critic Roger Fry claimed that in order to get at material beauty it is "necessary to respect the life and quality of the material itself" (Fry quoted in Hiller 2003: 68–69). The emphasis on "truth to materials" in modernist discourse reached its apotheosis in Clement Greenberg's formalism and his claim that the character of an art form – painting, for example – should be determined by its material medium:

The unique and proper area of competence of each art coincided with all that was unique in the nature of its medium. The task of self-criticism became to eliminate from the specific effects of each art any and every effect that might conceivably be borrowed from or by the medium of any other art. Thus would each art be rendered "pure," and in its "purity" find the guarantee of its standards of quality as well as of its independence. "Purity" meant self-definition, and the enterprise of self-criticism in the arts became one of self-definition with a vengeance (Greenberg 1960).

However, Mary Kelly points out the problem with of modernist formalism, as an incoherent contradiction: on the one hand it trumpeted materialism through its claim to "truth to materials," but on the other, it championed notions of the artist's genius, originality, self-expression and the Kantian notion of taste (Kelly 1981: 47–48). Adding her voice to Kelly's insistence that art always takes place within specifiable conditions of production and consumption, Griselda Pollock took modernism to task for its failure to "take into account the social character of artistic activity" (Pollock 2003: 80). Kelly and Pollock's feminist materialism drew on *historical* materialism to frame art "as textual politics – an interrogation of representation as a social site of ideological activity" (Pollock 2003: 80).

The framing of art as primarily social and ideological in nature occurred within the broader context of the rise of cultural studies, post structuralism and postmodernism. Where formerly art had been understood in terms of art history and aesthetics and was vested with the power to re-present reality, it now became framed (along with a multitude of other social and cultural activities) as a "cultural production". Art like identity was no longer a given, but was produced through social discursive practices. Michel Foucault coined the term "discursive formations" to elaborate the categorical grid that is placed over "reality" by different disciplines (or bodies of knowledge) in order to comprehend it. Central to the constructivist position was the assumption that discourse prescribes what can be thought and what can be represented – whether in writing, speaking, acting, dancing, music, filmic or visual representations. That is, our making and experience of art is always and necessarily socially and culturally mediated. Thus, through the colonization of the arts by cultural theory, arts' very materiality has disappeared into the textual, the linguistic and the discursive. According to this conception, art is constructed in and through language. There is nothing outside of discourse and language is its vehicle.

Whilst materialist feminist theory has struggled to disentangle matter from discourses on matter, it may be argued that the art *is* a material practice and that materiality of matter lies at the core of creative practice. Dance, theatre and fashion, as embodied practices, engage the matter of bodies. In music, the material bodies of composers, musicians and singers co-collaborate with instruments and other technologies in spaces that allow music to emerge. Film-making is an intense relationship between a myriad of human and non human actants – lights, cameras, editing machines, actors, editors, film-makers and directors, to identify just a few in this complex assemblage. Finally the visual arts engage all manner of material processes in making and assembling art. The material facts of artistic practice appear so self-evident and integral to our understanding of art that it may seem unremarkable to frame them in terms of the material turn. What then are the stakes for art in invoking the material turn?

Andrew Poe proposes that the work of art may be the necessary limit to the materiality thesis. In his review of Diana Coole and Samantha Frost's book *New Materialisms: Ontology, Agency, and Politics* (2010), Poe questions if the new materialism can actually account for the aesthetic dimension of things. His thinking is twofold. Firstly he proposes that the human creation-of-things needs to be accounted for, querying whether or not creation and assembly can be considered identical. Secondly, he argues that in aesthetic experience “we, as those who perceive the aesthetic, are thereby always doing work in the world, even without labouring on it” (Poe 2011: 161).

Poe's question provides a fitting challenge for this anthology of new materialist thinking in the arts. However, before setting out how the contributors respond to the challenge, it is worth unpacking the assumptions that underpin Poe's questioning. Firstly, in framing art as the “human creation-of-things”, Poe reinforces a particular view of art that sees humans as the active creator in the creation of things. Here matter becomes “the substrate and field for the artist's formative action” (Heidegger 1977a: 52). Martin Heidegger calls this the form-matter synthesis. The form-matter synthesis seems to make such common sense that it is difficult for us to think outside this framework. The danger of the form-matter conception that Heidegger identifies is that:

If form is correlated with the rational and matter with the irrational, if the rational is taken to be the logical and the irrational the alogical; if in addition the subject-object relation is coupled with the conceptual pair form-matter; then representation has at its command a conceptual machinery that nothing is capable of withstanding. (Heidegger 1977b: 153)

The form-matter synthesis reinforces the “muteness” and “dumbness” and “irrationality” of matter. Matter is the “dumb” “mute” “irrational” stuff on which humans act.

In his essay “The Question Concerning Technology” (1977a), Heidegger questions the proposition that matter is the substrate for the artist's actions. Through this essay, concerned with the formulation a different relation between humans and technology, Heidegger undoes the Aristotelian notion of causality. In the place of

our accepted notion that things are a means-to-an-end, he proposes a relationship of co-responsibility and indebtedness. He cites the example of the making of a silver chalice to suggest that the matter of the silver, together with “aspect” (chaliceness), “circumscribing bounds” (the purpose for what the vessel is to be used) and the “artist” are co-responsible for bringing the silver chalice into appearance. For Heidegger, the artist doesn’t create the silver chalice nor is the chalice formed matter. Instead, he proposes that the silversmith is co-responsible for and indebted to other co-collaborators for the emergence of the “thing” as a silver chalice (Heidegger 1977a: 6). Together with the silversmith, these ways of being responsible, bring-forth something into appearance. The “re-distribution of power” posited by Heidegger’s reconceptualization of createdness has much in common with neo-materialist positions on political agency. Heidegger’s rethinking of createdness and his re-interpretation of causality shifts our understanding from the “form-matter” thesis to a notion of care and indebtedness between co-responsible elements. In sum, art is a co-collaboration, not a form-matter synthesis and matter as much as the human has responsibility for the emergence of art. In other words matter has agency.

A number of essays in this anthology foreground the agency of matter in artistic and cultural practices. Kaisa Kontturi’s essay “From double navel to particle-sign: toward the a-signifying work of painting” focuses at the molecular level, demonstrating that the actions of the matter are essential to the creative process. Brian Martin’s essay, by contrast operates at the macro level, demonstrating that in Indigenous Australian culture, the spiritual or immaterial is a continual practice created by the reality of land. The focus on agential matter is also taken up in the section on “The Matter of Film”. Here Nicholas Chare and Liz Watkins’s essay “The matter of film: *Decasia* and *Lyrical Nitrate*” and Dirk de Bruyn’s chapter, “Recovering the hidden through found-footage films” examine how, in films made with found footage, the materiality of film stock erupts into the symbolic register producing powerful performative effects.

The notion of the performative power of materiality in and through art addresses Poe’s second objection to the new materialism. This objection concerns the aesthetic appreciation of art, where he proposes that “we, as those who perceive the aesthetic, are thereby always doing work in the world, even without labouring on it” (Poe 2011: 161). Here, Poe’s understanding of the aesthetic experience is underpinned by Kantian notions of the aesthetic and carries with it the assumption that it is the active human agent who is doing the work in the world. A materialist aesthetic doesn’t place the aesthetic experience in the human subject. Rather the “I” as an articulation of a material-semiotic actor, situates the aesthetic as a relationship “between” – between the human and non-human, the material and immaterial, the social and physical.

Julia Kristeva provides us with a way of apprehending this relationship in her conceptualization of the triple register of the artistic function. For Kristeva, aesthetic experience involves an intertwining of the physical, the psychic and the social. This intertwining (what Karen Barad (2003, 2007) calls *intra-action*) complicates assumptions about aesthetic experience either as reducible to questions of taste by a

disinterested Kantian viewer, or on the grounds that our experience of art is merely culturally coded.

In her essay “Giotto’s joy” (1980) Kristeva takes our experience of color to examine the dynamics of the artist function, or what Estelle Barrett refers to in her chapter as the aesthetic image. Kristeva begins her discussion of the triple register, evoked through an encounter with color, arguing that through the artistic function, instinctual drives insinuate themselves into the symbolic through the operations of pressure, excitation and *cathexis* (investment) that derive from the interaction between the organism and its outside world. In her tripartite schema, the first register involves an external excitation, the activity of matter impressing on the body. The second register revolves around the effect of this excitation on the body. This internal excitation or eroticization of the body under instinctual pressure occurs at the molecular level, at the level of matter rather than at the level of the subject. In this sense it is not so much a question of perception, but rather, as Deleuze and Guattari (1994) suggest, of sensation. Finally, Kristeva identifies the third register as the sign. This register involves the integration of this excitation as a sign in the system of representation in a given culture. It is this integration of color into the representational schema of a culture that has led to the assumption that color, and more broadly art, is culturally coded. However, through the elaboration of the triple register, Kristeva demonstrates that it is the operations of the triple register, whether in the use of language or in the visual or performing arts, that frustrates representation. The chromatic apparatus, like rhythm and texture shatters meaning and its subject into a scale of differences. This is the performative power of materiality that is invoked in the essays on found footage film discussed earlier, in Barrett’s notion of the aesthetic image and in Jondi Keane’s elaboration of *Æffect*.

Some of the writing around new materialisms has a manifesto quality that seems to want to throw out the contributions of the cultural turn. Kristeva’s discussion of the triple register demonstrates that humans are material-semiotic or material-discursive actors in a social world and the social character of art cannot be ignored. Thus it is not a question of negating the insights of the cultural turn. Nor are we seeking a return to modernism with its humanist assumptions about truth, autonomy, genius, originality and self-expression. The task, as Susan Hekman sees it is to be able to “account for the material reality of our social existence without losing site of the discursive dimension of that reality” (Hekman 2010: 90). The question that Hekman poses for feminism is similar to the dilemma facing the arts: How do we bring the material back into the equation “without losing the insights of the linguistic turn that characterized the last decades of the twentieth century?” (Hekman 2010: 107).

This brings us finally to the contribution of *Carnal Knowledge*. This anthology of writing around the theme of “new materialism” through the arts, negotiates the relations between the various bodies that enable art to come into being – the material bodies of artists and theorists, the matter of the medium, the technologies of production and the immaterial bodies of knowledge that form the discourse around art. Rather than providing a limiting case for the new materialisms, this anthology demonstrates the case for why the “new materialism” is relevant now. The essays

in this anthology are divided into four sections: Discourse, affect and material interventions; the matter of film; carnal insistence and finally materiality: the virtual and the real. Each section, represents an engagement with key themes concerning the new materialism, but seen from the point of view of the arts. The contributors to this collection are drawn from across the arts and bring to the anthology a range of experiences in scholarship and with the various art forms. This book articulates a conversation between the matter of art and its interlocutors, both scholars and scholars who work as artists.

Discourse, Affect and Material Interventions

Katve-Kaisa Kontturi's chapter, "From double navel to particle-sign: toward the a-signifying work of painting", argues for the creative and transformative power of matter in the artistic process. Kontturi draws from her engagement with a painting by the Finland-based artist Susana Nevado. Kontturi is captivated by the "appearance" of a curious double navel that marks a figure's stomach in the painting. She argues that the double navel was not the artist's intention, nor was it not consciously created. It emerged through the operations of matter. Through its emphasis on matter, this essay questions representationalist approaches that focus on meanings inscribed in the matter rather than on the workings of the matter. Konturri activates the Deleuzian concept of the particle-sign to draw attention to the vital connection and collaboration of matter and meaning in the creation of the new. According to Konturri's argument, the particle-sign breaks down traditional dichotomies between matter and meaning, human and non-human, and culture and nature through drawing attention to the mutual and reciprocal motion between the stratifying forces of signification and the destratifying material flows that operate in the work. Thus, Konturri suggests, that particle-sign encourages a shift in the practices employed in art history and an acknowledgement that the molecular actions of the matter are an essential part of the creative process.

In "Metaphors of the mind: art forms as modes of thinking and ways of being", Danielle Boutet extends insights that emerged through her interdisciplinary studio experience to argue for creative practice as an epistemic modality. She argues that studio practice is "not the kind of thinking where one finds *answers* to *questions*, but rather where one *contemplates* and *experiences* situations, themes or feeling complexes." Radically, Boutet argues that traditional art forms operate according to a generative grammar that draws on deeper "root forms", which she proposes may constitute universal structures. Drawing on intermedia and interdisciplinary art practices, Boutet argues that these generative principles correspond to ways of being and modes of thinking and that creations of a mind are also metaphors of the mind. Thus universal root forms in the arts actually epitomize *ways of being*, modes of thinking, as well as image formation principles.

Jondi Keane's chapter, "Æffect: initiating heuristic life" re-examines the relationship between affect and effect in formulating an art of living. For Keane, "affect and effect are systems of relations that underpin an experiential orientation through which the organism-person-environment is able to initiate change and hold the

world in place.” He terms this coordinated system of cognition, *Affect*. Though a discussion of works from Arakawa and Gins, Keane sets out to show how the functioning of affective and effective systems in the body allows, and even requires, paradoxical logics to coexist. He suggests that Arakawa and Gins’ employment of these paradoxical systems demonstrate that embodied cognition undoes the boundaries separating the organism from person and the person from environment and moves us toward what Arakawa and Gins term, “atmospheric intricateness”. Keane suggests that “atmospheric intricateness” may become the blocs of a new materialism for heuristic life.

Through “Materiality, affect, and the aesthetic image”, Estelle Barrett examines how affect and aesthetic processes generate images that are implicated in the production of new knowledge. She proposes that in aesthetic experiences, the materiality of images produces a body blow that triggers a production of new and emergent knowledge. According to Barrett material process, affect and feeling are crucial elements that constitute an “embodied knowing”. This knowing exceeds established codes. Drawing on the thought of Julia Kristeva, John Dewey and Antonio Damasio, Barrett demonstrates how discovery is predicated on material and sensory processes. She argues that knowledge is derived from the inextricable and necessary relationship between biological processes, affect and language. Through experience-in-practice, this interrelationship constitutes what she terms the *aesthetic image*. For Barrett, the aesthetic image is not the work itself, but a particular structuring of compositional or formal elements that allows the work to reveal unfamiliar or mutant forms of knowledge.

The Matter of Film

The existing language of film studies is frequently focused on predetermined questions surrounding representation and signification. As a consequence they are all too often unable to account for the contemporary spectator’s encounter with the underlying materiality of film. In their chapter, “The matter of film: *Decasia* and *Lyrical Nitrate*”, Nicholas Chare and Liz Watkins aim to redress this balance. Through the analysis of two films, *Decasia: the State of Decay* (Bill Morrison 2002) and *Lyrical Nitrate* Peter Delpout, 1990), they seek to account for the encounter with the materiality of film. *Decasia* and *Lyrical Nitrate* were both created from found film footage. Chare and Watkins propose that through the deterioration of the film stock, what lay concealed beneath the images that appeared on screen in early cinema, emerges and reshapes our encounter with it. Using a Kristevan analytical framework, they argue that *Decasia* disrupts the symbolic register of language to reveal its drive-invested semiotic underside. In their analysis of *Lyrical Nitrate* they demonstrate how the film traces a schism between the indexical qualities of the film and its narrative imperatives. Though the analyses of the rhythms generated by the shifting surfaces of the film-stock used in *Decasia* and of the vibrant color fields created by tinting and toning in *Lyrical Nitrate*, they demonstrate how Kristeva’s ideas about language enable the beholder to give expression to an encounter with the matter of film. Through this careful analysis, this chapter breaks free from modes of interpretation that focus

on the aesthetics of the cinematic image and on narrative form; instead it attempts to engage with and celebrate the excitations and intensities of the drive-invested underside of representation that comprises the matter of films.

Dirk de Bruyn's chapter, "Recovering the hidden through found-footage films", complements Chare and Watkin's writing through its focus on found-footage films. However, whilst Chare and Watkins celebrate the excitations and intensities of the drive-invested underside of representation, de Bruyn eschews psychoanalysis for a "direct materialism". Materialist cinema, with its focus the non-narrative and implicit perceptual processes, has often been criticized for its failure to address content and hence position itself politically and ideologically. In his essay, de Bruyn returns to this materialist cinema, arguing that the emergence of trauma theory has created a new situation that opens up alternative readings of materialist cinema. Working with the found footage films, *Alone Life Wastes Andy Hardy* (Arnold 1998) and *Dreamwork* (Tscherkassky 2001), de Bruyn suggests that such materialist films offer an understanding of trauma cinema. He examines each film in relation to trauma and, by placing these films in a historical and cultural context, demonstrates that the re-editing and layering shots and parts of imagery from original film stock creates a complexity and intensity that is affectively powerful and works performatively to actually perform trauma "directly". In linking materialist film to trauma, de Bruyn draws on Chris Brewin's neurological research into trauma, which identifies the relationship between parallel memory systems, one implicit and the other explicit, to demonstrate that materialist films create situation where memory is re-experienced in the present rather than recalled. Against the utopianism view of materialist film, de Bruyn argues that materialist film can re-perform the trauma of overwhelming experience for the viewer both historically and experientially.

Ilona Hongisto's chapter concludes the section on the matter of film. Her interest is in the "materialities of experience" in documentary video. In "Moments of affection: Jayce Salloum's *everything and nothing* and the thresholds of testimonial video", Hongisto focuses on the transitional moments between the interviewer and interviewee in Jayce Salloum's testimonial video *everything and nothing* in order to draw out a distinction between the discursive testimonial moment and moments of affection. Salloum's video revolves around an interview with the Lebanese resistance fighter Soha Bechara, a video made shortly after Bechara's release from the notorious El-Khiam detention center in December 1999. Through a close analysis of transitional and interstitial moments in the video, Hongisto demonstrates how the materialities of experience disrupt the expectations of the testimonial genre and strengthens our connection "to a reality that is deeper than its actual forms". She maps the moments of affection that emerge in the video arguing that these moments of affection come to replace the recognizable testimonial moment with a material experience of affection. Hongisto argues that this material experience is unique to video and to installation.

Carnal Insistence

Any essay on authorship raises the specter of Michel Foucault's famous essay: "What is an author?" (1984). His essay tends to have been interpreted as presenting the *prima*

facie textbook example to support the social construction of authorship, despite his material discursive understanding of discourse. In her chapter, “In the name of the author: toward a materialist understanding of literary authorship”, Kaisa Kurikka calls for a new notion of authorship, one that overcomes the binary system of either placing the author inside or totally outside the text. Drawing on Deleuze and Guattari’s concept of the “conceptual persona”, Kurikka proposes a materialist understanding of literary authorship that enables authorship to escape imprisonment in the “theater of representation”. Her materially insistent reading of the polyonymous authorship of the controversial Finnish author, Algot Untola (1868–1918), allows her to focus on the materialization of authorship. Here, Kurikka concentrates on matter, functions and effects instead of (transcendental) ontological aspirations of defining authorial existence.

The performative power of practice is at the core of Liza McCosh’s PhD extract, “The sublime: process and mediation”. In this extract McCosh proposes the concept of a “material sublime”, which she suggests, has relevance for understanding the artist’s creative process. Through a discussion of the inter-relation of nature, body and material practices, she argues for a co-emergence, whereby “the sublime experience traverses a liminal space through the negotiation between mind and matter.” She argues that artistic practice can never be fully known in advance, but operates at the borders of a subject’s pre-existing knowledge. Drawing on her own practice, McCosh proposes that the material sublime, as realized through creative practice, challenges binary opposition and collapses distinction between landscape, bodyscape and mindscape.

In the arts, carnality could be seen to be most insistent in the arts in relation to fashion design. Fashion, after all, is designed to be worn by the body. Paradoxically, however, as Llewellyn Negrin points out, despite the inextricable association of fashion with the body, the fashion industry in Western culture has disavowed this relationship. In her chapter, “Fashion as an embodied art form”, Negrin charts the history of Western fashion to demonstrate that fashion design has been characterized by designed forms more directly related to images of the ideal body than they are to actual bodies. In her critique of the fashion industry, Negrin argues that a more adequate approach to dress and fashion is one, which acknowledges the materiality of the body rather than seeking to negate it. She argues that instead of upholding as the ideal, clothes that transcend the body, we need to envisage new modes of dress that engender a sense of the body, as an active corporeal presence rather than as an object of spectacle. Her concern with a kinaesthetic awareness of real bodies rather than an image of the ideal body is exemplified through reference to the fashion designs of Issey Miyake.

Materiality, the Virtual and the Real

Cathryn Vasseleu’s “Material-character animation: experiments in life-like translucency” is concerned with digital light and raises questions concerning our preconceptions about the appearance of things through addressing simulation in appearance modeling. Whilst appearance modeling is normally associated with the digital

imaging, Vasseleu contrasts the material-character animation simulations of Henrik Wann Jensen, with light simulations in Rachel Whiteread's liquid resin installation *Water Tower* (1998). Vasseleu argues that, whilst Jensen's work gives the "appearance" of light interacting with matter, it is Whiteread's material practice that exceeds the appearance of things. She suggests that Whiteread's *Water Tower* produces a knowledge that eludes perception and thought. In developing her position, Vasseleu proposes that appearance modeling looks to material practice in order to exceed representation rather than merely simulating reality; the implication is that Whiteread's work enacts the aesthetics of translucency. She concludes that this strategy might include recasting the flow of virtual light to show that the material character of the world it models is returned to us in strangely altered forms.

Rose Woodcock's "Instrumental vision", also draws on material perceptual practices to open up the world of simulation. In her analysis of stereo-immersive virtual reality (VR) technology, Woodcock challenges the claim by VR advocates that stereo-immersive virtual reality (VR) offers a new world experience for an immersant. Woodcock argues that in their attempt to mimic the realness of ordinary perceptual reality, VR designers have created a perceptual redundancy. In simulated environments, she claims, we are returned to the known and the familiar rather than being offered new perceptions or new ways of perceiving. In order to open up the field and move beyond the "perceptual redundancy" of stereo-immersive VR, she argues that we need to turn to the visuality employed in VR, rather than focus on its purported "virtuality". Woodcock returns to the material experience of vision, and in particular the stereo-acuity of binocular vision on which VR relies. She proposes that by employing theories of pictorial perception, such as ambiguity, which are found in more traditional visual arts, we can open up the possibilities for rethinking the productive and creative practice of vision in stereo-immersive VR.

A caution for the new materialism relates to the "loss" of cultural and ideological positioning. In "Immaterial Land", Brian Martin addresses the question of the political economy of the new materialism through investigating the relationship between art, ontology and ideology in Indigenous Australian culture. His aim is to articulate a different premise *for* and understanding *of* cultural ideology. The important concept in his alternative framework is "land". Martin begins his chapter with the traditional acknowledgment to country. This act of acknowledgement introduces the key theme of the chapter: for Indigenous Australians, cultural ideology is premised on "country". The fact that people belong to country rather than country belonging to people forms the basis for Martin's argument that the immateriality of cultural ideology is constituted materially through this relation with the land. Against Althusser, he argues that cultural ideology is created through the real material conditions of existence and not by imaginary relationship with them. Here ideology is grounded. The spiritual or immaterial nature of Aboriginal culture is a continual practice created by the reality of land. For Martin, "cultural ideology' is the 'real' world."

In the final chapter, Jussi Parikka and Milla Tiainen join with Martin in addressing the issue of the political in the new materialism. They do not believe in a naïve celebration of creative practice, but rather are concerned with engaging with the

problem of a changing biopolitic of life, that art can explore. Their chapter, “The primacy of movement: variation, intermediality and biopolitics in Tero Saarinen’s *Hunt*”, engages with the Finnish dancer-choreographer Tero Saarinen’s dance performance *Hunt* in order to address with the problem of changing biopolitics in dance. Parikka and Tiainen argue for a radical empiricist ontology of movement and of the primacy of relations in grasping the body as an *event*. They argue that in *Hunt*, the original thematics of bodily capture and sacrifice can as such, be connected to the “modern biopolitical concerns revolving round the labouring and productive body”. Here, connections can be readily made to Marxist notions of capitalist production. However, Parikka and Tiainen, take *Hunt* on a different trajectory. For them the new materialism is a political economy where the intensities of movement and “creative” embodiment are closely aligned with immaterial labour, the creative economy and innovation culture’. In their analysis of *Hunt*, Parikka and Tiainen suggest that there is an open dynamism where dance as event is constantly refashioned in contemporary multi-media and informational environments.

Carnal Knowledge is a book that sits between a posthumanist neo-materialism and a corporeal materialism. It engages with the “matter” of things and at the same time acknowledges the material groundedness of cultural practices. This project, which began as a quiet conversation between a group of Australian and Finnish scholars, has grown into an interdisciplinary exploration of what has come to be known as the “new materialism”. Embracing artists and scholars across the world, it attempts to demonstrate the significance of the arts for articulating the epistemological underpinnings and significance of the “new materialism” in contemporary cultural contexts.

FROM DOUBLE NAVEL TO PARTICLE-SIGN: TOWARD THE A-SIGNIFYING WORK OF PAINTING

Katve-Kaisa Kontturi

Art engenders becomings, not imaginative becomings ... but material becomings ... in which life folds over itself to embrace its contact with materiality, in which each exchanges some elements or particles with the other to become more and other. (Grosz 2008: 23)

The starting point for this chapter is a curious double navel that marks a female figure's stomach in a painting by the Madrid-born, Finland-based artist Susana Nevado. But fascinatingly indeed, the emergence of the double navel cannot be attributed to the workings of the artist alone – this is what my field data gathered during the working process clearly reveals.¹ The double navel, or navels altogether, do not appear in any of the discussions I had with Nevado concerning the making of the piece recorded during a six-month period of observation of the process. Nor does the theme figure prominently in the photographs I took to document the process. The obscure navel features only in the very last pictures. Besides, Nevado usually studies her “subject in the making” extensively both visually and through literature, and this is what she did this time too, only her subject was not navels. Rather, it was concerned with creating something new at the crossroads of Catholic imagery of “incorporeal” holy women, scraps from the 1950s women's magazines and photographs of contemporary female bodies. And what is more, there simply does not exist a history for double navel as a signifier had she been looking for one; no signifying network, which would occupy various domains from painting to literature and so forth as it is the case with many iconographical signs we can study from handbooks. Thus, it becomes obvious that the creative process that gave birth to the double navel was not planned nor intended: the artist did not master the process. What the following study focuses on, then, is not the semiotic history of the double navel, its

meaning in the common sense of the word, but its ontological heterogenesis, and the autopoiesis of the process of art-making (Guattari 1995).

Regardless of how fascinating a pictorial puzzle the double navel might offer, it is not something this chapter seeks to solve. Here, the double navel serves as an entrance, not to the world of iconography, to the regime of representation, but to the *work* of painting in which matter acquires expressive qualities; in which matter is an active and indispensable participant that scrambles and goes beyond the conventional binaries of painting. The action I am interested in calls for new conceptualizations for in it, no image, no sign stands self-sufficiently, independently of “actual textures” of painting that are “always stamped with a mark of singularity” (Guattari 1995: 38). Thus beyond unifying and generalizing tendencies that uproot signs from their emergence in and through materials, this chapter hopes to attain the complexity of an art process in which binaries of form and matter, content and expression, and also human and non-human collapse continuously. Recalling this, I proclaim a kind of knowledge production that does not overpower or master singularities of a material process with socially and institutionally established discourses or approved images of the semiotic canon. Instead it attends to their transformative co-emergence.

This is what the concept of “particle-sign” in my title stands for; it refers to continuous interminglings of matter and meaning in the encounters of bodies, both human and non-human. Put differently, it stands for radical immanence (Braidotti 2002), which contests both the power of “transcendent signifiers” and reducibility of all being into “a univocal ontological foundation” (Guattari 1995: 52). Whereas the concept and its references (Deleuze & Guattari 1987; Zepke 2005) will be pursued in more detail below, suffice it to say for now that particle-sign alludes to corporeal, material existence and emergence of a sign, and also to a kind of embodied “knowledge production” in which signs can have a real and direct effect on bodies. The following will take us toward this understanding by both practical and theoretical means. So, in one sense, the double navel as particle-sign is both the starting point and the outcome of this chapter. Yet to fully understand what it evokes and entails – what kind of event it sets up – it is necessary to grasp the action of art-making, to assemble a cartography of the process of creation.²

However, the following does not tell a retrospective story of how this one piece of Susana Nevado’s painting installation *Honest Fortune Teller* (2005) was made; a detailed list of chronological events that led to the emergence of double navel (as particle-sign) is not offered. Rather, the chapter pursues a conceptual and analytical experimentation in which both the concepts and the work of art call for a new kind of art historical praxis that crystallizes itself in a new materialist ontology of art.³

In terms of –isms, this chapter will work *from* a materialist reading of power relations based on Marxist materialism so relevant to the development of feminist art historical analysis *to* a new feminist materialism which emphasizes the transformative force of matter and embraces connections between different modalities of expression such as the symbolic and the *material*. My collaborators in this move are Deleuze–Guattarian. What Gilles Deleuze and Félix Guattari criticize with their ma-

terial-affective approach to art is “the dominant method of ‘political reading’” (Lambert 2007: 12). They disapprove of the ideological critique that makes art a vehicle for certain ideologies and at the same time misses the artwork’s own political potential; potential that is tied to the (form of) expression (Manning 2009; Massumi 2007), or more inclusively to aesthetics (O’Sullivan 2006: 7). Elizabeth Grosz (2007: 253) clarifies this Deleuze–Guattarian stance: “Political art is art that produces new artistic forms and norms.” Although, in Grosz’s view the politics art expresses should be understood in relation to art’s history, to previous works, this does not presuppose that art would not have any influence outside its own realm. Actually, Grosz (2007: 256) daringly insists that it is art that can give us a new world, a new body. But this is possible only if we do not disregard “its ontological textures” (Guattari 1995: 41), and see art connecting to bodies directly, and not just through representations.

While ideological critique might be indispensable in terms of feminist empowerment, in making space for women in various institutions such as art collections, exhibitions, classes, and in creating more apt, more flexible representations, new materialist politics is devoted to the transformative actions of matter flowing through all these power structures. However, to enter the transformative flows of art-making does not necessarily indicate a consequent dumping of issues concerning gender representativeness. It is important to note that I evolve my argument with Susana Nevado’s art, for too rarely has Deleuzian thinking been elaborated in connection with a woman artist’s work.⁴ Instead, there has been a tendency – which could be labeled as oedipal even – to obediently revisit over and over again the work of (male) artists such as Francis Bacon, Paul Klee and Jackson Pollock considered by Deleuze himself.

To engage with Susana Nevado’s work of painting, I suggest a method of encountering rather than a conventional art historical act of interpreting. For one, interpretation as a method maintains the form/content divide this chapter seeks to surpass: put bluntly, in interpretation the work does not speak itself, it is something else that speaks through it (Lambert 2007: 14). In Simon O’Sullivan’s (2006) words, interpretation is a non-encounter. It does not allow the work of art to work but diminishes it to an object of recognition working within the realm of representation. To draw attention to how an art process *works* actively in itself, autonomously in terms of process, nevertheless connected to social reality, let us now turn to the painting through which I shall work out my argument.

Toward the Work of the Work of Art

The double navel painting belongs to an installation that comprises 35 paintings and small scale sculptures of varying size, material and texture. If glanced through, what an (art historically cultivated) eye probably first catches is a repetitive figure of a woman standing her arms wide open. In terms of iconography, this figure is known in Spanish as *María Madre de la Misericordia* (Mother of Mercy) and is widespread especially in Catholic popular prints such as pocket-size holy cards. It is the very tradition of holy cards presenting saintly women that Nevado wanted to re-work; contemporize within her installation.

In this particular painting, a girl in black underwear embodies the figure of the holy mother (see Figure 2 on next page). The painting shows her belly as round, swollen and oddly empty, that is, post-natal. The model of the painting had just given birth – the baby was with her when she posed for Nevado as María Madre de la Misericordia in October 2004. But what stands out the most is that there is a peculiar thing in the middle of her stomach: the girl–mother seems to have not one navel but two. Side by side, but *not* in perfect harmony, and *not* of similar shape, there exist two navels.

Interestingly, it is not only the figure of the girl that seems to be somehow swollen and strangely worn out but also the whole texture of the painting itself. The lead white–yellowish acrylic paint that covers most of the surface of the piece is thin and opaque. There seems to be too much underneath it for it to fully cover: blurred figures, swarming layers of paint and varnish. Clearly, then, the girl was not painted on a blank canvas. It could be even argued that materially, the painting is as pregnant as the girl was just a few weeks earlier.



2. Susana Nevado, a detail of *Honest Fortune Teller* installation in process, June 2005, mixed media, 30.5 x 23 cm

Whereas the recognizable figure or pose turns the painting into an object of recognition, the obscure double navel and the messy cracking surface produce wonder and amazement, both antithetical to recognition, for there is no history with which to contextualize the double navel. Thus, I would claim, the double navel as a part of this cracking surface rather turns the painting into an object of *fundamental encounter*. That is, to cite Deleuze (1994: 139), an object that forces us to think; something that challenges our habitual being in the world. In contrast to the object of recognition, which serves as a vehicle for the already known, the object of fundamental encounter produces something new in itself. I began this chapter with the contention that the

double navel was not Nevado's intention. Now I would like to be more specific and say that the double navel is a surprising expression of *the work of a work of art*.

In her book *Art Beyond Representation*, Barbara Bolt (2004: 5, 111–122) contrasts the concept of *work of art* to that of *artwork*. She claims that whereas artwork is clearly a noun, work of art is rather a verb. Also, artwork refers more straightforwardly to the object quality of art rather than to its processual nature. Bolt writes:

Artworks don't seem to provide us with the same problem as that of the "works of art." We can identify artworks, classify them, interpret them and make evaluations according to criteria established by the discipline of Art History. We can exhibit artworks and study the reception of them. However, does this allow us to get any closer to the "work" of art? (Bolt 2004: 5)

The obvious reply that Bolt would call for is "No, no it doesn't." But what does then? To answer her question, Bolt (2004: 6, 130–131) refers to participant observation as it is exercised in social sciences, for example. Thus, by observing Nevado's processes of art-making and by participating in them via discussions for example, I have come to know that her works of art work in layers. So the figure of the girl as well as her peculiar double navel are inseparable from the layers of paint, varnish, and women's magazine scraps from the 1950s involved in the particular process under discussion in this chapter. In other words, the figure of the girl with its double navel emerges in and through those layers.⁵

Powers of Stratification

To tackle the emergence in and through the layers, that is, the *work* of layers, I suggest turning to the concept of *strata*, and also to a related method named *stratoanalysis* as they are introduced in Deleuze and Guattari's *A Thousand Plateaus* (1987: 39–74). While the term "strata" is nowadays widely employed in social sciences to analyze the sediments of class for example, it has its origin in *geology* in which it refers to rock formations taking shape in layers, which always come at least in double. According to Tamsin Lorraine (1999: 113–114), the eclectic array of disciplines including geology and cellular biology, that from which Deleuze and Guattari draw their ideas, allows for thinking in terms of the impersonal and pre-individual singularities out of which human and non-human worlds are constituted. The geological term "strata" is one such concept that they have deterritorialized to give voice to the non-human in humanity, to show how the non-human takes part in the becomings such as art, that are often understood as exclusively or principally human.

For Deleuze and Guattari (1987), there are three major strata that govern the world: the physico-chemical, the organic and the anthropomorphic, of which the first one can be connected to the organization of matter, the second with life and the third with the human (although it is actually never as simple as this). Thus, strata can be understood as that which regulates movement or halts it to an organic and organized whole, if not stops it altogether. No wonder that Deleuze and Guattari describe it as a belt! Identification is a form of stratification as are routines of every-

day life and art. When something is stratified, it becomes common and shareable, easy to communicate. Consequently, what was above called “artwork” – art as object of recognition and interpretation – is art that is stratified. The work of art, however, escapes from becoming stratified.

In Deleuze-Guattarian vocabulary stratoanalysis is a parallel to perhaps more renowned and also more widely applied practices of rhizomatics, schizoanalysis, nomadology and micro-politics (Deleuze & Guattari 1987: 43). These are all toolboxes for grasping the late capitalist society with all its singularities and generalities in terms of movement and positive difference. Thus, stratoanalysis does not look only for organized, stabilized nor petrified strata be it layers of society, signification or something else. Rather, Deleuze and Guattari explore strata in order to escape them (Goodchild 1996: 156). Yet this might evoke an overly simplified image for stratification and destratification always come together.

In fact, no strata should be understood as inherently negative. Humans need strata, the world needs strata – art history needs strata otherwise it would collapse into a state of incomprehensible chaos. To put it in more positive terms: “Stratification is like the creation of the world from chaos, a continual renewed creation” (Deleuze & Guattari 1987: 502).



3. Susana Nevado, a detail of *Honest Fortune Teller* installation in process, June 2005, mixed media, 30.5 x 23 cm

Let us turn to the process in which the double navel girl emerged in terms of strata. The process in itself lasted months, and this was mostly because Nevado was very discontented with the earlier layer or phase of the painting, now all but fully hidden under the present figure of the girl (Figure 3). It was because of this layer (that consisted of many other layers in itself) that the painting spent long times “in

arrest", out of sight, separated from other paintings all alone on the kitchen wall of Nevado's studio. As Nevado explained, she felt that at that earlier phase the painting did not breathe; it did not have a life of its own. To paraphrase Deleuze (2003), it was all "given" and nothing new.

At that time, the painting was strongly governed by the anthropomorphical strata; by the significance of images or more discernible scraps of pinup girls and a painted faceless woman all in their underwear. The pinup girl of the scrap in the lower part of the painting was laughing, her mouth open; another one was more bashful resting her arm close to her face as if to secure her bearings. Both figures posed arms above the chest line exposing their curvaceous upper bodies tightly controlled by white corsets. These fixed figures were obvious and perhaps all too recognizable as they are part of popular visual culture nowadays infinitely repeated and reiterated in advertising. Also, strong contrasts of light and dark offered an easy route to interpret the painting in terms of standards of "good" and "bad". A diagonal stretch of dark paint crossed the canvas, and another one to the opposite direction made from black shred of lace cut from cheap underwear. These divided the painting in separate segments: while the pinup girls with their "Pepsodent" smiles appeared on the sunny side, in the middle of the painting where the double navel girl now poses, was a painted figure of a faceless woman as if shadowed by the scraps.

In terms of signification, the whole painting was like a worn Marxist feminist maxim, visually declaring that advertising oppresses women by placing "real women" in the shadow and turning them faceless, while celebrating pin up girls eager to please the consumer. As mentioned, it was all given – common and already known – and there was nothing new, nothing surprising. That is, the painting at that stage was stratified. Hence something was needed to *destratify* the powers of representation and the related possibilities for critical feminist interpretation, so as to push the work toward something "new".

Forces of Destratification

When I visited Susana Nevado's studio on a Sunday afternoon on 20 March 2005, Nevado had, so to speak, stood against "the powers of stratification". I listened surprised as she announced quite dramatically:

See, I've rubbed this one [the painting] completely! This is the one I've been struggling with ... Today I came here and sandpapered it from top to bottom. It has annoyed me so much all the time. (Nevado 2005)

Nevado had rubbed the painting thoroughly with coarse sandpaper so that the figures had lost their obvious forms and referentiality; all that was fixed and easy to recognize was now gone. In other words the destroyed *artwork*, the object of recognition gave way to the *work of art*. The action of rubbing resulted in breaking the surface of the scraps, paint and varnish, and thus in making them more porous, more amenable, more open to new connections.

Before the scrubbing, it was images and their significance on the anthropomorphical strata that governed the painting. This coincides with Deleuze and Guattari's (1983 & 1987; Guattari 1995) claim that Western capitalist culture is based on the imperialism of language and, above all, on the imperialism of the *signifier*. What separates the anthropomorphical strata from other strata is its tendency to govern the other strata, to extend its own laws and legacies upon everything else (Deleuze & Guattari 1987: 62–63). It behaves like a despot. And in this sense it is a twofold despot. As Stephen Zepke puts it, “[s]ignificance implies the autonomy of meaning from materiality in a seemingly free circulation of signifiers, but this freedom hides another despot, that of the subject” (Zepke 2005: 120). This is so, for the signified–signifier binary always requires an individual subject that wants to express something.

Therefore, Nevado's act of rubbing was a way to give up the “givens” in two terms. It worked toward getting rid of the easily recognizable signs such as pin-up girl figures from the 1950s.⁶ Secondly, rubbing was used to get rid of paint – as a deconstructive, textual activity that plays games with signs thus mainly working within the anthropomorphical strata of “significance”.⁷

The smooth surface or space that resulted from the rubbing connected the paint and the paper scrap figures so intrinsically that they were no longer separate layers. In the heat of rubbing an *interstrata* was born. What happened was that now, the physico-chemical strata of the painting – the strata often seen as *substrata*, as a kind of raw material or ground for meanings to emerge – connected directly and reciprocally to the anthropomorphical strata, to the semiotics and significance of the painting. In this newborn assemblage “a semiotic fragment rubs shoulders with a chemical interaction ...” (Deleuze & Guattari 1987: 69). In other words, the “*semiotic* components are inseparable from *material* components and are in exceptionally close contact with molecular levels” (Deleuze & Guattari 1987: 334).

From Double Articulation to A-Signifying Semiotics

And where the material and the semiotic connect, and are indeed inseparable, content and form are not in a straightforward hierarchical conjunction either. The overcoming of this duality of content and form is part and parcel of what Deleuze and Guattari (1987) name *double articulation*. According to them, double articulations form strata. And as they often do, Deleuze and Guattari first offer an easy explanation of double articulation by separating the first and the second articulation but then add that beyond this obvious relation there are always more complex connections. Nevertheless, they contend that whereas the first articulation has to do with molecular flows and their ordering, the second offers a more systematic hold and a functional structure, an overcoding, hierarchization and totalization for example (Deleuze & Guattari 1987: 40–43).

It would be tempting to claim that the first articulation is about contents and molecular movement and the second about forms and molarities. This is where the representational thinking surely draws us. Deleuze and Guattari's trick is, however, to avoid getting stuck into this duality inherent in representational thinking by arguing that strata are not simply constituted of forms and contents, of certain styles

and images or figures, for example. Rather, the strata constitutes of content (first articulation) and expression (second articulation). But this is only half of the trick. Namely, Deleuze and Guattari claim further that there are not only forms of expression and substances of content but also forms of content and substances of expression. In Brian Massumi's (1992: 152 n. 36) words, "both content and expression are substance-form complexes". The paint made to form certain figures, for example, is *substance of content* that is overpowered by style of painting – a *form of expression*. Yet, the paint has its *form of content* too: it has a certain chosen order, it is a certain, known to be functional mix of water and minerals, oil paint, for example.

But the paint can also be *matter* – paint has its own chemico-physical *potentialities*; it is also a bundle of *indeterminate affects*. Notably, this "active" understanding of matter resides only *outside* of the double-pinchers of articulation. Thus in Deleuze and Guattari's vocabulary, matter appears as something that is not chosen, tamed, stratified. Guattari's elaborated version of the above-described reversible quadripartite model welcomes matter along, which for its part makes it possible to fashion semiotics beyond signification: a-signifying semiotics. In Guattari's model, matter and form connect directly; they do not need the mediation of formed, overpowered matter, that is, of substance. This, of course, entails a different understanding of both matter and form, an understanding that stresses their dynamism: matter as material flows and intensities and form as non-stable "diagram of a process of becoming", or rather as function (Massumi 1992: 14; Massumi 2007: 72). A-signifying semiotics operates by directly and *non-communicatively* "transmitting ideas, functions, intensities, or sensations with no need to signify any meaning" (Watson 2008: 8). In a-signifying semiotics, "form interacts directly on matter", in other words, there exists "a reciprocal relation between material fluxes and the semiotic machine" (Watson 2008).⁸ It is this that we will now turn in order to arrive at the most intensive, and also the most practical stages of the emergence of the double navel as particle-sign.

Double Navel as Particle-Sign

When Nevado began to paint a new figure on the smooth, destratified surface, the emerging figure connected directly to the earlier layers of the painting, to the freed flow of meaning-matter particles. This movement had its autonomy. At a certain phase, certain material layers reacted to each other, and also rejected each other, and a particle of pink paint from Nevado's brush stroke intended to compose the girl's skin did not fasten well enough but fell away, and consequently the girl acquired a second navel, a double navel. Without the field data presenting various stages of the painting process, this understanding would have been hard to acquire. Yet my material does not offer any straightforward answers either, rather different field materials direct attention toward different ends.

When I look at a digital reproduction of the painting, and this is what art historians often work with, the navel appears as a clear signifier, as a challenging detail, maybe a disruption – double as it is and not a regular navel – in the semiotic order of the painting. This calls for tracing its meaning. Elisabeth Bronfen (1998: 3) explains the all-embracing meaning the navel has acquired: "In the cultural repertory of Western

imagery, the navel is the firmly privileged representative for the origin of human existence.” Tracing the navel through the stories of bible and Freudian psychoanalysis, she concludes that in the common sense understanding the navel signifies lack and forever lost connections with the mother and the immanent pre-symbolical world of the womb. Would the double navel then mean a double lack?⁶

However, in the physical presence of Nevado’s painting, my perception is different, and the above-described tracing of meanings loses its purpose. The double navel does not stick out as such. Instead, the double navel is inextricable from the girl’s figure as well as from the whole painting pregnant with lavish, swarming layers of material action. For example, the pin-up girl’s once so seductive “Pepsodent” smile has transformed into a grimace that now gnaws the double navel girl’s pink pelvis flesh. Everywhere the pink painterly skin fuses with the blurred paper scrap figures, with the dense and intense layers of acrylic paint and varnish. As implied earlier, there are no self-standing layers, no independent representations or signs any more but an autopoietic assemblage – an assemblage that creates itself in its own movement. In this assemblage, the double navel is not so much a symbol with a general(izable) meaning as it is a singular expression of destratified matter-meaning flows, of the a-signifying work of a work of art.

Put differently, the double navel is a *particle-sign*. Deleuze and Guattari (1987: 142, 145, 224) employ this two-part term in plural (particles-signs) to emphasize its non-unitary be(com)ing. While not wanting to belittle this aspect, I will stick with the singular so as to grant the term more conceptual status, and also to make the (positive) difference between (semiotic) sign and particle-sign more apparent. In it, “particle” stands for material, molecular movement, to the destratified content and “sign”, for sign-expression, for meaning, but importantly, not in any common sense meaning of the meaning. Sign in particle-sign does not point toward a signifier, or toward representation. Rather, it has an ontological status, it connects directly to material qualities, to the matter in movement. Particle-sign, then, is not something that would dwell on the surface of an artwork as a separate, independent sign; rather it is integral part of material becoming of that work of art. Therefore, sign is dependent on particles, which for their part, are not eternal in their constellation but are subject to change.

Stephen Zepke’s definition of the concept is worth citing here. For him, particle-sign expresses:

[D]estratification, a radical break with, or a line of flight from the strata that introduces something new. ... [It] will appear in painting both as its destratification, as what escapes the stratifying articulations of content and expression, and as a new reality they construct. No creation without destruction. (Zepke 2005: 122–124)

And this is what has been at stake in the cartography of creation that my chapter has sketched: “No creation without destruction.” But whereas in Deleuze’s (2003) understanding, art emerges only when the “given” has been destroyed or overcome,

what my encounter with the work of the work of art suggests, is that both “givens” and their destruction are essential to creation. There would be no double-navel without the givens.

As Zepke claims, the particle-sign is not only an expression of destratification; it is a creation, a construction of a new reality as well. What is then this new reality, or in Elizabeth Grosz’s terms, new world, new body that the double-navel as particle-sign offers?

Against the lack and the lost connections that the single navel signifies, the double navel as particle-sign offers an entrance to the abundance of material movement: to the world of *radical immanence* where images connect with human bodies directly without the mediating work of representations – as suggested by that pelvis-flesh biting pin-up girl, who does not reside inside or outside the double-navel girl’s body but within it, as immanent to it. In the same manner, the pose, the figure of María Madre de la Misericordia is incorporated into the double navel girl’s body.⁹

In an assemblage like this, the human and the non-human, the material and the meaningful intermingle directly: paper scraps and other media “representations”, rhythmic brush strokes, acrylic paint, Christian iconography, varnish and the shred of black lace, they all work “on the same level as the real” (Deleuze & Guattari 1987: 141). If Guattari’s *new aesthetic paradigm* is a study of (new) contact zones between bodies and sensibility, and an effort to create aesthetics of how human and non-human “bodies perceive each other in the social field” (Berardi 2008: 32–33), then Susana Nevado’s double navel painting incites us to do so in material terms of painting. What the painting does is to direct us to stop the navel-gazing, that is sign-gazing/interpretation that so often involves making molar and moral judgments. Instead, with its vital layers that intensively extend beyond the double navel itself, it rather calls for attending to the molecular action of art. This kind of (r)evolutionary politics of painting, that is the *new materialist* politics of painting, is underpinned by peculiar materialist dynamics that is neither Hegelian nor Marxist and that epitomizes itself in the work of particle-sign.

METAPHORS OF THE MIND: ART FORMS AS MODES OF THINKING AND WAYS OF BEING

Danielle Boutet

Each tradition is itself an organized habit of vision and of methods of ordering and conveying material. As this habit enters into native temperament and constitution it becomes an essential ingredient of the mind of an artist. (Dewey 1934: 265)

The ideas and intuitions in this chapter come from creative practice: for a few years now, I have written from and about insights that came to me in the context of my studio experience. Beyond the potential value of these insights as such, my larger goal is to explore creative practice as an epistemic modality. With the premise that art is a way of knowing and grounded in a phenomenological approach, I ask: What can I know through art practice? How do I think during the creative process; what noetic paths does my thinking process follow throughout the realization of material form; how are those paths different from deductive reasoning paths? And can this “artistic thinking” be a valid kind of reflection on experience – in the sense that reflection on experience is fundamental to creating knowledge? And if actual knowledge does come out of that process, then what kind of knowledge is that? How does it – or does it – compare to other forms of knowledge?

I am a classically trained music composer, an interdisciplinary artist (with an MFA in Open Media), and a writer: my MA explored the interdisciplinary phenomenon in the arts – especially the interdisciplinary creative process – and my PhD was on “artistic epistemology”. By “artistic epistemology” I mean looking at art practice as a form of knowledge and what kind of knowledge that is, compared to scientific, philosophical or spiritual knowledge. I am a university professor and I see being an artist working in the university as a consequential position: when the art world can be viewed as a site of cultural construction, then the university is a site of knowledge construction. So for someone interested in art practice as a form of knowledge, it makes sense to work in the university, sharing knowledge primarily, before sharing

artworks. Like scientists seeking knowledge in university laboratories, and philosophers in university libraries, artist researchers may seek knowledge in university studios (while other artists or scientists work in the cultural or technological industry). In a way, this reminds of the ancient division between monastic and secular life; and indeed, I have been more interested in the revelatory potential of art making than by its expressive potential, making art as a form of contemplative and spiritual discipline. I spend my time in the studio more like one spends time in a dojo or a laboratory: to practice and learn, more than to express, to share knowledge, more than an aesthetic production.

When I talk about art as a form of knowledge, I am not talking only about self-knowledge, as when one uses art modalities as *active imagination* techniques. I mean, rather, a kind of knowledge epistemologically comparable to gnostic *knowledge* as in hermetic traditions. Indeed, I have been led to compare creative practice to the practice of alchemy: interrogating materiality, reality and spirituality through formal and physical operations. Both the alchemist and the artist think through matter, by way of aesthetic/symbolic operations. This is not a thinking process where one finds *answers* to *questions*, but rather where one *contemplates* and *experiences* situations, themes or feeling complexes (or ways of being). In that sense, I invite readers to hear me not as a scholar, but rather as an heir to Renaissance philosophy, for my alchemical discipline of thinking through practice somewhat makes me state things in ways that might appear closer to ancient forms of thinking: integral, metaphorical, using duality and synthesis.

Bateson's Question

My premise is that art making is a way of knowing the world, different from science in the sense that we don't study the same things or the same side of things, or at the same level (and of course with no comparable methods): where science studies the facts of nature, art studies its meaning. Science and art should be complementary, not contradictory – and any culture with a genuine quest for knowledge should integrate both. To be more precise at this point, I would go further and say that art is a way of *becoming more conscious* of/in the world – and of oneself (Boutet 2008).

To ground my discussion, let me start with a series of remarks by Gregory Bateson. These are from a short text originally intended for Mary Catherine Bateson's book (M. C. Bateson is Gregory and Margaret Mead's daughter), *Our Own Metaphor*, and published posthumously (Bateson 1991). Generally speaking, Bateson wrote little about art, yet his insights on the subject are often compelling: they have helped me more at least, than many philosophical writings on art. Bateson states a perennial question as a general problem: he asked "What can we know?" Bateson looked at it not as a philosopher, but as the transdisciplinary scientist he was, bringing together the integrated perspectives of biology, anthropology, ethology and systems theory:

If epistemology must always come between me and my organic perception of the world, and similarly must always come between me and any understanding of myself; if my epistemology is the organizing principle of all my understand-

ing; then I can never know anything. My machinery and processes of knowing simply constitute one enormous blind spot. (Bateson 1991: 226)

These processes and machinery, Bateson explains, “shape and limit” what we receive from the external world, to a point where it becomes virtually impossible to recognize what belongs to the world and what is our own interpretation or fabrication. Ultimately, we come to realize that all experience is subjective, that our epistemological machinery is “systematically fallible.” But having stated that, Bateson underlines the word “systematically”: if there is anything systematic about our subjective perceptions of reality, then another pathway is possible, other than renouncing forever the possibility of our knowing anything:

There is the interesting possibility that we might attach meaning to the word ‘systematically.’ If the ‘self’ as a perceiver were randomly fallible, then there would be no hope of any knowledge or understanding. But I am (personally) sure that neither perception nor even dream or hallucination contains more than a very small random element – and that random component always only indeterminate within a limited subset of alternative possibilities. (Bateson 1991: 227)

Thinking that our failures to know the objective world do not occur randomly, but systematically, leads us to that other, fascinating, possibility: What if our inner world is “our microcosm; and our microcosm is an appropriate metaphor for the macrocosm?” (Bateson 1991: 227). Bateson sets here a most fundamental question in my view; more fundamental than our “to be or not to be”, or than Leibniz’s: “why is there something rather than nothing?” Before any other question, indeed, shouldn’t we ask, “Can we know anything?” ... “What can we know?”

There are two possibilities, says Bateson: one is that our sensory apparatus limits and prescribes too much of what we are able to see of the world, and so we cannot know the world in any real sense. The other possibility is that our sensory apparatus being the creation of nature, it is a reflection of it – it is “of nature”, to Nature’s own image. Our senses and our mind have evolved from natural processes; we are wired according to those processes. So we can know the world because our perceptions are not hopelessly random, they are in metaphorical correspondence with the world.

Later in the text, as if to anchor that point, Bateson adds: “And now, it begins to look as if the natural history of that oak tree is the natural history of *me* (and you). Or at least as if there is a macrocosmic natural history with which all the little natural histories are so conformable that understanding a little one gives a hint for understanding the big one” (Bateson 1991: 227). In other words, any one of those “little natural histories” would be a metaphor of the larger natural history.

Bateson’s thinking, here, brings to mind the opening phrase of the Emerald Tablet, attributed to the legendary Hermes Trismegistus: “That which is below is as that which is above, and that which is above is as that which is below.” This phrase states one of the founding principles of hermetic knowledge, that is, the essential

correspondence between the natural realm and the higher spiritual planes. Bateson's thinking also resonates with that of the Renaissance philosophers and scientists (e.g. G. Bruno, Von Nettesheim, Paracelsus) who hypothesized the homology or correspondence between the natural world, the heavenly plane of stars and planets, and the spiritual plane of soul and God – all three planes, or “cosms”, being reflected within the human person, as the microcosm containing all three (Copleston 1958). This hermetic principle is at the root of ancient science and knowledge, and of divination systems such as astrology, the I Ching, the Tarot and so forth. It has been dismissed by Cartesian thinking and scientific positivism, but remains as the mode of knowing in poetry, art and alchemy. It also continued through nineteenth century Romanticism – when Schelling, for instance, explicitly establishes the correspondence between Mind and Nature. “Nature should be Mind made visible, Mind the invisible Nature” (Schelling 1989: 42), a phrase reminiscent of an idea often applied to art: that art spiritualizes matter and materializes spirit.

The hermetic correspondence between the macrocosm and the microcosm, and Bateson's own version of it – the little natural history corresponding to the larger natural history – does not mean that our experience of the world IS the world. More logically, it means that the form of our thinking is metaphorically related to the form of the world.

Creative Production as Metaphor of the Author's Mind

At first, when I began studying what I am able to know through art, I was not interested in whether the discoveries and visions occurring in the process of art making were “true” in a scientific or philosophical sense; I was happy with poetic or artistic *relevance*, with a *feeling* of meaningfulness, of some vision being “right” – which are truth in an experiential sense, as opposed to truth in a factual sense. Then Bateson's epistemological considerations made me think that, in fact, there could be some actual significance to the visions and experiences happening through studio activity. If understanding a microcosm could help understanding the macrocosm, then maybe there was something to learn about the world and about the mind in the microcosm of the studio. Reading Bateson (especially his latest works: *Mind and Nature*, *Angels Fear* and *A Sacred Unity*), we understand that in his view, the creations of a mind are metaphors of that mind. The creations of nature are metaphors of nature, because there is “mind” in nature as well. Studying the creations of a mind can inform us on the nature of that mind. And since that mind is a creation of the world (Bateson says Nature), then knowing the mind through its creations can inform us on the world.

This is to be compared to French theater artist Antonin Artaud's observation:

Here where others offer up their works I pretend to nothing more than showing my mind. [...] I recognize myself as much in a letter written to explain the intimate shrinkage of my being and the insane castration of my life, as in an chapter exterior to myself that seems to me like an indifferent pregnancy of my mind. (Artaud and Hirschman 1965: 26)

It is a relatively obvious proposition to say that a given work of art is a metaphor of the artist's mind at the moment of its creation (Saint-Martin 2009). An abstract painting, for instance, could easily be understood as a representation (representation being defined, here, as a materialization of an invisible idea or making it happen in the world in form and matter) of the internal spatiality of the artist's psyche or mind. Likewise, instrumental music can be understood as unfolding an interior landscape – and that landscape would be metaphorically related to the composer's psyche at the moment of creation. Any creation, for that matter – artistic or not – is a reflection, even a metaphor, of its author's mind.

We are also familiar with the notion that one can study one's works to understand oneself. Jungian “active imagination” techniques, used widely in therapy and in self-development, are based on that idea. But art is different from active imagination, and here, I am really talking about “fine art,” music, theater, performance, installation and such – which I look at as epistemological processes. Not only are they methodologies for knowing the self of the artist, but for exploring the world as well (its nature and its meaning), through knowing the correspondence between world and mind. In a revelatory process rather than a conceptual or rational one, we encounter and experience dimensions of the world and dimensions of ourselves; and this is a way of knowing – closer, as I said, to ancient, Hermetic forms of knowledge, closer to spiritual and experiential knowledge, but knowledge nevertheless.

Metaphors and Correspondences: Hidden, Underlying Forms

When we talk about metaphors – creative work as metaphor of the author's mind, or the mind as metaphor of nature, etc. – we are talking about form, more than about content: two things, or systems, are formally similar or, at least, comparable. The form of my work reveals the form of my mind; and when Bateson says one is a microcosm, he means that the form of the human mind reveals a hidden or underlying organizing principle in nature. This means that it is not enough to compare the form of a given painting to the form of a given emotional or experienced state of the painter's mind: one also has to wonder about the form of painting itself, as metaphor of an organizing principle in the human psyche. Here I am going beyond Artaud's statement that works are metaphors of their author's mind by looking at the forms and mediums structuring those works. Those forms – art forms – are common throughout whole series of works – not only mine, but other artists' as well. Many are ubiquitous throughout cultures and times.

Those art *forms* operate as archetypes, or generative templates. They look like deep structures – or “deep forms”, structurally organizing the creation of single works: a traditional form as “an organized habit of vision and of methods of ordering and conveying material” (Dewey 1934: 265). If a single work is a metaphor of an artist's mind at a given moment, classical or traditional art forms are metaphors of human mind in given situations of human life – hence why I am tempted to use the term “archetype”. In that sense, art forms are models of the psyche: they are forms of thinking, ways of being, forms of experience, and forms of the psyche's life.

Let us look at a few examples, starting with the obvious example of photography. I think we can all agree that taking a picture actualizes a specific way of seeing. Broadly, without defining it in any absolute way: one notices something, stops, frames it and keeps it for memory. This allows us to see photography as an archetype or model of that particular way of seeing. Framing a subject and taking a snapshot, to, then, remotely in time contemplating it, is actually a common operation of the mind – the mind does that even without a lens or a camera, as in abstract thinking. So photography as an art form is a metaphor of a certain form of the mind: a certain way of being in the world, or of being present to the world – noticing, paying attention.

Other examples would be weaving and gardening, universal activities often used as potent metaphors. How much of our daily thinking processes are metaphorically related to one or the other of these creative forms? One often uses weaving to speak of community and relations; one uses gardening as metaphor in education, we talk of cultivating our mind, etc. These activities, as forms, actually are models for ways of organizing our mind and our actions.

Puppets and Deity Presences

I remember a period of great relational stress at work, when I was having the hardest time relating to some of my colleagues. During that same time, I was inspired to create a puppet theater play. I had never made a puppet before, but at that moment, the project was compelling. Not knowing any puppet-making technique, I had to contemplate the idea for some time, to see if there was a way for me to actualize the vision without too much technical complication. It is during that suspended time that I was struck by the meaning of puppets in my particular situation. As if they had come to me in a dream, it became obvious that they were symbols of my need to control my relational situation. Understanding this, I was less compelled to go about and learn the art of puppets, yet the project continued to evolve by itself in my mind, as if it had its own notion of where it needed to go. Having let go of the idea of a whole cast of several puppets, I remained haunted by a single one, a character calling himself The Vizier. Still looking at it as if it had come in a dream, I understood that I wanted someone, a wise being, to speak with about my situation. So I made a small mask, then from it, a puppet with an old face and a long black robe.

Throughout that time, the thought of puppets stayed with me and I was able to witness in real time how the symbol actively helped me deal with my situation. I had already hypothesized that this whole idea of puppets was speaking to my need to control my situation. But beyond that psychological consideration, I thought that a puppet was sign that somewhere in my mind, something believed in souls and sprites, gods and goblins, since puppets manifest externally and materially the *mode of being* of imaginary spirits; the whole concept resting on the idea of enchanted matter.

Similar considerations can be made about theater, opera and other forms of embodiment of characters. Each of these are about different types of embodiment, but in some way they seem related to the profound and ancient notions of souls (gods or the dead, or mythic or legendary characters) speaking through human beings, reminding of shamanic possession (psychokinetic control by a spiritual being). My

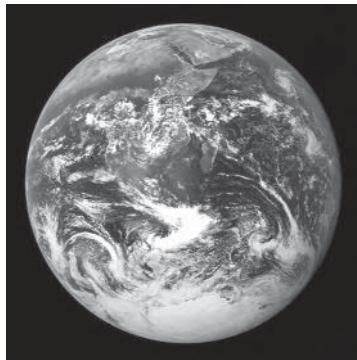
point here is not to define an ontological meaning of theater or puppets, but rather to show that these forms do have meaning, that they actualize certain conceptions (most often unconscious) and are significant in terms of our own ways of being.

The Curve

Let me now go through a detailed example. It is about an artwork, titled *The Curve*, combining two classic forms: mosaic and landscape. As a matter of fact, it is the project that originally inspired the thinking in this chapter. My PhD project was to examine the cognitive processes happening during artistic practice and I made my studio the laboratory for this examination, through a series of five different works involving literature, visual/material arts, and music. Among the series, this particular work was certainly the most influential in the actualization of my reflection.

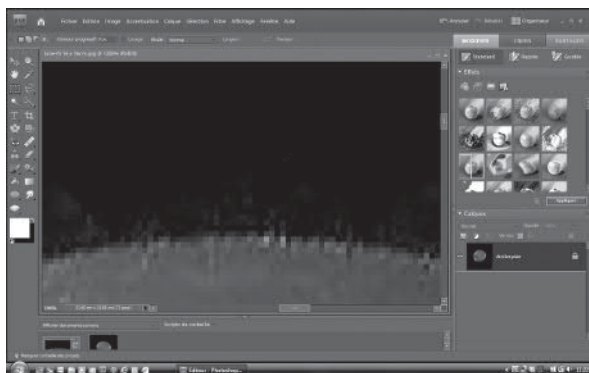
I had been working with the notion of landscape for some time when researching on the Internet serendipitously brought me to the famous photograph, called “the Blue Marble”, of the planet Earth taken by the Apollo 17 crew in 1972¹. Previous works had entertained me in an intense reflection around notions of human consciousness, its origin, its cosmic dimension. In particular, I had been interested in French theologian Pierre Teilhard de Chardin’s ideas, especially his explanations of the concept of *noosphere* (Teilhard de Chardin 1955), a word crafted on the model of the word “biosphere”, to mean the layer of human mind enveloping and transforming the surface of the planet Earth, as biosphere designates the layer of living matter enveloping the surface of the Earth. To me, this picture of the Earth taken by the astronauts was a representation of the *noosphere*: not only the graphic form itself, but the very fact that the image had been taken by human technology and seen by (potentially) all humans around the globe in one defining moment in history, and the visible and invisible effects it thus had on our collective concept of being one humanity living on the planet Earth.

Decisions made in the studio, however, are not always (to be honest, not often) based on reflection and logic. I had a problem with this image: I didn’t like the color



4. *The Blue Marble*, photograph of the earth seen from Apollo 17, 1972

palette. I had been working for some time with a palette of blues and greens which literally fascinated me and in comparison the pale blues, whites and browns of the *Blue Marble* felt subdued and unattractive. So I looked around for an image with nicer colors and found another one, from the Observatoire de Paris: also an image of the planet taken from a spacecraft, but with a lot more vivid green and blue tones on black background.



5. *The Blue Marble*, computer zoom image, print screen

Zooming into it with Photoshop, the pixels of all shades of blue and green looked like an abstract mosaic. Moving the mouse, the curve of the horizon appeared, separating the image between blue-green pixels in the lower part and black sky in the upper part. I was captivated by that “horizon”.



6. Danielle Boutet, *The Curve*, 2007, 8-panel mixed media, 122 x 732 cm

It stayed with me for days, I would return to it constantly on my screen, until I knew what to do with it: a mural polyptych, composed of eight 3'x4' panels, representing the curved horizon using small squares painted with various hues of blues, greens and violets. This mix media work is titled *The Curve*.

The theme in this work is the co-presence – and co/respondence – of macro and micro visions, which generates a series of paradoxes. First, it is a huge enlargement (a few pixels blown to 24 feet) of a microvision – a zooming into a small digital image. Second, it is the largest horizon possible, representing a portion of arc of about 3,600 to 4,000 kilometers (2,500 miles) or one tenth of the planet's circumference, yet I arrived at this image by enlarging 1600 times a small portion of the original 320-pixel image.

The Curve is paradoxical in another way than juxtaposing cosmic and infinitesimal realities. It is a figurative work, and yet it represents a philosophical abstraction. Indeed it does, literally, what abstraction means: walking backward far enough so as to embrace the totality of the horizon (the whole planet Earth) and finding an abstract landscape image – as does generalization in philosophy and science. Yet, at the same time, it is a zooming into the image (moving forward) – as with a microscope. Interestingly, zooming into pixels creates an image of how our brain recreates images from pixels; and here we find the age-old mosaic principle of image making, replayed in digital image making. There is coincidence of mind and world, when the image represents the creation of image by the brain, at the same time as it represents an arc of the Earth's circumference.

In a highly abstracted way, *The Curve* is a landscape painting. What defines landscape as an art form, I think, is the *axis mundi* schema: the horizontal plane of life on earth (the horizon line) and the vertical line of human presence (the presence contemplating/constructing the horizon). The work can be seen as representing the mind (through the mosaic form) and representing a cosmic vision (through its aspect as landscape).

From a *poiesis*, or “process”, point of view, it is interesting to note that at first only form and color guided the production of this work. It was only after the fact that I began thinking about the philosophical content of *The Curve*, and realized that the intuitive selection of aesthetic elements and the combination of two classical forms (mosaic and landscape) had led somewhat naturally to an illustration of the Hermetic principle about the correspondence of the Below and the Above, as well as its counterpart of the correspondence between the inside world of the imagining psyche and the outside world of nature. I had not intended that subject-matter originally, I simply followed a fascination with landscape, with a certain image, and with a palette of blues and greens. The questions I asked during the making process had nothing to do with philosophical or hermetic ideas – I never thought of how to best represent ideas of that kind – but rather, were related to how to actualize that fascination, how to make a strong image of it, i.e. an image that would work formally and artistically.

Thinking Through Form and Matter

The generative process that I just described is unmistakably a thinking process, even if it does not use deductive/inductive paths. It is aesthetic thinking: constructing an image first, based on intuitive criteria of aesthetic coherence, and if necessary, using metaphor to interpret its meaning. (Interpretation is not necessary to experiencing the image – it is only an additional, optional step in our contemplation of an image.) When our mind thinks aesthetically, it thinks in metaphor. It is interesting, actually, that we have two large categories of ways of thinking (deductive/inductive reasoning and metaphor) and that each of them leads to one or the other of Bateson's alternative.

Our creations do not emerge entirely from scratch, but according to certain structures, or traditions, or art forms, operating somewhat like metaphors of our mind – as if they were archetypes, or deep structures in the mind. Those archetypes or deep structures are actualized in the mediums and traditional forms that we use – which act as aesthetic organizing principles. More importantly, those forms correspond to ways, or modes, of being. Forms such as puppetry, theater, impersonation, masks, mosaic (as image forming principle), hologram, landscape, self-portrait, counterpoint and the sonata form, weaving and tapestry, and so forth, each make into material existence attitudes, ways of being and interacting in the world. In fact, they manifest not only our thoughts and emotions, but beyond that, the very structure of our thinking patterns and of our experiencing patterns (our emotions), the form of our ways of being and of being in relation.

Note that I do not say that they “express” our thoughts, emotions or attitude. This verb, “express”, although not entirely erroneous, is misleading. “To express” evokes an idea or image all formed in my mind and which I “represent” on paper or in matter so as to communicate it to others. This is not how it works: usually, it is an informed energy aspiring to become a full-fledged idea or thought or emotion, and the action and materialization makes it happen, calls it into existence. And often (though I suppose, not always) that happens through pre-existing forms, classical/traditional art forms, acting as organizing principles – or through variations and transpositions of those forms.

As exemplified in my puppet example, or when we see the vertical human presence standing before the horizon in a landscape, art forms work as models of modes of being. They are materialized metaphors of how we are in the world; and individual artworks made in those art forms are like different versions in a common category of experience, or ways of being. The main power of those “metaphors of the mind” is their generative power: they act as templates allowing infinite variations. For example, a puppet by itself may manifest the idea/feeling of a god's presence, which is already something, but what is more important is what it allows the puppet to say, to reveal of those invisible presences, and the fact that an infinity of different puppets form an infinitely diverse scene of goblins' actions and experiences. The sonata form, a complex architecture of interacting musical motifs that was prevalent in European music throughout the late Baroque and Classical era, was not only a formative principle able to sustain and carry the immense inspiration of composers such as

Haydn, Mozart and Beethoven, but it also had the power to morph into even more complex forms and grandiose musical works. Nineteenth century Romantic music's complexity owes much to the earlier forms that led to it, complex and fecund forms like the sonata form. As with Jungian archetypes, those forms have inexhaustible generative power, and new ones are created (most often from old ones) throughout history – the examples that come to mind, here, are the hologram, conceptual art, interactive computer art, electronic music (sequencers, synths and samplers). When that happens, it is like a real evolution in culture, almost “as a new character to a species”, to borrow from Bachelard.²

Postscript: A Remark on Truth

One major difference between science and art as complementary epistemologies is how to envision “truth”: science seeks *factual truth*, observable, quantifiable and repeatable, while art and poetry seek to give form and material existence to *experiential truth*: its criteria being relevance and significance. So what is the value of metaphor as a form of knowing/thinking? There is no objective truth with metaphor: we can never say that any one given metaphor is “true” in any objective sense – we can say “We are all puppets”, as a student once wrote me; we may compare life in the world to a chess game, a battlefield, a classroom, a stage (as did Shakespeare), a monastery, or to driving a car. It is not that this or that metaphor is true; it is, rather, that all metaphors together are true. What is true is metaphor itself.

ÆFFECT: INITIATING HEURISTIC LIFE

Jondi Keane

The value of positing a new materialism will be found in the way matter and materiality, newly configured, intensify the world-forming capacities across the human-non-human collective. Increased capacity will be the test of a new materialism's *newness* – literally the degree to which the reconfiguration of ideas and things, thoughts and feelings, intensive qualities and relational structures expand the potential for action.

This chapter explores the ways in which an ongoing practice can be produced that links matter and materiality with the human organism to form feedback loops that can be described as emergent cognitive systems. Clark and Chalmers's (1998) important essay on the theory of extended mind suggests all components that move across the barrier of skull and skin should be recognized as part of cognition. This observation can be taken further by examining the role of art and art processes in the coupling and decoupling of thought and the environment.¹ The dispersal of cognition across material processes allows the proposal of a new materialism to push the extended mind theory toward extended life and vital materiality.² When human cognition can be re-positioned as part of environmental information, it becomes necessary to recognize thought as an event that is perceivable and therefore available to all actants in the human–non-human collective.³

To indicate a newly configured relation of thought to matter, I have coined the term *Æffect*, which correlates the concurrent measures of affect and effect so prized in research (in the arts and sciences) and persistent in experience (of non-linear and causal events). *Æffect* allows the emphasis I have placed upon embodied, situated and distributed cognition's role in the organism–environment relation [see Information Box 1 below] to enter the descriptive language of a new materialism. It is my contention that *Æffect* will bring into focus the blind spot that each mode of measure – formed by research values – has for its *other*.

It is a matter of convenience to assume that a causal chain of events can be followed like the logic of an argument. If for example, we follow a set of propositions: 1. the body shapes the mind; 2. the environment shapes the body; 3. the person shapes the environment – it is possible to derive a relational set of causes. However,

it is more accurate to follow a series of “con-sequences” that are dispersed across the event-space of an organism-person-surround, which impact directly on causal links and indirectly on non-linear conditions of linking. A new materialism would require the inclusion of cognition as a component of the event-fabric in which the intensive affects of the infinitesimally small acts of initiating – in language, perception or action – become shared environmental information. Discrete measures of affect and effect prove inadequate to describe the relationships across domains so crucial to human perception and action.

Information Box 1: embodied, situated and distributed cognition

In the *Cambridge Companion to Situated Cognition* (2009), Smith and Conrey’s essay identifies four principles that interrelate and overlap in the descriptions common to social psychology and situated cognition (Smith and Conrey 2009: 454–66):

1. **Cognition is action and task-oriented**, adaptive and regulatory, resulting in the close connection between, cognition, motivation and action (2009: 456).
2. **Embodied cognition** refers to the notion that the body is the vehicle of action producing perception and action loops and sensorimotor aspects of mental representations. These actions, considered as cognition, are not isolated but extend throughout the nervous system and into its relation with environment (2009: 457–8).
3. **Situated cognition** rejects the ‘picture of autonomous, context-free inner process in favor of a view of an organism as involved in intensive moment-to-moment [embodied] interaction with its environment’ (2009: 248–9).
4. **Distributed cognition** posits that cognition is not only distributed throughout the body and therefore ‘not contained within minds, but implemented by systems that link [embodied] minds with aspects of the physical and social environment’ (2009: 461).

This understanding of cognition as both efferent and afferent (incoming and outgoing) as well as direct and indirect proves useful when a person attempts to perceive and purposefully interact with the components of each instance of cognitive coupling and de-coupling on the fly. A proposal such as extended mind (Clark and Chalmers) takes Radical Empiricism – that experiences themselves are experientiable (James 1967) – as a starting point for understanding how it is possible for humans to join and separate segments of the world by making and then acting upon distinctions.⁴ The value of a new materialism, in my view, will be measured by the degree to which cognition – the stuff of the new materialism – can be put in service of learning to construct extended life and not simply enhance a concept of extended mind.⁵ A new materialism can no longer make assertions about matter, but must become part of the material world by realizing that observing relationships is a way

of generating meaningful consequences. Clark and Chalmers give methodological importance to the power of making distinctions:

Thus in seeing cognition as extended one is not merely making a terminological decision; it makes a significant difference to the methodology of scientific investigation. In effect, explanatory methods that might once have been thought appropriate only for the analysis of ‘inner’ processes, are now being adapted for the study of the outer, and there is promise that our understanding of cognition will become richer for it. (Clark and Chalmers 1998: 12)

In the decade following the publication of Clark and Chalmers’s 1998 essay, there have been notable new accounts of the organism-environment relationship. The enactive theory or *enaction* (Varela, Thompson, Rosch 1991; Thompson 2007; and Stewart, Grahame & Di Paolo 2010) posits that it is through precarious autonomy and materiality that human organisms are able to construct meaningful interaction with their world.⁶ To a practitioner immersed in material processes, this implies that persons are able to couple and decouple purposefully to literally construct the features of the world and their value and meaning. This is how practice leads to an heuristic life through a reformed method of engagement with matter and materiality – Æffect – that leads through the arts back to increased human and non-human capacity [see Information Box 2, below].

The materiality term Æffect acts as a mnemonic device. Instead of situating the mnemonic externally in the materiality of a string on the finger, the letters of an acronym or the rhetorical placement of ideas into an existing space, this mnemonic device is located in the mouth as a bodily mnemonic for remembering an embodied measure. Within the time it takes to decide how the mouth will move from “a” to “e” in the pronunciation of the diphthong Æ (writ large), a moment of hesitation insinuates between conceptual and perceptual processing. This moment is long enough to notice a shift from the difficulty presented in voicing a word-image to the heuristic moment that the hesitation highlights about the moment-to-moment production of perception and action. Looking for Æffects becomes a methodology for examining the complexity of material life distributed across that which, as I will discuss later, Arakawa and Gins refer to as the organism-person-surround.

Information Box 2: Effect + Affect = Æffect

Æffect attempts to correlate two historical modes of information capture and research value: affect in the arts and effect in the science. This correlation, that Æffect represents, foregrounds the importance of measure when ascertaining facts or perceiving events. This approach emphasizes the ‘continuous interactions performed upon the systems of affective and effective measure, reapplied to events and things’ (Keane 2007: 442). It invites a multimodality assessment of

the plastic potential of thought, feeling and sensation to be reconfigured. Ecological psychologist James J. Gibson evaluates the clinical data against his multimodal ecology of perception: “There is no correspondence between nerves thought to be specified and the senses, e.g. the optic, auditory and olfactory nerve. And there is no distinct nerve for touch or taste. The 12 bilateral pairs of cranial nerves have mixed functions. There are not 12 corresponding senses. Incoming (afferent) and outgoing (efferent) fibers are found in each bundle ... The units of anatomy are not the units of function” (Gibson 1966: 42). The term *Æffect*, together with the histories that collide within it, combines concurrent specificities and takes aim at the institutionalization of concepts and the bureaucracy of measure.

Effect, requires identities to remain solid and static, autonomous and persistent, in order to measure interactions that maintain identity distinctions and maintains subject–object relationships. The search for invariance requires preconditions or parameters that model portions of the world in order to apply the results generally – that is, to the greatest number of users.

Affect emphasizes the impersonal character of our common condition. Affects in Gilles Deleuze are ‘becomings that spill over beyond whoever lives through them (thereby becoming someone else)’ (Deleuze 1995: 137). For Deleuze and Guattari (1987) affect and affectability are modes of individuation ‘very different from that of a person, subject, thing or substance’, such as ‘a season, a summer, a date or an hour’ (1987: 258, 261). For Brian Massumi (2002) following on from Deleuze and Guattari, affect, or affective intensity, emerges as the critical point between ‘multiple and normally mutually exclusive potentials’ (2002: 31–32) of the virtual. According to Massumi, the virtual and its affective dimensions constitute a lived paradox (2002: 30), forging unanticipated connections that make viable the shortfalls of the impossible. This means that in terms of embodied cognition, impossibility *Æffects* the relationship within the organism-person and action potentials with the surroundings.

Carnal Knowledge in Marx

Before we examine what a new materialism ushers in through the arts and in particular the work of Arakawa and Gins, which offers a host of *Æffects* to explore, it would be useful to review previous configurations of materialism. My contribution to a history of materialism, in tandem to the other trajectories that this volume will include, is a pragmatic view of the embodied cognitive dimension of a new materialism.

It is curious that materialism, like vitalism, is getting a new lease on life in the sense that they are both being revived to retain the ethos of a world not centered solely on human interests, and reconfigured to provide connections that are not possible

without a more inclusive view of life (Bennett 2010, Thompson 2007). In Bertrand Russell's 1925 preface to Friedrich Lange's *History of Materialism* (1860/1957), he traces materialism back to the beginning of Greek philosophy and associates materialism with scientific advances, which have seemed, in certain epochs, almost synonymous with a scientific outlook. Russell states that "a system of thought which has persistent vitality must be worth studying, in spite of the professional contempt which is poured on it by professors of philosophy" (1957: v). Whereas materialism has undergone many transformations since 1925, it is important to note Russell's comment as a turning point in the revisions of materialism toward empiricism and the naturalizing of philosophical inquiry within science (see Petitot et al. 1999).

The reimagining of materialism is not new. Marx asserted that his "new" materialism was a challenge to Hegel's speculative dialectic materialism, famously stating in his *Thesis on Feuerbach* (1845) "philosophers have only interpreted the world in various ways; the point is to change it" (Marx: HREF). Marx's 9th and 10th theses offer succinct connections across body, individual and society. "[IX] The highest point reached by contemplative materialism, that is, materialism which does not comprehend sensuousness as practical activity, is contemplation of single individuals and of civil society. [X] The standpoint of the old materialism is civil society; the standpoint of the new is human society, or social humanity." (Marx: HREF) Clearly Marx considers the limits of Hegel to be the way Hegel understands the dispersal of human activity in the ecology of perception and action.

Rather than argue for the absolute newness of a new materialist position, it is more productive to find nascent in Marx, an embodied, cognitive, informational and even fictive character of materiality within his practical conception of materialism. To connect political economy with social cognition will more adequately address the categorical separation of thought and action embedded in the mis-construal of "inner" and "outer" life. The integration of thought as the movement that joins, separates and configures material connections is the basis of a new materialism conceived as distributed cognition and practiced from an embodied perspective.

In order to make the seeds of a cognitive turn in Marx evident, I will pinpoint two moments in Marx from which to extract the barest indication of radical empiricist and ecological psychologist tendencies (for example, passages that suggest experiencing experiences to be indications of relationship to the surroundings) that bridge political economy and an embodied understanding of distributed cognition. These two moments, thirteen years apart, link Marx's materialism with a new materialism of embodied and distributed cognition.

In 1844 Marx observed that: "the senses have become theoreticians in their own right" (1970: 139). This phrase pertains to the way Marx feels the activities of sensing and judging (interpreting and acting/changing) coming together in perception and the act of perceiving.⁷ This initial observation of 1844 is arguably a phenomenological one reflecting the embodied experience of insight. Marx seems to be acutely aware of the difference between passively attending to events in the world and being attuned to particular connection and configuration of activities, material processes and social realities.

Then, in 1867, Marx uses a phantasmagoric description of a wooden table to describe the experiential force of commodity. Here, the commodity becomes the ulcerated expression of Marx's earlier feeling, in which the organism-environment is overflowing with social-political-economic reverberations:

It is as clear as noon-day, that man, by his industry, changes the forms of the materials furnished by Nature, in such a way as to make them useful to him. The form of wood, for instance, is altered, by making a table out of it. Yet, for all that, the table continues to be that common, every-day thing, wood. But, so soon as it steps forth as a commodity, it is changed into something transcendent. It not only stands with its feet on the ground, but, in relation to all other commodities, it stands on its head, and evolves out of its wooden brain grotesque ideas, far more wonderful than 'table-turning' ever was⁸. (Marx 2000: 50–51)

Marx's account of the table integrates the affective, analytical and topical registers of signification and intuition. The use of poetic imagery "stepping forward" "standing on its head" and evolving "grotesque ideas" from its "wooden brain" anthropomorphizes the table by endowing it with a conscious life and acrobatic ability. The grotesque ideas awarded to the table a reflexive imagination which is the expressive image that animates Marx's thought-feeling-intuition that the senses have become theoreticians. Marx implies the table already knows the impact of its own commodification through the way its materiality partakes in a system of relations of its extended materiality. For example, when looking at a pine tree, one might perceive the possibility of a pine table afforded by the materiality of the wood. However, to perceive in the pine tree an entire system of relational forms for tables and for the efficient organization of labor and distribution (for example, IKEA) requires perceiving the pine trees as an *abstract machine* connected to a much larger set of formal relations in which a table has many recombinant lives. Marx's embodied perception, when considered as a form of action, changes the world because it proliferates the number and type of *affordances* that can be directly perceived in the materiality of the pine tree.

Commodification considered as an emergent set of relationships shares characteristics with James J. Gibson's *affordances*, but also with Deleuze and Guattari's notion of *abstract machines*. Gibson coined the term "affordance" to characterize the animal-referent description of objects and events (Warren & Shaw: 11). Shaw was among many not satisfied with the exclusion of the perceiver in Gibson's suggestion that affordances, as environmental information, were available to be perceived directly by all perceiving agents, not just humans. Affordances are different from physical properties because they are measured relationally, with respect to intentional acts (Warren & Shaw 1985: 11). Shaw's notion of "effectivities" (Turvey & Shaw, 1979) emphasizes the perceiver-specific capabilities that activate the category of potential encounters that an affordance names (Warren & Shaw 1985: 12). Warren and Shaw remark that, "in sum, every disposition of an animal for some action co-implicates

a disposition of some environmental structure to support that action” (1985:12). Robert E. Shaw and William Mace, prominent ecological psychologists, describe Gibson’s key notion of affordance as invariant environmental properties that provide specific causal informational support for a potential goal directed activity. To explain an agent’s interaction with invariant properties, they suggest that control-relevant task-constraints, or effectivities, must match the environmental affordances (Shaw & Mace 2003: 202).

While affordances explain the way in which commodities might be directly perceived in the environment, Deleuze and Guattari’s notion of abstract machines takes direct perception to a new level of abstraction, where relationality can be perceived in the same way as functionality. “Abstract, singular and creative, here and now, real yet nonconcrete, actual yet non-effectuated – that is why abstract machines are dated and named (the ‘Einstein abstract machine’, the ‘Webern abstract machine’, but also the ‘Galileo’, the ‘Bach’, or the ‘Beethoven’, etc.) Not that they refer to people or to effectuating moments, on the contrary, it is the names and the dates that refer to the singularities of the machines, and to what they effectuate” (Deleuze & Guattari 1987: 637, 511). “A machine may be defined as a system of interruptions or breaks” (Deleuze & Guattari 1983: 36). They go on to say:

Our criticism of these linguistic models is not that they are too abstract but, on the contrary, that there are not abstract enough, that they do not reach the abstract machine that connects a language to the semantic and pragmatic contents of statements, to collective assemblages of enunciation, to a whole micropolitics of the social field. A rhizome ceaselessly establishes connections between semiotic chains, organizations of power, and circumstances relative to the arts, sciences, and social struggles. (Deleuze & Guattari 1987: 7–13)

Manuel De Landa (2004) suggests that Deleuze’s materialism differs from the old dialectic materialism based on Hegel in two significant ways: first, that the world is independent to our minds; and second, matter has a morphogenic charge, making the interaction with material/s and material processes a progressive and affirmative form of co-selection.⁹ This is consistent with Jane Bennett’s recent claims to “pursue a materialism in the tradition of Democritus–Epicurus–Spinoza–Diderot–Deleuze more than Hegel–Adorno–Marx” in an effort to acknowledge the vitality of materials and the “ubiquitous intensity of impersonal affect” (2010: xii, xiii). Bennett expresses a “willingness to theorize events as encounter between ontologically diverse actants, some human, some not. Though all thoroughly material” (Bennett 2010: xiv). These thinking styles use local information and intensive qualities to emphasize differences at each level of the process.

Across these thinkers we can see materiality becoming distributed across an increasingly abstract set of relations, only to land, as I will argue, within the materiality of cognition, if cognition is understood as all the activities of the nervous system [see Information Box 1]. In order to argue this, we need to add an idea of embodied cognition supported by our creative impulse to this repertoire of new materialist

thinking. This can be found in an approach that Arakawa and Gins call biotopology, or biotopological thinking – an art-science of emphasis. *Biotopology* – the material surfaces of life and the surfaces from which living emerges – does not point to the lack of concreteness, but to the inconclusiveness of position, where surfaces are up on the air, decidedly undecided, open to valences and tentative configurations. “The gist of biotopological thinking is that no less attention should be paid to the atmospheric component of the architectural body (the bioscleavic atmospheric surround¹⁰) than to the body proper” (2006: 205). Biotopology closely studies related but notably different degrees of emphasis (Gins & Arakawa 2006: 209). Imagine if one’s attention could be shifted – as when concentrating on two hands touching – to include segments of the world (which they call “bioscleave”) in the parsing and apportioning of connection.

Through the Arts

When considered in light of the *Æffective* measure a new materialism utilizes, it is not such a big leap from the commodity and commodity fetish, born fully grown from the brain of a wooden table, to the readymade of Duchamp a half-century later, born from the thigh of the environment. While the commodity ushered in a political economy, the ready-made snuck a perceptual ecology in the back door via the context rather than the labor of a person.

Duchamp famously stated that he was not interested in ideas *per se*, or visual products, but in putting painting back into the service of the mind. At the time this was a gambit that revitalized art by ironically asserting the situatedness of meaning contingent on body and embedded experience. It is ironic because one might assume that the sensuality of painting is an utmost bodily activity, but by Duchamp’s assessment, an uncritical, unthinking body is locked into an optical unconscious. In a reversal of the production of meaning, Duchamp skewered the loaded materiality of a surrounding context with his readymade objects rather than allow the work of art to be given meaning by its context. Readymades mobilized the context-environment, getting the environment to perform gymnastic feats of connection usually left to commodified objects unlike Marx’s wooden table. The materiality of one object or feature of the environment can no longer be considered the benchmark by which the material qualities of another feature are determined. The artwork – the work of the work of art – has gone into orbit, tumbling with cognition toward re-entry and the realization of living.

In response to Duchamp statement about the role of painting (and art), a new proposition is needed: to put cognition back into the service of the organism-person-surround *through* the arts.¹¹ This is where Arakawa and Gins make their contribution. They do not reverse Duchamp’s reversal, instead they break down the barrier between contexts, using each context tactically as a tools for building all capacities that coordinate with the surrounding and not just “mind” in its autonomous or extended versions.

Arakawa and Gins arrive on the New York art scene in the early 1960s. They are interested in adding to the climate of change, at a time when everything is up in the

air, unhinged and trembling with uncertainty. Rather than ground the intellectual project of art in their exploration of “the mechanism of meaning” (1971, 1979, 1988), Arakawa and Gins choose to take advantage of indeterminacy at the site in which self-organization meets co-selection. In a world where all previously established relationships are made tentative, the texture of entire system changes due to the re-attunement of all the constituent parts. Transformations become more accessible the longer tentativeness is sustained. Any movement – including the movements of thought – thicken, attenuate, dilate or amplify the *atmospheric intricateness* of a world that is always separating and joining – cleaving – as bioscleave (Gins & Arakawa 2003a: 25). The Duchampian gambit of recontextualization and appropriation no longer suffices to charge the connective tissues that flex thought-matter into various constellations.

Environment as the Excluded Middle

The Aristotelian principle of the law of the excluded middle anchors the formal character of the logic of identity. If there is no value accorded to a position that is between true and not true (logic) or between A and not-A (identity), then all claims to truth are forced into a one side or the other: either/or – either you are for us or against us. Historically the context or surrounding environment is not part of the equation and acts as the backdrop by which truth effects and identity distinctions acquire significance. The door in Duchamp’s Paris studio, positioned in a corner to open one passage as it closed another, was a refutation of the Cartesian proverb “A door must be either open or shut” (Descartes in McEvelley 2005: 22). Once again Duchamp activates the context, transforming it into a large scale slide-rule that recalibrates truth effects, identity and the relative attributes of an object, in this case the door as open or shut.¹² What is excluded by the *law* of the excluded middle is the conditions under which identity is established. What the law ignores is the transformative involvement of the environment. The readymade transforms the environment in the same way that an environmental context inflects the meaning of an object. It is the movements of an actant across the terrain of the excluded middle (moving the door into ambiguously open and shut positions) that become the mechanism of meaning. Duchamp’s “standard stops”, “mile of string” and “corner door” are the modest beginnings of architecture made tool in service of life. The process of including the environment as all that is missing in the transition from one identity to another, produces an infolding movement of conscious reflection upon “a state of receptivity is produced that is so pure that it can only be conceived as a third state, an excluded middle, prior to the distinction between activity and passivity: affect” (Massumi 2002: 32). In other words, the activated context/environment attenuates and dilates the suspension of judgment while amplifying attention and perception.

Arakawa and Gins’ *blank* and *forming of blank* (1987) as the basis of a biotopologist daily research¹³ cultivates the state of receptivity to which Massumi refers. It is only in this tentative state that particular possibilities can be initiated. Therefore research, as it is traditionally formulated, pre-states the conditions in which an object of study can be identified and observed, for example the parameters by which to measure

light as a particle or as a wave. Arakawa and Gins work against the imperative to pre-state the conditions in which to solve X. X is the way to mark the spot of interest, intersection, desire or most probable site for yielding knowledge. Arakawa and Gins push the object of study and observation – a designated X – away from its tacit agreement with a given identity toward an open or blank state. For Arakawa and Gins identifying an X for observation and study entails selecting an “underplayed X that, if used more fully, could significantly alter what it means to be alive” (2003b: 12). The invention and assembly of architectural procedures aims to build questions that require embodied modes of investigation (for example, built-environment as a three dimensional question asked of or directed to the process that cannot be identified as organism, person or surround).

If all relationships are tentative or can be made tentative, selecting an X to study gives one pause because the surrounding context is no more formed than the selected X. As we will see in the barest and most restrained operations of Arakawa’s prints for Edgar Allan Poe’s *Eureka* (to be discussed in the next section), building architectural procedures used for studying the interactions between body and bioscleave have an observational–heuristic purpose, while those devised for transforming body and bioscleave have a reconfigurative purpose (2006: 207). Arakawa and Gins propose a more adequate name for describing the world-forming process of which there is no part that is not constantly cleaving – joining and separating – thus, “bioscleave” (2006: 208).

From uncertainty and indeterminacy, Arakawa and Gins (unlike Duchamp) choose not to cultivate indifference, deciding instead to construct purpose from all that melts into air. Art theorist Thomas McEvilley calls the erosion of the distinction between art and life by art projects such as Duchamp’s, “anti-art”. Arakawa and Gins erode the distinction further by calling art-science projects the science of emphasis – for example biotopology, the basis of their procedural architecture. The pathway *toward a new materialism through the arts* might best be achieved by following Arakawa and Gins *through* the processes made available in art in order to re-apply them to perception and action, selection, self-organization and the recursive movements of learning.

The inclusion of non-metric, non-linguistic and asignifying qualities within the ongoing embodied processes of measuring preserves the complexity of our understanding and experience of the world within our attempts to observe, study and transform the world. *Æffect*, as an embodied condition assigning value, invites the distinct registers of measure to rub up against each other as embodied cognitive processes. The ensuing calibrations, calculations and biochemical projections are *effeent* (outgoing) and *afferent* (incoming), which is to say that the body initiates paradoxical and conflicting modes of activity that both inhibit and prompt action. The configurations that impact upon how persons attend, perceive, select, decide and judge are distributed into and also organize the environment. The stakes of embodied practice (assigning, implementing and interpreting value) initiate a shift from pre-reflective everyday life to daily research of an heuristic life, where an emphasis on learning about one’s own modes of learning impacts upon all manner of activities. For example, any activity

can become heuristic tool. The heuristic approach I am proposing presupposes the aim to be one of increased human capacity across the board. If you choose to run a marathon, the benefits of this activity will be many and vary across health, concentration, community, etc. The way you go about training to run a marathon would involve cross-training, partner work, perhaps meditation, yoga, etc. To train and run a marathon would increase your capacity as a human. But does it increase your capacity in other ways? This depends on your degree of openness to the qualities of experience – to ones playful and inquisitive engagement with those qualities – when moving through that process of training. Arakawa and Gins talk about this in terms of tentativeness, which applies to both the “what” and the “how” of engaging with the world. Gins and Arakawa “tentative constructing toward a holding in place” procedure (2002: 23) holds the world open while attempting to hold the modes of learning open as well. When running a marathon, a highly directed focus on the goal of winning, or on achieving a new time record, for instance, can override such an attentiveness through which one learns about the process of increasing one’s capacity (learning about learning). It is for this reason I would suggest that some activities, by their design, are more conducive to a heuristics. Art, as an activity that searches the contours of consciousness for new modes of engagement, can supply a plethora of tools and tactics for heuristic application toward increased capacity. In other words, an attentiveness to the qualities of experience within processes of rigorous activities, such as can be the case in sport or in art production, leads us toward the more omnidirectional *Æffectivity* of an heuristic life.

Arakawa and Gins’ research practice offers insight into how this can occur through art practice, providing examples of how this attentiveness, or degrees of freedom, is formed in the excluded middle of identity formation. As such, the second half of this chapter will be devoted to a discussion of the specific modes of Arakawa and Gins’ inclusive research practice. I have chosen to characterize their contribution as “discovery” – a scientific term – because it implies a coordination with and emergence from the biosphere rather than opt for a term such as creation, laden with hubris.

Arakawa and Gins: Initiating Discovery

If the notion of *Æffect* disposes a person to become aware of embodied processes by which connections to the environment can be explored, a practice is still required to prompt and guide newly discovered dispositions and unfamiliar sensations. A life that organizes itself around discovery rather than accumulation of knowledge, a heuristic life, begins by constantly re-entering all modes of interaction between body and environment and learning, reconfiguring and coordinating the emergent modalities. Gins and Arakawa state that an architectural surround that is “procedural” enables movement within and between its own modes of sensing (2002: 58).

Rather than discussing Arakawa and Gins’ large-scale installations, houses, parks, apartments or proposed communities, which have attracted a lot of attention and scholarship, I have decided to discuss a little-known set of prints by Arakawa for E.A. Poe’s prose poem *Eureka* from 1991. In my view, these prints demonstrate a

moment of reflection and consolidation of thought and energy from prior works such as the panels for *The Mechanism of Meaning*, the text of *To Not to Die*, and their plans for the *Bridge of Reversible Destiny/The Process in Question*, which resonates with and impact upon their subsequent large-scale works (*Ubiquitous Site* at Nagi, the *Site of Reversible Reversible Destiny* park at Yoro, the *Bioscleave House* and *Mitaka Lofts*) and written discourse (*Architectural Body*, *Making Dying Illegal*). The prints demonstrate that, at every scale, even within the seemingly immobilized activity of looking at/reading a book (what does one call a person experiencing the materiality and scale of the pages of a book?), the process of reconfiguring sensing, perceiving, deciding and judging can become the heuristic tool for initiating change and developing the daily research of one's own capacities. The prints enact the barest movements and experiential rumblings that resonate to all scales of action, discovering embodied connections by experimenting with the material processes that may best afford activity across perceptual and conceptual processes.

In 1991, Arakawa returned to an influential book from his youth and published a series of relief prints inserted into the reprinting of Edgar Allan Poe's *Eureka: An Essay on the Material and Spiritual Universe* (originally published by George Putnam in 1848). In the new edition, Arakawa selects specific passages with which to interact and from which to extrapolate, not overwriting Poe's text but interacting with Poe's thought by interrupting, re-sequencing and diagramming its movements at certain key points. This work on Poe's *Eureka* occurred at a time when Arakawa and Gins were making the decision not to produce any more art or poetry. Arakawa's last painting exhibition was at Ronald Feldman Fine Art in New York City in 1990. It was called *Paintings for Closed Eyes*, and consisted of several wall-sized paintings with ramps of equivalent size for viewing placed a few meters in front of them. These paintings explored the same layout of marks and symmetrical and non-symmetrical division of canvas/page as his prints for *Eureka*, but without Poe's text. During the period in which Arakawa was re-examining his formative ideas, Madeline Gins had been writing *Helen Keller or Arakawa* (1994), which was a re-entry into the history of Arakawa's works and their collaborative works. This crucial period marks the transition from the body-sized painting-puzzles of *The Mechanism of Meaning* (1963–1973) to immersive built-environments such as *Bridge of Reversible Destiny/The Process in Question* (1973–1989) and subsequent invention of architectural procedures. This transition, in 1991, was no less that re-entry into the moment/site/possibility of initiating. Arakawa's prints for *Eureka* are not well known, and they provide an insight into a moment of reflection and projection which gathers momentum internally – in the work, in the collaboration and in the *Reversible Destiny* project – sufficient for escape velocity into work that does not stand in relation to art. Given the project's premise – they ask “Who would not want to live a world built to serve the body to the nth degree?” It attempts to address the crisis condition through contentious and polarizing statements such as “We have decided not to die”, which Arakawa and Gins suggest is an “open challenge to our species to reinvent itself and to desist from foreclosing on any possibility, even those our contemporaries judge to be im-

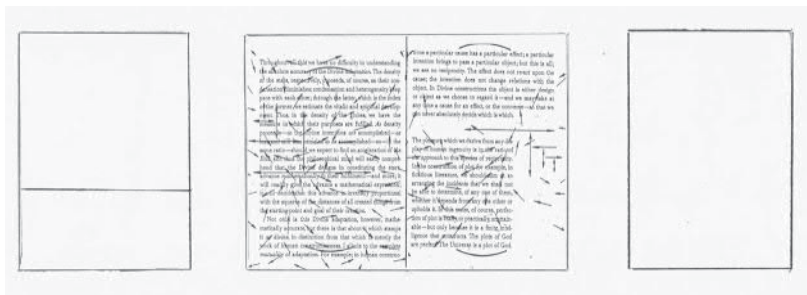
possible” (Gins & Arakawa 2002: xviii). To achieve this, artifice must be dragged *through* art into the *Æffects* of heuristic life.

Poe’s *Eureka* combines aesthetic philosophy, prose poem and essay in an irrepressible oration on the role of intuition in human knowledge being as valid as deduction and induction. It also demonstrates a desire for reconciliation of science and art; plays through the nineteenth century’s preoccupation with the macabre; contains a letter from the year 2848, and exhibits a form of serious irony that overflows from brushes with the infinite in thought (Todd, in Poe 1991: iii). Poe’s science is certainly suspect by today’s standards, and betrays a vitalist view of the universe. However, his use of science is not for proof, but rather a supplement to his vision, inductions and deductions. Knowledge is ultimately useless; one reaches the point where only “intuition ... of which the processes are so shadowy as to escape our consciousness” can lead one to the truth (1991: ix). These are tendencies that are shared by Arakawa and Gins, who demythologize them to be redeployed as environmental information. The benefit of demythologizing the romance of unknowing is to reconfigure even the most fantastic of intuitions into the set of ideas available to the organism–person–surround for the production of new relationships.

Poe’s passions and the circumstances of his personal life (the death of his mother and wife, and his reactions to these events) show the degree to which he is embedded in his cultural and personal context. The latency of the world does not haunt Arakawa, who views imagination and perception as actions. This is why he engages *Eureka* in a diagrammatic material encounter. The interactions are severe, but they are also respectful of Poe’s naive vision and extraordinary foresight. For example, through Arakawa’s reworking, the “will” of Poe has given way to Arakawa’s complex interaction and configuration of *blank*, *tentativeness* (Arakawa & Gins 1987) and the activity of *landing as a site* (Gins & Arakawa 2002: 5–22).¹⁴ In other words, mind over matter is replaced with a notion of mind formed by the materiality of cognition, which includes sensing, perceiving, feeling and thinking. The holding of thing in place through the power of the will is dissolved into a much more pragmatic idea of “sited awareness”, literally that the body is that “which initiates pointing, selecting, electing, determining and considering ... an organism/person, a sited body lives as one site that is composed of many sites” (2002: 5). Sited awareness is the activity that involves the entire body and its relation to the surrounds to “deposit awareness as a result of the very bodily activity of landing (read attending, selecting, apportioning emphasizing) everywhere around oneself” (2002: 7).

Poe’s “sorrow” has given way to the tools of procedural knowing which combat desperation and historical defeatism through what they call a *crisis ethics* (2002: xvi–xviii). Procedural knowing – as a process of keeping the connections across the organism–person–surround open – combats defeatism because it is the embodied process by which one resist foreclosing on an idea no matter how seemingly impossible (2002: xviii). This sorrow extends to the entire set of conceptual limitations placed upon persons with which the organism may not, and should not have to,

comply. Arakawa and Gins' *reversible destiny* is not a method of overcoming irreversible loss, but utilizing the changeability of direction. The basis for this possibility according to Poe – dreaming as the foundation of reality and science – proceeds by intuitive leaps (1991: 7): “The simple truth is that the Aristotelians erected their castles on a basis far less reliable than air, for no such things as axioms ever existed or can possibly exist at all” (1991: 8). The ironic construction of Poe’s simple truths supplies the switching mechanism between perception and imagination, precisely because, as different orders of embodied processing (top-down conceptual processing and bottom-up perceptual processing), they can interact with a built space that requires the two processes to conflict, then conflate then remix. The lived experience of this reconfiguration would be a tracking of conceptual effects and bodily affects. By proposing that the middle can no longer be excluded, a series of *Æffects* gives texture to thoughts and perhaps allow a proprioceptive or kinaesthetic imagination to blossom.



7. Layout of each set of Arakawa’s prints for Poe’s *Eureka* comprised of four pages, 1991

There are eight sets of prints in all, all of which contain four pages. I will only discuss the first, third and fourth sets. Each set of four pages follows a sequential program. The first page is always a right-hand page opposite the Poe text, and has a single line that in subsequent sets moves incrementally up the page. The second and third prints form a double (centerfold). The last “print”, which is left blank, leads again to the Poe text. The interruption that the prints perform constitutes a living response to Poe’s vision. On the first page of each set of prints, a single line marks a horizon or watermark level. In each subsequent set of prints, the line on the first page moves incrementally higher. The next two pages form a diptych: two facing pages, which constitute a single print where Arakawa interacts with Poe’s text. The last page follows the diptych and always falls on a left-hand page that is blank. The text of *Eureka* resumes on the subsequent right-hand page.

This blank (page) is loaded with all that Arakawa has invested in his “blank” paintings from the 1960s. The blank page presents a state of depletedness and readiness – not the projective blank page of Olson’s *Maximus Poems* (1983), where the page becomes the sea, but an actual living blank of tentative yet active possibility attached to and kept separate from the text Arakawa is re-forming. The linear logic of the book is shifted into a non-linear logic of *Æffects*.

To reiterate, Arakawa uses passages from Poe's text in each set of prints. The text that appears in the prints is never chosen from the place in the text where the prints are inserted. Arakawa uses portions of text before they appear in Poe's text. These selected texts are therefore anachronistic within the original sequence of the prose poem, and do not comment on or engage with the flow of Poe's ideas. Arakawa's prints are diagrams from the future and the future *Æffect* of Poe's thoughts on him. In the letter transcribed from 2848, he also invokes an anachronistic and synchronistic universe where, although logic denies it, Poe challenges the veracity of one of John Stuart Mill's axioms: "Contradictions cannot both be true – that is, cannot co-exist in nature" (1991: 14). Mill says the tree cannot be and not be, because it is impossible to conceive it so. Poe obviously thinks that a tree can be and not be. Arakawa and Gins not only agree, they make this double thinking "say one think two" the connective tissue of the architectural body (from section 7 panel 3 of *The Mechanism of Meaning* in Arakawa & Gins 1997: 77). Suddenly we are back in Duchamp's studio, at the doorway of the excluded middle.

From this a paradox ensues between logic and experience, science and art. Logic victimizes intuition. The paradox is the result of judgments made from this position, which does not logically allow a single thing to have two (contradictory) values. Poe's indirect answer to Mill is: "Infinity is the representative but of a thought of a thought" (1991: 17). The intuition on which Poe relies prefigures Arakawa's notion of the blank as a reservoir of regenerative possibility. The middle cannot be excluded. The surrounds impinge at every point in every scale of measure.

Just as Duchamp changes Kant retroactively (De Duve 1998), so Arakawa's selection of text makes it difficult to hear Poe's words without Arakawa and Gins' blank in one's ears. Poe writes:

Let us now endeavour to conceive what matter must be, when or if, in its absolute extreme of simplicity. Here the reason flies at once to imparticularity – to a particle – to one particle – a particle of one kind – of one character – of one nature – of one size – of one form – a particle therefore 'without form and void' – a particle positively a particle at all points – a particle absolutely unique, individual, undivided, and not indivisible. (Poe 1991: 17)

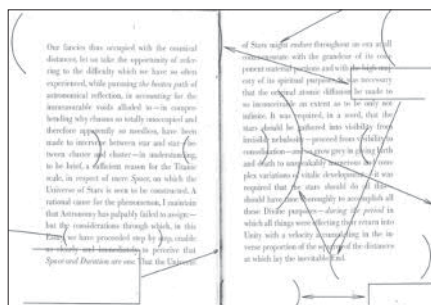
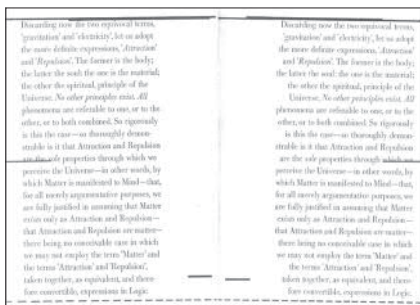
In a corresponding paragraph from *To Not to Die* (1987), Arakawa and Gins push past the meditation on matter to place matter squarely within the activity of forming blank:

Instantaneously and repeatedly, blank serves as a station for our senses, making possible an impression of continuance. Simultaneously < I > of a fiction of place comes to be formed in much the same way. And so, blank may be found thoroughly interspersed throughout < I >, forming an integral part of any act. When trying to bring it into focus, it must be remembered that blank is widely dispersed, capable of behaving in many different ways at once, and itself plays a fundamental role in the act of focusing. (Arakawa & Gins 1987: 38)

In *Architectural Body* (2002), Gins and Arakawa go further to suggest the relation of material practices such as written discourse and constructing built-environments, stating:

Discursive sequences of tactically posed surrounds, constructed as built propositions, marshal existing logical connectives and position newly invented ones into the ‘real’ steering, regulating and guiding interactions between body and bioscleave through three-dimensional THEREFORES, BUTS, ORS, ANDS and built-up WHATEVERS. What will need to be studied is which types and combinations of bodily movements are most conducive to an optimal *tentative constructing toward a bolding in place*, and which constructed discursive sequences best constrain them. (Arakawa & Gins 2002: 58–59, emphasis in original)

The movement in and out of idea and material, nothingness and form, is only possible because of bodily actions that allow an organism to interact with its own descriptions as if they were external. Perception of potentially useful features in an environment allows an ecological niche to influence the “co-selection” of an organism’s features. This process runs its course. Initiating, however, requires another order of movement – the insertion of an additional feedback loop that deflects the configuration of the organism through one more possible qualifying circuit. This is Arakawa’s gambit in *Eureka*. The placement of the excerpted text requires a correlation between the literal and implied space of the page. The texts float and are anchored – that is, they participate in the implied infinite space “behind” the surface of the page and they adhere to the material texture on the page in the way the ink is pressed into paper in the printing process. The operation of orienting oneself to the relationships on the page recapitulates the observation in Poe’s text that repulsion proceeds from coalescence or condensation, an idea that Poe equates with heterogeneity. Arakawa’s breaking up and repeating of texts sets up ratios and scales, perspectives and condensations, depending on the local diagrammatic system for each page. They are literally finite and infinite spaces in Poe’s sense of being at the



limits of the conceivable. Arakawa subtly plays with the way his diagrammatic lines intervene with Poe's text (see Figure 8). Lines that initially appear to underline important phrases also seem to cross out the same phrases when repeated. These are hesitations materialized in the vernacular of embodied cognition, in its processes of perception, selection and action. These materializations are the movements of thought, which are not proper to discrete modes of cognition or values of affective experience or evaluations of effects and are therefore the *Effects of the thresholds between values and sensations*.

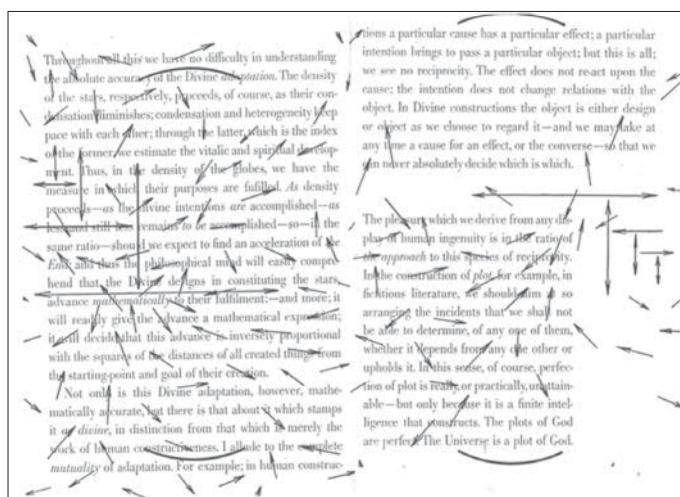
In the first set of prints (1991: 8–13) (on the left in Figure 8), the same text is repeated on the facing page with diagrammatic lines in seemingly symmetrical arrangements. Offsetting the two small dashes near the spine of the book breaks the vertical symmetry, left to right. There is no horizontal symmetry. On closer inspection, the double line at the top of the page changes from left-hand page to the right-hand page. On the left-hand page, the extra thickness is added to the bottom edge of the continuous line. On the right-hand page, the thickness is added to the top edge of the line. Although the text is repeated, it is justified left on the left-hand page with wider margin to the spine and justified right on the right-hand page with the wide margin again to the spine. The alignment is mirrored without the text being reversed. These adjustments and repetitions echo and comment on the symmetry and asymmetry of the shifting dualisms that Poe is discussing in his text. The small adjustments appear in various guises as the duration of perception opens to, or engages with, literal space. What seems to be duplication or mirroring or symmetry changes from moment to moment in the time of embodied cognition. In the passage Arakawa uses, Poe discards the equivocal terms of gravitation and electricity for “more definite expressions of attraction and repulsion. The former is the body the latter is the soul: the one is the material; the other is the spiritual, principle of the universe. No other principles exist” (1991: 9). Poe's universe requires the interaction of properties apparent in both the universe and perception of the universe. He names these operations *matter*.

In the third set of prints (1991: 32–35) (on the right in Figure 8), there are different texts on the facing pages with exactly the same spacing and placement on each page, and justified on both sides. Arakawa's diagrammatic interactions are not symmetrical, but play off repeated motifs, implying inclusion or groupings through the use of brackets, and connection through the use of arrows, which activate the pages – unlike the more static lines of previous prints.

Poe's text anticipates Bergson and Einstein declaring that “space and duration are one” (1991: 90). The dramas of the universe are what happen while all things return to unity and an inevitable end. Poe is not attempting to dismantle the inevitability, but to reanimate the dead by attenuating the effects of living matter so that they can be felt and interacted with long after their death. After the death of his child bride, Virginia Clem, Poe was in a state of physical collapse for months and his body began to deteriorate rapidly. The central concern of Poe's *Eureka* is wrapped up at every level with truth's relationship to beauty, and beauty is tied to the changing (transmuted) forms of animation. “Poetry and truth are one”; therefore his theory

of the universe should be regarded as coming into being through mythopoiesis. This sentiment marks a strong divergence between Poe's desire and Arakawa and Gins' pragmatism of impossibility. The *Reversible Destiny* project not is based on longing or reanimation, but upon *Æffecting* the conditions of automaticity and inevitability.

Through the extent to which the elements in Poe universe impinge upon each other, he exhibits a late romantic porosity that Todd suggests anticipates Duchamp's *infrathin* (an *Æffective* modality), where the ways in which we are tied into the environment allow us to attach and fixate upon certain persons and things to draw from them what the entire environment might normally provide (Todd, in Poe 1991: vi). The old-fashioned romanticism Poe exhibits might, in time, lead to the postmodern melancholy of Lyotard (1988: 80), or the condition of being caught between the motivation of unrelenting regret and an insistence upon innovation as beginning, and beginning again, in the twentieth century. Poe sought a universal answer to the meaning of death. Arakawa and Gins seek an answer to sustaining life in the very operations that initiate growth and change. Poe entertained the idea that humans could triumph over death, not through immortality but by coming back to life. For Poe, as Todd suggests, "That life and death could synchronically exist in the same body is the basis for Poe's prototypical story, 'Ligeia'" (Todd, in Poe 1991: vii). The poetry of Poe was aimed at writing so as to bring back to life. Arakawa and Gins do not exhibit, or concur with, the tropic approach or romantic, melancholic motivations. Instead, they counter Poe's metaphysical irony with sincere irony – non-ironic irony – exemplified in the statement "We have decided not to die" (Arakawa & Gins 1997). Non-ironic Irony exemplifies the complex relation between causal change and contextual inflection signaled by the inclusive register of *Æffect*.



9. Fourth set of prints by Arakawa for Poe's *Eureka*, 1991

The fourth set of prints (Poe 1991: 44–47) offers an asymmetrical field operating on zones of attraction rather than conventionally framed pages (Figure 9). Arakawa's more chaotic coordination of diagram and text, bodily attention and ideational positioning breaks with the formality of the page to imply a stronger connection to the reader's world/universe. In this set of prints, there are remnants of previous repetitions – perhaps a last battle between the infinite and finite, except that now Arakawa's more complex perception of the world begins to reposition how we may continue to understand Poe's *Eureka*. This idea of staging a critical point in a program or series of interactions is consistent with much of Arakawa and Gins' architectural approach, in which reassembly often begins to occur at the midway point after a process of disassembly.¹⁵

Because the diagrammatic schema in the fourth set of prints is denser and not aligned as much with the grid of the text, the traces can be seen on the other side of the page, through to the blankness of the last print of the set. This kind of control is very deliberate as a material consideration that has critical overtones, in the sense that, through Arakawa's discoveries, we witness the ability to contain the categories of our interaction and attentions becoming less conceivably possible, or perhaps more conceivably impossible. The distinction between effects and affects are no clearer than the status of Duchamp's door as open or shut. When the compartments of the world are held open, allowing effects and affects to mix in a Brownian motion of *Æffects*, the excluded middle rushes in and overwhelms processes of identity and all established systems to determine, position, distance, relation, density and boundary. For example, the use of brackets, parentheses and arrows acquires a symbolic meaning in relation to the text on which they are superimposed as well as spatial meaning in relation to the literal page. They even imply a relation to the space of the viewer as they divide spaces that carry over, across and outside the book space. Arrows are the most curious diagrammatic indications that designate here (where the arrow is located) and there (where the arrow points). The arrow is also “not here” but indicates “a” here, purposefully transparent as things on a map, but also indicate a literal mapping movement. Arrows are exemplary indices of *Æffect* participating in the actual located materiality of the diagram (a line in the sand) while also participating in the virtual movement of the materiality of cognition (the perception and action of distinctions). At the intersection of these two spaces – as when daydreaming while driving – emergent areas of the world clamor and dissipate as the effects and affects of each realm are made to mingle. It is through the repeated placement of arrows on a textured paper of a particular size (torso-size) in relation to a text, that the observation, study and transformation of the world come together. This realization of the material basis of cognition is the heuristic component of practice.

The reprinted *Eureka* with its newly inserted prints is an oversized book. The vertical dimension of each page is larger than A3-sized paper. When standing/sitting in front of the open book (two pages across), it takes on a relation to one's torso and begins to connect into the surrounding perceptual environment. The words and markings (arrows, dashes, brackets, parentheses, border-like lines) are no longer constrained by the page. The book is no longer a book. It is the tip of the iceberg

of the environment inserted like a conic section into the normally sequestered space of reading. This link to the excluded middle-fore-background is not activated unless the book behaves differently, either indirectly pointing toward or directly prompting perceptual connections that extend from the book-as-discrete-yet-complex object to the book as intensified segment of the environment. Initiating can and must take place at every level of abstraction and scales of action. The excluded middle in the organism-person-surround is not “person”, but the flow of initiating movements across the practices of embodied cognition to the processes of *atmospheric intricateness*, a threshold point at which *architectural body* *Æffects* a reorganization of its co-extensiveness with *bioscleave*.¹⁶

Conclusion: Initiating Heuristic Life

Æffect as verb, *Æffect* as noun, *Æffect* as tongue twister and *Æffect* as paradoxical proposition all emphasize the dynamic aspects of attending, perceiving and perceptual learning. These lived paradoxes of measure rife with hesitations mark the beginning of a practice of embodied cognition. However the goal of proposing *Æffective* measuring is to engage with the conditions that would allow connecting to flourish at the threshold of initiating action – where top-down conceptual processing and bottom-up perceptual processing, afferent and efferent projections of embodied cognition as well as the organism and environment play out the selection process. By *Æffectively* dilating the thresholds separating the organism from person and the person from environment, re-entry and constant delimitation can be encouraged. A new materialism depends upon a collective ability to move toward *atmospheric intricateness* (Arakawa & Gins 2003a: 25) and toward the ongoing learning that each new distinction/relation provides.

Æffect is a relational/co-relational tool devised to help one learn how to negotiate the material processes of self-organization. A *practice* of embodied cognition – the distribution of the “mind” throughout the body and into the environment – *Æffects* the construction of self through the recognition of two modes of expansion/inclusion: first, the recognition of the role of the environment in the co-selection of the organism-person-surround (e.g. cognition as perception and action); second the role of abstract relationships in the coordination of the organism-person-surround (e.g. cognition as attention, emphasis, and the production of value-based distinctions). Books of discovery such as Arakawa’s revision of Poe’s *Eureka* (through the insertion of his prints), Gins’ *Helen Keller or Arakawa* or Gins and Arakawa’s *Architectural Body* enact these modes of inclusion, re-positioning the book as both object and environment, indication and context. This mobilization ensures against the rehearsal of soundproofing the text from ambient conditions and the quarantining spatiality from contagious proximities that dampen the proprioceptive perturbations of thought. New modes of relation must be initiated.

Æffect promotes initiating new distinctions. Practice advances re-entry. The movement from one position to another enables continuous learning. With varying degrees of freedom, minimal movements of position bare indication within the body-book-environ persist across scales of action to the maximal movements of architecting the

features of an organism/person/surround. Both sets of movement initiate (____) – fill in the *blank*: attention, perception, action, selection, decision and judgment). Each operation is a different type of motivated shift from reflective life of observation to a heuristic life of collaboration and futurity, through attentiveness to the qualities of experience within processes of rigorous activities purposefully designed to develop re-entry into one's own modes of perceptual learning and increase human capacity omnidirectionally. Gins and Arakawa refer to these movements as the “engineering of atmospherically intricate domains” to amass forming power (2003a: 25).

William James (1967) offers insight into where efforts at initiating might be focused:

The world experienced (otherwise called the ‘field of consciousness’) comes at all times with our body at its center, center of vision, center of actions, center of interest ... the body is the storm center, the origin of co-ordinates, the constant place of stress in all that experience-train. Everything circles round it, and is felt from its point of view. The word ‘I’ then, is primarily a noun of position, just like ‘this’ and ‘here’. (James 1967: 86)

To be *Æffected* then is to be dispersed, to be both “I” and “not-I”. To be in two places at once, three places at once, many places at once. However, even the phrase “to be in” complies with the model of a self that must maintain continuity of the “I” of experience, even when “I” am in several places at once. But, if as James suggests, “I” is a positional proposition, then in Arakawa and Gins’ notion of organism-person-surround, the “I” is an expression of the expanded field; a zone of emphasis. In this mode of openness, or *Æffection*, learning (heuristics) cannot be distinguished from the production of life. In order to include the environment and avoid speaking for it, or turning the environment into the Hegelian picture show, a new materialism must be built on the subtle difference initiated by embodied reality sensitive to *Æffects* and prompted by *atmospheric intricateness*.

Becoming atmospheric occurs when the readymade object transforms into the ready-to-be-made event already moving toward the extended life of cognition where material and thought share events but not extents. This ready-to-be-made event no longer belongs to an abstract *space* but constitute a *place* in which the internal movements build up momentum sufficient to initiate action (make distinctions that produce relation between material forms) and initiate change (recontextualize distinctions). A new materialism is a pro-position of *Æffects* that enables a practice of embodied cognition, through mobilization of the environment, to discover again and again (eureka!) the science of our own fiction.

MATERIALITY, AFFECT, AND THE AESTHETIC IMAGE

Estelle Barrett

In this chapter I will examine how art practice and aesthetic experience operate as a mode of knowledge production that is rooted in material and subjective processes and how this articulates the epistemological dimension of what is referred to in the title of this volume as the “new materialism”.

The notion of artistic practice as a mode of enquiry and knowledge production constitutes a challenge to assumptions that underpin the privileging of the science in the knowledge economy. Drawing on the work of Julia Kristeva, John Dewey and Antonio Demasio, I will first examine the material and experiential dimensions of creative production that give rise to what I call the “aesthetic image”, a structural aspect of the artwork that emerges as an outcome of the grafting of affect to the symbolic through artistic practice. Unlike images that operate via established symbolic codes and that serve to communicate information, the aesthetic image is “performative”; it emerges through sensory processes and gives rise to multiplicity, ambiguity and indeterminacy. Out of this, meanings that fall beyond the codes of a given sign system (visual or verbal) may be accessed. This will be demonstrated in my analysis of two creative arts research projects at the end of this chapter. First, however, I will examine the notion of “performativity”, its relation to the emergence of the aesthetic image and how this furthers our understanding of the “new materialism” as a critique of science.

The Problem with Science

Canonical social science and humanities research has tended to privilege assumptions about research and research training derived from science and the idea that distanced, impersonal observation is the only reliable method for the production of “truths” or knowledge about the world. However, more recently there has been a general shift in the arts and humanities toward a second mode of discovery and learning, which Iain Biggs (2009) describes as a mode of knowledge production that emphasizes a context of application, heterogeneity, a reduced reliance on hierarchy and an acknowledgement that the production of new knowledge is potentially a

transformative act; that is to say that the production of knowledge is experiential and performative. I suggest that in such an approach, distanced observation is replaced with aesthetic awareness and knowledge occurs as material process through interaction and action. The key to unlocking this alternative mode of production can be understood via Julia Kristeva's account of experience-in-practice and her view of creativity as material process that involves a disruption of the established codes of language in ways that allow new objects of thought and understanding to emerge.

Kristeva's critique of science is based on the idea that it is founded on methods that were first used capture the truth about nature by classifying and encoding phenomena through a system of formal language that "represses the process pervading the body and the subject" (Kristeva 1984: 13). Knowledge in such a framework is viewed as static and passive. Kristeva has shown that the notion of objective, empirical observation as a possibility – and truth as universal – cannot hold if we are to take account of subjective processes: the interaction between, the body as non-human (as matter), embodied experience, language, and thought operating in social contexts.

The logic of science is a logic that supposes a direct correspondence between naming and describing the phenomena to which it refers. This logic presumes that the signifier designates the fullness of referent and is flush with the referent. Kristeva exposes this logic in her account of proper names, which she describes as abbreviations of descriptions that describe not particulars, but systems of particulars. Science not only abbreviates, but by collecting all the possible descriptions of phenomena into its formulae, symbolic representations, concepts and laws, it erases indeterminacy (Kristeva 1986b: 234). Aesthetic experience or art, on the other hand, operates through the particulars *and* indeterminacies of embodied experience-in-practice and interaction with objects in the world, bringing into play an alternative logic that the logic of the *discourse* of science forecloses (Kristeva 1986b). My emphasis on "discourse" in the previous sentence is deliberate since many scientists have acknowledged that all "real" discoveries in science have come about through processes that are ultimately, aesthetic and subjective. Indeed, philosopher and scientist Michael Polanyi asserts that "Complete objectivity as usually attributed to the exact sciences is a delusion and in fact a false ideal" (Polanyi 1958: 18). This is so because the subject, or the one who observes is, "heterogeneous" – both a subject of rational thought and the symbolic as well as of biological processes (Kristeva 1984).

Artistic practice is often motivated by emotional, personal and subjective concerns; it proceeds not only on the basis of explicit and exact knowledge, but also on tacit and experiential knowledge. Experience operates within in the domain of the aesthetic; knowledge produced through aesthetic experience is always contextual and situated since it involves direct sensory engagement with objects in the world. Artistic experience therefore, occurs as a continuum with normal processes of living and is derived from an impulse to handle objects and to think and *feel* through their handling. What emerges from this process is the aesthetic image – an image that is heterogeneous in that it permits a *knowing* that exceeds what can be captured by the symbolic.

Kristeva's ideas on the relationship between biological processes, affect and language help us to understand that thinking is predicated on our material experiences and interaction with the world. Her account of materiality acknowledges that matter itself is not static or mute, but is in constant interaction and intra-action with the living organism resulting in causal and differential production. Thinking emerges through signs and symbols that stand in for the things that were *materially* and emotionally felt or experienced (Kristeva 1984). Affect and sensation give rise to internal images that are forged through encounters with objects in the world; these images that are born of aesthetic experience, occur prior to the emergence of signs and symbols. In this sense, thought and language can be said to have an aesthetic and material dimension. This dimension is often masked in everyday communication because such communication is reductive, operating through codes that fix meanings or confine them to the rules of the system.

The Heterogeneous Subject

Kristeva's account of the "speaking subject" (Kristeva 1986a) as *heterogeneous*, allows us to conceive of a layering of subjectivity that emerges through material process and contradiction. She tells us that experience-in-practice puts the subject in "process/on trial", a condition in which subjective responses are predominantly determined by biological processes and drives so that an alternative logic is at work. This is the logic of material process and of the unconscious where there is "no time" in the sense of linear temporality, and where the binaries and contradictions of the symbolic do not hold. The knowledge or reality brought about by direct experience is then, an "a-signifying apprehension of a new heterogeneous object" (Kristeva 1984: 202). In creative practice the subject can be viewed as a passageway where there is a struggle between conflicting tendencies or drives in response to external stimuli and matter as it is felt. Moments of resolution or stases are rooted in affective relations – the positive or negative values that emerge through the encounter with objects in the world. This point is crucial to understanding why and how Kristeva places the subject and forms of subjective agency, rather than mechanistic or automatic processes at the core of revolutionary practice. The key term is "affect" – positive and negative affects – that originate in pleasure and displeasure. The centrality of the subject and human agency in meaning-making processes does not negate the agency of objects. Drawing on Sigmund Freud, Kristeva explains that pleasure can be understood as the removal or absence of displeasure. There is a constant struggle between the drive-ridden processes of the *Id*, the pleasure principle and the *Ego*, the reality principle. Negativity and rejection, aspects of the living organism's response to encounters with objects and material forces give rise to sensation. Accompanying raw sensation is *affect* experienced first as fluctuating intensities. On the basis of pleasure or displeasure, these take on positive or negative valency and emerge as feelings that attribute value(s) or that "color" conscious meanings once they are grafted to the symbolic. Hence, the aesthetic image emerges: first as the a result of the flow material forces that occur through the organism's interaction and intra-action with the world, and subsequently as coalescing of subjectivity according to the pleasures

and displeasures of encounters with objects in the world; the aesthetic image is not the artwork *per se*, but a materialization of the object as it is felt and a structuring of symbolic elements in ways that disrupt customary codes. Kristevan psychoanalysis provides an account of thought/language, subjectivity and meaning making as an infinite dialectic – *material process* that involves a two-way movement between matter, biological processes and discourse. The subject as biological organism, capable of thought and language, is a “filter” through which objects and matter pass as raw sensation and are then transubstantiated, if you like, into language.

Antonio Damasio’s account of the relationship between biological processes, emotion feeling and conscious or discursive knowledge in his work, *The Feeling of What Happens: Body and Emotion in the Making of Consciousness* (1999), is also useful here. Although culture and learning may shape the emergence and expression of emotions, emotions intrinsically constitute an embodiment of the logic of survival. They are dependent on brain devices laid down through the evolutionary process and emerge from homeostatic processes that regulate and maintain the efficient functioning of living organisms. Damasio’s explanation demonstrates how such processes are inchoate precursors of conscious thought and language. Dispositional arrangements for the regulation of internal states and that maintain the organisms boundary and living processes – what Damasio refers to as *core consciousness* – are biological antecedents of a sense of self (Damasio 1999: 136). Emotions are complicated collections of chemical and neural responses, which can be triggered by transient changes in the internal and external milieu. Internal states, which occur naturally along a continuum of pain and pleasure, become the unwitting non-verbal signifiers of goodness or badness relative to the organism’s inherent values. Sensory patterns that constitute emotions can be induced non-consciously, but may nevertheless result in particular affects and responses. What we often think of, as the intuitive or non-verbal dimension of aesthetic experience and practice may be more readily understood through Damasio’s elaboration of emotion. Emotions form the substrate of bodily representations, neural patterns and images, which may evolve as feeling and subsequently, as conscious thought. Indeed, emotions together with other processes of core consciousness may be understood as non-language maps of related events that follow the logic of narrative (Damasio 1999: 184). Hence we may argue that affect is “knowing” differently coded. Creative practice allows us to access such codes via the aesthetic image and to make visible knowledge that everyday use of language and discourse hide.

Experiential Knowledge and the Aesthetic Image

John Dewey contends that knowledge is by definition experiential, whether it emerges from art or from science. In illustrating the link between materiality, experience and knowledge, Dewey’s thinking, like that of Kristeva, broadens our understanding of the value of the artistic product beyond the confines of traditional aesthetics. He tells us that knowledge, gained by the artist through everyday living and activity, is captured in the artwork or aesthetic image through creative practice. Though more intensified, the process of artistic enquiry can be likened to what is described

by Dewey as: the flow of experience that runs its course to closure or fulfillment through processes of adjustment to our environment and objects in the world. In artistic experience, as in everyday experience, “action, feeling and meaning are one” (Dewey 1980: 35). Experience involves interaction as a response to discord or lack of adjustment. In the flow of experience occurring in the everyday, or in artistic practice, a problem can be said to emerge when there is a lack or absence of adjustment to surroundings. Initially, such experiences are aesthetic, since they involve sensory responses that are then qualified with emotion, a process by which positive or negative value is attributed to experience.

Scientific knowledge, which is predicated on description or naming, can only constitute *partial* truth about reality. The artwork or aesthetic image ruptures and transgresses the rules and codes upon which naming or the fixing of meaning relies. Its structure or *style* is polyvalent and opens on to new horizons of meaning by short-circuiting accepted codes and ways of looking. This production of knowledge is dependent on interactive experience – a fluid movement between the viewer’s feelings, thoughts and the art object within given social context. Sutherland and Acord describe this as “thinking with art”; they suggest rather that it gives rise to “experiential knowing” (Sutherland and Acord 2007: 125).

The moment of practice implies testing to what degree experiential knowing that emerges from material processes corresponds to, or deviates from what we already know. It is in the heat of creative practice, the workings of material process orchestrate what is laid down as an unconscious mark in a painting or a movement in dance. For this reason, artists themselves, and audiences are often bewildered by work that is “revolutionary”. Kristeva draws on her experience in psychoanalysis to explain that the space of interpretation within creative practice, can be configured in terms of the space of psychoanalysis, where there is potential for a transfer of “knowledge” between the artist as analysand and the work. In the process of interpretation the *work*, or aesthetic image is *itself* posited as the “site” or figure of the analyst reflecting back to the artist perspectives or understandings of the world that may not have been available prior to the encounter. A similar relationship is set up between viewers and the artwork where the work of interpretation or “testing” proceeds as a creative practice in its own right. Through this process, the heterogeneous language of the artwork becomes the site of inter-subjective exchange predominantly controlled by the *structure* of the artwork or aesthetic image. Interpretation and reflection permits the transfer of knowledge to be extended and applied in further practices that occur beyond the original site of making. This will be shown in the final part of this chapter in my discussion of specific case studies. Before proceeding, let us examine more closely, what is meant by “performativity” and how this concept is fundamental to articulating a rationale for creative arts research and for a revealing the limits of the scientific method.

In his work *How To Do Things With Words* (1962), J.L. Austin identifies a class of utterances that he calls “performative”. He explains that these utterances are unlike what he calls “constative” utterances – those statements that are limited to describing phenomena. Performative utterances do not describe or report actions and

events, but actually perform the actions to which they refer. Austin uses the example of the words “I do”, spoken in marriage ceremonies to illustrate this. Hence, his claim that in certain contexts making a statement of an action is equivalent to carrying it out. Austin’s framework suggests that, in some cases, there is no gap between language and the reality it represents. However it does not capture the idea of inventive production that is implied in Julia Kristeva’s theory of creative textual practice. The user of language in Austin’s account does not generate something new, but is a conscious ego who employs words that have socially agreed meanings.

Brad Haseman draws on Austin thinking to make the following observation on creative practice as research:

When research findings are presented as performative utterances, there is a double articulation with practice that brings into being what, for want of another word, it names. (Haseman 2007: 150)

Haseman’s comment does suggest that performative utterances have the capacity to bring forth something new. However it draws some correspondence between artistic research and science in its implication of an unproblematic relationship between ready-made language and the complex processes of production that occur in creative practice. His observation, glosses over the issue of “naming”, one that is at the heart of Kristevan thinking on creative production and which recognizes that there is a gap between what language can describe or signify and reality. Art operates within this gap; this is what underscores Kristeva’s critique of science.

Kristevan thought highlights material processes in creative practice as *pre-linguistic and intra-linguistic processes*. In her account, aesthetic processes occur in relation to both what is known and an *as yet* unsymbolized other (Kristeva 1984: 203). This leads to the conception of a more radically transgressive performativity: performativity as the bringing into view of the as yet unimaginable through the *extension* of language. It is only through the productive material *alteration* of language itself, or the forging of the aesthetic image – that this “unimaginable” may be accessed. What is significant here is the idea that if we cannot consciously know an object without adequate symbols or language, there is a necessity to produce *new* forms of language in order to externalize our experiences or aesthetic encounters. This runs contrary to notions of performativity, either as a speech act, or as a citational practice emerging solely from an already constituted discourse.

We can now assert that performativity in creative production involves an interaction between the subject (artist) as material process, as *being and feeling* – and the subject as *signifying process*, as *sense-making*. This results in a renewal and alteration of both the subject and language. In the making and viewing of art, experience-in-practice materializes or makes available to consciousness, a new object of knowledge. Knowledge then, is not a static entity, but is what Ian Sutherland and Sophia Krzys Acord describe as an interactive *in situ* encounter. This shifts our understanding of knowledge from a passive to an active ingredient of social life. (Sutherland and Acord 2007: 126).

The two case studies to be discussed below will provide more concrete illustration of some the ideas presented here.

Lucas Ihlein, Blogging as Art

Lucas Ihlein's recently completed PhD research, entitled *Framing Everyday Experience: Blogging as Art* (2009), investigates the way in which the practice of blogging can be conceived as an art form, and as such is capable of deepening engagement and attention to everyday experience. Ihlein's research is influenced by the work of Kaprow who pioneered the participatory art form known as "Happenings" that explored the relationship between art and every day life. Though recognized as artistically innovative and politically significant, the interactive and participatory works of Kaprow and others such as the Situationists of the 1950s and 1960s were nevertheless ephemeral staged events that are largely frozen within the time and place of their occurrence. One of the objectives of Ihlein's project, *Bilateral Petersham*, is to extend on the practices of his antecedents by devising a method of recording and reflecting on participatory art and his own art as everyday practice whilst at the same time maintaining the integrity and spontaneity of the process. Ihlein frames his research with the following questions: "If life and art can be successfully integrated what new knowledge might emerge in the process? How does the method of bilateral blogging work to produce aesthetic experience and new insights within the flow of everyday life?" (Ihlein 2009: 5–6).

A requirement of the site-specific project was that the artist should remain within the boundaries of Petersham, a suburb of Sydney for two months and maintain a blog to record his everyday activities and interactions over that period. By devising a new way of making relational artworks and producing an experiential document of the particular environment in which the artwork is situated, Ihlein's project uncovers a new approach to understanding the functioning of aesthetic experience as an integral aspect of the research process.

His method involves application of a spatial frame – the constraints of the specific site of the investigation; a temporal frame, the duration of the investigation and the synchronous unfolding of the blog, and a material/technological frame – the hardware and apparatus required to maintain the blog. These framing devices are used as a method to draw attention to the minutiae of daily life and to provide a record for further reflection, analysis and practice. The latter included an interactive gallery installation and the production visuals and a book of the blog narratives. Crucial to the project is the interactivity afforded by the blogging, which results in a performative and co-emergent production by the artist and other participants from the community. What emerges is not a simple recording or communication of events but a textured patina or impressionistic "portrait" of people places and events. Ihlein's project demonstrates the aesthetic dimension of blogging uncovering aspects of Petersham that are not contained in institutional archives, maps and other institutional records. His research also reveals how blogging can be used as a finely honed instrument for social research.

Beyond articulating a form of practice that blurs the distinction between art and life, Lucas' reflections within the blog and the meta-reflection afforded by the exegesis produced insights and outcomes that have broader application and significance. For the purpose of this chapter, I will focus on one component of the project. A series of walks, which involved the artist's attempt to trace the boundaries of Petersham by walking its borders. What emerges is an *experiential* and embodied map of the suburb disclosing the arbitrary nature of the lines on the official map. Ihlein observes that over the years, local council allocations of land and property development have carved up the terrain in ways that are alienating to human activity. He discovers that the official boundaries indicated on the official map pass through fenced off properties and tenements, cross railway lines and cut through inhospitable highways. At times this necessitates transgressing the imperative to remain strictly within the locale and these deviations reveal uncharted boundaries, interstices and connections. A recording of the experiences of the border walks and chance encounters along the way results in an aesthetic "remapping" of Petersham along *experiential* lines. Significantly, the spatial narratives produced from the walks reveal a different relationship to place: one of belonging rather than ownership of it. This insight is sharpened in an encounter with Aboriginal Elder Uncle Lester who recounts an alternative history of the occupation of Petersham. In Ihlein's own words, the resultant blog "creates a body of evidence that reclaims suburban space as having inherent value" (Ihlein 2009: 109). Institutional rules and static entities such as lines on a map are constantly being written and re-written by actual *behavior*, experiences and activities of everyday life and in this way, experiential knowing transcends institutional knowledge. It is this polymorphous and unruly "knowing" born of experience, that art rather than science is able to reveal.

Valerie Ingham, "Multimodal Research on the Fireground"

The second case study to be discussed examines the broader application and significance of how images function and how aesthetic awareness informs human behavior and the capacity to innovate. Valerie Ingham lectures in Emergency Management at Charles Sturt University and is also a practising artist who presented reflections on her recently completed PhD research in a paper entitled "Multimodal Research on the Fireground" at the Deakin University *Material Inventions: Applying Creative Research Conference* in December 2009.

Ingham's research investigates the role of aesthetic awareness in time-pressured decision making of emergency workers such as fire fighters. Her work indicates that this form of awareness is crucial to decision-making and is the source of transgression from which innovative practice emerges. Her experience as an artist led her to connect two previously disparate fields in developing her research question:

What is the relationship between risk perception, decision making and aesthetic and somatic forms of awareness in Incident Controllers on the fireground? (Ingham 2009: 3)

The research involves interviewing fire fighters to examine their responses and decision-making during critical incidents. On the basis of collected data from incidents Ingham develops the theory of multimodal decision-making, which suggests somatic awareness is simultaneous, holistic and inseparable from the ability to think rationally and make decisions. Those interviewed often observe that their actions and decisions are made intuitively and without consciously thinking. Ingham observes that there are some parallels between the experience of fire fighting and art practice:

In the practice of an artist, spatial awareness may take the form of encompassing multiple images with one sweeping glance, or a concerted 'look'; there is a sorting and comparison between patterns, objects and movement, some of which will appear in the image under construction and some which will not. This capacity to compare and sort elements of images in parallel, resulting in a selection and an understanding of contrast, I argue is comparable to the process of sizing-up for an Incident Controller. In size-up not only are visual pieces of information placed side by side and meanings ascertained, but conflicting verbal reports are also visualised and reviewed in instants of a second. (Ingham 2009: 6)

The process of sizing up described here relates to the impact of matter and sensation; flows of forces between human and non-human agents produce the aesthetic image that opens up new possibilities for action. In studio practice and within the heat of art-working, such action may involve finding the means to externalize, at least in part the internal mental image or the knowing it conveys. In this way, the structure of aesthetic image reveals new possibilities that exceed or transgress established codes.

Because incident controllers recount their experiences after the event, there is a tendency to use rational and chronological narrative to recount their experiences and actions. An objective of Ingham's research was to capture the multimodal and holistic essence of the scene in its entirety by engendering repeated and sustained reflection of the events through retelling. Understandings of arts-based practice emphasizing the artist as both practitioner–researcher and interpreter led to the development of her multimodal research method. By placing the Incident Controllers in the position of artist–practitioner, she was able come closest to their first-hand view of the scene.

Ingham argues that there are aspects of the recognition process which are non-verbal and aesthetic in nature and which cannot be isolated or easily articulated. This underpins her views that aesthetics is vital to risk perception and decision making processes. Amongst incident controllers she interviewed, those with a greater level of aesthetic awareness, or who were able to forge situated and experiential aesthetic images – rather than relying on established procedures and codes – were most successful in making decisions and arriving at unexpected and innovative responses.

In her account of one incident involving a fire at a glass factory, Ingham clearly articulates this:

I suggest that what the Glass Factory Inspector sees is comparable to what Matisse saw when he looked at the seventeenth century Dutch still life painting by Jan Davidsz de Heem. Matisse did not see the photo-perfectionism, he was not sidetracked by the realism and he disregarded the traditional values and rules of representational painting. Instead he cut to the core, went straight to the stark outline of the incident and exposed the composition's true structure. (Ingham 2009: 25)

It can be suggested that Matisse was "struck" by the painting as if by a material object. Similarly, for the Glass Factory inspector, confronted by the awesome splendor of 200 tons of molten glass, a kind of selective vision takes hold, one that focuses on the lines and the raw elements of the scene: this is the aesthetic image or the scene as it is felt. Undeterred by the heat and danger, the Factory Inspector discovers possibilities for action that have not been hitherto envisaged and "wants to go into action with a plan that nobody feels very motivated by – he wants the pump driver to forsake his safe post and get into the action where his skills will be put to greater use; in essence, the Inspector wants to break the 'rules'" (Ingham 2009: 25).

Conclusion

In this chapter, I have examined Kristeva's concept of creativity as a mode of knowledge production that is made possible through biological and material processes. What Kristeva has theorized as the materiality of language, allows us to conceive of *beterogeneous* subjectivity. In the context of creative practice as material process, there is a movement between human and non human elements and between what we may conceive of as the subject or knowing self (the socially constructed "I"), as opposed to the "I" that is articulated by drive or excitability. This process underpins the production of the aesthetic image and its relation to the "transgressive" dimension of creative practice.

An understanding of Kristeva's account of materiality is essential to the critique of science presented here and to the notion of aesthetic experience as an alternative epistemology or ontology of knowledge. The focus on subjective processes elaborated through Kristevan aesthetics, must also be understood in relation to the "agency" of materiality itself. Where traditional and Marxist accounts of materialism ultimately place consciousness at the center of the apprehension of reality, Kristeva's work acknowledges the agency of "brute" materiality or matter. In creative production, there is no opposition between inside and outside: consciousness and materiality are mutually constitutive, enfolded and emergent. It is in this sense that we can begin to speak of a "new materialism".

THE MATTER OF FILM: *DECASIA* AND *LYRICAL NITRATE* Nicholas Chare and Liz Watkins

In his introduction to *Silent Cinema: An Introduction*, the re-titled, revised and expanded edition of Paolo Cherchi Usai's classic work on the early years of the moving image, *Burning Passions*, David Robinson describes Cherchi Usai's interaction with nitrate film in carnal terms. He writes of film stock that Cherchi Usai "will fondle (within the limits of archival propriety) this fabric that can be softer and suppler than the beloved's skin, and draw it to his face to savor the vaguely narcotic fumes. The gestures are sensuous, even sensual" (Robinson 2000: x). Film here is desirable flesh: a body that *matters*.

The underlying materiality of film is usually disavowed in studies of cinema. This is because the existing language of film studies is frequently too focused on predetermined questions surrounding representation and signification to account for the contemporary viewer's encounter with the substance that undergirds the image. This stuff, which until the beginning of 1951 was usually made of cellulose nitrate, is typically composed of a base comprising of a very thin gelatin adhesive layer, and a light sensitive emulsion consisting of a suspension of silver salts in gelatin which adhered to the base (Cherchi Usai 2000: 1–2).

Old film is therefore organic matter; it is as Paolo Cherchi Usai explains, composed of "animal bones crushed and melted into a semitransparent layer intermixed with crystals of silver salts" (Cherchi Usai 2001: 113). These natural forms that combine to constitute celluloid gradually decay over time. Film stock decomposes. This becomes apparent when old motion pictures are screened in the present. This chapter will analyze two films, *Decasia* (Dir. Bill Morrison, USA, 2002) and *Lyrical Nitrate* (Dir. Peter Delpout, Netherlands, 1990), which were created out of found footage of early films that manifest this deterioration. The footage now reveals a materiality, originally far less perceptible, concealed behind the ethereal, evanescent images that appeared on screen.

Drawing on Julia Kristeva's conception of language, it will be argued that the substance of this footage, the decaying nitrate, now actually functions to undermine the grammar and syntax of the films, their symbolic aspect, disrupting cinema's narrative

function and revealing instead something of its carnality, permitting the thing that is film to materialize alternative, often disavowed, meanings. This carnality, the fleshiness of film, resides both in the substance that is the film stock and in the sensual effects of the film upon the spectator. The chapter will constitute a fresh approach to the study of motion pictures, which breaks free from modes of interpretation that are restricted to focusing on the aesthetics of the cinematic image and on narrative form. Instead, the excitations and intensities, the drive-invested underside of representation that comprises the matter of films, will be examined. This disavowed element of cinematic experience (ever present but usually passing unnoticed) is only recognizable if a materialist ontology is embraced and a theory of knowledge developed that looks beyond symbolic language.

Michel Serres has written that “being enclosed in language [...] overwhelms the things which compose our world, and causes them to vanish” (Serres 2008: 88). It will be suggested here that Kristeva’s conception of poetic language provides a theoretical framework able to explain how *Decasia* and *Lyrical Nitrate* resist an overwhelming of the kind described by Serres, instead permitting the thing that is film to materialise. We will begin with an exploration of how Charlotte Delbo’s ideas about “deep” and “intellectual memory” can productively be related to Kristeva’s idea of language’s “symbolic” and “semiotic” aspects in order to explain the significance of the bubbling and shifting surfaces of the film stock used in *Decasia*. The chapter then moves on to a consideration of how the vibrant color fields created by tinting in *Lyrical Nitrate* provide a means for the spectator to access what is usually suppressed in the cinematic experience; the matter of film.

Decasia is, as Ursula Böser emphasizes, “both a film and a symphony” (Böser 2007: 305). It is not an example of silent cinema. The found footage, drawn from numerous archives and of sundry subject matter, was originally screened in Basel in 2001 and was accompanied by the Basel Sinfonietta playing Michael Gordon’s *Symphony of Decay*. The cinematic release of *Decasia*, in which the film and symphony were again synchronized, took place a year later. The musical aspect is noteworthy in that Gordon asked the orchestra to perform out of tune. He also encouraged a continuous sliding of pitches from one note to the next to create a sense of instability. This instability is, as will be discussed later, mirrored in the visuals of the film.

The audience reception of *Decasia* is insightful. In a radio interview about *Decasia* which is included on the US DVD release of the film, its director, Bill Morrison, revealed that people who saw the film frequently said that it reminded them either of the concentration camps, of Hiroshima, or of 9/11.¹ *Decasia* does contain some military imagery including a submarine and some planes dropping parachutists. It also depicts a scene of several figures in a Japanese interior. This *content* partly explains these audience responses to *Decasia* at least in relation to the atomic bombings of Japan. It is, however, also the *form* of the film which contributes to the kinds of reactions the work generates.

In form the film is a collage, an assemblage of fragments, of diverse parts, distant pasts, spliced together, reused to promote a message usually at odds with their initial use and intention. The repeated motif of weaving in this film of films, of a

man spinning, of women working at a loom, reminds the spectator that the motion picture they are watching is itself an act of combining, a working together of separate reels, shots, clips, into a whole. The interlacing also connotes the fates spinning the thread of life, the patterns of bubbling, troubled, troubling images signaling a destiny of decay. There are figures to be seen, the weavers and other characters, children, couples, dancers that endure beneath, behind, within this dissolution. They are perceptible albeit sometimes briefly, momentarily emerging from the morass of corrupted stock only to rapidly submerge again beneath the nightmarish biochemical degradation of the filmstrip.

This makes *Decasia* a film that is, at times, not looked at but rather seen through. The spectator strives to see beyond the decay induced interference, ostensibly superficial, in order to discern the images which are seemingly present behind it. The desire is to make sense out of the senseless swirl of celluloid, to find the figure, the form, the feature the film was made to record. The desire is to make the mess mean, to resolve the shifting textiles on screen into a coherent, a stable text. In *The Pleasure of the Text*, Roland Barthes writes:

Text means Tissue; but whereas hitherto we have always taken this tissue as a product, a ready-made veil, behind which lies, more or less hidden, meaning (truth), we are now emphasizing, in the tissue, the generative idea that the text is made, is worked out in a perpetual interweaving; lost in this tissue – this texture – the subject unmakes himself, like a spider dissolving in the constructive secretions of its web. (Barthes 1990: 64)

Here the desire to retrieve a hidden meaning, to reveal the so-called true image, is continually thwarted. The tissue asserts itself.

The spectator, whilst endeavoring to make meaning out of the disintegrating footage, also at times finds themselves participating in a process of their own unmaking, undoing. Through its poetic language of disintegration, *Decasia* provides a point of access to the semiotic (in Kristeva's sense of the term) aspect of language. It is this (more than the residual, primary content), which causes the work's viewers, (beholders is a better word to use here carrying, as it does, connotations of touch, of intimacy, of closeness) to find it evocative of the horrors of the Holocaust and the atomic bombings. Viewing the film disturbs the boundaries of self in a way that echoes, however faintly, the assault on identity and the accompanying breakdown in language experienced by concentration camp inmates and survivors of the atomic bombings.² As will be discussed in more detail later, the film fosters feelings of abjection in many of its viewers.

Decasia does not contain footage of events such as the atomic bombings and the Holocaust although such footage does exist. Rienhard Wiener, for instance recorded members of a Jewish forced labor detachment being shot in 1941 (Hirsch 2004: 1). There are also newsreels of the liberation of the camps and there is stock of the aftermath and after-effects of the nuclear attacks on Hiroshima and Nagasaki. It does, however, possess an abject quality which makes it such a powerful form of

testimony to horror. This property resides in the material base of what André Habib calls “the flesh of the film” (Habib 2006: 129).

The matter of the film is palpable, corporeal. The scarring and scoring on the footage in *Decasia* are like crow’s feet, those wrinkles which form as the skin ages, spreading out from the corners of the eyes. Film is like skin, or, as Laura U. Marks suggests in *The Skin of the Film*, film can be a kind of skin in that it is impressionable and conductive (Marks 2000: xii). It is a skin that goes the way of all flesh. The decay of film is accompanied by an increase in its tactile qualities, both film and video “become more haptic as they die” (Marks 2000: 172). Marks argues that “every time we watch a film, we witness its gradual decay: another scratch, more fading as it is exposed to the light, and chemical deterioration” (Marks 2000: 172). The grain of degenerating film stock, its increasing texture, encourages a close encounter, one engineered through another skin, the screen which Marks argues is a membrane that “brings its audience into contact with the material forms of memory” (Marks 2000: 243). The memories which are held in the film stock function to activate other preconscious memories in the spectator that will be examined later in relation to abjection.

The marks, Marks refers to, the scuffs, the scratches, each is a mnemonic trace of a coming together, of an impact, caused either by human hand or by the rub of the projection apparatus. These scratches are not, however, like those occasional minor incisions human flesh forgets. Old film is skin that never heals: a palimpsest of perpetual cicatrices. Its dermis continually accrues damage, amassing enduring remembrances of brief encounters. In her book *Days and Memory*, Charlotte Delbo, a survivor of Auschwitz, famously writes about “the skin of memory” (Delbo 1990: 1). Delbo conceives of two kinds of memory, intellectual memory, a kind of remembrance which she describes in *Days and Memory* as “memory connected with thinking processes,” and deep memory which “preserves sensations, physical imprints” (Delbo 1990: 3). It is deep memory that is persistent, that is “unalterable, precise” (Delbo 1990: 2). Intellectual memory is driven by the desire to understand, to *make sense* of things, to give shape to experience, make it signify. The two forms of memory could usefully be aligned with Kristeva’s semiotic (deep) and symbolic (intellectual) aspects of language.

The predominance of binary thinking, which privileges mind over body, reason over feeling, means that intellectual memory is frequently unconcerned with deep memory, with archives of physical feeling. As a *form* of remembrance allied to powers of reasoning, intellectual memory fashions recall, crafts reminiscence through the construction of a historical narrative. This narrative is usually conceived of as objective, as detached from the event in the past which it describes, and as seamless. The possibility of blind-spots in such stories of the past is seldom acknowledged. Hayden White explains, however, that the flawless character of intellectual memory is illusory: a historical narrative is “necessarily a mixture of adequately and inadequately explained events, a congeries of established and inferred facts, at once a representation that is an interpretation and an interpretation that passes for an explanation of the whole process mirrored in the narrative” (White 1985: 51). Intellectual memory, however, despite its deficits, is invaluable in that it is within intel-

lectual memory that conflicts of understanding are played out, that viewpoints are challenged and new perspectives offered.

Delbo's *Days and Memory* provides an example of this kind of memory. It describes an archive that resides in the flesh, in the writer's body, using words upon a page, ink imprinted on paper. This quality of recollection as a thing inscribed, however, reinforces the reliance of memory upon material, upon the materiality of the signifier. Language is material in that it is drive-invested. The drives function as representations of bodily excitations in thought. They operate at the borderline between the psyche and the soma. Kristeva calls them "energy charges" as well as "psychical marks" (Kristeva 1984: 25). They are formed from stimuli within the body that seek satisfaction without. These marked energies – composed of ideas and affects – act as representatives of bodily pressures within the psychic apparatus.

Through the materiality of language, a wrought, highly thought out language that privileges the drives, Delbo *represents* deep memory to us. Representation suggests something "coded in rhetorical, textual or pictorial terms" (Pollock 1988: 6). It acts as a form of mediation between the individual and the world that surrounds them. It is through representation that a sense of separateness from the world is achieved. The mirror-stage, as described by Jacques Lacan, in which the child misrecognizes itself as its reflection and then seeks to live up to this ideal image of itself forms an early part of this process of separation, which is also one of alienation, a process which must be accomplished in order for a sense of self to emerge (Lacan 1989: 1–7). If the process is not navigated successfully then there will be no mediation between self and stuff, there would be no self, a person would be "too much of the Real: infantile or insane" (Orton 1996: 108).

As we will return to later, the "deep memory" that Delbo describes forms part of the Real here understood as Jacques Lacan's first register of psychic reality. The Real is the stage of psychic life at which the figure of the mother and the child are as one. It is therefore, at least for Lacan, beyond difference, before the recognition of discretion between things that comes with the entry into language. The Real is, for Lacan, "absolutely without fissure" (Lacan 1991: 97). For others, however, the Real is like history in that it does contain gaps even if these are repressed. Kristeva, for example, poses the question as to how an infant is able to reach the stage at which it can misrecognize itself in the mirror. Her answer is that "even before being *like*, 'I am not but do *separate, reject, ab-ject*'" (Kristeva 1982: 13). This process of rejecting, of abjecting the mother, is absent from Lacan's account although it is what provides the child with its initial sense of boundaries.

Kristeva distinguishes between this process called abjection and the abject. The abject causes us to experience abjection. It causes us to push it – the abject – away. Some examples of abject material include the shit, spit, mucus and tears. These are waste products which traverse the body from the inside to the outside. They cross boundaries. They trouble the notion of the body having a boundary working to remind us of the time when we did not possess stable contours, spurs to the most troubling of memories, a memory that threatens our very identity as a self. *Decasia* can also be considered abject material in that it can be seen to assume the status of dis-

eased skin. It looks like damaged epidermis and causes the same psychic reaction as actual distressed skin in the spectator. Skin diseases, which Kristeva discusses in *Powers of Horror*, visibly break the boundary between the inside and outside of the body, impairing the cover that guarantees our corporeal integrity (Kristeva 1984: 101–103).

Part of the role of the mother in early childhood is to help establish this border through a constant caressing and touching of the infant. The mother maps out the child's body for it, presents its skin to it. Wounds on the skin threaten this established corporeal integrity. They cause us to remember the time before our body was felt as separate from the mother. The skin of the film shown in *Decasia* prompts similar memories in the spectator. This is what lends the work its horror. This is what provides it with its ability to powerfully evoke the concentration camps and Hiroshima, both places in which visible injuries to the skin were commonplace.

At Hiroshima, terrible burns were caused which led to keloid scarring. These scars form what, in his book *Death in Life: The Survivors of Hiroshima*, Robert Jay Lifton (1991) has called A-bomb Stigmata. He writes that this scarring, deep-burns memorialized in skin tissue, has come to symbolize *hibakusha* identity.³ The opening sequence of *Hiroshima mon amour* (Dir. Alain Resnais, France, 1959) in which the ash-covered skin of victims of the bombing are replaced by the unblemished skin of lovers demonstrates how important the cutaneous is within efforts to figure the event. The bubbling present in parts of *Decasia* is also visually reminiscent of the blistered roof tiles from ground zero at Hiroshima.⁴ In the context of the Holocaust, Primo Levi and Leonardo de Benedetti attest to the various skin diseases caused by life in the concentration camps in the report they co-authored for the Russian authorities in the spring of 1945 (de Benedetti and Levi 2006: 48–61). Their report provides a powerful sense of bodies in decay, of marasmus, of the emaciating effects of the camp environment, its physical horror. This materialization of the horror, its inscription on the skin, is echoed by Robert Antelme in his testimony of his experiences at the concentration camp at Gandersheim. In *The Human Race*, Antelme writes:

There's blood on my shirt, there's blood on my chest, which is red from scratched bites; scabs are starting to form, I pull them off, and they bleed. I can't stand it any longer; I'm going to scream. I'm nothing but shit. It's true: I'm just a piece of shit. (Antelme 1992: 114)

This challenge to Antelme's corporeal integrity, his bleeding, diseased skin, causes him to feel abject, to no longer feel himself as a self, to feel he is nothing more than shit. The decomposing film stock in *Decasia*, a kind of impaired skin, "reminds" spectators of the camps and Hiroshima because it possesses this abject quality. The form taken by this reminiscence is visceral, the viewer feels queasy, nauseous. This physical repulsion at the psychically troubling material that is *Decasia* constitutes a memory trace of the child's loathing toward the abject mother. It is therefore a response the origins of which can be traced back to those provisional separations, which set the infant on the path to selfhood prior to the mirror stage. What happens during encounters with the abject is that a memory of this period of psychic life

is activated. The self is momentarily carried back to the Real and reminded of its fragile foundations. This is why spectators are disturbed, troubled, repulsed by the seething maelstrom of the footage. It threatens to envelop and engulf them. Their nausea forms a kind of protection, a pushing away of this troubling material.

Antelme's account, however, implies that if the encounter with the abject is too powerful then the process of abjection designed to expel it fails and the self comes close to collapse. At times like these, the individual perceives themselves to be abject. Antelme states he is shit. Delbo describes instances when "deep memory" breaks through in the present as creating moments where "...the suffering I feel is so unbearable, so identical to the pain endured there, that I feel it physically, I feel it throughout my whole body which becomes a mass of suffering; and I feel death fasten on me, I feel that I am dying" (Delbo 1990: 3). Deep memory here is the memory of being abject. In a time such as this the "I" is close to death: the self has been brought to the brink of breakdown, it threatens to dissolve like the reader described by Barthes earlier. The experience of this psychic disintegration is also intensely physical.

The sense of abjection a beholder of *Decasia* experiences demonstrates that the boundary between "deep" and "intellectual" memory, between body and mind, is permeable. The two aspects of memory are interlinked such that, to borrow from Elizabeth Grosz, "through a kind of twisting or inversion, one side [can become] another" (Grosz 1994: xii). In *Decasia* "deep" rather "intellectual" memory is brought to the fore. This means that the film privileges the semiotic or drive-invested aspect of language over the symbolic aspect. The drives, as will be returned to later, form a hinge between psyche and soma.

Cinema usually contains a greater symbolic rather than semiotic emphasis. This is because, even in its earliest manifestations, it often has a documentary or narrative function which aligns it the symbolic aspect. Kristeva understands the symbolic to be allied with clear communication, with grammar and syntax, for instance, which operate in the service of transmitting meaning, making sense, as plainly as is possible. The filmic equivalent of grammar and syntax might be conceived of in terms of arrangement of shots, actions of framing, cuts, close-ups, fade-outs, as they all interact, at least in mainstream cinema, to serve as vehicles for particular messages.

Thomas Edison's *Electrocuting an Elephant* from 1903 provides a good example of a film acting as a historical document. It also demonstrates, as will be discussed shortly, how film's documentary capacity can come to be undermined. This short shows Topsy, a Coney Island fairground elephant who had killed three people in as many years, being executed for her crimes. As Mary Ann Doane explains in her analysis of it, the film then was made "to act as an indexical record of Topsy's death" (Doane 2002: 146). She suggests that the "camera simply substitutes for the spectator who cannot manage to be at Coney Island at the appropriate moment" (Doane 2002: 159). It is a form of testimony, a visual account of an event. In her reading of the footage, however, Doane suggests that "the film spectator *also* sees something that the Coney Island spectator cannot see – a break in the film where the camera is stopped and then started again".⁵

It is a break in the act of filming which she interprets as, “functioning to elide time that is perceived as ‘uneventful’ – the work of situating the elephant and binding her in the electrical apparatus” (Doane 2002: 159). The time excluded from the film, the non-event that is the break, reveals a conception of the event as having a “clear-cut and inherent structure” (Doane 2002: 160). It requires decisions to be made about what constitutes the before and after, the inside and outside of an event. The break in *Electrocuting an Elephant*, however, reveals even as it conceals, provides even as it elides. It shows that there is an exterior – a time outside the time of the event – hidden within it.

This stitch in time, a fastening together of other times through a time’s exclusion, an exclusion which becomes a time in itself, a visible temporal jump, foregrounds the filmic experience. It betrays the apparatus of representation, the representative, mediated status of what is being viewed. We are looking from a particular perspective, watching a set of documented decisions, about what is of import, about what is to be exposed. We are seeing intellectual memory in action, seeing the shaping of an event, it making as representation. It is an exercise in intellectual memory. If we could enjoy the footage in *Decasia* as it was first projected, clear if not pristine, we would see similar operations being carried out.

The stock of *Electrocuting an Elephant* as it can be viewed in the present is usually not as distressed as the different footages which feature in *Decasia*. It carries scarring but not to a degree which overwhelms the celluloid. The figure of the elephant is, however, gradually obscured from view by smoke caused by the electrocution. This smoke provides an example of what Michel Chion calls a visual micro-rhythm. He describes these as “rapid movements on the image’s surface caused by things such as curls of smoke, rain, snowflakes, undulations of the rippled surface of a lake, dunes, and so forth – even the swarming movement of photographic grain itself, when visible” (Chion 1994: 16).

The smoke here, the visual traces of burning, and the shadowy figure who passes in front of the camera during the execution of the filming, serve a similar purpose to the scoria that layer much of the footage in *Decasia*. The smoke and shadow work in a similar way to the visible grain of the image carrier in *Decasia* in that they rupture intellectual memory, undermine the film’s documentary function, and provide an opening to physical memory, to a memory grounded in early interactions between mother and child, a memory carefully policed in later psychic life. This is, as discussed earlier, the memory of abjection. The notion of transference, in which infantile occurrences are re-experienced in the present, could provide a means of theorizing what happens to the spectator in these circumstances. The spectator’s feelings toward the film, like those of a patient toward their analyst, express unconscious anxieties that have arisen from early childhood experiences.⁶

Habib has contended that distressed cellulose forms a “new poetry of ruins” which “arises from the history of the archive and the preservation of films, but operates a major inversion: what creates the emotion is not the objective history of film, but time’s presence, which the archive – fragmented, in ruin – can re-present” (Habib 2006: 136). We want to argue, however, that the wracked film in *Decasia* does

not solely act as a record of the impact of time but also performs a revolution in filmic language similar to that discussed by Kristeva in relation to avant-garde poetry in *Revolution in Poetic Language* (Kristeva 1984). Here, Kristeva explores the ways in which poetic language can destabilize the symbolic aspect of signification.

As discussed earlier, Kristeva distinguishes between two aspects of language, the symbolic and the semiotic. Although, John Mowitt explains, they can “be separated only for the purposes of exposition” (Mowitt 1992: 114), the two aspects aggravate and sustain each other in continual oscillation and thereby animate language. The disappearance of either term would be disastrous. The absence of a symbolic dimension to signification would cause the self to descend into psychosis whilst the removal of the dynamic semiotic element would result in stasis and totalitarianism. *Decasia* operates to overturn conventional narrative structures in film and liberate it from culturally concretized laws about the cinematic experience. It does not hide the matter of film, suppress it beneath the imaging of other worlds on screen, but instead foregrounds it.

Through this foregrounding the film also poses a challenge to traditional historical narratives. The black and white footage as seen in the present, signals a relationship to the past. *Decasia* is a historical document. It is, however, not a work of history that conveys facts about the past as seamlessly as possible in the manner that, as White explains, most histories are written. *Decasia* does not unfold across the terrain of “intellectual memory”. The film instead allows the beholder uneasy access to “deep memory”, it forms a physical history that does not endeavor to make sense of a past event but instead to bring into being something of its sensation, which here is abject. This is why *Decasia* is associated with such distressing events from history. It provides an insight into their psychic tenor, their abject, horrific character, what they felt like in the flesh, even as it refuses to represent them.

Decasia disrupts the symbolic register of language and reveals its drive-invested semiotic underside. *Lyrical Nitrate*, however, traces a fault line between the temporal and spatial dislocation of the film as photographic index and the dissolution of image composition through the erosion the narrative imperatives denoted by Delpeut’s intertitles (“Love”, “Death” and so forth). Although *Lyrical Nitrate* provides easier, which is not to suggest less complex, viewing, as a film it also privileges the semiotic but to markedly different ends. Returning to Habib’s sense of “time’s presence, which the archive – fragmented, in ruin – can re-present” and which the decomposition of image and film material can evoke, it is important to note that the effect of colors on the perception of the image in *Lyrical Nitrate* elide a summary interpretation of the narrative logic that could be inferred by the film’s thematic chapters. *Lyrical Nitrate* is created from out of 37 films of the collection, which formed part of Desmet’s legacy and had previously been stored in an attic above the Cinema Parisien until his death in 1956. The selected fragments were duplicated through frame-to-frame transfer and re-edited into ten chapters according to a thematic of Cinema, Body, Passion, Dying and Forgetting. The process of selecting fragments from an eclectic archive is replayed in each film still, through the erosion of image composition as the material disintegrates. The transfer of each frame includes the

imprint of decay and becomes an index of an instant in the continual process of photochemical deterioration.

Through the spectator's interpretation of the arrangement of color in *Lyrical Nitrate* and its narrative reliance on the marks of deterioration, the film foregrounds pleasure rather than pain. The shifting color fields of tinting and toning disturb the spectator's fascination and summon an attention to the film material; the residual marks of decay testify to the fragility of film within the vitality of color in a play on restoration, silence and the past. This film is assembled from disparate fragments to establish a logic and narrative from the thematic associations of image and content. This film-making practice is informed by questions surrounding the preservation of color film from the silent period.

In *Disorderly Order: Colors in Silent Film*, the documented conversations of filmmakers, archivists and historians orientate around the variations in color practice to be found in early cinema. Although certain associations between colors and meaning, such as blue for night could be traced, this pattern, as indicated by Peter Delpout, was not consistent:

Watching these colored silent nitrate prints over four or five years in the archive, my very disturbing experience was that I could find no recipe, no hidden theory, no codes that applied to all the color films I saw [...] Every film is a new experience and any code you find in one film is broken in the next. (Delpout quoted in Hertogs and de Klerk 1996: 23)

The variform color practices of early cinema are further destabilized by processes of deterioration; color fades and individual prints bear the marks of the conditions of storage. Dyes alter as they are repeatedly exposed to the heat of the projector lamp and the chemical composition of the changes affecting the film image as its colors and celluloid base decay. Such marks and scratches offer an index of the temporality of the film material itself, whilst the differences that be perceived between prints in a period when both a colored and black and white version of a film could be circulated, testify to the instabilities of the film archive. For example, in referring to his work as a film-maker and archivist who has engaged with the Jean Desmet collection (1905–1920) held at the Filmmuseum in Amsterdam, Delpout invokes the color and material of film where “the same yellow will have faded differently in different prints. On a print kept in a hot attic the color will have faded more than on a print kept in a cellar” (Hertogs and de Klerk 1996: 67).

Color in *Lyrical Nitrate* is non-indexical. Both stencilling and tinting involve the application of dye over a photographic image, whilst toning refers to the manipulation of a chemical reaction in the silver nitrate to produce a monochrome effect. Toning is characterized by a reduction in contrast which reduces the resolution of fine detail of the photographic image and so tends toward silhouettes and shadows within a monochrome color field. Delpout comments on the affectivity of color fields in reference to the production of another found-footage film *The Forbidden Quest* (Dir.

Peter Delpout, Netherlands, 1993), which constructs as fictional polar expedition from tinted and toned found documentary footage.

Cutting from the archival monochrome to the black and white shots, we'd get the monochrome colour sort of persisting in the black and white. That made me realize that monochrome colour has a very strong physical effect. Maybe you forget it, but it's still there, something is happening between your body and the screen. There's more to colour than something we can interpret as we watch a film, there's something physical too. (Hertogs and de Klerk 1996: 60)

Color is significant for the *way* it is introduced as a chromatic field and evokes a response in the viewer that cannot be entirely explained at the level of the symbolic. The color field registers, is touched by decay, which affects both the dye used to produce it and the film stock itself. Within a found-footage film which infers new meanings by drawing a film fragment into a new context, Habib's play on "time's presence" can be discerned at the levels of process and the spectator's interpretation. Color fields are layered over the photographic index, whereby each film still registers as an instance and is then veiled by the cinematic illusion of movement, thereby summoning a response that is both physical and intellectual. Color draws attention to or instigates a movement toward the film material and process, the apparently unchanging monochrome image as an illusion of stillness foregrounds the minutiae and flicker of detritus and decay. The specificity of the way color is introduced offers an additional layer of significance, which is supplementary to the symbolic function of color itself.

This practice has significance for both historical and political discourse, indicating the necessity of preservation whilst foregrounding the complexities of an archive which is in flux. Penelope Houston, for example, reveals in her comment that "archives influence taste by what they choose or are able to make available; even more by what they put their weight behind" (1994: 131). The processes of (de)selection are inherent in archival processes; even storage has conditions. There are monetary implications in abeyance which are similarly implicit in the formation of knowledge for which access to archival materials plays an integral part. As Cammaer has noted, the problems encountered and the questions inherent in archival work inform Delpout's film-making practice (2009: 371–373).

The spectator's interpretation of *Lyrical Nitrate* encounters the economic implications of the film archive, where the ongoing costs of storage and preservation impact on access to the film and the object of study. In *Lyrical Nitrate* the marks of decay are drawn into a film-making practice where the associations between image resolution, decay and pictorial content, participate in and elide the formation of meaning at the level of the narrative. The film formulates new scenarios and texts in which the shadows of the photographic are reactivated and reframed in way which indicates *Lyrical Nitrate* as a comment on the film archive itself. The spectator's interpretation of the film in decay and of details hitherto concealed from view is drawn into an engagement with the film archive itself, where both the materials and content are of

significance. For example, in the chapter titled “Passion” a shot of woman stepping back, her arm stretched out and her hand pressing against the back of a chair as she sinks into it in grief is replayed as details are fragmented into new framings of the same image. At first the area of the frame tracing the gesture of her facial expression and then the pressure of hand wracked against the upholstery are replayed in slow motion, altering the spectator’s perception of the sequence beneath the wash of music. Similar manipulations which play on the “silent” film era can found in the music which accompanies the journey of a steam train through the color fields of landscape which turns into the crackling rhythm of a record rotating; the sound of movement alludes to that of the locomotive gradually slowing down against the rails. The lyricism and rhythm of fragments edited and reviewed in Delpeut’s film opens the possibility of new meanings which are evocative of a Kristevan poetics as the undoing of the symbolic.

In order to understand the chromatic significance of *Lyrical Nitrate* it is useful to turn to another work by Kristeva, the essay “Giotto’s joy” (Kristeva 1981: 210–236).⁷ This work provides an extended analysis of Giotto’s fresco cycles at the Basilica of St Francis in Assisi (c.1297–1305) and the Arena Chapel in Padua (1305–1308). Kristeva is interested in tracking the way in which the drives are translated into colored surfaces. The drives are, as mentioned earlier, described by Freud as at the borderline between the psyche and the soma (Freud 1981: 83).⁸ Kristeva shares this view, calling the drives “energy charges as well as psychical marks” (Kristeva 1984: 25). They are formed from pressures within the body that seek satisfaction without. These marked energies – composed of ideas and affects – act as representatives of bodily pressures within the psychic apparatus. They speak for the body, for matter, in thought. They speak for the body through what Kristeva calls the genotext in literature. The semiotic and symbolic aspects to language manifest themselves in literary texts through the phenotext and genotext. The phenotext is allied to the symbolic whilst the genotext is associated with the semiotic. The phenotext describes that part of the text concerned with communication whilst the genotext refers to the style by which that communication is actually carried out. The genotext contributes to meaning but is in excess of meaning or an excess to meaning. These forms of text can be transposed to the visual field where color usually contributes to the genotext.

For Kristeva, in “Giotto’s joy”, the drives speak for the body through color in painting. Color carries the energetic pressure of the drives within it. The same can be said of film. Color provides cinema with a carnal dimension. Although in a painting color carries the form of the composition, in early cinema, as previously discussed, color is layered over the photographic imprint. The affectivity of color in *Lyrical Nitrate* however is evoked through the assemblage of the film as it draws attention to the process of its production. Through the processes of tinting and toning color appears to be formless revealing the surface of the film material. The layering of dye over a detailed composition in certain sequences evokes the incompleteness of the representation. In the chapter entitled “Dying” a series of shots traces the careful steps of woman through a conservatory corridor, the play of light through windows, like sound through glass is dulled by the monochrome blue layer; the music has

ceased. As she opens the window, bubbles of deterioration consume the discernable composition of the image. This sequence, unlike much of *Lyrical Nitrate* is not accompanied by the music which for Cammaer is presented in “a way that expresses decay” (2009: 374). For example, the film continues without sound as a man bearing a gun traces a similar journey which is inter-cut with the “silent” image of a dog barking through the viscous layer of the color blue. The editing suggests the dog is to be silenced, whilst conversely “noise” manifests in the scratches and marks that scar the dermis of the film.

For Fredric Jameson, dislocations of sound and image operate both as a tension and affinity; a calling of one sense by another in which the privileging of the visible is disputed. This synaesthetic relation is not metaphorical, but a characteristic of differentiation and contamination, that is “a more tenuous form which prolongs touch” (Jameson 1995: 4). Jameson initially offers the literary examples of “green so delicious it hurts” and “the peculiar dullness of a drop of water” (Jameson 1995).⁹ The juxtaposition of sound and image in film invokes a similar affinity that is, offers a configuration of cinematographic elements that are proximal to interpretation. The spectator’s response to the incompleteness of the depiction of the dog’s anxiety, to the narrative interference in the film, his “silent” bark in which the image in decay appears to function as visual noise, constitutes such a proximal point through which the film reveals its lyricism, its genotext.

The illusionary stillness of the monochrome color fields in *Lyrical Nitrate* are drawn into a rhythm of editing which makes the process of (de)selecting material visible. By intermittently drawing instances of decay into a thematic narrative, the rhythm and lyricism of editing plays on the “silence” of a film and archive hitherto concealed. Each image registers an instance in color fading and the variables of image (de)composition that is of an internal movement which occurs at level of the film material. The film substrate, which is integral to and yet more usually concealed within the cinematic image, continues to disintegrate. In this sense the gesture of incomplete representations created through color and sensation in the image of decay summons a response to the film material and reveals the fractures of a film narrative and logic in flux, a process which for Kristeva, traces the drive invested underside of representation.

In conclusion, therefore, both *Decasia* and *Lyrical Nitrate* provide distinct ways for accessing the semiotic register of language, that material aspect which comes closest to bring a listener/spectator into contact with the undifferentiated stage of psychic life that is the Real. The semiotic privileges the drives (representatives of the body in thought) and therefore provides a carnal dimension to language. In moments where the symbolic breaks down, the drives, the body impresses itself upon consciousness. This experience can either take the form of extreme psychic pain, abjection, or an overwhelming pleasure, joy. In *Decasia* the emergence of the semiotic via distressed film stock is of a disturbing nature. In *Lyrical Nitrate*, by contrast, the poetic play of the fields color created through the practice of tinting is pleasurable for the spectator but tinged with melancholy. Both these experiences are, however, united in their capacity to make film *matter*.

RECOVERING THE HIDDEN THROUGH FOUND-FOOTAGE FILMS

Dirk de Bruyn

In the early 1970s materialist experimental film was cogently rejected by feminist theorists for its inability to deliver a feminist counter-cinema addressing its political agenda. The concomitant development of feminist psychoanalytic readings of “dominant cinema” against its grain also discounted such work. This split is marked by Peter Wollen’s formulation of “two avant-gardes”, one narrative and explicit about its political position and the other non-narrative and focusing directly on implicit perceptual processes. Materialist film’s fixation on structure jettisoned content, and extended post-war painting’s essentialist move to pure abstraction manifest in abstract expressionism and minimalism. The emergence of trauma theory and the recent explosion of moving image digital media with its non-linear bias and the complex layering of “technical images” have created a new situation opening up alternate readings of such discounted materialist practices. As well as a historic precursor for digital media, it is suggested that a materialist cinema, represented here by the found footage films: *Alone: Life Wastes Andy Hardy* (Arnold 1998) and *Dreamwork* (Tscherkassky 2001), signposts a belated return for materialist film within the context of trauma studies. This materialist turn rescues such experimental film from its traumatic excision and extends an understanding of what has been termed a “trauma cinema” by Janet Walker. Rather than pure, abstract or visionary such practice is read here through trauma theory as performing implicit mechanisms of denial and erasure. It is no wonder that materialist film was avoided in the seventies by a feminist project concerned with moving out of the debilitating “victim” positions women identified as their lot. In the seventies women had had enough of direct experience of denial and erasure. The view of “content denied” enables the practice of materialist film to speak to the laying bare of the debilitating traumatic flashback experience in a way not available to the narrative approach of Wollen’s second avant-garde. To further legitimize materialist film’s link to trauma, correspondences are drawn between Wollen’s two strands of implicit and explicit film-making practices and Brewin’s neurological research into trauma which clarifies the interaction be-

tween two parallel memory systems; one implicit (Situational Accessible Memory) the other explicit (Verbally Accessible Memory).

Historical Background

The history of artists working with the moving image forms a significant dialogue with technology that has been variously described as an experimental, avant-garde or non-narrative or more specifically, as a formalist or materialist practice. (The more specific term “materialist film”, that has evolved through Peter Gidal’s practice and theoretical writing (Gidal 1989) will be the favored term used in this chapter.) John Hanhardt suggests the pre-reflective or non-narrative space has been a traditional area of interest for avant-garde film: “The avant-garde continues to explore the physical properties of film and the nature of perceptual transactions which take place between viewer and film” (Hanhardt 1976: 44). Maurice Merleau-Ponty’s phenomenology’s perceptual emphasis is particularly suited to frame and describe this materialist practice: “a movie is not a thought; it is perceived” (Merleau-Ponty 1964: 54).

As one of its practitioners and theorists, Malcolm Le Grice points out that 1970s materialist film’s “action on the automatic nervous system seeks to create a nervous response which is largely preconscious” (Le Grice 1977: 106). Such interest corresponds to the primacy Merleau-Ponty assigns to perceptual processes and that pre-reflective space before thinking occurs. The implicit register of this work is linked to recent neurological research into the effects of overwhelming experience on memory processing as discussed later in the body of this text.

The specific area of found and stolen images and their re-processing within such an experimental practice can be experienced as transforming the originating material and emphasizing aspects previously hidden. According to William C. Wees such avant-garde found-footage films “are intended to ‘undo’, deconstruct, or ‘detourn’ images produced and disseminated by the corporate media” (Wees 2002: 4). Such an intervention empowers a marginalized and impoverished practice to work with images produced by a commercial film and television apparatus to which it has no economic access, but to which its practitioners are nevertheless subjected to in daily life. Such a “detournement” performed on historic material is examined in the recent film work of Peter Tscherkassky, *Dreamwork* (Austria 11 minutes 2001), and Martin Arnold’s *Alone: Life Wastes Andy Hardy* (Austria 15 minutes 1998).

Alone: Life Wastes Andy Hardy reprocesses sequences from the *Andy Hardy* series of black and white features (1937–58). This series, designed for mass consumption by a general audience, reflected back family values and family life to nineteen forties and fifties Middle America. In 1939, 1940 and 1941 this MGM series made Mickey Rooney the most popular Hollywood star despite its comparatively low budget and low-brow register. The series charted Andy Hardy’s growing up years through childhood and adolescence. Judy Garland starred in three of these films as Andy’s love interest. Tales of small town morality are resolved with man-to-man discussion between Andy and father Judge Hardy. The films pioneered a formula that later became the staple of general television viewing. Arnold samples this resource and focuses in on perceived ideological cracks in this material.

The second film to be considered, *Dreamwork*, re-edits sections of the nineteen eighties horror color feature film, *The Entity* directed by Sydney J. Furiein (1982), whose title sequence asserts that its story is based on real events. *The Entity* follows the drama of a woman (played by Barbara Hershey), a single mother, who is raped by an invisible assailant and is not believed. The assailant is later identified as the devil. In her struggle to be believed Hershey's character comes into contact with a cluster of skeptical psychiatrists who dismiss her as a hysteric plagued by her "single" status. There is also a group of ghost-busters who try to set an elaborate but misguided trap for the entity. *Dreamwork* is Peter Tscherkassky's compacted re-configuration of Hershey's relationship to a home environment saturated with para-normal activity and a recurring rape experience. By re-editing and layering shots and parts of imagery from the film over the top of each other he creates a complexity that overwhelms and challenges the viewer. In both *Dreamwork* and *Alone* narrative is fractured to uncover the hidden from within the originating Hollywood melodrama. Each film is examined in turn in relation to trauma and placed in a larger historical and cultural context.

Alone

Martin Arnold identifies his practice as one of revealing the suppressed:

In the symptom, the repressed declares itself. Hollywood cinema is, as I have already said, a cinema of exclusion, denial, and repression. I inscribed a symptom into it, which brings some of the aspects of repression onto the surface, or to say it in more modest words, which gives an idea of how, behind the intact world of being represented, another not-at-all intact world is lurking. Maybe this is my revenge on film history. (MacDonald 1994: 11)

In *Alone* Arnold repeats obsessively sampled extracts and gestures from the Andy Hardy series of films. Arnold's repetitions are sometimes as little as a couple of frames long, turning the actors into tic-riddled puppets with accompanying stuttering soundtracks reminiscent of scratch video and other sampling strategies employed in contemporary experimental music. Judy Garland's song, in its repetitions and looping movements, unearths new meanings and moves the performance from the lyrical into the fragmentation of concrete poetry. Through similar technical manipulations a Rooney and Garland kiss is transformed into a long and tentative primal grimace. Similar sexual innuendo is unsettlingly teased out of a kitchen scene between Hardy (Rooney) and his mother as they rub up against each other ad infinitum while washing and drying the dishes.

The back and forth scrubbing of gesture evident here is not comedic, but hones in on a melancholic moment of fracture in the gesture itself. It is through such fissures that the hidden obsessions of the characters emerge to be inscribed or re-inscribed into this obsessively stretched out, repetitive performance of selected sequences from the film. The archaic technique of the cut becomes the film's central character. Though its directness and as a trace of the amusement parlor of pre-cinema and a

cinema of attractions where “an attraction” aggressively subjects the spectator to “sensual or psychological impact” (Gunning 1990: 59), Arnold’s hyper-editing unleashes a psychoanalytic joyride inside a phenomenological framework.

Arnold has stated his interest in this obsessive repetition as a way to bring back or reveal hidden elements. These repetitions perform as tics and stutters in the film. In Arnold’s psychoanalytic reading, a tic moves over, works against an opposing movement at the visceral level of the body itself, in stuttering “a message that is in conflict with what is being said wants to be expressed” (Arnold in MacDonald 1998: 362). For Deleuze stuttering “puts language in perpetual disequilibrium” (Deleuze 1998: 111). As well as stigma and anxiety, stutterers tend to have “greater emotional reactions and more problems with flexibly controlling attention and emotion” (Prasse and Kikano 2008: 1272). Tics and stutters offer rich sites of ambiguity, conflict and multiple meanings. Through such breakdowns of expression the body performs the unspeakable. Through such repetitions, tics and stutters, *Alone* can be read as the baring of an unspoken trauma resiliently embedded under consciousness, endlessly replaying as some undecipherable situation. Arnold’s technique is about the obsessive location, repetition and making visible such scarring moments.

As a filmic field of a thousand cuts this trauma is read productively as a societal trauma rather than a personal or individual one. Kirby Farrell suggests that the “contagiousness” of personal traumas can lead to a general social impact “when particular social conditions and historical pressures intersect” (Farrell 1998: 12). It is such a trajectory from the self to the social that Maggie Humm identifies for feminism in screen studies in the seventies when she points out that “understanding the personal as political – how identity is constructed and represented – is the task of feminist theory” (Humm 1997: 179). It is this discursive formation that in recent times has accommodated trauma theory within its discourse, initially through the work of Shoshana Felman, Dori Laub and Cathy Caruth (Radstone 2001: 188).

Caruth’s view on repetition offers some insight on the perceptual operation of *Alone*. The flashback repeatedly returns trauma’s suffering. In normal processing of an event, the instant of “seeing” becomes “knowing”; yet in trauma this instant is seen but remains unknown, returning over time repeatedly to an unknown or “unspeakable” gap. These repetitions suggest “a larger relation to the event which extends beyond what can simply be seen or what can be known and is inextricably tied up with the belatedness and incomprehensibility that remain at the heart of this repetitive seeing” (Caruth 1997: 208).

Correspondences can be drawn between Arnold’s found-footage practice, the 1970s feminist use of textual analysis and their shared mobilization of psychoanalysis. For one, Arnold trained in psychoanalysis at its Viennese source. In his film practice he translates into visual and performative form the strategy of textual analysis developed via feminism in screen studies. Generally textual analysis operates to expose the unconscious or repressed through identified fissures in the smooth operation of a particular ideology. As Annette Kuhn points out, within feminism it was used to uncover the work of patriarchal ideologies in mainstream cinema and show that “such disjunctures are discernible within the text in the form of ‘symp-

toms' – cracks, ruptures and so on. These symptoms provide us with clues as to what, ideologically speaking, is going on" (Kuhn 1982: 86). Arnold has described his practice in constructing *Alone* in very similar terms at the beginning of this section. Within screen studies, such a psychoanalytically grounded textual analysis arose in the nineteen seventies out of an appropriation of French critical theory that included Lacanian psychoanalysis. As one of the initiators of this tactic Laura Mulvey stated that "psychoanalytic theory is thus appropriated here as a political weapon, demonstrating the way the unconscious of patriarchal society has structured film form" (Mulvey 1975: 6).

Textual analysis's migration and metamorphosis from written into visual and performative form re-emphasizes the pre-reflective moment. *Alone*, in its tics and stutters continually and obsessively returns to the immediacy of the perceptual event from which its analysis is read. As viewers we seem to be in the same immediate space in which the critic or theorist (or the artist) is sitting in front of the film-editing machine. In this moment and space of genesis, the critic is actively stopping and starting the film, re-inspecting frames and sequences, creatively grazing. He or she is at the point of constructing critical understandings on the film being "analyzed".

We have moved from one way of thinking to another. Merleau-Ponty's phenomenology's ability to theorize and describe this pre-reflective realm of perception accurately recognizes this space:

My field of perception is constantly filled with a play of colors, noises and fleeting tactile sensations which I cannot relate precisely to the context of my clearly perceived world, yet which I immediately place in the world, without ever confusing them with my daydreams. Equally constantly I weave dreams around things. (Merleau-Ponty 1962: x)

This view produces insights unavailable to psychoanalysis. The viewing experience is one in which the repeated performance of these cracks and ruptures impact directly as a pre-reflective stun. Does not such a stun itself produce anxiety, fear and deliver back the trauma rather than the memory? There is an unresolved tension between a re-constitution and a stun or stutter. What comes back in this break? How easy is it to think, to negotiate when you have just been "stunned"? Analysis comes after.

Dreamwork

Tscherkassky's title of *Dreamwork* references the psychoanalytic method. Like Freud's *Interpretation of Dreams* (Freud 1971/1900), Peter Tscherkassky articulates his technique as unearthing something underneath narrative:

It's like digging. You have something like a landscape, and you know that there is something covered, and you dig it out- in that sense 'archaeologist'. It's not primarily about ideology, but you have that sense of uncovering new meanings. (Blumlinger 2002: 25)

As well as this digging into this material, Tscherkassky's film moves beyond *Alone* by charging a new visual complexity into the original material. The multi-layered imagery of *Dreamwork* suggests Merleau-Ponty's world of colors, noises and fleeting tactile sensations. Having identified "Arnold's" fissures and cuts in the narrative, Tscherkassky layers these sequences on top of each other, creating a compact multi-layered field of movement and re-presentation. We see images of a room, a window, Hershey's face and body, we glimpse snatches of movement, see strips of film and other textured artifacts (rayograms of nails and grains of sand) often as afterimage, flickering and in negative and all at the same time. This layering suggests that the film and its narrative content is being materially shattered before us.

Through his technique of "manufacture", "contact printing and the simultaneous manipulation of the elements" (Horwarth 2005: 44), the scene of the rape is re-directed back into the audience's face. The aesthetic cacophony of multiple meanings being presented concurrently or simultaneously mirrors the psychic disorder of the rape experience. This is a strategy of construction that recreates the force of the originating event directly and imparts a sense of overwhelming trauma. Given the nature of the originating material, Felman and Laub's approach to trauma as articulating an "an event without a witness" (Felman and Laub 1992: 75) is pertinent here.

Is what is presented here a compact narrative of dreams as the title suggests? As with *Alone*, it can indicate a trace of something more immediate. This is not an imagined rape that is being communicated but something more real, direct and immersive. It is more about the "here and now" than the processing of dreams. *Dreamwork* can be read as a perceptual performance that re-enacts the rape directly by bringing us back to the pre-reflective cacophonous moment of its initiating, implicit bodily impact.

In describing his strategy of folding the narrative back on itself Tscherkassky appropriates Maya Deren's concepts of vertical and horizontal editing:

What I really try to do is convert the horizontal structure of a narrative film into a vertical structure. This was something that Maya Deren pointed out... well if you take the narrative structure of prose, you have the story unfolding on a horizontal line. And you have poetry, where you may have, within every single word, several multiple meanings, in terms of connotations. (Blumlinger 2002: 26)

Deren uses the horizontal structure of a film to refer to its linear cause and effect progress, its narrative or story. But she is also acutely aware that an event can be read in multiple ways, has various levels of meaning so that an event can be returned to repeatedly to reveal these layers. This Deren describes as an event's vertical structure:

The poetic construct arises from the fact, if you will, that it is a 'vertical' investigation of a situation, in that it probes the ramification of a moment, and is concerned with its qualities and its depth, so that you have poetry concerned, in a sense, not with what is occurring with what it feels like or what it means.

A poem, to my mind, creates visible or auditory forms for something that is invisible, which is the feeling, or the emotion, or the metaphysical content of the movement. (Sitney 1970: 173–4)

Deren demonstrated and applied such a relationship between the vertical and the horizontal in her *Meshes of the Afternoon* (Deren 1943, USA, 15 minutes). Through its similarities *Dreamwork* can be viewed as a contemporary response to *Meshes*. Stated very simply, *Meshes* shows a woman entering and moving through her flat. This situation is serially re-iterated from different points of view. These views are represented by her multiple selves that finally appear together onscreen. *Meshes* has been read as unpacking the traumatic thinking and dissociation of a suicide. Kaplan observes that Deren's film:

Produces a visual correlative to the subjective, emotional and visual experience of trauma, leaving the situation uncertain and to be deduced by the viewer. That is, the entire world of the film is 'inside' the traumatic experience. (Kaplan 2005: 125)

The physical difference between Tscherkassky and Deren is that Deren places her multiple readings of the one horizontal or linear event in series, one after the other, whereas Tscherkassky gives them to you overwhelmingly compacted and layered, all at once, as a complex field of moving images. The experience of the vertically charged moment for the viewer is "here and now" and has to be graphically unraveled. In *Dreamwork* is as if the pre-reflective moment of, in this case, the rape is re-constituted re-assembled like a raw flashback that is experienced as REAL.

From a neurological perspective the traumatic flashback takes "the form of relatively brief sensory impressions such as images, sounds, tastes or smells which are accompanied by the original emotions that the individual experienced at the time of the event" (Kleim, Wallott and Ehlers 2008: 222). Through this compartmentalized, fragmented nature, traumatic memories are implicit, unspeakable. "In some people the memories of trauma may have no verbal (explicit) component at all; the memory may be entirely organized on an implicit or perceptual level, without an accompanying narrative about what happened" (Van der Kolk 1997: 255).

Maureen Turim makes the connection between the flashback in film and trauma. She notes that in melodrama the cinematic flashback breaks settled narrative to mark the visceral and unexpected return to the senses of a significant or traumatic experience. She identifies twenties European avant-garde films as instrumental in developing its form: "similar abrupt flashbacks mark 1920s avant-garde films" (Turim 2001: 207). These are the same films claimed by experimental film-makers as initiators of a non-narrative tradition (Michelson 1978: 175–6). In *Dreamwork* Tscherkassky pays particular homage to twenties avant-gardist Man Ray, whose Dadaist work *Retour à la Raison* (1923) is considered an important contribution to this cinematic form "which can function essentially on the psychophysical rather than the psycho-interpretive

level" (Le Grice 1977: 106). Tscherkassky explains: "I added some of Ray's more famous objects- needles, tacks and coarse salt- to my found material and interpreted these objects as sexual metaphors within the framework of the plot of the film" (Tscherkassky in Horwarth and Loebenstein 2005: 158).

Neurology

Deren's concept of vertical and horizontal editing and Tscherkassky's use of it are ways of thinking about the relationship between implicit and explicit memory systems that predicts recently developed models of memory processing. Recent neurological research summarized by Bessel van der Kolk (Van der Kolk, McFarlane and Weisaeth 1996) into the processing of trauma, dissociation and the flashback throws some light on the nature of these interactions. He points out that Pierre Janet, working with the victims of shell shock in the late 1800s showed that such shock or trauma is precipitated by severe emotional responses and that such responses effect how memories are stored in a fragmentary manner: "Intense emotions, Janet thought, cause memories of particular events to be dissociated from consciousness, and to be stored, instead as visceral sensations (anxiety and panic), or as visual images (nightmares and flashbacks)" (Van der Kolk 1996a: 214).

Brewin Dalgleish and Joseph's (1996) clinical research into memory processes in post-traumatic stress proposes a dual processing model in which there is a dialogue between two memory systems – verbally accessible memory (VAM) and situationally accessible memory (SAM) – to help explain such traumatic responses. Verbally accessible memory (VAM), also referred to as declarative memory (Squire and Zola-Morgan 1991, Van der Kolk 1996a: 285) involves the "encoding and storage of conscious experience" (Brewin 2001: 161). Verbally based, VAM enables narrative with retrieval upon request. Because it is linear and consequential in assembly its process speed is limited. This has the same effect as the impact of low bandwidth in computer technology. VAM exhibits a strong sense of time. The hippocampus is involved in the formation of conscious memories, of building up a unified "cognitive map" (Van der Kolk 1996b: 295) that allows flexible access to these memories. These neurological structures enable objective or reflective thinking.

"Situationally accessible memory" (SAM) or implicit memory allows no retrieval upon request and displays no sense of time. You do not have "at call" access to your own stored experiences. It is the situation that triggers the memory that is re-experienced rather than recalled. This describes the unexpected flashback triggered by external cues or thoughts in traumatized individuals. SAM is "unable to encode spatial and temporal context" (Brewin 2001: 161). It focuses narrowly on risk and is detail rich. According to Hellawell and Brewin (2004), SAM consists of "the exclusive automated mode of retrieval, the high level of perceptual detail, and the distortion of subjective time, such as the event is experienced in the present." (Hellawell and Brewin 2004: 3) This memory system is aligned with Merleau-Ponty's notion of the pre-reflective and with subjective experience before thinking. These memories are processed through an older part of the brain known as the amygdala.

The amygdala's functions are not flexible and are concerned with attaching affect to incoming cues and the "establishment of associations between sensory stimuli" (Vander Kolk 1996a: 230).

How do "samming" and "vamming" interact when these two systems are operating normally in parallel to each other? When you have a conversation, for example, or travel from A to B, information is laid down in both memory systems. You may recall or explain what you have done and seen (vamming) but there are certain gestures, impressions that somehow do not fit in and are not part of the recounting. Such moments may come back to you uneasily or unexpectedly (samming) and belatedly. These flashbacks may pre-occupy you. You may talk to someone about them until, somehow, you re-work or reconsider these impressions, integrating them into a cohesive "story" with the events already understood. In this way visual replay (flashback) is rehearsed and then inserted into narrative, moving from the SAM to the VAM memory system.

Metcalf and Jacobs (1998) have identified that high levels of arousal (trauma) breaks down hippocampus functioning and inhibits vamming so that, when the situational memory returns there no narrative for this flashback to be inserted into. Like a broken record it has nowhere to go and is destined to try again later. Rich in detail with no temporal context, such "affect fragments" periodically redial into a network that was never built and jolt the receiver as if they are happening right now.

This model can describe how *Dreamwork*, for example, as a "pre-narrative" film can be experienced with its immediate narrative context stripped away. It is in the shock of the perceptually rich performances, in their immediacy and directness that the film is encountered (or more accurately perceived) in samming mode. The film has been constructed by working backwards, folding back to the pre-reflective moment. In Tscherkassky's technique of construction its integrated vamming (explicit) memory is broken down into its high definition samming (implicit) fragments. The perceptual and making sense process of being in the world is retraced back to the pre-reflective perceptual moment of "first contact". We experience this moment of first contact with an overwhelming immediacy.

This relationship between vamming and samming also re-calls the relationship between verbal and visual thinking. Brewin's model is also a much more systemic and dynamic model than the left brain/right brain oppositions that Small (1994: 6) employs in his argument in *Direct theory*. Here Small argues that the visual reflexivity that occurs in experimental film as a form of theorizing with a right brain emphasis. It is also more developed than the old brain/new brain dichotomy used by Len Lye to talk about his "doodling" film work as old brain work (Horrocks 1979: 33). As previously emphasized, Maya Deren's thinking on vertical and horizontal editing also resembles Brewin's dichotomy. The body-centered samming and the cortex-centered vamming can also be applied to the perennial mind-body split present within Western culture. Traumatic experience itself in fact has been conceptualized as an extreme mind-body dissociation.

The Two Avant-Gardes

Peter Wollen's influential article "The Two Avant-gardes" (Wollen 1982), first published in *Studio International* in 1975, places the avant-garde into two camps that uncannily resemble the two memory systems that Brewin's Dual processing model identifies as SAM and VAM. Wollen's dichotomy is performed at the point of emergence of feminist film practice and its critical textual analysis of dominant cinema. Partner and film collaborator Laura Mulvey's influential feminist text on male scopophilia in cinema also performed an initiating function in these developments. At this stage Wollen and Mulvey had already collaborated on *Penthesilea* (1974), which according to Al Rees "attempts to construct feminist discourses in a triangulation of Marxism, semiotics and psychoanalysis" (Rees 1999: 91).

Wollen's text splits the avant-garde in two. The first comprises non-narrative and formalist (materialist) work residing in the multi-voices and collective emphasis of the artisanal "co-op movement" that emerged out of the fine arts (painting, sculpture) and the second embodied a more politicized narrative practice employing psychoanalytic and Marxist ideologies in its analysis, with a relation to literary criticism, theatre and the margins of commercial cinema. One focuses on vision, perceptual processes and the image and the other emphasizes the social implications of the text and language.

Wollen observes that a similar split can be seen in the twenties between the Cubist influenced cinema of Leger, Richter and Man Ray to extend the scope of painting and a Russian avant-garde in which Wollen includes, Eisenstein, Dovhenko and Vertov (Wollen 1982: 24). This Cubist cinema identifies those same films that Turim suggests display the characteristics of the flashback in melodrama and contains the work of Tscherkassky's exemplar, Man Ray.

Interestingly, the contrast in qualities between Wollen's two avant-gardes, the visual and visceral, direct impact nature of materialist film and the referential, intertextual, verbal and narrative emphasis of avant-garde's political arm match the implicit and explicit qualities of SAM and VAM in Brewin's dual processing model developed through neurological research of traumatic experience.

Wollen identifies the work of Jean-Luc Godard as the contemporary exemplar of the political arm of this avant-garde. Through semiotics, he describes this political avant-garde as focusing its "work within the space opened up by the disjunction and dislocation of signifier and signified" (Wollen 1982: 98). There is a dismissive tone for those working in the materialist arm of the first avant-garde: "it is as if they felt that once the signifier was freed from bondage to the signified, it was certain to celebrate by doing away with the old master altogether in a fit of irresponsible ultra-leftism and utopianism. As we have seen, this was not so far wrong" (Wollen 1982: 99). Though there is an appearance of evenhandedness, clearly Wollen prefers the second avant-garde in which his collaborative practice with Mulvey is placed.

Wollen's criticism of ultra-leftism and utopianism may well have been directed at his contemporaries Malcolm Le Grice and Peter Gidal. Gidal's didactic texts in favor of materialist film were placed at a further extreme end of any negation of the status quo than any feminist counter-cinema: "Without a theory and practice of radically

materialist experimental film, cinema would endlessly be the ‘natural’ reproduction of capitalist and patriarchal forms” (Gidal 1989: xiii). Gidal identifies materialist film as influenced by abstract expressionism and minimalism and stresses a post-Warholian stance so that “to intervene crucially in film practice, the un-thought must be brought to knowledge, thought” (Gidal 1976: 15). For Rees the breaking point between these two camps rested upon “the issue of narrative, realism and representation, which *Screen* analyzed critically but which the avant-garde rejected as primary goals for film” (Rees 1999: 93).

Rather than utopianism, the freedom of form from content that Wollen identifies, can today also be read through trauma as “content denied”, as an erasure, or the signified’s absolute gutting. A viewer could receive a materialist film as a trauma inducing dissociative experience. This is the belated rejoinder that both *Alone* and *Dreamwork* make to this debate from thirty years earlier. Caruth has underlined the importance the belatedness in the experiencing of trauma that not only informs the thirty-year gap in materialist film’s response, but also the unknowability of the experiences addressed in these films at the heart of Wollen’s dismissal:

Traumatic experience, beyond the psychological dimensions of suffering it involves, suggests a certain paradox: that the most direct way of seeing of a violent event may occur as an absolute inability to know it, that immediacy, paradoxically may take the form of belatedness. (Caruth 1995: 6)

From this new perspective, rather than utopianism materialist film re-performs the trauma of overwhelming experience for the viewer both historically and experientially.

The critical splitting made visible through Wollen’s text was a decisive strategic move that sets up a space from which a critically and politically aware feminist counter-cinema can be asserted. As argued by Wollen’s then partner and collaborator Laura Mulvey:

Women cannot be satisfied with an aesthetic that restricts counter-cinema to work on form alone. Feminism is bound to its politics; its experimentation cannot exclude work on content, and it has, in this way, more in common with the avant-garde tradition that has tried to radicalise the signifier. (Mulvey 1979: 9)

Constance Penley’s critical textual analysis of the writings of Le Grice (1977) and Gidal (1989, 1976) on structuralist/materialist film in the *Imaginary and the Avant-garde* (Penley 1977) enacts this split by operating on materialist film with the same scalpel of psychoanalytic theory mobilized to unpack melodrama. A closer look at her arguments there are productive when re-examined within the emergence of trauma theory, the digital media explosion and the speed-up of information flow associated with it. Penley argues that in avant-garde film there is an overemphasis on the technical and as a predominantly male, “narcissistic” enterprise using a politically ineffective language devoid of fantasy and desire. She enlists Metz’s understanding of film viewing as an “authorised scopophilia, legalized voyeurism” (Penley 1977: 13) against

films that are “presented as near scientific investigations of scientific process” and reads this latter position as containing a tendency “to suppress a knowledge of the imaginary” (Penley 1977: 13).

After her marking of materialist-structural film as “fetishist impasse”, Penley reveals her own preference for a political avant-garde, in line with Wollen, as the preferred response for the “current” socio-historic situation for a feminist cinema. She observes that:

The films of Godard have consistently taken into account this work of language on image, as have those of Straub-Huillet, and Mulvey and Wollen. These film-makers realize that images have very little analytic power in themselves because their effects of fascination and identification are too strong. This is why there must always be a commentary on the images simultaneously with the commentary of and with them. (Penley 1977: 24)

Penley is not concerned with the “advanced solutions to formal problems” (Penley 1977: 25) or “perceptual processes” (Penley 1977: 8) identified in materialist film but about working at the limits of narrative and fiction that she finds in the works of Chantal Akerman, Marguerite Duras, Yvonne Rainer and Babette Mangolte, that same area of feminist practice more recently identified by Kaplan (2001, 2005) and Walker (2005) twenty years on, as fertile for articulating aspects of traumatic experience under the banner of “trauma cinema”. Such a “trauma cinema” includes at its vanguard that same group of films that Penley identified. Of these examples of trauma cinema Walker observes that:

These films are characterised by non-linearity, fragmentation, non-synchronous sound, repetition, rapid editing and strange angles. And they approach the past through an unusual admixture of emotional affect, metonymic symbolism and cinematic flashbacks. (Walker 2001: 214)

Kaplan similarly identifies those “independent cinematic techniques (that) show paralysis, repetition, circularity- all aspects of the non-representability of trauma and yet of the search to figure its pain” (Kaplan and Wang 2004: 204).

Commenting at the time on the tenor of the face-off between these two avant-gardes, Turim has observed that in terms of abstract film “the avant-garde has been engaged in a rhetorical arsenal aimed at granting or denying these films power as tools or weapons in an ideological struggle” (Turim 1980: 144). In this seventies period, we may well be at that cultural moment when the “resistance” politics of May 68 is being domesticated to move off the streets back into theory and the academy. The time had come to transform banner slogans like “the technological is political”, “less is more” and “the personal is political” back into cultural capital. This is a contested space of academic careers, of prominence and priorities which leaves Turim to wonder: “How many critical tautologies can be set up which circumscribe with their own assumptions their own conclusions” (Turim 1980: 144).

The New Situation

Read through the contemporary prism of “trauma theory” and Brewin’s neurological model Penley’s criticisms of Gidal and Le Grice’s writings on materialist film read like the deficiencies that define implicit memory (SAM) through its graphic intensity, immediacy and lack of story or context. Penley could just as easily be describing situationally accessible memory (SAM) in noting: “the first tactic of structuralist/materialist film is the emptying from the cinematic signifier of all semantic, associative, symbolic, representational significance” (Penley 1977: 8). This is what trauma does to memory. Mulvey’s description of Gidal’s strategy of “rejecting all content and narrative” (Mulvey 1979: 9) articulates Laub and Auerhahn’s trauma’s “not knowing” (Laub & Auerhahn 1993: 290).

Nor do Gidal or Le Grice situate or understand their seventies anti-illusionist materialist practice in any relation to trauma or VAM. Yet this practice, as Penley’s criticisms suggest, may be read productively today through trauma as “content denied” and as making evident the mechanics of erasure at the heart of traumatic experience rather than a sublimely radical position that frames cognition and the apparatus of film as its intrinsic de-facto subject. There is a trace of a traumatic view discernible in the practices of abstract expressionism and minimalism available to support this view, given that Gidal sourced these New York art movements in developing his arguments for materialist film. Guilbaut has made this point: “avant-garde art became an art of obliteration, an act of erasure” (Guilbaut 1983). With reference to Frank Stella’s minimalist black paintings with such Nazi invoking titles as *Die Fabne Hoch!*, *Reichstag* and *Arbeit Macht Frei*, Anna Chave suggests that “minimalism generally, might well be described as perpetrating a kind of cultural terrorism, forcing the viewer into the role of victim” (Chave 1990: 49). I am also reminded of Robert Rauschenberg’s erasure of a Willem de Kooning drawing gifted to Rauschenberg in his “*Erased de Kooning Drawing*” (1953) over one month with forty erasers (Solomon, 1997: 238). Gidal’s “un-thought” (Gidal 1976: 15) describes trauma. His description of his self-reflexive practice as “a filmic practice in which one watches oneself watching” (Gidal 1976: 10) brings to mind dissociation’s out-of-body experience. By Le Grice placing the “flux of perception” (Le Grice 1977: 10) at his practice’s core, he focuses on the pre-conscious realm of the traumatic flashback experience.

Alone and *Dreamwork* mark a return of the first avant-garde to critical scrutiny within the context of trauma, a scrutiny not only dismissed by Penley but also Mulvey and Wollen. Neither did its proponents envision this return or predict its new shape. It comes back by transforming the method of textual analysis used to discount its political relevance into a performative visual and perceptual practice.

Thomas Elsaesser utilizes trauma theory in screen studies to suggest that a general traumatizing has occurred of “the moving image in our culture as the symptom without a cause, as the event without a trace” (Elsaesser 2001: 197). Vilem Flusser identifies this as a world of “technical images” pre-occupied with surface and a new inherent magic not penetrable by those linear forms of critical thinking available through a written text. This world of magic is a

World in which everything is repeated and in which everything participates in a significant context. Such a world is structurally different from a linear world of history in which nothing is repeated and in which everything has causes and will have consequences. (Flusser 2000: 9)

Prensky suggests that, with young people today, their “thinking patterns have changed” through the impact of digital technology on their daily lives. They “think and process information fundamentally differently” (Prensky 2001: 1). Digital Natives are used to receiving information really fast. They like to parallel process and multitask. They prefer graphics before their text rather than the opposite. They prefer random access (like hypertext). On the other hand, Digital Immigrants place more emphasis on the type of reflection enabled by literate culture. Here again we have the VAM and SAM split with an identified shift to SAM like processing.

Bellour (2001), Penley (1977) and Mulvey (2004) have also identified in their more recent writing this digitally enabled shift, situating a preferred “pensive spectator” for a “fetishistic” one. They point to the computer’s desktop manipulation, freezing, slowing down and speeding up the moving image as indicators of a more sophisticated and interactive consumption. Mulvey observes that, “As the spectator controls the unfolding of the cinematic image, so the drive of the narrative is weakened and other, previously invisible or unimportant details come to the fore” (Mulvey 2004: 90). Mulvey’s “invisible and unimportant details” have always been the subject of Gidal’s materialist practice. For Bellour this shift marks an end for a textual analysis that as early as 1985 in “Analysis in Flames”, Bellour marked as “an art without a future” (Bellour 2001: xii), as technological developments allowed a form of textual analysis to be performed by the multitudes on the VCR. This has since expanded further to home computer use. This is an incisive insight given Bellour’s remarks about the rise of the critic in the nineteen sixties and his own participation in this rise through his practice of textual analysis in such publications as *Screen*.

For Bellour there was an important shift in the role of the critic in the nineteen sixties, which up to then depended on sitting in the darkened theatre scribbling notes with eyes fixed on the screen. In this posture:

A vertigo and a hysterical trembling remained: the vertigo of not being sure of my text, and with it, by the relative impossibility of doing so, a different kind of profound vertigo by what the implications would be if I were someday able to be sure of it. (Bellour 2001: 3)

This changed with access to the then new technology of the film editing machine (e.g. Steenbeck), and the film analysis projector. The vertigo migrated elsewhere out of the body of the critic. The critic could privately stop, start, freeze and replay the film outside of its theatrical screening context, creatively transforming the possibilities of criticism and analysis. Bellour could be sure of it.

The connection has previously been made between seventies textual analysis and recent found-footage films. When Humm outlines textual analysis she could be de-

scribing Tscherkassky or Arnold's practice: "By examining condensations, ruptures and excess stereotypes in the filmic text, critics aim to reveal a hidden sub-text which may structure a film's 'identity'" (Humm 1997: 15). In discussing the film critic Bellour's textual analysis, Belton notes its transforming effect:

Bellour, like all critics, necessarily rewrites the film he is analyzing to reveal or make clear its operations. For the original text he substitutes, out of necessity, his engagement with the text and this becomes the subject of his analysis. (Belton 2006: 243)

This viewer position of replay that emerged for the critic in the sixties and for the general public in the eighties resembles the pre-reflective and performative space that materialist film presented to the viewer. It is discernible in the found footage work of the period, for example, Le Grice's *Little Dog For Roger* (1967, 12 minutes) and *Berlin Horse* (1970, nine minutes, one and two screen versions) and, of course, the two contemporary films identified here. This realization opens up reading the face-off between the two avant-gardes as a contestation over ownership of the production of cultural meaning between the artist and the critic. Who owns the space in front of the film-editing machine? In the 1970s the critic's claim prevailed.

With the advent of the digital the situation changes. *Alone* and *Dreamwork* perform for the viewer those image manipulations that provided the source and re-configurations of Bellour's value adding work. In so doing the critic's position is bypassed and perhaps rendered redundant. That the ground has shifted, perhaps even inverted, is clear. When Penley asserts that "now we can do research on cinema history and theory by quoting images rather than by turning images into language" (Penley in Bellour 2001: xiii), she displays an unexpected affinity with the materialist position she so effectively dismantled thirty years before. Penley's earlier assertion on the necessity of "a commentary *on* the images simultaneously with the commentary *of* and *with* them" (Penley 1977: 25) reduces in authority.

The importance of this new prevalent accessibility is also part of Mulvey's reading of this new digital multi-platform situation:

I have tried to evolve a different kind of spectatorship, one driven by curiosity and a drive to decipher the images unfolding on the screen. However the curious spectator was, by and large, the product of feminism, of the avant-garde, and of a consciously alternative relation to cinema...this spectator is the ancestor of the one formed by new modes of relating to the screen image now immediately accessible to anyone who cares to experiment with the equipment available. (Mulvey 2004: 89)

I begin this summing up with a series of questions. Is Mulvey's pensive spectator not that same spectator created by materialist film in the seventies? Is it not its history that has to be belatedly re-inserted here? Is this not the wound that flashes back as a trace in the work of Tscherkassky and Arnold, in their repetitive, cyclical and

layered sampling strategies? Do not these technical operations excavate the pre-conscious of textual analysis? Although the film text itself may prove an “unattainable text” for the critic as Bellour suggested in *Screen* in 1975 (Bellour 1975) this may not be the case the other way around for a new materialism that, in a display of its ascendancy, visually lays bare the critic’s own practice and recontextualizes the search for the sublime in painting and an essential perceptually based film-making practice in terms of its shadow; the articulation of the mechanics of denial. Trauma theory enables a re-integrating of feminist criticisms made of seventies formalist materialist film-making practice, retooling that practice for critically unpacking the digital image in an emergent visual culture where, according to Nicholas Mirzoeff, “the world-as-a-text has been challenged by the world-as-a-picture” (Mirzoeff 1998: 5).

MOMENTS OF AFFECTION: JAYCE SALLOUM'S *EVERYTHING AND NOTHING* AND THE THRESHOLDS OF TESTIMONIAL VIDEO

Ilona Hongisto

One of the most striking audiovisual forms one encounters in art galleries and museums today is the talking head. This seemingly blunt audiovisual composition is striking for at least two reasons. To start with, one is often perplexed by the presumed objectivity in the uses of the form: the visual frame is frequently used as a simple mediator of the experiences verbalized by the person in front of the camera. The relationship between testimonial speech and the talking head frame has become a pervasive “moment of truth” recognizable across media, ranging from confessional videos posted online, television talk shows to video art. It is a relationship that alerts the viewer to bear witness to a significant testimonial moment.

The proliferation of talking heads in media culture has, however, also generated striking artworks that weigh the conditions of the testimonial moment in their expression. In a recent installation, the Turkish artist Kutlug Ataman displays a series of silent talking heads in an ascending spiral of second hand televisions. Citizens from the remotest corners of Turkey stare mutely at the camera in the familiar testimonial frame. The power of *Column* (2009) lies in the fact that it shows the children, youngsters, adults and the elderly as silent in a frame that comes with the association of speech in contemporary media culture. Muteness foregrounds the citizens’ inability to make their voices heard, to tell their stories.¹

Whereas Ataman’s piece challenges the simple objectivity of the audiovisual form with the series of television monitors and mute testifiers, Jayce Salloom’s *everything and nothing* (2001) experiments on the conditions of the testimonial moment with a more explicit take on the talking head frame. The video foregrounds the work of the frame at the intersection of the recognizable testimonial moment and “moments of affection” that engage the viewer to bear witness to emergent sensations captured in the video.

This chapter focuses on the experiments performed on the testimonial moment in *everything and nothing* and maps the implications of these experiments for how the video is experienced. The chapter argues that by specific uses of the talking head frame, the video inaugurates moments of affection that function beyond the call of truth that comes with the traditional testimonial moment.

Thresholds of Testimony

Like many other testimonial videos, *everything and nothing* is actually an interview.² It is an interview with the Lebanese resistance fighter Soha Bechara taped only a short while after her release from the notorious El-Khiam detention center in December 1999. Bechara was twenty-one at the time of her arrest and spent ten years at El-Khiam, of which six years were in complete isolation. She was arrested after a failed attempt to assassinate the leader of the South Lebanese army, Antoine Lahad. The assassination attempt and the captivity made her a national hero, a spokesperson of her people. When Salloum was in Beirut working on another project in 1992, he noted that posters of the woman were everywhere; they graced the walls of public spaces and in private homes, they were placed next to martyrs of war, in places of honor (Salloum 2005: 1; 2007: 164).

The interview takes place in Paris where Bechara relocated to study international law after her release. The resulting video coincides in part, with the general interest directed at Bechara's experiences in captivity, but it simultaneously refuses to simply mediate the rather stratified view of the political figure repeated in other media. Salloum remarks that he was hesitant to ask for the interview in the first place as Bechara was being "interviewed to death by the European and Arab press about her captivity ..." (Salloum 2007: 168–169). Put differently, the video is reluctant to position itself as another window to war experiences and torture. It aims at avoiding what Salloum calls the "gratifications of immediacy" that come with the simple objectivity of the testimonial frame (Salloum 2005: 1; 2007: 176).

In the beginning of the video, Bechara sits on a bed with a serious look on her face and recounts in Arabic for the camera who she is, and why she has accepted to do the interview. Framed in a medium shot – the testimonial frame – she tells about the struggle for Lebanon and her arrest. She speaks solemnly yet softly, convincing the listener of her dedication with her calm and reflective articulation. The beginning of the video ties the woman to a testimonial discourse in which the boundary between personal experiences and the public domain is blurred (see Ahmed & Stacey 2001: 1–6). Bechara adopts a public position of speaking through which her personal experiences are filtered.

The interview space adds to a sense of mediated testimonial discourse. A white wall and a closed door on the background frame the interview space. Even though the space is actually Bechara's room, there are no personal elements in sight. From the viewer's perspective, the enclosed and tightly framed space can easily be associated with an anonymous setting in which Bechara speaks as the voice of the resistance. A fixed camera records her testimony and as the video unfolds, it is not difficult to imagine that Bechara has spoken of her capture and release and its historical context



10. Jayce Salloum, *everything and nothing*, 2001, frame enlargement of video

a thousand times before. She speaks in a reflective and convincing manner, directing her eyes to the floor submerged in thought or occasionally looking straight at the camera as if to summon the viewer to the cause (Salloum 2001).

A few minutes into the video, the testimonial frame changes. The talking head is suddenly replaced by black frames that interrupt the recognizable testimonial moment. The rupture is followed by a close-up of Bechara's face. Then, the camera zooms out to a medium shot only to zoom back in, this time to a tighter medium frame. The testimonial frame is restored, only at a slightly different angle. In a few minutes time, the camera zooms back to a close-up of Bechara's face. Over these first four minutes of the interview, Bechara explains that she accepts to participate in conferences and give interviews because it gives her a chance to talk about the choices and possibilities available in the post-release period. Also, it gives her a chance to remind people of the long line of human rights violations inflicted on Lebanon and elsewhere that should be documented and remembered in order for a collective future to become possible. After Bechara has contemplated on the need to act as a witness in order for us to know where we "who are a small drop in the world" are headed, the testimony is interrupted with black frames yet again. From the visual disruption of the testimonial moment, the frame returns to Bechara who now sits on the bed in the same position as before and looks at the interviewer on the side of the camera. She rests her head on her hand with an expectant look on her face, smiles and asks in French "You don't understand?" which Salloum answers from behind the camera, also in French, "I don't understand but it's ok."

Salloum's entrance into the testimonial frame follows the abrupt zooms and disrupting black frames and yet it comes as a surprise. His sudden audible presence, as well as the fact that he does not understand Arabic reorient the viewer's relationship to the video. Although Bechara's statements are subtitled for the viewer, the interviewer's explicit entrance to the frame changes the orientation of the testimonial. What began as a recognizable testimonial moment passes into a compound of questions and answers in which the "failed transmissions" are clearly foregrounded. From this point on, the questions that Salloum poses in broken French are audible

on the soundtrack. Sometimes Salloum resorts to English when he cannot express himself otherwise. Once Salloum asks from behind the camera if Bechara has finished answering because he has no idea what she just said.

Bechara's account becomes more personal as it becomes related to Salloum's search for words. She talks about what she left behind when she moved to Paris ("everything and nothing"), how she experiences the distances between Paris, Beirut and El-Khi-am and why she never puts flowers in water. When Salloum has difficulties in finding the words, Bechara comes to his assistance and compliments his "very sweet questions". The speaking positions become more fluid as Bechara takes charge of the situation and Salloum openly reflects about the questions he poses. Addressing Bechara's survival methods in El-Khi-am, Salloum states that, "I don't want to ask you directly about that." Instead, he wants Bechara to tell the "little stories" of her captivity.

According to Michael Allan (2002: 166), *everything and nothing* is a conceptual site for considering the problems of representation, mediation and translation. What I have said about the video thus far supports Allan's argument. The piece criticizes the way in which war and torture experiences are typically mediated with the sudden changes of angle, the overt *in-situ* contemplation of the questions and answers as well as the linguistic threshold between the two parties of the interview. As the video confuses the speaking positions and thus also the viewer's expectations of the talking head testimonial, it takes a critical stance regarding the recognizability of the testimonial moment. The testimonial frame in a sense breaks with itself in order to call into question the testimonial moment of truth.

The disruptions in the relationship between the talking head frame and the participants, direct attention to what is outside the testimonial moment. More precisely, as the testimonial frame breaks with itself it makes the viewer aware of the arrangements involved in the construction of the testimonial moment of truth. In her recent work on media frames and war, Judith Butler argues that "[w]hat happens when a frame breaks with itself is that a taken for granted reality is called into question, exposing the orchestrating designs of the authority who sought to control the frame" (Butler 2009: 12). By de-framing the testimonial moment, *everything and nothing* transgresses the thresholds of testimony and puts forth a critique of the modes with which war experiences are typically mediated.

Frames of Emergence

Another way of describing the de-framing that takes place in the video could be re-orienting what is already in place. The testimonial frame breaks with itself in order to reorient the associated ways of speaking, listening and viewing. As the video undoes the positionalities that typically form the moment of truth, it points to the other possible positions that the testimonial moment could embrace. In this way, it points to an outside of the testimonial frame that simultaneously determines what is inside the frame and holds the possibility of other framings.

However, the work of the frame in *everything and nothing* extends beyond the reorientation of speaking modes and positions. In *Cinema 1. The Movement-Image*, Gilles Deleuze points to this other operation when he claims:

In one case, the out-of-field designates that which exists elsewhere, to one side or around; in the other case, the out-of-field testifies to a more disturbing presence, one which cannot even be said to exist, but rather ‘insists’ or ‘subsists,’ a more radical elsewhere, outside homogenous space and time. (Deleuze 1986: 15)

The main difference between these two outsides can be clarified with the concepts of possibility and potential. Whereas possibility is variation implicit in the testimonial moment, potential is undetermined variation. In Brian Massumi’s terms, it is “the immanence of a thing to its still indeterminate variation” (Masumi 2002: 9). In *everything and nothing*, possibility refers to the variations that can be evoked with the breaking out of the testimonial frame and potential refers to variation that “insists” or “subsists” but does not belong to the testimonial structure as such. Whereas possibility lines up with reorientations of what is already in place, potential has to do with emergent variation.

There are two particular instances in *everything and nothing* that foreground emergence over reorientations of the emerged. The first instance appears after Bechara’s lengthy description of her political work and commitment. After she has finished speaking, she looks at Salloum in anticipation. The brief moment feels longer than its couple of seconds – perhaps because it breaks the question-answer cycle so thoroughly. The feel of the moment could be described as something of a “leaning toward”. As the testimonial speech comes to its end, the speaker begins to step out of her position but as the position of comprehending the question is not yet available, she leans in-between.

The other instance is equally fascinating, only this time the leaning is reversed. As Salloum searches for the words with which to formulate his question, Bechara looks at him attentively. The suspension of the testimonial exchange creates a disorienting in-between space to the video. The waiting, the searching and the pending understanding that come with the “failed transmissions” break with the recognizable logic of testimonial truth.



11. Jayce Salloum, *everything and nothing*, 2001, frame enlargement of video

The in-between moments framed in the video are the “moments of affection” with which *everything and nothing* engages its viewer. These are disorienting moments as they exceed the norms and functions of the testimony and induce an experience that is not conditioned on recognizing a moment of truth or becoming aware of the orchestrations involved in its construction. Instead, these moments create a “disturbing” sensation of presence to the testimonial exchange.³

Instead of pointing to what remains outside the frame, these moments emphasize what emerges within the frame. As the camera lingers on Bechara while she is about to understand Salloum’s question or right after she has finished answering, the frame captures the woman stepping out of her role as the spokesperson of her people and thus expresses her in what could be called a state of indeterminate variation or potential.

Ontologically speaking, indeterminate variation has to do with an understanding of reality that consists of actual forms and virtual forces. Whereas actual forms are those that have already emerged and that can therefore be reoriented, virtual forces are indeterminate and therefore emergent. Elizabeth Grosz – drawing from Gilles Deleuze and Félix Guattari – takes this ontological stance as the point of departure for a definition of the work of the frame in the arts. Grosz claims that an artistic frame expresses virtual forces as sensations (2008: 12). Virtual forces do not belong to the frame as such, but the creative act of framing ensures the becoming expressive of the virtual, which then again accords the actual forms in the frame with an intensity that exceeds them. Framing harnesses virtual forces by constituting a territory and that territory may become expressive of sensations that exceed the frame itself (Grosz 2008: 11).

This is significant for a new materialist take on the arts because it defines the frame as an active material agent in the creation of sensations. The frame does not mediate sensations, but creates them. In *everything and nothing*, Bechara is framed in a manner that creates a tangible feeling of presence that exceeds the testimonial structure. This feeling insists that there is more to the Lebanese resistance fighter than the testimony itself can convey. This coincides with Salloum’s remark that he wanted to avoid the “overexposed” manner in which Bechara is typically portrayed and her testimony mediated (Salloum 2005: 3; 2007: 169).

Amodal Affection

The distinction between a testimonial moment of truth and moments of affection brings the discussion to how these distinct yet related areas of *everything and nothing* are experienced. Following the Italian film-maker and theorist Pier Paolo Pasolini there seems to be a double reality to the piece that bears on how it is experienced. Pasolini argues for a double reality of cinema that consists of a grammatical and a pregrammatical level (Pasolini 1988: 170–171). The first level is the level of cinematic signs (*im-signs*) that can be arranged in a number of ways. Pasolini insists that the grammatical outline of cinema is an indispensable pretext for pregrammatical expression. Pregrammatical expression, on the other hand, is an unrealized film that unfolds beneath the grammatical ordering of images (Pasolini, 1988: 173).

The double reality of cinema – its grammatical and pregrammatical dimensions – can be taken further in the realm of spectatorship. The grammatical and the pregrammatical levels of expression entice the viewer with different means and thus require a conceptualization of the viewing experience that is equally double. Raymond Bellour (2002) suggests conceptualizing the double reality of the viewing experience with the notions of modal and amodal perception. Bellour draws his outline from the developmental psychologist Daniel Stern, who discusses the distinction in the context of an emergent sense of self in an infant. Stern argues that an infant is simultaneously aware of the process of an emergent organization as well as its product. The sense of self is emergent because the objects and compositions the infant perceives are tied to an emergent process in which the modally perceived objects are tied to an amodal process of organization. Stern notes that amodal experiences are elusive – hard to seize upon as such and they seldom fit to existing lexicons of experience. He calls them “vitality affects” that are better described in dynamic and kinetic terms such as “fading away” and “bursting” (Stern 1985: 54). Bellour argues that the spectator in cinema is comparable to Stern’s infant in the sense that the cinema as an experience is composed of the dual perception of an emergent process and a more categorical dimension of objects and their relations (Bellour 2002: 109). In *everything and nothing*, this distinction is apparent in how the video first relies on clear speaking positions and modes of speaking and then moves to creating a sensation of presence that surpasses these assigned positions.

Bellour’s insistence on the role of amodal perceptions in the cinema coincides interestingly with William E. Connolly’s outline of “materialities of experience”. For Connolly (2010), materialities of experience coincide with the rich history of inter-sensory circuits that contribute to our perceptual habits.⁴ These sensory circuits form an amodal history of perception that grids the conditions of modal perception. It is this level of materialities – sensory relations of movement, affect, touch, sight and smell – that Connolly sees as politically most timely. He speaks in favor of media experiments that challenge the simple objectivity of mediation with audiovisual means and in this way expose and address the ways in which perception is inflected and gridded in the media-saturated society. He argues that this kind of audiovisual exposing may – eventually – lead into a more affirmative existential attachment with the world.

Put differently, Connolly’s take on the materialities of experience, Bellour’s amodal perception and Stern’s vitality affects all imply that reality and thus an experience thereof is deeper than its actual forms. Equally, they argue that the amodal level of perception or experience is not conditioned on discursive relations although the amodal or inter-sensory level does contribute to how perceptions are formed in discursive networks. This argumentation has wide political consequences as the political event is hereby transposed from weighing the conditions of perception and experience in their discursive determination to looking at the potential of transforming the conditions of perception and experience on the amodal level of inclinations and inter-sensory circuits. To this background, the moments of affection in *everything and nothing* foreground an experience that confuses the habitual ways of experiencing a testimonial

moment and this connects the audiovisuality of the piece to a process of emergence that traverses the actual positions of the testifier, interviewer and the viewer.

Although the moments of affection in *everything and nothing* disrupt the testimonial order, they nevertheless do not abolish the testimonial function altogether. The moments are followed with yet other reflections on the importance of speaking out and telling about the situation in Lebanon and El-Khiam. Bechara foregrounds the importance of democracy and a regime that asserts itself without attacking. With the moments of affection, the charge of the testimony is transposed from a moment of truth to an affective encounter between the participants. In this transposition, the frame as a material agent in the creation of sensations is of primary importance.

IN THE NAME OF THE AUTHOR:
TOWARD A MATERIALIST
UNDERSTANDING OF
LITERARY AUTHORSHIP
Kaisa Kurikka

Only expression gives us the *method*.
Gilles Deleuze & Félix Guattari

Orthonymity, anonymity, pseudonymity, antonymity, homonymity, mononymity, allonymity ...¹ As this listing of the diverse naming-practices shows, the recording of the author's name on title pages and on covers is never a matter of mere conventionality or ordinariness. To sign a work with or without a proper name, with one's legal name or with a fictional one, is always a matter of choice. This choice belongs, nevertheless, to the category of "regulated liberties"; the freedom of signing is regulated for example by prevailing rules of publishing. Before appearing on book covers the signature of the author works as a piece in a manifold set of negotiations, which are both artistic, commercial and social in nature. As such the author's name functions as a site for thinking and re-thinking the terms of authorship. Could it be that the proper name of the author appears also as a site for using and, especially, *abusing* the conventions and regulations governing the territories of authorship? Or: What can be done in the name of the author, is there such a thing as the materiality of authorship?²

Gilles Deleuze and Félix Guattari write in *Anti-Oedipus*:

The theory of proper names should not be conceived of in terms of representation; it refers instead to the class of 'effects': effects that are not a mere dependence on causes, but the occupation of a domain, and the operation of a system of signs. (Deleuze and Guattari 1996: 86)

Indeed, if we take up the polyonymous authorship of *Enten/Eller* (*Either/Or*, 1843), a work of philosophy examining the ethical and aesthetical stages of existence, we

enter a peculiar territory marked by authorial signatures; it is written by five different authors – none of whom is Søren Kierkegaard (1813–1855), but instead there are Victor Eremita, the aesthete named only as A, Johannes the Seducer, Judge William, and finally an unnamed pastor. At least according to Kierkegaard, the writings of these authorial names do not belong to him: “Thus in the pseudonymous books there is not a single word by me ...” (Kierkegaard 1992: 626). And if we consider the heteronyms of the Portuguese poet Fernando Pessoa (1888–1935), we are rather dealing with total operations of authorship and creative actions than with questions of identity and reference or other “terms of representation”. The logics of representation working on the predicates of the same, the similar, the analogous and the opposed (Bolt 2004: 37–40), seem no longer plausible.

To schematize the author’s name as a representation of authorship based on rules of equivalence, analogy and similarity between the name and the person seem to coalesce with the production of a portrait-photo of the author, a transcendental author-image, which mediates imposed identity and reduces the author to an interactive construction (see Benedetti 2005: 54–56) disregarding the singularity, and thus materiality, of each authorship. If the notion of pure difference is maintained instead of negative difference (different *from*), the singularity of authorship can be reached by turning to materiality instead of *interpreting* the author in question with the terms of representational logics, that is by projecting a pre-given understanding of authorship (which differs from other kinds of authorships) onto the author at hand. The celebration of singularity and difference makes materiality matter.

Author-name belongs to a special category of proper names – as this chapter later shows – but nevertheless it belongs to a power structure, “a regime of signs” with both discursive and non-discursive dimensions (Deleuze & Guattari 1988: 111–148). In this respect language becomes a special kind of action, which according to Ronald Bogue (2003: 82) “presupposes a complex structure of actions, a regular pattern of socially sanctioned practices that inform the transformations of bodies effected by words”. The relationships between “words” and “bodies”, “signs” and “things” has not to do with representation, but rather signs intervene in things while both preserve their functional independence as a formalization. Understood this way, the interconnections between the proper (author-) name and the related personal body have little to do with correspondence or conformity. Deleuze and Guattari rule out the identifying function of a proper name with an ego by saying that:

There is no ego that identifies with races, peoples, and persons in a theater of representation, but proper names that identify races, peoples, and persons with regions, thresholds, or effects in a production of intensive qualities. (Deleuze and Guattari 1996: 86)

To avoid the assimilation and homogenization of the proper author-name into a simplified image or symbol of (mentally constructed) authorship, a totality of the territory of creativity calls for consideration. In this chapter I sketch an understanding of an author’s name by drawing on the ideas of the materialist epistemology as

expressed by Deleuze and Guattari in their various writings. This experimentation escorts the author outside the stages of a theater of representation and outside the walls of galleries filled with portrait-photos of authorship, where the writer (the historical personality) is turned into a “portrait-image” of the author and the real experience of the writer is reduced to a photo. This, as Lambert has noted, “impresses the meaning of this experience into a mental representation – the snapshot of a blocked, oppressed or oppressive, neutralized desire with a minimum of connection” (Lambert 2008: 39).

The problematics of studying authorship were materialized to me when thinking about one particular author, Algot Untola (1868–1918), a paradoxical figure in the history of Finnish literature. Untola’s first novel was published in 1909 and the last one came out eight years later in 1917. Untola is unknown to international readers (due to lack of translations, since only a couple of his prose texts have been translated into Russian), nevertheless he may well hold the record of authorial signatures: Untola never published anything with his “real” name (a paradox in itself, because he was born Algoth Tietäväinen) but instead he published both prose and drama with three different proper names: the male name of Irmari Rantamala and the female name of Maiju Lassila were accompanied by the genderwise neutral J.I. Vatanen.³ He also wrote newspaper editorials, articles and political columns with more than forty different nicknames and initials. In addition, he wrote several thousand pages of unpublished manuscripts, which he signed with various author’s names, some of which are female. He also corresponded daily with his publishers never signing the letters with his legal name. This polyonymous authorship and the multiplicity of author’s names of Algot Untola, if anything, not merely problematizes the uses of author-names but also questions the regions of authorship by occupying the whole domain of literary creativity and (inter)connections of authorship. Algot Untola works as a mediator in my treatment of authorship – in this context it is not possible to *read* the works of Untola irrespective of the fact that detailed reading is, naturally, essential in studying authorship.

In what follows, I firstly propose to discuss in more detail the production of portrait-photos of authorship. Secondly, I fabricate an alternative entrance to literary authorship not by following the post-Barthesian⁴ legacies of treating the author as a web of signifying practices without material links, nor by following the biographers impulses to reduce authorship to the subjectivity of the actual, historical writer but rather by accentuating the immanent materiality and singularity of authorship as a force pointing at the new. Thirdly, I discuss in detail the functions of author’s names not as representation of something but as *expression* and as *gestures* of authorship. By suggesting the double-articulative nature of author’s names and by emphasizing their force (“puissance”) of producing new connections this chapter wishes to point to the future of studying authors: the potentiality of authorship.

Nameless Faces, Faceless Names

In the social domain of language, two formalizations are encountered: one of content and one of expression; one of the seeable and one of the sayable. Each has a

form and a substance – because of this the relationship between them is not representationalist. (Deleuze & Guattari 1988: 43–45, Deleuze 1988: 32–35, 47). In terms of author (ships) the body of the person or the body of writing can be designated as “content” and the author-name as “expression”. If one is to produce the above-mentioned photo-portrait of the author, “an author-image”, the complex linkage of content and expression is reduced into a relation of reference, in which both the person and the author-name act as mere surfaces of the inner depths of his/her writing – writing is interpreted to represent whatever is projected onto it.

The operations of producing an author-image can be viewed as symptomatic of faciality (*visagété*) following the reasoning Deleuze and Guattari develop in *A Thousand Plateaus* (1988: 167–191). The head and the body of the author has been over-coded by the face, facialized to a surface, where the head has ceased to be a part of the body, no longer having a multidimensional corporeal code. This facialization follows the semiotics of signification and subjectivation, in which a single substance of expression is formed to “ensure the almightiness of the signifier as well as the autonomy of the subject” (Deleuze & Guattari 1988: 181). Deleuze and Guattari outline as “despotic regime of signs” a special organization, in which the biunivocal correlation between signifier and signified, the dominance of the signifier over the signified and the identification of the signifier with law and authority are maintained. The despot becomes the unifying source of all signification and all signs refer back to the despot and they are registered on his face (Bogue 2003: 83–83).

The canon of Western literature seems to have been constructed in the manner of this signifying despotic regime. The “despotic author”, the “Master Author” has the face of a white man and since, for example, the body has been over-coded by the face, women authors have been excluded (Felski 2003: 57–64). The canon consists literally of faces of power. The other operation of faciality, subjectivation, takes place according to the workings of “postsignifying passional regime” (as Deleuze and Guattari call it): the face is turned into a personal and discernible entity in which various affective energies ranging from fear to passion or collectively shared traits of social power are invested. The great authors of Romantic period gained their subjectivity in relation to the self-expressing aspects of their art, or in the nineteenth century Europe, under the era of building nation-states, the face of the author was namely the face of the nation-to-become, of the national author. When taken to the furthest limit, the canonized author appears merely as a face with no need of an author-name – a picture of the face of, say, Ernest Hemingway, can be regarded as the ultimate portrait-photo of the author, a nameless face; Ernest Hemingway has undergone the process of firstly becoming-author by gaining his author-name under the rules of institutional practices of literature and publishing. Secondly the author-name has been written over by the face. Canonized authorship is as much visible (the face) as it is sayable (the author-name).

The operations of faciality are at work also in the very materiality of books. The book-covers can be regarded as faces, which attain their subjectivity with the printed letters forming the author-name. As such, the covers of books function as show-cases of authorship with the purpose of gaining attention in order to attract buyers.

The author's name on the cover of the book works as a signal for a certain kind of literature, it becomes a guarantee of a standard since it is the mark of subjectivity – the author-name is transformed into a trademark, a brand like any other commercial commodity in capitalist marketing. The author-name as a brand needs no “real face”, since the material body and head of the author, overcoded by the name-as-face, disturb the semiotics of capitalist economics based on surface values.

Do Kierkegaard, Pessoa or Untola need a face? It seems they all operate inside the frames of the regime of faciality, because in order to publish anything at all one has to be facialized to a certain extent. These authors, however, make use of it by drawing a line of flight. All of them have several author-names, as was mentioned above, but none of them have *visible faces* – they appear as authors of faceless names. To be more precise: Kierkegaard, Pessoa and Untola or rather their author-names do have faces but only through *the sayable*: Kierkegaard's author-name Johannes achieves a face through *The Diary of the Seducer*, Pessoa describes the looks and manners of his heteronymic authors and Untola's female author-name Maiju Lassila manifests as a character of a novel. These sayable faces draw lines of flight under the ruling of despotic signifying practices, since they break the laws of the master signifier: it is no longer possible to identify “contents” with “expressions”. This brings us back to the materiality of author-names, authorship and language. It turns us also to the intensities and forces of affect that are at work both in expressing and studying authorship and studying.

Author as Conceptual Persona

In 1911, Maiju Lassila, the author of 20 novels and plays categorized as humoristic realism in Finnish literary history, was granted a state award for the best novel of 1911. However, she refused to accept this honor. In a letter published in the main newspaper of Finland, *Helsingin Sanomat* (*Helsinki News*), she accounted for her refusal the following way: “I never try to write a master piece in order to be awarded ... If I should ever succeed in publishing anything in the future, I want to be excluded from all kinds of nominations.”⁵ Today her plays are performed in theaters all over Finland and despite her desire she has been nominated to the Finnish canon of humoristic realism.

In 1915, Irmari Rantamala, whose first novel had been published six years earlier, wrote to his editor, who had suggested some minor changes to his forthcoming novel. His first novel is a massive *fin-de-siècle* writing of 1803 pages covering the anxieties of its protagonist attempting to write “an opus magnum” while taking part in the revolutionary movement against the Czar of Russia. Anticipating a similar fury the first novel raised among Finnish readers, the editor proposed changes in style and, tellingly, in length. Rantamala answered to his editor thus: “That might be right from your point of view, but, anyways – my writing is wrong: I do not write right.”⁶ Nowadays this author of false writing is regarded as a notorious example of decadent literature in Finland.

These two proper names, Maiju Lassila and Irmari Rantamala, belong to a network of several other names. They are literary signatures of Algot Untola, but to define

them as “pseudonyms” would be an understatement, since pseudonymity is usually understood as a means of hiding one’s “true identity”, as a way of separating the private from the sphere of publicity. As a concept, pseudonymity works on the level of representation without self-evidently acknowledging the potentialities of the proper name: the quest for searching the (psychological) motives⁷ of the writer to adopt a nom-de-plume persists on reductive strategies of research since the act of studying follows the path of return to face the “authentic” moment of creative decisions. The concept of polyonymous authorships suits better Untola’s “namology”, because his authorial names are connected to each other in various ways. Polyonymity deterritorializes conceptual frameworks and replaces them with material forces and actions. As such, it moves away from movements of the mind to sensations and it tunes the expressions of language into their material potentiality.

Conceptually, polyonymity also stresses the material network under construction between the names of Algot Untola: the “element” connecting them is, of course, Untola as a singular person, but even more importantly, writing as a form of material action. In his letters Irmari Rantamala is constantly commenting on the novels authored by Maiju Lassila and vice versa, sometimes in first-person, sometimes in third person in such a fashion that finally it becomes impossible to determine, who is saying “I”: I, Maiju Lassila equals I, Irmari Rantamala etc. The “I” is never singular, but plural, a collective subjectivity under ceaseless proliferation, in which there is no original point of reference or representation. The “I” is both a lingering swarm and a temporary position of subjectivation in need of being read in its immanent materiality.

The afore-mentioned Danish philosopher Søren Kierkegaard (1992: 625) denies that the point of pseudonymity is to hide the true identity of the author – he himself produced the “pseudonymous” texts, but he is not their author. Louis Mackey has stated, that “a Kierkegaardian pseudonym is a *persona*, an imaginary person created by the author for artistic purposes (...)” (Mackey 1971: 247). The etymological root of “persona” refers to a mask used by an actor: the purpose of this persona/mask is not to hide the actor/author, but conversely to make a point in itself in drawing attention to the actions and acts of the actor-author. In the cases of Kierkegaard and Untola, polyonymity manifests itself as a question of personae not of disguises, but in conceptual frames, which seem not to insist on the strict separation between the actual writer and the “outcome” of creative actions (that is, a persona, which by no means is to be understood as a human personality) since choosing a polyonymous authorship is a material act of desire. When pondering on the polyonymous authorship an attempt at a sense-making question appears as “what can an author do?” instead of “what is an author?”. Rather than on the mere materialities of actions taken, the focus is set on the potentiality of authorship.

In *What is Philosophy?*, Deleuze and Guattari introduce “conceptual persona” (*personnage conceptuel*) (2005: 61).⁸ “Conceptual persona” refers to the “subject” and body of thinking and the becoming of philosophy engendering philosophers and writers. A conceptual persona is situated alongside with “the plane of immanence” and “a concept” sometimes materializing as a named figure or as an aesthetic, historical, mythological and fictional character. Conceptual personae manifest territorial-

izations, re- and de-territorializations of thinking, and they appear as a means of confronting chaos. Author as a conceptual persona, seems a plausible way of constructing the cartographies of literary authorship. On a given plane the personae are composed by different genres of traits (Deleuze & Guattari 2005: 68–71).

Maiju Lassila's refusal to accept the literary award for example, expresses antipathy toward institutional regulations concerning the appreciation of art (*traits pathiques*). The polyonymous authorships of Untola and Kierkegaard as well also engender relations between the various author-names (*traits relationnels*) and complicate the juridical questions concerning authorship (*traits juridiques*): ever since the making of copyright-legislations (the end of eighteenth century England) writers fulfill a double-bind position before the Law and the Capital. On one hand, writers are responsible for their writings since they are the rightful owners of their published words (and as such writers are governed by the Market), but simultaneously on the other hand, they are also always potentially guilty (of blasphemy, plagiarism, etc.) because of their writings. Also in the cases of polyonymous or pseudonymous authorship, the actual writer is the one to face the Law. Institutionally the separation between the writer and the author persists, although the creative thinking of writers such as Untola and Kierkegaard and of critical study aims at resisting such a two-folded categorization either focusing on defining authorship without reference to the empirical personality or shifting the focus from essence to actions of authorship.

The notion of author as conceptual persona also allows for taking into consideration biographical aspects of the writer. Deleuze and Guattari (2005: 70–71) write about “existential traits” (*traits existentiels*) referring to biographical details of the philosophers and writers in question. These existential traits can be characterized as “biographemes” as suggested by Roland Barthes in *Sade, Fourier, Loyola*. Barthes writes:

...were I a writer, and dead, how I would love it if my life, through the pains of some friendly and detached biographer, were to reduce itself to a few details, a few preferences, a few inflections, let us say: to ‘biographemes’ whose distinction and mobility might go beyond any fate and come to touch, like Epicurean atoms, some future body, destined to the same dispersion ... (Barthes 1976: 9)

I suggest that author's names belong to the category of “existential traits” as biographemes expressing the desires and experiences of the writer not as a reductive vehicle, but as a means of pointing to dimensions of the new, as potentialities of creating authorship. Interestingly enough, Deleuze and Guattari never use the term of “materiality” while taking up these different traits – however, I would suggest that, in particular, “existential traits” are linked namely to the materialities of authorship whether in terms of the body of the author or in terms of the bodies of books (circulation, publishing, etc.). On the other hand, in terms of the materiality of Deleuzian epistemology as a whole, all of these traits open up into material dimensions.

Author as conceptual persona has emerged in multiple mutations and it reappears in various forms in changing times and different contextual and conceptual frames. Ever since the printing press was invented special weight has been invested on the name of the author. In a sense it peaked with Romanticism when the author's name began to function as a password or code to enter the reign of author-god as the source of signification. The bourgeois and capitalist notion of modern authorship as the legal owner of one's writing is guaranteed by a single proper name. Polyonymity can be regarded as an attempt to de-territorialize the powerful position of the particularity of signature (Frow 1996), since the plurality of names already in itself opposed the powers of One.

Deleuze and Guattari (2005: 62) briefly take up the question concerning the proper name of a philosopher by designating it as the pseudonym of conceptual persona. Conceptual persona on the other hand appears as the "heteronyms" of philosophy – the philosophers mention the problematics of proper names in connection to conceptual persona only through these comments, so a look elsewhere is needed. "Heteronymy" is the concept Fernando Pessoa used to characterize the personalities he invented. For Pessoa, heteronyms have a personality of their own and they reside in separation from their creator. Pessoa's heteronyms acquired identificatory characteristics by reference to their looks and upbringing typical of biographemes. Pessoa explains the differences between pseudonymity and heteronymy by saying that in pseudonymous writing:

An author [appears] writing as himself; a heteronymic work is by an author writing outside his personality: it is the work of a complete individuality made up by him, just as the utterances of some character in a drama would be. (Pessoa cited in Lisboa and Taylor 2003: 133–134)

It seems that Pessoa's heteronyms can be understood at least in two ways: they seem to resist the authorial power of the actual writer by dispersing authorship to fictional dimensions. Simultaneously, however, they manifest the authority of the writer: he is so powerful, that he is able not only to create characters but authors as well. More importantly, Pessoa's heteronyms strongly underline the materiality of each author-name in question.

The proper names of Algot Untola also appear as heteronyms, because they are manifested as characters in novels and manuscripts written by various author-names of Untola. One novel by Maiju Lassila, called *Love* (*Rakkautta* 1911) is an "as-if" autobiographical novel telling the "tragic love stories" of Maiju Lassila, a young woman of 17 years of age, who wants to be loved by as many young men as possible. Through this novel, the proper name of Maiju Lassila turns into a heteronym. A lifestyle is offered to Maiju Lassila, and thus the author's name gets a point of reference. In the context of Untola's authorship this reference is depicted in humoristic tones, because the "face" of Maiju Lassila belongs to a superficial girl whose interests lie on matters of heart, not on such serious stuff as art and creation. By creating "life" in connection to a proper name Algot Untola is mocking the notions of

literature as patriarchal self-expression: the depiction of Maiju Lassila as a superficial girl mocks “serious” male authors, because the novel makes it clear that this girlish Maiju is a famous author of several idealistic novels – usually written by great (male) authors. This kind of reading is made possible by treating all Untola’s author-names as materially connected to each other.

This author-name and other female author-names used by Algot Untola also point to becoming-a woman-author, a threshold for becoming-other. The proper name of Maiju Lassila is not so much a mark of identity; it is an event, it refers to operations of becoming-multiple. The multitude of author’s names is an act of de-nomination effectualizing more and more productive intensities, since in the first instance there is no such person, no “Maiju Lassila”, to whom the proper name could refer back to, at least not in the outside of the texts written by Maiju Lassila. The name Maiju Lassila is a sign referring to pure potentiality in its material intensity.

Names as Gestures

When entering the domain of author’s names from a different standpoint, the dimensions of expressing potentiality can also be defined in terms of a modulating the molar authorship into the molecularity of author-names. In discussing stratification in *A Thousand Plateaus*, Deleuze and Guattari (1988) create the concept of a “double articulation”. Each stratum is constitutive of double articulation, of which the first chooses and deducts from unstable particle-flows (*substances*) upon which it imposes order of connections and successions (*forms*). The second articulation establishes functional, compact, stable structures (*forms*) and constructs the molar compounds in which these structures are actualized (*substances*). Each articulation has a corresponding type of multiplicity: one type is supple and more molecular, merely ordered, the other is rigid, molar and organized. All in all, both of these articulations are materially constructed, and as a conceptual tool for thinking they bring about the multidimensional inter-connectedness of words and things, the sayable and the seeable (Deleuze 1988: 40–41).

If author’s name is understood as such a stratum, it has two separate functions. The first of them works on molar level, namely as an order-word (Deleuze & Guattari 1988: 79), a form of communication, which provides a set of information in a way typical of any other trademark or label. As a molar order-word, author-name situates the body of the author under a socially hierarchical ruling: a male/female author, an English/a Finnish author, and so on. Gérard Genette (1997) has offered a taxonomical account of the literary conventions concerning the author’s name. His taxonomical reasoning can be named as “the molar strata of author-names”. Recording the name on the title page and on the cover have various paratextual functions, as Genette calls them, which go beyond strict designation, such is the indication of sex, nationality et cetera. Genette emphasizes that the author’s name has an ability to produce a certain kind of personality, since the name of the author with its paratextual features functions as a threshold of interpretations, connecting the inside and the outside of the text.

According to Michel Foucault (2002) the author's name has more than indicative functions. The name does not merely refer to someone, it is also a description and it assures a classification of grouping certain texts, defining them, differentiating them from other texts and establishing a relationship among the texts. In other words, Foucault is also pointing to the molar, classificatory level of the functions of the author-name. The name of the author as an order-word seems to work on the level of representational logics. As an order-word it brings forth the incorporeal transformation: as soon as Untola is named as "Maiju Lassila", he is transformed. The molar stratum articulates such contents as identification, categorization, definition of certain texts, and the form of the content is the proper name of the author. As such, for example, the name of Maiju Lassila groups together 20 works of fiction and a bunch of letters, the novels of Maiju Lassila can be defined as realism and so on.

But the name of the authors functions on the level of molecularity, and understood in this way the materiality of author-names opens up to intensive forces capable of entering new dimensions. The deterritorializing movement enabled by the author-name's double-articulative nature makes it possible to move away from the despotic regime of representation and to get closer to the zones of intensive affectivities that are founded on the material processes and production of names. In *Dialogues*, in a note on "Dead Psychoanalysis: Analyse", Deleuze writes on the function of the proper name:

The proper name, here, is precisely not a reference to a particular person as author or subject of enunciation; it refers to one or several assemblages; the proper name brings about an individuation by 'heccicity', not at all by subjectivity. (Deleuze & Parnet 1987: 120)

Here Deleuze is referring among others to the name of Virginia Woolf, "which designates a state of reigns, ages and sexes", not a person (Deleuze & Parnet 1987: 120). Deleuze is also asking, why does the proper name make an assemblage into a particular regime of signs according to a transformational component. Deleuze writes: "What is fascinating in medicine is that the proper name of a doctor can be used to designate a group of symptoms: Parkinson, Roger..." (Deleuze & Parnet 1987: 120) Could it be possible to talk about the symptomology of Algot Untola, of Kierkegaard? At least it would indeed be proper to talk about Operation Algot Untola, in which the molecular stratum of the author's name opens up to a series of becomings, it brings about "non-personal individuations". The polyonymous operation appears as materially and potentially becoming-other.

This non-personal individuation takes place also on materially visible means. Maiju Lassila as the name of the author is printed on different title pages and book-covers: the name is repetitious. It is in this form of expression, iterability of the singular, in the repetition of the proper name that Maiju Lassila becomes an author. The singularity of the name is what enters a series; a series is composed of these singularities. Paradoxically the name of the author is simultaneously a sign of singularity and a citation, repeatable in the context of the *oeuvre* of the author. It might be proper to

talk about a Maiju Lassila, with an indefinite article, because through these repetitions the proper name seems to flee the rules of definitions.

Earlier, I introduced two citations written by Maiju Lassila and Irmari Rantamala. Maiju Lassila renounced the position of becoming an officially acknowledged author by refusing to accept the literary prize. Irmari Rantamala declared proudly to be an “incorrect” author – that is, an author who is not following the rules of “correct” writing. Rantamala’s desire is to write himself into the margins of literary life in order to become a minor author. According to these citations and other writings by Untola, the prevailing literary institution of Finland in the early 1900s is clearly hierarchical. Authorship, and writing on the whole, has been institutionalized into a system of inclusion and exclusion. There are Great Authors, who dwell on artistic and aesthetical dimensions; they write masterpieces acknowledged highly by the literary elite. In Untola’s words they are the Real Authors with capital letters: they master the diverse rules set by the literary circles. Real Authors themselves have the power to manipulate and govern the systems of writing. Untola himself is as scribbler, who writes incorrectly and who wants to remain in the margins of the centralized institution of authorship.

The polyonymous authorship of Algot Untola is in itself, not following the rules of Real Authorship based on self-expressing subjectivity. On the contrary the polyonymous authorship paradoxically functions as an escape from subjective personality and as such it is constantly deterritorializing the informative and molar functions of author’s names. This operation of polyonymous authorship works as a way to criticize the state of authorship in the early twentieth-century Finnish literary institution, where author-names are taken as signs of genuine, authentic representations of personality and as the sign of ownership in a capitalist and “middle-class” ways. In this sense another component can be added to the Operation Algot Untola: the politics of polyonymous authorship is aiming to reside in the margins of literary life. It is also denying the power of one name to act as a signal of possessing creative actions and products.

Foucault emphasizes that “the trace of the author is found only in the singularity of his absence; he must assume the role of dead man in the game of writing” (Foucault 2002: 10). Algot Untola is doing precisely this: by signing with multiple names he is becoming-imperceptible – the author’s name is the site of “the singularity of absence”. Using various names can be defined as material gestures of authorship as proposed by Giorgio Agamben. He writes: “... the author is present in the text only as a gesture that makes expression possible precisely by establishing a central emptiness within this expression” (Agamben 2007: 66).

It might be plausible to claim that the image of the author is situated in the name of the author, not as a representation but as an image-in-movement, as a gesture. In an essay titled “Notes on Gesture” Agamben comments on Deleuzian concept of image-movement in connection to “gesture”:

There are no images, only gestures. Every image, in fact, is animated by an antinomic polarity on the one hand, images are reification and obliteration

of a gesture (it is the *imago* as death mask or symbol); on the other hand, they preserve the *dynamis* intact. (Agamben 2000: 55)

This *dynamis* can be traced back to the definitions of “gesture”. The *Oxford English Dictionary* defines gesture as “a manner of carrying the body”. The author-name as gesture is a bodily movement, but also a corporeal rhythm that has become expressive in and through language (see Väliaho 2007: 17). Agamben goes on define gesture in the following way:

What characterizes gesture is that in it nothing is being produced or acted, but rather something is being endured and supported. The gesture, in other words, opens the sphere of *ethos* as the more proper sphere of that which is human. (Agamben 2000: 57)

The name of the author as a gesture of authorship lies in the process of making a means visible as such. As a gesture the author-name is essential on discursive level but it is not irreducible to language: non-discursive intensities, namely materiality accompanies the verbiage of author-names. As a gesture the name of the author unfolds faciality and reinforces power relations and extends through visions of authorship: it redirects the notions of authorship toward potentiality.

During the Civil War of Finland in 1918, that is, the war between the whites and reds, Algot Untola became a literal “dead man walking”. He was imprisoned and sentenced to death because of his writings as Irmari Rantamala, the editor of a left-wing newspaper *The Worker*. The principle of Law linked to the names of author became materialized when he was shot dead. Algot Untola was convicted and accountable for what Irmari Rantamala wrote. It does matter, who is speaking. However, it is not always evident, who that “who” is.

THE SUBLIME: PROCESS AND MEDIATION

Liza McCosh

This chapter expounds on the concept of a “Material Sublime” which I suggest is relevant to the artist’s creative process and to the development of artistic practice. I maintain the creative process is generative and highlights the connection between nature, the body, and latent forms of knowledge that is revealed through material interaction significant through the activity of painting and the process of a material sublime. These ideas, drawn from my experience as a painter and realized in my creative output, are more fully detailed in my doctoral thesis, *The Sublime: Process and Mediation* (McCosh 2003).

The sublime is a subjective, sensory experience; what is sublime to one person may not be felt in quite the same way by another. A subject’s preconditioning plays a role in determining what is sensed as sublime. The experience has a manifold of sources; it can occur fleetingly in everyday experiences or be activated through engagement with objects or activities that inspire awe or are substantial in magnitude (Burke 1757; Kant 1763). The sublime encounter generates feelings of awe and wonder within a person as initially, the subject cannot fully comprehend the object or experience. The faculty of reason is challenged, causing an indeterminate relationship between the subject and object perceived. This relationship is restored as the subject gains a new understanding in relation to the source of the experience. This process implies that the sublime experience operates at the outer parameters of a subject’s existing knowledge and functions as an experience through which new understandings may emerge. As a revelatory experience, the sublime is relative to creative practices, generating feelings of elation and wonder in the artist when new possibilities for creativity emerge and are realized. I propose that the interaction of matter is inherent to this process, exemplifying that the sublime is a materially based experience.

Painting and other forms of creative practice rely on the interaction of matter to produce new form. The process of the sublime results from this material interaction and is the vehicle through which new insights are revealed to the artist. This material perspective positions the creative process as significant in the formation of new knowledge.

Historically, interpretations of the sublime have largely been based on Idealism which gives primacy to the mind by maintaining that the real is of the nature of thought, and perception is merely a presentation of ideas. In this context, matter and material processes have no existence independent from mind (Rummel 1975). Rene Descartes (1596–1650) viewed knowledge as being derived from mind, rather than the physical experiences of body (Armstrong 1999: 10). Matter, in the form of body and environment, were regarded in terms of “other”. In rationalizing the sublime Immanuel Kant (1724–1804) also emphasized the elevation of the mind through its capacity to “overcome with reason” the object perceived (Kant 1763, Goldthwaith 1960). Idealist views largely discounted engagement with the material world in favor of privileging the mind in the formation of knowledge. Alternatively, Materialism maintains that knowledge is derived through the senses and material interaction. Mind is therefore produced by matter and cannot live without it (Novak 1965: 4–6). The notion of a material sublime holds no distinction between mind and body; both are viewed as matter, co-dependent in and for existence. Rather than viewing the material world as “other”, a material sublime emphasizes a subject’s development through the interaction and co-emergence of matter and this is readily exemplified through creative practice, which in turn, relies on interaction with the world.

Traditionally, the sublime was defined in relation to the notion of the transcendental, rather than material aspects of existence. Rudolf Otto suggested this was due partly to the ineffable nature of the experience; it cannot be anticipated or represented which gives it a mysterious dimension (Otto 1923: 41). Eighteenth-century Romantic painting attempted to evoke a sense of the sublime by depicting vast, expansive landscapes that were aligned with the notion of spirituality. In attempting to represent the infinite, artists emphasized aspects of mystery and the spiritual as evidenced by the work of German Romantic painter Casper David Friedrich (1774–1840). The interpretation of Romantic art works was based on an idealist view that now proves problematic. Nature was the subject of these paintings but interpretations focused on the idea of transcendence beyond the earthly realm. If a painting is viewed in this context without consideration of the earthly materiality of its making, the meaning of the work pivots on the notion of transcendence beyond the material world. Romanticism influenced the mystico-religious aspirations of much art to follow as illustrated by the work of Kandinsky (1866–1944), Malevich (1878–1935) and Mondrian (1872–1944). Such artists “sought to evoke the Absolute through the metaphysical metaphors of pure abstract shape” (Cubbs 1994: 81). Defining art purely through a transcendental framework ignores the implications of matter in the making of meaning. Such interpretations can be viewed as dualistic in structure by privileging mind and conscious thought above body, matter and experience and by failing to recognize the co-emergent process that operates in creative practice.

Edmund Burke’s *A Philosophical Enquiry into the Origin of Our Ideas of the Sublime and the Beautiful* (1757), examined emotions produced by engaging with objects. His theory was complex, but put in simple terms it states that experiences arousing a sense of delight in a subject are sublime, whereas those inciting love are beautiful. Burke focused on the emotion of terror; the interrelation of terror and pleasure termed by

Burke as “*tranquility tinged with terror*”, produced the sublime feeling (Burke cited in Boulton 1987: xxxiv). The following passage offers further explanation:

When danger or pain presses too nearly, they are incapable of giving any delight, and are simply terrible; but at certain distances, and with certain modifications, they may be, and they are delightful, as we everyday experience. (Burke 1757: 39–40)

Burke’s “*delight*” arises when pain, danger or fear is suddenly withdrawn from the subject. The Burkean sublime can be viewed from a material perspective as emotive experiences derive from engagement with the material world and the body mediates these experiences through the senses.

Moments in my studio practice resonate with Burke’s ideas on the sublime. Apprehension felt in the developmental stages of creativity, when as an artist I am unclear of my direction, often precedes what I sense is a sublime experience. When I am at this stage of the painting process, I don’t always understand where my actions will lead or what results will be achieved when particular materials interact. At times mishaps may occur during the studio process, leaving the resolution of the work in doubt and leaving me with an uncomfortable feeling. This experience is not grounded in terror but is accompanied by a distinct feeling of apprehension. However, when new ideas, understandings or painterly effects emerge from this process, I feel a sensation of delight and wonderment just as Burke has described. This became evident to me whilst painting the *Altitude* series (Figure 12) and when I reached a point of despair regarding the progress of the work. I had been using a sponge to softly apply pigment and without consideration. I squeezed the remains of the sponge onto the surface of the painting in an action resembling the wringing of a wash cloth. The effects from this type of application amazed me. As the pigment was forced through the sponge its volume and consistency changed. The paint retained air bubbles, taking on a whipped texture and the appearance of porous rock. The effect did reduce throughout the drying process, but in some areas of the application a semblance was retained. New possibilities for my painting practice had emerged simultaneously with the sensations of delight and wonder that Burke had earlier identified with the sublime. The studio process demonstrated the material basis of the sublime and also revealed the sublime in the context of its relevance to epistemology. The “new” was illuminated and materialized through the interaction of matter and the process of making.

Matter informs the artist throughout the creative process. When painting in oils, I have often achieved desired results by employing my pre-existing skills and knowledge as a painter. However there are times when my practice seems to lack challenge and then the work stagnates rather than progresses or develops in fresh directions. This can be likened to what has been referred to as a state of *habituation* (Kreitler and Kreitler 1972: 50). Habituation occurs when tension is not evoked by new, surprising or complex stimuli and in my case, I felt this state occurred more when I painted in oils and used more conventional techniques of application. When I began to paint in



12. Liza McCosh, *Altitude*, 2003, acrylic on canvas, 75 x 37cm

acrylics I did not have the same control over the material; acrylic paint has a quicker drying time and usually offers less scope for manipulating the imagery. This time was fraught with struggle as my boundaries of knowledge, determined in part by my pre-existing skills in oil painting, were challenged. Occasionally, and sometimes in desperation, I haphazardly applied more materials to the canvas and more water to the materials without direction or due consideration for the mark or its placement. Sometimes this resulted in an unworkable mess but at other times the interaction of materials unexpectedly produced new and wondrous effects that I had previously not been able to achieve. Accidents occurring with the mixture and application of pigments have, at times, brought similar results; paint would ripple or separate in unfamiliar ways causing particular effects and leaving me wondering about the possibilities of reproducing the effects in future works. The activity of material dictated challenge and change within my practice and offered fresh knowledge in relation to creative effects that could be achieved. I felt a heightened tension within my practice, which to some extent disallowed the state of habituation. Rather than controlling the paint as I had previously done with more conventional methods of application, the material was suggesting other ways of achieving new effects. It became obvious to me that my creative practice was a co-emergent activity rather than one of material being mastered. The notion that artist and material co-emerge through the practice of

painting challenges the binary subject/object division, identified as “mastery” in traditional forms of painting and in interpretations of the sublime based on Idealism.

Drawing from Kant’s schema, Thomas Weiskal defined the sublime as “that moment when the relation between signifier and signified breaks down and is replaced by an indeterminate relation” (Weiskal 1976: ix). He views the process of the sublime in three phases. Firstly, the mind is in a state of normal perception, in a determinate relation to the object. Nothing interrupts representation and there is a smooth correspondence between inner and outer, or the rational mind and object perceived. In the second phase this smooth relationship breaks down and is replaced by a feeling of awe or incomprehension overtaking the subject. In the third phase, the rational mind recovers some understanding and a balance between the inner and outer is regained (Weiskal 1976: 23–24). This corresponds with moments of the sublime that I feel within my studio enquiry when new insights are revealed through the interaction of matter, producing feelings of wonder and elation. Balance is something I achieve when I realize exactly how the effects have resulted and how I may use the particular process in future paintings that I produce.

Traditionally, the third stage of the Kantian sublime privileged ideas and mind above the material world, of object and experience. A form of *mastering* is implied through the mind’s attempt to reason matter; nature “sink[s] into insignificance before the ideas of reason” (Kant quoted in Hitt 1999: 609). This Cartesian-based view alienates matter from the lived experience from which knowledge of our world is formed. In creative practice, matter, artist and world are engaged and co-emerge through the work of art. The matter of painting reveals the relationship between a knowing subject and the environment and this interaction is mediated through the experience of the sublime.

Various contemporary theorists have challenged binaries implicated in the Kantian sublime. Christopher Hitt has identified the third stage of Kant’s process as problematic in terms of reinforcing binary opposition. He calls for an “ecological sublime”, maintaining that “the discourse of the sublime has operated to confirm the authority and autonomy of a subject over and against a threatening other”. He suggests the concept of a sublime wilderness “depends on or re-inscribes the notion of nature’s otherness, of the separation between the human and non-human realms” (Hitt 1999: 603). Hitt suggests that a modification of Kant’s third stage of the sublime would address this. He maintains an ecological sublime “ultimately yields a heightened understanding” in relation to nature, rather than a mastering over it and “offers a unique opportunity for the realization of a new, more responsible perspective on our relationship with the natural environment” (Hitt 1999: 604–11). A material sublime functions through the co-emergence of matter rather than the mastering of matter. Its interpretation therefore opposes the dualism that Hitt identifies in the Kantian sublime.

Hitt puts forward a redefinition of the “mastering” stage of the sublime by addressing a section of writing by Thoreau, taken from “Ktaadh”, an essay from his collection, *The Maine Wood*:

... I fear not spirits, ghosts, of which I am one, – *that* my body might, – but I fear bodies, I tremble to meet them. What is this Titan that has possession of me? Talk of mysteries! – Think of our life in nature, – daily to be shown matter, to come in contact with it – rocks, trees, wind on our cheeks! The *solid* earth! The *actual* world! The *common sense*! *Contact!* *Contact!* *Who* are we? *Where* are we? (Thoreau quoted in Hitt, 1999: 615)

Nature is not represented from an anthropocentric perspective in Thoreau's writing; it is not mastered or dominated. Drawing from the writings of David Robinson (1997) and Max Oelschlaeger (1991), Hitt suggests Thoreau uses "matter" as a general semiotic term to denote aspects of nature and that the "contact" with the earth he refers to is "beyond the realm of language". Thoreau reveals a presence concealed by language; the true meaning of wilderness is based in living nature and in the relation of human consciousness to that world (Hitt 1999: 616). Hitt explains:

Thoreau's description of the moment of 'Contact!' with nature suggests a new way of imagining 'transcendence'. For this *is* a kind of transcendence – but not a transcendence of the physical world. (Hitt 1999: 616–17)

Hitt suggests that in Kant's concept, the discovery of reason abrogates the natural world whilst in the writing of Thoreau "the discovery of nature abrogates reason" (Hitt 1999: 616–17). By drawing on Thoreau's writing, Hitt demonstrates that the aspect of mastery within the sublime can be redressed through creative strategies.

Barbara Freeman and Patricia Yaeger also present ideas that support the concept of a material sublime. Both writers apply their arguments to literature. However, their concepts can equally apply to painting and other forms of art. Their enquiries are based on a feminist revisioning of the sublime, therefore a material sublime that challenges binary opposition has relevance to their aims. Freeman attempts to re-evaluate traditional interpretations of the sublime by challenging past studies, which she says have conceptualized the experience as a "struggle for mastery between opposing powers, as the self's attempt to appropriate and contain whatever would exceed, and thereby undermine it" (Freeman 1995: 2). She believes that within the metaphysical theories of sublimity, there lies another account of the sublime "that does not attempt to master its objects of rapture"; rather her notion of a feminine sublime "neither possesses nor merges with the other but attests to a relation with it" (Freeman 1995: 3–9). Freeman's revision of the sublime is relevant to my creative practice. The painting process that I employ allows for a degree of autonomy of material in forming imagery; the material process directly informs me. A relationship emphasizing co-emergence, rather than mastery over an "other", results. My painting process is compatible with Freeman's objectives for a more equitable relationship between artist and materials in evoking experiences of the sublime.

The feminine has traditionally been understood as the creative principle. However, Freeman uses the term not to specify a particular feminine subjectivity or mode of expression, but to contest binaries, thereby calling such categories into question. To

consolidate her ideas she argues against the preexisting construction of the “feminine” by applying Luce Irigaray’s idea of the feminine self as a “no-thing” for which “no identity can be defined”. This “contests the logic of identity that conceives of the self in exclusive terms of presence or absence” (Freeman 1995: 9). In this context, the feminine “becomes one name for a residue that disrupts the oppositional structure male/female and thereby calls for a radical re-articulation of the symbolic order”. She calls for the consideration of a feminine sublime, as it takes “a position of respect in response to an incalculable otherness” (Freeman 1995: 10–11). I suggest Freeman’s notion of otherness alludes to bodily processes and the material world.

In a feminine sublime a subject enters into a relation with the excess experienced, rather than neutralizing it as suggested in transcendental notions of the sublime. Otherness is embraced rather than subordinated. A feminine sublime recognizes the sensate body; it links to the materiality of existence rather than the notion of the metaphysical and therefore works against the binary framework inherent in traditional interpretations of the transcendental sublime. Freeman’s model is useful in advancing the concept of a material sublime which challenges the binaries of mind and body, human and nature, spirit and matter that have dominated Western thought.

Challenging binaries is an attempt at equalizing. Bearing this in mind enables us to conceptualize what Patricia Yaeger terms a “horizontal sublime...that expands toward others, spreads itself out into a multiplicity”, rather than “the sublime of domination” which she views as vertical and associates with the ego and masculinity (Yaeger 1989: 191). Yaeger’s notion of a “horizontal sublime” suggests interconnection and illustrates the aspect of co-emergence characteristic of creative practice and the material sublime. In the context of painting, the experience of the sublime assists the artist in the generation of new form. Although always connected with the artist, the work develops several existences somewhat apart from its creator. An artwork is engaged by a multiplicity of viewers, who have a plurality of views and experiences that they relate and share with the work. The artwork becomes more than a mere object, it is something that is inter-relational and bound to experience.

“Materialism” in this context, is a holistic concept; it determines that body is not distinct from the objective world and that mental processes result from physical activity. The distinctions between mind, body and environment are therefore eliminated. Inherent to any artwork or creative process, is the combined action of matter in the form of artist’s body, environment and creative materials. Creative processes implicate matter in the formation of new knowledge, which for the artist may comprise hitherto unfamiliar creative effects which provide fresh insights into the production and development of their work. This is a materialist perspective that identifies the epistemological relevance of matter in which the sublime experience is grounded.

My studio process allows for the autonomy of matter in making meaning. In my acrylic paintings I apply large washes of heavily diluted pigment over the surface of the canvas, which is laid flat on the studio floor. I prime the surface of the canvas to prevent absorption. This lets the medium gather, flow or form pools upon the face of the painting thereby allowing the evaporative processes of nature to come in to play. Materiality is further signified by engaging natural processes in the making of the work;

the autonomous flow of the liquefied material interacts with the natural evaporative processes of the water cycle, and this directly informs the emergence of the image.

Multiple layers of pigment are applied to the canvas with the images slowly emerging after long drying periods. Applying the paint is often messy as the pigment is heavily diluted and sometimes runs quickly from the paintings surface the floor of the studio. There are times when I am immersed in paint and water, and other times that require long periods of waiting and watching for the pooled areas of liquefied paint to evaporate. This method contrasts other more traditional forms of painting as the watery material and its natural cyclic processes largely dictate the painting process and development of image. It is a significant painting process in terms of interaction as the materials retain autonomy within the creative process thereby reducing the notion of mastery and binary subject/object opposition. Instead it emphasizes the act of painting as a participatory process that signifies material co-emergence.

The painting process that has developed within my practice has a strong connection to the place where I live. Warrnambool is located in a rugged coastal region, on the southern coast of Australia. Here I constantly witness the changing face of the ocean and am exposed to the changeable weather patterns of the sky, a feature for which the area has gained renown. The emphasis on liquidity within my painting process identifies with the transitory effects of the water cycle visually prominent and felt within my environment. When working in my studio I feel as though I am immersed in these elements. Applying the paint is like dabbling in the shores of the ocean, while waiting for the pooled areas of pigment to dry and the images to emerge is like waiting for early morning mists which appear in the valley below my studio, to clear. My painting process and subsequent images result from my connection to this environment.

My painting practice has assisted in revealing the connection that exists between me, my environment and my creative practice. This inter-relationship is vital to my practice. The sublime experience is pivotal to the development of creative practices, as it operates at a junction between the making of a subject and the world. Paul Crowther elaborates on this idea:

The shifting field of relations wherein experience is constituted, is stabilised by human embodiment, and the standpoint of free agency. These two factors coincide in traditional insights and decision-making. What is fundamental is not this or that moment of self-consciousness, but the urge to create or discover meaning that is embodied in every such moment. (Crowther 1995: 17)

For Crowther, the sublime “is experienced as profoundly and inseparably connected with – indeed, as *called forth and projected from* – finite being’s struggle to launch itself into and articulate the world” (Crowther 1999: 167). Painting is a means of demonstrating this, of articulating the world and our relationship therein. Crucial to the creative process of the painter is the notion of human embodiment that Crowther mentions.

Merleau-Ponty identified the integration of body and environment as basic to our existence and understanding of the world. He maintained “the body is in the world

as the heart is in the organism” (Merleau-Ponty 1978: 203). In his essay, “The visible and the invisible” (1968), he refers to the lived body and the world as flesh, not inert matter, but “perpetual pregnancy, perpetual parturition, generativity and generality, brute essence and brute existence, which are the nodes and the antinodes of the same ontological vibration” (Merleau-Ponty 1968: 115). His preliminary notes state that body is made of the same flesh as world that the flesh and body are shared by the world which reflects this (Merleau-Ponty 1968: 248–9). His notion of flesh is inclusive of the sensible:

The flesh is the body inasmuch as it is the visible seer, the audible hearer, the tangible touch – the sensitive sensible: inasmuch as in it is accomplished an equivalence of sensibility and sensible thing. (Lingis cited in Merleau-Ponty 1968: iv)

My paintings exemplify my relationship with the tangible and intangible aspects of the environment I experience. My awareness of nature and place is heightened through the materials and creative process of painting. The paintings become mediatory objects inscribed with the interactive relationship between artist and world; a co-emergence manifested in a new form via the materials of creative production.

Anne Whiston Spirn puts forward the idea that there is a *landscape language* that speaks to us. She suggests that “materials are sensory” and that through sight, smell, hearing and touch knowledge of the world is gained. Gravel crunching underfoot, grass swishing or the hollow noise of a wooden bridge can provide knowledge. “Materials arouse senses, carry meaning, pose limits. Deployed deliberately, to modify process and form and extend their meanings, materials furnish precision and nuance” (Spirn 1998: 97–101).

Spirn implies that landscape experiences inscribe and mold the perceiver. She further suggests that “landscape metaphors modify perceptions, prompt ideas and actions”, therefore molding the landscape, in turn (Spirn 1998: 24). She views material interaction through landscape experiences, as action that embodies elements of the landscape in the subject and in turn, elements of the subject in the landscape. This view aligns with Merleau-Ponty’s ideas on the reciprocal relationship and embodiment that occurs between subjects and their world. Spirn states, “to recover and renew the language of landscape is to discover and imagine new metaphors, to tell new stories, and to create new landscapes” (Spirn 1998: 25). This has relevance for all artistic processes. When the world is experienced through the senses, it is transformed in various ways through the materials of production. For me, the landscape I experience is renewed in my paintings, by the material process that I employ. The paintings therefore are embodied with the knowledge of my landscape experience, realized through the interaction of paint, water, canvas, body and tools.

Merleau-Ponty posits a subject and world in union, enmeshed and co-emergent in development. He examined the connection between perception, the visible and the body, suggesting a “primordial perceptual encounter” as an act of revealing a relation between ground, figure and world (Roberts 1998: 130). Crowther observes that

Merleau-Ponty makes no distinction between seeing and knowing or visual and tactile perception: “fundamental contact with the world is prereflective and involves our sensory, motor, and affective capacities, operating as a unified field”, all aspects of experience are therefore considered to contribute to one’s full perception (Crowther 1993: 107). This is relevant to the practice of painting which draws on stored or embodied knowledge of one’s experience in the world. I feel that my landscape experiences are with me in the studio; the experience of seeing, knowing, the visual and tactile perception of the environment becomes one with the process of painting.

According to Merleau-Ponty, perception of the world involves the act of embodiment which spans the invisible dimension of perception, a space between the “seer” and the “seen”:

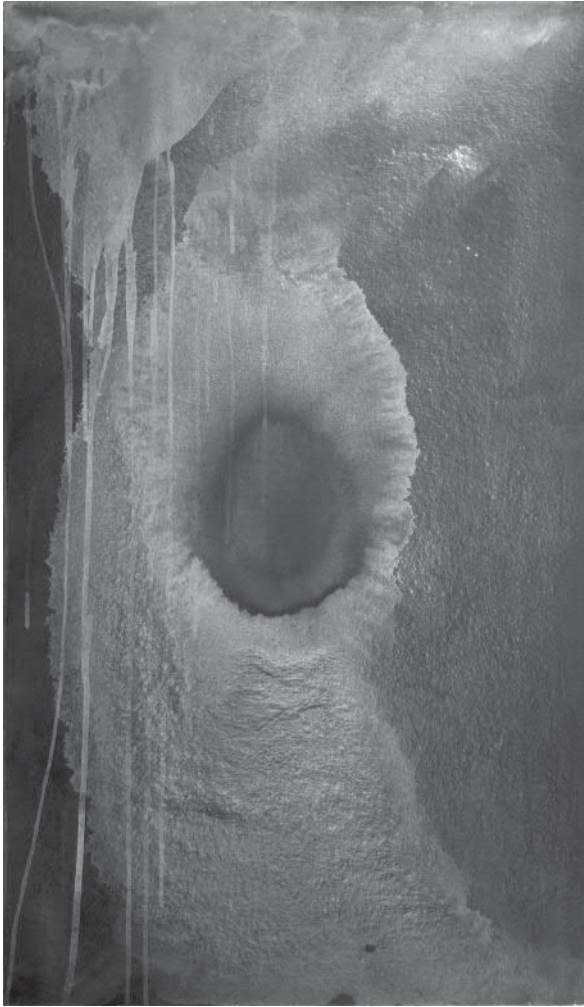
There is a human body when, between the seeing and the seen, between touching and the touched, between one eye and the other, between hand and hand, a blending of some sorts takes place – when the spark is lit between the sensing and the sensible ... (Merleau-Ponty 1964: 163)

Merleau-Ponty suggests that when body and world engage, roles become interchangeable; “seer and seen are capable of reversing their roles as subject and object” (Johnson 1993: 47). This was revealed to me whilst painting *Threshold* (Figure 13).

The liquefied pigment and the longer drying period, partly enforced by the cold winter weather, resulted in creative effects of rippling and separation of the paint. The flow of pigment gathered in the center of the stretched canvas, which sagged under the weight of the fluid. As the pool dried and the pigment separated, a radial pattern that resembled a vortex-like structure emerged. The autonomy of material had largely created this unique effect that could not be reproduced by other means. This process provided insight into my practice and moderated subject/object relations through shifting the emphasis in painting from the artist as master and creator, to the artist and materials actively co-emerging in the process of painting.

Merleau-Ponty’s suggestion of “seer” and “seen” interchanging, implies an invisible, liminal space where negotiation between subject and object takes place. Elizabeth Grosz suggests Merleau-Ponty’s perception operates midway between mind and body, that he “attempts to take up and utilize the space in between, the ‘no man’s land’ or gulf separating oppositional terms” and that this space “makes possible the binary terms insofar as it precedes and exceeds them, insofar as it is uncontainable in either term” (Grosz 1994: 94). Through sensory interaction, the sublime can be understood as a threshold experience that bridges the gap between the subject and object by mediating between the visible or the known, and latent forms of knowledge. The painter’s materials play an important part in this process. Materials make visible latent forms of knowledge. Matter therefore is active. Materials provide a bridge for the painter between the subject and object, making experience visible in a new way.

Merleau-Ponty emphasizes the invisible but this should not be interpreted as transcendental in terms of a separate reality or a spiritual world apart from this one. In *Eye and Mind* (1961) he stresses that “the lines of visible things are doubled by a



13. Liza McCosh, *Threshold*, 2003, acrylic on canvas, 117.5 x 70cm

lining of invisibility that is *in* the visible” and in doing this he “decenters the aesthetic task and desire away from the pursuit of an invisibility that would be a separate reality” (Johnson 1993: 52–3).

Radhika Mohanram points out that Merleau-Ponty’s concept of perception through the act of embodiment “reverses the Cartesian subordination of the body” by locating it “as primary for the perception of the world and the formation of knowledge” (Mohanram 1999: 16–17). Merleau-Ponty says that perception is a nascent logos; that it is through primordial perceptual encounters that our understandings of the world emerge (Merleau-Ponty 1964: 25). In a literal sense, I can understand this as

over the years my practice of painting has been affected by my bodily position in relation to the world. The practice of locating my body in a particular space is revealed to me through my practice of painting.

Merleau-Ponty maintains the body constantly changes its perceptual positioning in relation to the world thereby organizing the world in different ways. Thus, perception for him is a creative experience, as Crowther puts it “a constant and ever-renewing process of structuration” (Crowther 1993: 104). A simple formula can be deduced from the philosophy of Merleau-Ponty that shows:

Practice > revealing > perception > knowledge and development

This formula demonstrates a process whereby knowledge derives from the field of practice; the sublime experience overlays a space between material practice and perception encapsulating hitherto unrevealed knowledge. My experiences of the landscape are transformed into my paintings. I believe that my initial decision to use aqueous mediums was partly influenced by my interaction and experiences in landscapes that emphasized sky, mists and water. These experiences of nature were embodied; they became part of my body and part of my painting.

When I pick up a paintbrush loaded with pigment, I feel sensations through the implement that relate to my body, the surface of the painting and the pigment. When I touch the canvas with my brush, a sensory response is felt through the brush to my body. This sensory response informs me, conveying if the paint is either thick and sticky, or thin and runny. Using board or paper surfaces delivers a different sensation, as no rebounding of the surface will occur. Particular things that guide me are revealed through the instrument, which acts as a conduit while at the same time is an extension of my body. Specific painterly effects may be achieved by holding the painting implement in particular positions and previous experience can identify the sensations conveyed through paintbrush to arm. These sensations mediate the artist's process of painting and assist in achieving particular effects. Until the body deciphers this *felt* information from the interaction of materials, an artist may not know whether the desired effect will be properly produced upon the surface. Therefore, the sensations felt through painting implements aid an artist's ability to handle the medium, the technique and the scale of the painting.

The knowledge that artists acquire through the sensations felt when implements and materials interact, accumulates and is stored in the body to be used subliminally in future practice. Painting has taught me that the body is a site of knowledge, mediating experience with the world. The world is embodied through the senses and revealed through the activity of painting. In the context of Merleau-Ponty's ideas, “the artist *is* the scene, the focal point of convergent experiences and events” and painting “virtually emerges out of perception and its presence to the world through the body” (Roberts 1998: 132–3). He viewed painting as a mode of disclosing our insertion and relation with the world (Crowther 1993: 107). The artwork then, becomes more than a mere projection of the artist; it reveals experience in and of the world.

FASHION AS AN EMBODIED ART FORM

Llewellyn Negrin

Fashion, by virtue of the fact that it is designed to be worn, is seen as one of the most embodied of art forms. Indeed, it is precisely because of this that some theorists (Hanson 1993) have recommended that greater attention be paid to it as an antidote to the suppression of the body that has occurred in traditional aesthetics. Conceiving of aesthetic experience as a mode of disinterested contemplation divorced from sensual desire, philosophers in the Kantian tradition have focused on those art forms that are most removed from the body, privileging the visual over the tactile. However, as Karen Hanson has argued, fashion's close proximity to the body highlights the unsustainability of the separation of the aesthetic response from all other forms of sensuous experience. As she puts it:

If, as Kant says, the satisfaction determined by the beautiful is unrelated to inclination, then one who would judge some fashionable dress beautiful will clearly have to cope with some difficult problems of desire ... Socrates' insistence on the philosopher's distance from physical desire – estrangement from an interest not only in sexual pleasures but also in 'other attentions ... to our bodies,' 'smart clothes and shoes and other bodily ornaments' (Plato 47) – is a constraining legacy. (1993: 235)

Paradoxically however, while fashion has been embraced as an embodied art form by theorists such as Hanson, its evolution in Western culture has to a large extent been predicated on a disavowal of the body. The trajectory of fashion, as Anne Hollander describes it (1993: 311–314), has been characterized by the devising of forms more directly related to images of the ideal body than they are to the physiognomy of actual bodies – a process which has become particularly pronounced in the twentieth century. Just as modernist artists abstracted from reality, so fashion has become increasingly abstract in form, being shaped less by the corporeal nature of the body and more by formal aesthetic considerations.

While Hollander, among others, has regarded this endeavor as liberating, I will argue here that a more adequate approach to dress is one which acknowledges the materiality of the body rather than seeking to negate it. Instead of upholding as the ideal clothes that transcend the body, we need to envisage a new mode of dress, which engenders a sense of the body, not as a visual image, but as an active corporeal presence. Before turning our attention to this alternative conception of dress, let us examine the shortcomings of the notion of fashion as transcendence of the body.

Fashion's Disavowal of the Body

Although all cultures engage in some form of body adornment in order to signal the individual's accession to sociality, what distinguishes Western dress is that it is predicated on a radical distinction between nature and culture. Thus, while in tribal societies the forms of body adornment often mimicked animals or birds as a means of indicating people's kinship with them (Boas 1951: 250–1), clothing in Western culture has been designed to cover up the body and its functions so as to distance us from our animality. This has been the case particularly since the rise of Christianity, with its conception of the naked body as the source of all those sinful desires that threaten to distract us from the path to spiritual salvation. As Soper writes:

In Christian mythology, we acquire our clothes in losing our 'natural' innocence and coming into knowledge of good and evil. Clothes are in this sense definitively cultural objects, closely bound up with a sense of shame and their primary purpose is to conceal the organs of those functions which have been deemed to degrade us by tying us too closely to a bestial nature. (Soper 2001: 17)

Thus, the Western concept of dress, from the time of Christianity, has been defined in opposition to a naked or "natural" body which is seen to pre-exist culture and to pose a constant threat to it (Mascia-Lees and Sharpe, 1992: 3). Clothing and our self-consciousness at the naked body were born together, each presupposing the other. As Barcan puts it: "Nudity (like modesty) does not precede clothing; rather, it comes into being with the invention of clothing" (Barcan 2004: 1). By contrast, in tribal cultures, which lacked the Christian sense of shame and guilt associated with the naked body, there was seen to be no necessity to cover the body with clothing. Body adornment was thus designed to enhance the body rather than to obscure it (Flugel 1950: 26 and 57).

While Western dress has been predicated on a disavowal of the body, this is not to deny that there have been occasions where it has enhanced rather than suppressed the erotic allure of the wearer, particularly in the case of female fashions, which have generally exposed more parts of the body than has been the case with male garb. James Laver however (1969: 14), overstates the case when he argues that the primary purpose of women's dress has been to heighten their sex appeal. According to him, while the main function of men's clothes has been to enhance their status, sexual attraction has been the main motivation behind women's fashions. In his view, the changes in women's fashions have been driven primarily by an imperative to eroticize the female body by constantly shifting the focus of sexual interest from one part of the body to another.

Although it is undeniable that there is an erotic element in much female attire, this has been contained within strict rules of propriety. Sensuality in dress has been acceptable only where it has been sublimated rather than overt. Where it has been too explicit, it has been subject to social censure. Women, more so than men, have been admonished for dress that oversteps the boundaries of modesty and probity.¹ Tseelon traces this concern back to the Old Testament, where it was Eve who was held responsible for tempting Adam from the path of God (1995: 9–14). In Christian theology, women were portrayed as being more susceptible to the temptations of the flesh than men, using their beauty and finery as a vehicle to seduce men away from the path of spiritual salvation. Christian doctrine taught that redemption lay in the renunciation of decoration and modesty in dress, a moral duty born of Eve's guilt. Thus, at the same time as female clothing has often enhanced the attractiveness of the wearer, women have had to be careful not to dress in too seductive a manner for fear of compromising their respectability.

Paradoxically, the more female attire has disavowed the body, the more tantalizing it has proven to be. Because women have been more closely associated than men with the body, sexuality and sin, their clothing, no matter how modest, has consistently been interpreted as a mode of sexual attraction. Since the whole of the female body has been sexualized, any exposure of the flesh, no matter how slight, has been seen as more highly erotically charged than exposure of the corresponding part of the male (Steele 1989: 44).

Male dress by contrast, has seldom been perceived primarily in sexual terms. Although at various times in the history of Western fashion, aspects of the male body have been accentuated such as the shoulders, chest, legs, buttocks and even the penis, masculine garb has been seen first and foremost as a marker of status and/or occupation (Steele 1989: 44). Especially since the advent of the business suit in the nineteenth century, male dress has been designed to de-emphasize the body, so conferring on the wearer the qualities of rationality and sobriety. In foregrounding the cerebral over the bodily, this mode of dress has been taken to be indicative of the assumed superiority of the masculine over the feminine – a superiority which is seen to rest in men's supposed greater capacity to engage the intellect independently of interference from the body (Polhemus 2000: 47).

Generally speaking, the more disembodied the clothes, the more respect and authority is accorded to the wearer (Harvey 2008: 63–4). This applies not only to the male suit but also to the dress of other figures of authority such as the robes of the clergy or the garments worn by the judiciary. It is not surprising then that as women have acceded to positions of power, they have adopted masculine-style dress designed to de-emphasize the body.²

This endeavor to transcend the body, which underlies the evolution of Western fashion, has been interpreted by some as a liberating process. Georg Simmel, for instance, in his essay "Adornment" argues that the more "evolved" forms of body adornment are those which are furthest removed from the contours of individual bodies, that is, those which have a "supra-individual" form that transcends the particularity of the individual. As he argues, the role of adornment is to intensify and en-

large the personality of the wearer – something which can only be achieved through the addition of that which goes beyond the specifically individual. To quote him:

[The] ...accentuation of the personality...is achieved by means of an impersonal trait. Everything that 'adorns' man can be ordered along a scale in terms of its closeness to the physical body. The 'closest' adornment is typical of nature peoples: tattooing. The opposite extreme is represented by metal and stone adornments, which are entirely un-individual and can be put on by everybody. Between these two stands dress, which is not so inexchangeable and personal as tattooing but neither so un-individual and separable as jewelry, whose very elegance lies in its impersonality...What is really elegant avoids pointing to the specifically individual; it always lays a more general, stylized, almost abstract sphere around man ... That new clothes are particularly elegant is due to their being still 'stiff'; they have not yet adjusted to the modification of the individual body as fully as older clothes have, which have been worn and are pulled and pinched by the peculiar movements of their wearer – thus completely revealing his particularity. (1950/1908: 340–1)

For Simmel, the transcendence of particularity, which is made possible by the wearing of clothes that do not bear any bodily traces, "ennobles" the individual by connecting them to a realm that exceeds them. The crucial element in clothing that enables the individual to connect with the general, is "style". As Simmel argues, style is not determined by the idiosyncratic features of the body of the individual wearer but, rather, "is always something general. It brings the contents of personal life and activity into a form shared by many and accessible to many" (1950/1908: 340). He continues that:

It is the greatest mistake to think that, because it always functions as the adornment of an individual, adornment must be an individual work of art. Quite the contrary: because it is to serve the individual, it may not itself be of an individual nature. (1950/1908: 341)

Hollander also regards positively, the quest of fashion to exceed the constraints of the body through the development of stylized forms derived independently from it. She conceives of fashion in Western culture primarily as an aesthetic project involving the constant experimentation with new visual forms which are more directly related to images of the ideal body (either in art or, more recently, in the mass media) than they are to actual bodies. Rather than the forms being determined with reference to the physiognomy of "real" bodies, fashion designers seek to reshape the body of the wearer in accordance with their own aesthetic ideals, which have in turn, been influenced by existing pictorial representations of the body. As Hollander writes:

[T]he new representational force of fashion [in the sixteenth century] allowed a retreat from the actuality of breasts into the abstract, emblematic figure dress

could make them a part of. It was with the development of such devices in fashion that dress became a modern art. (1994: 32)

For Hollander, fashion is primarily a *visual* art with the primary concern being the achievement of a certain *look*. In this context, changes in artistic portrayals of the body have been a crucial stimulus in fashion change since, without the visual demonstration that pictures provide, certain ways of looking could not be seen as more desirable than others. Such images have served as a template for what is considered attractive in the clothing of the body (1993: 350).

In the twentieth century, as modernist art freed itself from the necessity to imitate reality and became increasingly abstract, so the visual forms of fashion became further removed from actual bodies, evolving according to their own autonomous logic, thereby significantly expanding the creative possibilities of fashion designers. As Hollander puts it:

Clothes could [now] aim at an ideal shape of their own, to which a body was truly subordinate – a box, a cylinder, a pyramid ... Actual bodily shapes, always apt for distortion, now had the further task of turning themselves into detached patterns in their own mind's eye. (1993: 337)

This abstraction from the body was further encouraged by the fact that couturiers increasingly began to create designs not for particular clients, but for their own collections (Hollander 1993: 351–3). No longer constrained by the necessity of having to design with a particular person's body in mind, fashion designers could give free rein to their artistic impulses. Indicative of this was the closer association between fashion designers and artists that began to occur in the early decades of the twentieth century. Exemplifying this trend were the fashion designs of Paul Poiret. Influenced by the flat patterns of Art Deco and Art Nouveau and by the flattening of the picture plane that occurred in Cubism, Poiret created garments which treated the body as a two-dimensional surface rather than as a three-dimensional form (Hollander 1993: 334 and Martin 1998: 16).³

Hollander's positive appraisal of this abstraction of fashion design from the corporeal nature of the body is reiterated in her praise for the male suit. As she argues in her book *Sex and Suits*, the male suit is superior to female dress because of the simplicity of its form. Unlike female dress, which draws attention to specific parts of the body, the male suit provides a "complete envelope for the body" in which "the lumps and bumps of the individual body's surface are harmoniously glossed over, never emphatically modeled." (1994: 8) She continues that: "Decorative elements are integrated with the overall scheme, so that nothing sticks out, slides off, twists around, gets bruised, goes limp or catches on anything ... It is universally flattering because it does not insist on specific bodily detail" (1994: 9).

Likewise, Umberto Eco praises male dress insofar as it transcends the body. He argues that while the constricting nature of women's attire such as corsets and high heels induces a greater body-consciousness, deflecting attention away from more

intellectual pursuits, male dress, which tends to be less body hugging and looser fitting, allows the wearer to “forget” his body and thus engage unimpeded in rational thought. His preference for clothes that allow the mind to live an existence free from an awareness of the body is indicated in the following quote:

[T]hinkers, over the centuries, have fought to free themselves of armor. Warriors lived an exterior life, all enclosed in cuirasses and tunics; but monks had invented a habit that ... left the body ... completely free and unaware of itself. Monks were rich in interior life ... because the body, protected by a habit that, ennobling it, released it, was free to think, and to forget about itself. The idea was not only ecclesiastic; you have to think only of the beautiful mantles Erasmus wore. (1987: 194)

Critique of Fashion’s Disavowal of the Body

While this attempt to transcend the materiality of the body has been seen as liberating, at the same time it is premised on the highly problematic privileging of the mind over the body, which has been a long-standing feature of Western culture. As many theorists have pointed out (Hanson 1993: 235), underlying our attempt to transcend the body is the vain hope of immortality. Despite our attempts to deny our carnal nature however, ultimately we cannot escape our inevitable demise. Furthermore, implicit in this distrust of the flesh, is a denigration of the feminine (Bordo 1993; Hanson 1993), since women have traditionally been more closely associated with the body than have men. To quote Bordo:

If, whatever the specific historical content of the duality, *the body* is the negative term, and if woman *is* the body, then women *are* that negativity, whatever it may be: distraction from knowledge, seduction away from God, capitulation to sexual desire, violence or aggression, failure of will, even death. (1993: 5)

Rather than seeking to deny our corporeality, it should be acknowledged as an inextricable aspect of our being-in-the-world. As Merleau-Ponty has made clear, our bodies are not simply inert objects existing independently of our minds but rather, are the very means through which we come to experience and know the world. To quote him:

The perceiving mind is an *incarnated mind*. I have tried ... to re-establish the roots of the mind in its body and in its world, going against doctrines which treat perception as a simple result of the action of external things on our body as well as against those which insist on the autonomy of consciousness. (1976: 3–4) (my emphasis)

To presuppose that we can only truly be free if we transcend our corporeality is thus fundamentally misguided. Instead of regarding the body as the prison of the soul, it should be seen as the means through which to achieve self-realization.

Gail Weiss elaborates on this point in her critique of Iris Marion Young (1999: 43–50). In her essay “Throwing like a Girl,” Young argues that women are less able to act in the world than men because they are hindered by a greater self-consciousness about their bodies. Because young girls are socialized to focus on their bodily appearance, this inhibits their intentional engagement with the external world. Young suggests that one can only act freely once one loses one’s self-consciousness of one’s body. To quote her:

As human, [woman] is a transcendence and subjectivity, and cannot live herself as mere bodily object. Thus, to the degree that she does live herself as mere body, she cannot be in unity with herself, but must take a distance from and exist in discontinuity with her body. (1990a: 155)

Weiss however, questions whether bodily self-awareness necessarily undermines one’s capacity for action and suggests that Young’s argument ends up “in the unpalatable situation of supporting an entire patriarchal philosophical tradition that seeks to render the body and its contribution to everyday experience invisible” (1999: 46). As she argues, Young, in this essay, shares with Sartre and de Beauvoir, the problematic assumption that activity directed toward the external world necessitates a complete suspension of bodily self-consciousness. For Sartre and de Beauvoir, transcendent action is only possible insofar as one is able to escape the strictures of the body and its demands. Contrary to this, however, Weiss postulates that somatic awareness as such, is not inherently negative and does not inevitably lead to immanence. While consciousness of one’s body is problematic when it is dominated by a preoccupation with how one appears to others, there are other forms of body awareness that are enabling rather than inhibiting. Thus for instance, there are many tasks such as playing sport that require paying close attention to one’s body prior to, as well as during, the activity. In these circumstances, consciousness of one’s body enhances rather than detracts from one’s capacity to act effectively.

Young, herself, moves toward this in her essay “Pregnant Embodiment” where she challenges de Beauvoir’s argument that pregnancy is an inescapably immanent state since it is impossible for a woman not to be conscious of her body at this time and to be beholden to its demands to a large extent. Against this, she argues that the greater sensitivity that one develops to one’s body as it changes during pregnancy does not so much undermine as heighten one’s sense of self by acting as a constant reminder of the embodied nature of our existence. Even the most intellectual of activities cannot occur independently of our bodies which exercise an influence over our mental processes, whether we consciously acknowledge this or not. As Young puts it:

Pregnancy roots me to the earth, makes me conscious of the physicality of my body not as an object, but as the material weight that I am in movement. The notion of the body as a pure medium of my projects is the illusion of a philosophy that has not quite shed the Western philosophical legacy of human-

ity as spirit. Movement always entails awareness of effort and the feeling of resistance. In pregnancy this fact of existence never leaves me. I am an actor transcending through each moment to further projects, but the solid inertia and demands of my body call me to my limbs not as an obstacle to action, but only as a fleshly relation to the earth. (1990b: 165–6)

Once it is acknowledged that awareness of one's body is not automatically objectifying or disabling, it becomes clear that emancipation does not necessitate a denial of the body. Thus, while the goal of many feminists has been to challenge the patriarchal identification of women with their bodies, Weiss proposes that the aim should be to envisage new forms of body awareness that increase rather than undermine women's confidence in their bodily abilities and potentialities. Instead of seeking to eschew the body altogether, new ways of experiencing it other than as a passive object of the gaze need to be imagined.

This has important implications in the realm of dress. Rather than upholding as the ideal, clothes that make us forget that we have bodies, what one should seek is a mode of dress that engenders a sense of the body as active agent rather than as object of spectacle. In this light, the problem with women's dress is not that it induces a heightened consciousness of the body, as Eco, among others, has argued, but rather, that the nature of this bodily experience has been one which has given rise to a feeling of passivity rather than of empowerment.

Similarly, the supposed superiority of male dress also needs to be revisited. While it has been widely assumed that male attire is more "liberated" than female dress because of its relatively disembodied nature, Polhemus suggests that it has resulted in an estrangement of men from their bodies. As he laments:

From the Age of Colonialism until the present day, to be taken seriously on the world stage, a political or business leader must first renounce his physicality and hide his body behind that now universal uniform of the Invisible Man – the bland, ill-fitting suit. (2000: 47)

For him, one of the key problems of masculinity today lies in the fact that men in positions of authority are unable to take pride in and express themselves through their bodies but are required to present themselves as "disembodied intelligences". Only those who are in more marginalized positions, such as working class youth, blacks and homosexuals have the freedom to express their physicality more overtly. He interprets the growing popularity of tattoos amongst young males as an attempt by men to re-engage with the physicality of their own bodies. Because of the visceral nature of their acquisition, tattoos serve to remind the wearer of their corporeality.

Sweetman argues similarly, that integral to the appeal of tattoos is the physical process involved in their acquisition. Unlike other items of dress or adornment which can be easily put on or removed from the body, tattoos, which are indelibly incised into the skin, are seen as a way of providing a secure sense of identity by anchoring it permanently in the body itself. As he writes:

[T]he tattooed or pierced body cannot, or should not be reduced to pure text [since] the *process* of becoming tattooed or pierced ... is central rather than peripheral to many contemporary body modifiers' motivations and experience. Physical or corporeal affect, in other words, is key to the lived experience of many such 'subculturalists', and while socially and culturally mediated, the affectual cannot, in this case, be reduced to a purely discursive effect. (2001a: 187)

While most analysts of the sartorial behavior of young men in sub-cultures have focused on fashion as a semiotic system in which the symbolic codes of dress are manipulated to convey certain meanings, Sweetman draws attention to the importance of the bodily experience produced by the wearing of certain modes of attire and other forms of body adornment. As he points out, many of the activities engaged in by members of sub-cultures tend to focus on the body and so in order to understand the motivations behind those involved, attention has to be paid to the corporeal dimension rather than simply interpreting subcultures as signifying systems. To quote him once again:

Such groups are linked not only through their shared adoption of particular signifiers but also through their shared experience of the feel of the garments in question and the restrictions and possibilities that their materiality entails. (2001b: 67)

Thus, for example, the skinhead's adoption of shorn hair, boots and braces was not just an attempt to symbolically resolve the tensions inherent in the loss of traditional working class community, but also produced and reflected a particular orientation to the body.

This desire expressed by some men to re-connect with their bodies suggests then, that rather than seeking a mode of dress that minimizes our consciousness of the body, we need to consider clothing that re-engages with the body in new ways. How then might we imagine a new mode of dress which embraces rather than denies the body?

The Return of the Body

One way in which men have sought to re-engage with the body in recent times has been through the adoption of more body-revealing modes of dress. While women have traditionally worn more revealing clothing than men, especially since the beginning of the twentieth century with the shortening of hemlines and the baring of backs and midriffs, now there is a growing trend amongst some men to adopt clothing that accentuates rather than disguises their bodily contours. This is evident for example amongst homosexual men who wear body-hugging jeans and T-shirts and who "work-out" to heighten the sensual appearance of their bodies. Polhemus advocates such an approach in his article "The Invisible Man: Style and the Male Body" where he praises the uninhibited displays of male physicality evident in black and youth subcultures since the 1960s as well as in gay culture (2000: 47).

However, the problem with this “reclamation” of the body is that it continues to treat the body primarily as an object of the gaze – the body as passive, as visual representation. As women are only too well aware, this form of body consciousness, far from being emancipatory, is ultimately alienating since it is determined primarily by how one appears to others rather than enhancing one’s awareness of the body as active agent. Polhemus’ espousal of what he calls “the right of men to become sex objects” (2000: 51) is thus highly problematic.⁴

The attempt to reclaim the body by increasing its exposure is ultimately doomed to failure since the body is still conceived of as an image rather than as a corporeal reality. Richard Shusterman makes a useful distinction between two types of body consciousness, which is applicable here (2000: 142). On the one hand, there is the awareness of the external form of the body in which the focus is on its surface appearance. This form of body awareness, which converts the body into a visual representation, is the one that predominates today in our media-saturated environment where the image reigns supreme. On the other hand, there is another mode of somatic awareness, which Shusterman dubs “somaesthetics”, where the primary concern is not with how the body looks, but rather with how it feels. It concerns the body, “not in its external form, but in its lived experience, its kinaesthetic sense of itself”, as he puts it (2000: 149). While the first form of body consciousness induces a narcissistic preoccupation with one’s physical appearance, the second enhances one’s awareness of the body as a corporeal presence through which one engages with the world. Drawing on Shusterman’s distinction then, I would argue that for a mode of dress to be truly emancipatory, it needs to activate this second mode of body awareness which goes beyond the presentation of the body as object of spectacle.

Young makes a start toward a somaesthetics of dress in her essay “Women Recovering our Clothes” where she draws attention to the sensuous pleasures that clothes can provide women which are not structured around the male gaze. In particular, she identifies a pleasure that lies in the tactility of the garments. While most attention has been paid to the *look* of female fashions and how they appear to others, there has been far less focus on the haptic qualities of clothing and how these are experienced by the wearer. It has been assumed that women’s pleasure in dress is a secondary one which derives from them vicariously placing themselves in the same position as the putative male spectator: i.e. they only experience pleasure through their internalization of the objectifying male gaze, seeing themselves as others see them.

However, Young suggests that beyond this there is a female pleasure in dress that is not dependent on the male gaze – the pleasure of touch. This is a pleasure that can be experienced directly by the wearer without the mediation of the male spectator. Unlike sight, which is the most distancing of the senses, in which the subject is separate from the object, with the sense of touch there is no separation between subject and object since they are in direct proximity to each other. As she describes it: “Less concerned with identifying things, comparing them, measuring them in their relations to one another, touch immerses the subject in fluid continuity with the object, and for the touching subject the object touched reciprocates the touching, blurring the border between self and other” (1994: 204). Touch encompasses not just the feel

of fabric on skin but also “an orientation to sensuality as such that includes all the senses” (1994: 204). Thus, for instance, there is a pleasure that inheres in the way that fabric hangs and falls around the body as one moves.

However, at the same time that Young identifies a sensual pleasure in dress that is not determined by sight but by a more all-encompassing sense of the body, her subsequent discussion of the fantasies that women engage in regarding clothes is curiously disembodied and again seems to return to the primacy of sight. As she argues: “women take pleasure ... not just in wearing clothes, but also in looking at clothes and looking at images of women in clothes, because they encourage fantasies of transport and transformation” (1994: 206). She continues that:

Part of the pleasure of clothes for many of us consists of allowing ourselves to fantasize with images of women in clothes, and in desiring to become *an image, unreal*, to enter an intransitive, playful utopia. There are ways of looking at oneself in the mirror that do not appraise oneself before the objectifying gaze, but rather *desubstantialize* oneself, turn oneself into *a picture, an image, an unreal identity*. (Young 1994: 208) (my emphasis)

Here, Young recapitulates the conception of the body as adversary, which has to be transcended in order to attain true freedom. Instead of imagining new ways of re-engaging with the body other than as an image or visual representation, her utopian vision repeats the disavowal of the body that has underpinned the Western conception of dress.

Weiss argues in a similar vein that Young supports a concept of fantasy regarding clothes that is not sufficiently grounded in the materiality of the body. As she writes:

To claim, as Young does, that ‘the clothing image provides the image of situations without any situatedness’ implies that the clothing image is itself disembodied, a phantom that frees us from our own corporeality. It seems here as if Young has accepted a view of the body as a material prison that our fantasies can allow us to escape. In contrast, what I would emphasize is that when I am trying on clothes before the mirror, the clothing images that I create and fantasize about are images of my clothed body. Furthermore, I would argue, the self-transformations I explore through these fantasies offer so much narcissistic pleasure precisely because they are not unreal or desubstantialized but rather, because they hold out the possibility of resituating myself within my existing situation. (Weiss 1999: 59)

As is evident here, Weiss advocates a re-conceptualization of fantasy where the subject imagines herself, not as bodiless, but as experiencing new forms of embodiment that resist the current “techniques of corporeal inscription and normalization” (1999: 67).

The works of fashion designer Issey Miyake provide an example of a mode of dress, which evokes such an embodied fantasy. His aim is not so much to produce a certain “look” but rather, to stimulate an active interaction between the clothes he

designs and the body of the wearer. While Western fashions, particularly those worn by women, have generally tended to constrain the body, molding it in accordance with externally imposed aesthetic ideals, Miyake's clothing is designed to respond to the body of the wearer and its movements. His forms are not conceived of as static, visual representations but rather as dynamic, three dimensional sculptures which are constantly being re-formed by the body that inhabits them. As Jay Cocks comments: Miyake "likes to say that his clothes are unfinished, that a design isn't done until someone puts it on. And then, of course, it gets changed" (Cocks in Calloway 1988: 51). This is encapsulated by the Japanese concept of *fusoku-shugi* where expressiveness inheres in what is omitted or left out. The incompleteness of Miyake's garments invites the active engagement of the wearer who re-configures them in ways which suit their own bodies. To quote Miyake: "My clothes can become part of someone, part of them physically. Maybe I make tools. People buy the clothes and they become tools for the wearer's creativity" (Miyake quoted in Calloway 1988: 16).

In contrast to Western clothes, whose shape is fixed by the often elaborate tailoring involved in the making of them, Miyake's garments draw from the Japanese tradition of wrapping fabric around the body, as exemplified by the kimono, as Martin points out (1995: 215). This allows for a much more fluid and organic relationship between the fabric and the body in which the garment is constantly changing its form in response to the movements of the body. Rather than the body being constricted by the fitted garment as occurs in Western fashion, Miyake's often voluminous designs allow for a great range of movement. Martin describes the difference between the two approaches thus:

With a layering, cloaking propensity offered as a fundamental alternative to tailored clothing comes as well a completely different body *vis-à-vis* clothing expression, subverting or at least posing an option for dress beyond the simple erotics and mechanics of underlying and visible body. Willful scorn of tailoring is also indicative of a nonchalance and preference for the irregular and unconstrained with clothing composed on the body in a kind of informal pastiche. (Martin 1995: 215)

Thus, rather than being constructed around an aesthetics of revelation and concealment, Miyake's clothes invoke a kinaesthetic sense of the body in motion, which take them beyond the Western conception of fashion as a primarily *visual* art form. His primary concern is to design clothing "that allows the body total freedom of movement whilst at the same time being beautiful and practical" (Sato 1999: 38).

In keeping with the embodied nature of his garments, Miyake works not from two-dimensional sketches but rather from wrapping the fabric around his own body (Miyake in Tsurumoto: 103). Typically, he begins with a single, often irregularly shaped piece of cloth or a long piece of sash-like material, which he flings over the body. The resultant form is dictated by the way the fabric falls around the body so that "both the fabric and the body are able to assert themselves, enabling the space

between them to wave, flutter and fold with each movement of the body...Through movement, fabric and body become one" as Isozaki writes (in Koike 1978: 56). Often, Miyake's garments are seamless, hollow forms, which appear as shapeless sacks when not being worn. Once inhabited by the body of the wearer however, "their personality emerges" (Ishoka in Tsurumoto 1983: 32).

The capacity of Miyake's garments to metamorphose themselves is heightened by such features as the use of pleats, which increases the ability of the garments to expand and contract in concert with the movements of the body (Holborn 1995: 81–2) and also by the use of soft, lightweight, flexible fabrics such as polyester jersey and Pewlon – an acrylic knit (Holborn 1995: 30 and 36). Miyake is known for his innovative use and development of new materials, some of which have been derived from traditional Japanese fabrics (for example, sashiko quilting) while others have been developed from modern materials such as vinyl. These materials – chosen primarily for their feel rather than their look – heighten the tactile experience of his garments. In contrast to the stiff, unforgiving nature of the fabrics typical of much elegant Western dress, the materials employed by Miyake often have the feel of being well worn, for example, his adaptation of the traditional fabric *onioryu* – a cotton crepe made from dipping the cloth in the cold waters of northern Japan which is then twisted and crumpled by hand (Sato 1999: 35). The Japanese concepts of *sabi* and *wabi* encapsulate this quality of Miyake's clothing as Cocks points out. "Sabi" conveys the dull sheen of posterity, the finish, mystery and allure acquired by an object that has been well worn [while] 'wabi' suggests the use of humble material for a higher purpose" (Cocks in Calloway 1988: 51).

Although Miyake's garments are often worn by a wealthy clientele, he is also concerned to democratize his designs. As he has declared, his aim is "to create clothes that anyone [can] wear" (Miyake quoted in Sato 1999: 18). To this end he launched *Plantation* in 1981 which he called "clothes for real life". As Holborn describes them: "These were 'living clothes' designed for action with pronounced simplicity" (1995: 50). Similarly, his collection *Pleats Please*, launched in 1993 was aimed at a wider audience, being made out of the relatively inexpensive fabric of polyester, which was practical, lightweight and could be rolled up for storage without crushing. In 1999 he unveiled his *A-POC* (acronym for A Piece Of Cloth) clothes that consist of a long tube of jersey from which one can cut, without wasting any material, a large variety of different clothes, thereby enabling the wearer to have a greater input into the fashioning of their own garments (Sato 1999: 60).

Because of the sympathetic relation between his garments and the body, Miyake's clothing can easily be worn by people of all shapes and sizes as well as catering to people of all ages. Thus, he often uses elderly models to indicate the lack of age specificity of his designs (see for example Holborn 1995: 130–1, 142). Furthermore, many of his garments are far less gender specific than is the case with most Western dress. Equally adaptable to both male and female bodies, his clothing evokes a more dynamic sense of the body, which moves beyond the objectification of the body as image.

Conclusion

In conclusion then, despite the inextricable association of dress with the body, there has been a concerted effort to disavow this relationship in Western culture. Indicative of this has been the greater value accorded to those modes of dress that have transcended the body to the greatest extent. In general, male dress since the nineteenth century has been considered superior to female attire because of its more disembodied nature. Underpinning this, however, is a highly problematic privileging of the mind over the body. As I have argued, this fails to acknowledge the inescapably embodied nature of our existence as well as perpetuating our patriarchal denigration of the feminine.

While it is true that women's greater preoccupation with their bodily appearance has been detrimental to their ability to engage actively with the world, the answer should not be to seek a mode of dress that disavows the body altogether, but rather, one which engenders a dynamic sense of the body as an active agent. The fashion designs of Issey Miyake offer an example of how this more kinaesthetic awareness of the body may be achieved.

MATERIAL-CHARACTER ANIMATION: EXPERIMENTS IN LIFE-LIKE TRANSLUCENCY Cathryn Vasseleu

The transfiguration of radiant light into liquid matter is an art known to children in Gabriel García Márquez' story, "Light is like Water" (1978) (Márquez 1994). With their parents out for a night at the movies, two brothers close up the house and break open a glowing lamp globe. A cool jet of golden light pours out of it and pools to a depth of several feet before the boys turn off the current. Launching a rowing boat onto the stream, they navigate among the islands formed by the flow of illumination through the house. With these magical actions Totó and Joel have animated light. It now embodies the fluidity of water and the power to convey movement.

To be able to connect the physical characteristics of light and water in this way suggests some prior knowledge of the nature of the cosmos. The children's intuition of light's magical physical powers springs from a metaphor used by their father to explain how the light turns on by just pressing a switch: "Light is like water ... You turn the tap and out it comes" (Márquez 1994: 158). What the children do with this illuminating metaphor is unimaginable to either their father or the reader. Totó and Joel take the approximation between light and water literally, grasping its power in material rather than abstract terms.¹ In doing so they realize the improbable (uncanny resemblance between water and light), and then exploit the confusion to excess. Having taken to completely flooding the house with light and then scuba diving through it, the boys unintentionally cause a swirling mayhem of animated domestic items. A river of illumination eventually escapes the house and inundates the entire city.

At issue here is a sensibility that enables Totó and Joel to turn the entire physical world into life. A metaphor becomes the key to their discovery of a technique that releases an occult, energetic matter. Film and computer animators cannot do without technological tricks and techniques to invoke this animistic ability. They cannot bring things to life in their animated worlds merely by willpower or magic. Likewise, artists and scientists invent technical and figurative practices to summon forth the animate beings of a world composed of dynamic substance.

A case in point is the way light is rendered in simulated photorealistic depictions of physical spaces and objects. For example, in 3D computer animation, all the light in a scene is calculated by algorithmic formulae (also known as models). Light rendering programs have become an integral part of computer-generated imaging processes. Creative arts and architectural design have eagerly incorporated the perspective given by the invention of computer methods of handling the intermingled energies of matter and light. Among these are models for simulating the look of translucent substances (Jensen et al. 2001; Frisvad et al. 2007).

Translucency itself has many associations with magical and supernatural qualities. Unlike transparency, the subsurface glow of translucent substances implies interiority, or the evanescent presence of an embodied soul. The ability to simulate such a phenomenon raises issues that are not readily framed within existing analyses of computer-generated images. With eyes still trained on modernism and its signature aesthetics of transparency, Postmodern aestheticians initially focused on the giddy superficiality and seductive artifice of simulacra, the waning of affect, and the digital medium's perceived overcoming of the various material limitations imposed by other media. These included Baudrillard's articulation of the play of appearances of the "superficial abyss", Jameson's analysis of the "depthlessness" of readily assimilated second generational images, and Brummett's interrogation of the "prophylactic whiteness" or occluded interior of electronic images, devices and systems (Darley 2000: 58–77).

Transparency also continued to figure as a key feature of theories that contested the framing of media in this way. These include "Supermodernism", or an aesthetics of disappearance identified by architectural theorist Hans Ibelings (1998) in his study of architectural erosions of postmodern aesthetics. Supermodernism conceived of architectural space as an "empty medium", purposefully striving for transparency. Rather than communicating a message, Supermodernist architecture works like an invisible, enveloping medium that creates a unique sensory experience. Lev Manovich extends this aesthetics to the dissolution of personal technological devices into ambient objects in the era of ubiquitous computing. Machines disappear as they become implanted inside other objects, materials and surfaces (Manovich, forthcoming). Here transparency (and translucency) is understood as the aesthetic signature of the boundless flexibility of the computer as an all-in-one meta-machine.

There is no hidden appearance or inner depth to a simulation of translucency, yet somehow an image of "inner light" arises within its missing interior. Translucency has a thickness that is related to material consistency, not spatial depth. When viewers experience an image of light interacting with matter they attribute the reality of a carnal presence to it. Any difference between the simulation and physical enactment of the visual effect is, intentionally, undetectable at a conscious level. The image behaves like a voluminous object with communicable physical properties. How might we begin to pose what is unseen and unthinkable in that?

We could make a start by comparing the perspectives offered by a computer model and a physical model of translucency. Henrik Wann Jensen, a computer graphics scientist, and Rachel Whiteread, a sculpture-installation artist, have each developed

a way of working through the dynamics of light interacting with matter in the phenomenon of translucency. Jensen uses algorithms that simulate the flow of light to generate the appearance of photorealistic translucency. Whiteread casts translucent sculptural forms of unseen spaces out of liquid resin. On first impressions, an investment in the phenomenon of translucency appears to be their only connection.

Jensen describes his modeling technique as the recovery of the optical properties of matter. From this description are we meant to understand that the likeness distilled in Jensen's model refers to a refractory substance known only by its empirically defined contours? Surely we grasp the appearance of translucency in a more substantive way – just as Totó and Joel comprehend likeness – by responding to the image in physical terms, rather than as an abstract identity? This is where an art practice that emphasizes material process as a source of knowledge can pose a perspective that is unseen in the illusory appearance of simulated translucency. Unlike Jensen, who writes computer programs that render the illusion of photorealistic translucency, Whiteread works translucency as a material quality to render uncanny physical objects.

Despite these differences of media and method, both Jensen and Whiteread produce artworks that exhibit phenomenally lifelike behavior. Each perspective contributes to a heterogeneous practice that hinges on the elaboration of images of physical objects, animated in such a way that viewers are attuned to their material character. This is not the same way that cartoon-character animation works. Character animators create animated figures with psychical attributes. However, just as Totó and Joel inhabit a universe rendered animate by material transferences of energy, Jensen and Whiteread enliven built images and structures using processes that impart physical attributes. I will call this practice “material-character animation”.

In the commercial animation industry, there is a recognized distinction between character-animation (the art of imbuing animated figures with individual personalities or “souls”) and effects animation (which includes the movements of “non-character” scenic elements such as light and water). However, it is the nature of animated worlds that all elements in them have the capacity to become individual characters. It is par for the course in animated cartoons for cups and saucers, or flowers and trees to grow faces and limbs and break into song and dance. In becoming animated, a figure acquires a singular life by conveying a mutable life to others, not simply by embodying its own. Material-character animation renders the material world in similar terms. Here animation involves a transformative process that attunes us to a world composed of energetic matter. All aspects of this animated world evoke sensations that we equate with carnal life, because any element can embody material attributes that reflect, interact with and communicate an accentuated physical reality.

Transparency and Translucency

In order to elaborate the sensibility of algorithmically formulated translucency, it is first necessary to separate translucency from aesthetic movements where it has been understood in relation to transparency. In those instances transparency is defined as either an optical or a perceptual quality. A transparent substance is one that is capable of transmitting light as though there is no intervening matter. Transparency, under-

stood as a sensibility of both eye and mind, is the condition of optical surety and abstract clarity, secured by light passing freely though the concrete. As such transparency has aided idealist philosophy in its articulation of metaphysical vision, the spiritual soul and metaphoric illumination (through a light that illuminates matter but is refractory to incorporation). In modern thought, matter's capacity to transmit light in this way became a key factor in the explication of aesthetic concepts in painting, photography, film, sculpture and architecture.² It was also an emblem of the social, corporate and political ideals of structural and operational transparency, or openness.³

Architectural transparency was, in its material sense, a guiding construction principle of modernism. At its simplest, a fascination with the space-defining properties of transparency was translated into arrangements of glass and other materials that transmit light in such a way as to be able to see through them, or designs that involved unobstructed flows between interior and exterior spaces. The painter Robert Slutzsky and architect Colin Rowe famously elucidated a distinction between the architectural interpretation of transparency in this literal sense, and phenomenal transparency, where transparency is elaborated corporeally and conceptually rather than optically (Rowe 1976 [1955–6]: 160–162).⁴ As well as being an inherent quality of a substance, it could also be a quality of organization. Material explorations of phenomenal transparency extended the optical characteristic to space, or the simultaneous perception of different spatial points within a dynamic field of view. The spatio-temporal complexities of phenomenal transparency were (and still are) interrogated in the exploration of fluctuating perceptual phenomena such as emerging and receding figures and ground, reflections and shadows, and implied or revealed surfaces, shallows and depths. Here transparency can apply to opaque objects, and to the semi-opacity of translucency. Both are implicated in phenomenal interpretations, as levels of transparency.

The optical characteristics of translucency are different from transparency. Translucency is an incomplete transparency, or partial opacity. It is a liminal quality, existing on the threshold between clarity and obscurity. Optically, a translucent substance is one that is capable of transmitting light, but also causes sufficient diffusion of that light to prevent the perception of clear and distinct images through it. Instead of being entirely permeable to light – as one sees things through a clear glass window, a translucent substance is suffused throughout with a shadowless, diaphanous lucidity. The passage of light creates, simultaneously, opacities in our field of vision. We can see only partly, not fully, through translucent matter. Light is reflected within the material before being either absorbed or transmitted. The internal scattering of light gives rise to a soft appearance, rather than a luminous opening.

In the stained glass windows of Gothic cathedrals, translucency was used to create the impression of the unveiling of veiled truths. The iridescent beauty of partially opaque panels at once shrouded and revealed an ineffable, divine light (Von Simson 1962: 120–22). Elsewhere the ambivalence between opaque surface and transparent opening has been employed to create a concept of a multi-natured fluid space. Translucent paper screens, a characteristic feature of Japanese architecture, were

used in this way long before the concept was adopted by Western architecture in the early twentieth century. Presenting a glowing surface when closed, the screens allow direct access to nature when open (Kaltenbach 2004: 7).

As outlined above, the ambiguous physical properties of translucency have been understood and elaborated in terms of its optical behavior and its phenomenal or conceptual organization. However, there is another vital aspect to consider. The tendency of matter to dematerialize in translucency and become a thickness through which we can see is also associated with the appearance of an animated material universe. This association is made and developed in philosophical accounts of perception by Merleau-Ponty and Bergson, and in artistic renderings of a tactile sensibility.

Merleau-Ponty gives a unique inflection to the partial opacity of translucency. He regards matter as the invisible condition of seeing rather than an obstacle to perception. For example, looking into an ornamental garden pool and observing the tiling clearly visible below, Merleau-Ponty reflects: "I do not see it *despite* the water and the reflection there; I see it through them and because of them" (Merleau-Ponty 1962: 182). Here, the tiling materializes for the viewer in the intervening thickness of water: "If there were no distortions, no ripples of sunlight, if it were without this flesh that I saw the geometry of the tiles, then I would cease to see it *as* it is and where it is – which is to say, beyond any identical, specific place." A watery medium inhabits the pool, but it is not contained there or anywhere: "if I raise my eyes toward the screen of cypresses where the web of reflection is playing ... the water visits it, too, or at least sends into it, upon it, its active and living essence" (Merleau-Ponty 1962: 182). Merleau-Ponty's account of visual perception is anti-optical. It runs counter to a representational view of the empirical world, or a merely physical-optical relation in which vision is based on optical surety. For Merleau-Ponty translucency has a transparency that is unlike the immateriality of "see-through" things. Its transparency is of the order of the virtual image, or mirroring.

What transparency means for Merleau-Ponty is that:

Every object is the mirror of all the others ... I can...see an object in so far as objects form a system or a world, and in so far as each one treats the others round it as spectators of its hidden aspects and as guarantee of the permanence of those aspects. (Merleau-Ponty 1962: 68–9)

Thus translucency is the condition of the apprehension of the visible as a system or universe of beings that disclose themselves in and to each other as a whole (a world of implicated figures and grounds explored in phenomenal elaborations of architectural transparency). The visible is not a formless multiplicity of perceived objects-in-general that are foremost given over to abstraction and clarification. Any seeing of an object is a participation in the translucence, that is, the whole-in-one aspect of things.⁵ The "thickness" of translucence is not a geometrical dimension, as refined in the perspective painting techniques of the Renaissance to facilitate a more exact, artificial construction of the world. What is gathered *in* the intersecting

voluminous spacing of translucence is the dynamic unfolding of the fabric of the world. This is conveyed in the uncontained, infinitely dispersed radiation of the visible, whose freedom in self-arrangement can be described as “internal animation”.

Merleau-Ponty describes translucence as the shimmering clarity of a phenomenological vision anchored in the material world, but a simulation of translucent appearance is an image calculated by a disembodied model. Here translucency has a separate, non-phenomenological existence in a virtual form that is more readily aligned conceptually with Bergson’s understanding of the dynamics of perception. Like Merleau-Ponty, Bergson does not regard perceptual experience as a function of a physical-optical relation between a subject and a transcendently illuminated object-world. Instead, the appearance of the material universe is derived through our perception. In Bergson’s account, material objects have independent functions and real actions, but a living body participates in creating perceptions through the momentary delimitation and isolation of a light that would otherwise pass by and remain unrevealed. The world of perception manifests its spontaneity insofar as “Images detach from themselves that which we have arrested on its way, that which we are capable of influencing” (Bergson 1988: 37). In this schema, a perception is a phenomenon of the same order as a virtual image. It is an effect of light that has been impeded and arrested momentarily, reflected without otherwise affecting the totality and freedom of action of matter. Reflections can appear as isolated moments of intentionality (contracted by and limited to our interest), as when images “appear to turn toward our body the side, emphasized by the light upon it, which interests our body” (Bergson 1988: 36). In this way images can be perceived by us as living matters, or exist and act independently without being related to an individual, illuminating consciousness.

Bergson sets out in his vitalist philosophy to dismantle the notion of an immutable universe, which he associates with an intellectual viewpoint that renders matter inert by defining it in terms of properties that accentuate its materiality (Bergson 1911: 202). By way of contrast, he describes consciousness as a translucence in which the eye is elementally in things, universalizing a pure, non-human vision in an a-centered universe of images. Bergson’s mobile, iridescent consciousness is a translucence that dematerializes solidified matter and animates the universe of images and objects. The chaotic interaction of virtual influences is the source of movement in both these things.

Merleau-Ponty and Bergson make use of phenomenal and literal translucence to counter a representational view of the empirical world. Artists and architects manipulate the mutable characteristics of translucent materials to create a “tactile” sensibility, or a sensual play between surfaces that admit light and the smooth, soft appearance of inner substance. Involving a sense of matter as much as light, translucency gives luminous contours to the material passages of inner-space. These passages are not rendered invisible or obliterated, as they are in optical or conceptual transparency. Light blurs as it is incorporated and moved about within the substance. Illumination takes on a sensuous character, not the metaphysical clarity seen in light’s immaterial passage through material entities. If light figures as a connection between spirit and matter in virtually every culture, in translucency, it remains ambiguous

whether light passes through or is moved by matter. While unobstructed openness is a defining characteristic of transparent relations, there are interruptions of clarity in the opaque/transparent glow of translucency, where matter has the power to admit and also transmit light.

A Computer Model of Translucency

Appearance modeling aims to elicit the same visual response to a computer-generated image that a viewer has to the photographed appearance of the physical object it models. The appearance of a translucent object in a computer model and a photographed physical model are, to all intents, indistinguishable to sight. Inasmuch as both model and photographed object appear visually identical, appearance modeling is a transformative process that is not detectable at a sensory level. As will be discussed, any transformation that occurs applies to the sensibility of translucency.

Jensen's team test their algorithm's approximation of translucency by comparing the model's appearance with a physical model-box containing equivalent volumes and surfaces in a substantial form.⁶ The visual similarity is proof of the algorithm's ability to synthesize an image of an object whose translucent appearance is indistinguishable to sight from a physical translucent object. Unlike an animated character, which appears to embody life and express its character in its individual movements, we read the object's material character from how translucent it appears in the context of its surroundings. Milk, for example, is a translucent substance that Jensen's model can replicate. Synthesized in such a way that we can identify the familiar visual surface-texture of physical translucency, a substance looks milk-like rather than looking like white paint (Jensen 2006).

Semioticians ascribed an indexical role to photography, understood as the representation of idealized physical entities. Here indexicality refers to a physically enacted connection between an object and its traces in a photographed image. Computer modeling reinvents the age-old trick of seeing by mimetic illusion. The type of index that enters into the computation of translucent appearance is the "index of refraction", along with other parameters necessary to derive a computer model for a particular optical property. It is possible to devise models for handling all kinds of translucent substances, including the model already referred to, that can compute the parameters of milk-like translucence. Having the capacity to be fed descriptions of the fat and protein content of various milk products, it can handle the translucence of skim, regular, full fat, etc.

The experience we have of this model is regulated by an already worked-out make-up that is judged by the extent to which it achieves the intended effect – that is, a seamless visual approximation of familiar objects. Appearance modeling is based on the premise that images appear realistic when the model can capture, incorporate and reconstruct the essential optical-physical arrangement of the empirical world, modeled in accordance with psychophysical measurements of human powers of vision. In actuality, the model is an optical description that is designed for viewers to "fill in" (guided by a prior carnal knowledge of milk's translucency) rather than a purely optical description.



14. Half tone of color image: from left to right the glasses contain water, water and vitamin B2, water and protein, water and fat, skimmed milk, regular milk and whole milk.
(Frisvad et al. 2007: Fig. 1)

Translucency is, in varying degrees, a property of all non-metallic materials. A model that works for shiny, reflective materials assumes that light makes contact with and leaves the surface at the same position. In the model Jensen uses to replicate the appearance of translucency, light rays alter their position in their transmission through intervening materials. These include materials such as milk, marble and skin, whose internal structure and composition scatter light, thus contributing to their visual complexity. Optically, a significant amount of light is transported below the surface in translucent substances. Computer-models of translucent materials look completely wrong without the inclusion of this characteristic, known as subsurface scattering. The technique works by approximating the subsurface movement of light diffusing through translucent materials, without having to laboriously trace the individual photons.⁷

As a circumscribed visual rendering, the image is only a partial representation. An algorithm for simulating milk-like translucency is not indicative of the mastery of phenomenal complexity. It is indicative of the mastery of the model – just as we might experience milk in the Coca-Cola model way, as a product positioned in the company's global marketing strategy. In 2001 Coca-Cola proposed a new line of dairy drinks for children under the working title "Project Mother" (Brown 2001, Stephenson 2002). "Milk" was not conceived of primarily as a drink made of milk. It was a beverage that was designed to capture the breakfast market because its ingredients would be judged sufficiently nutritious to win maternal approval. In other words, Project Mother wagered that appearing to be nutritious would be more seductive than being the real thing.

A computer model of a light-scattering substance obviously doesn't affect our senses in the way a glass of milk does. We do not encounter the semblance as we would an "authentic" glass of milk, whose material character is expressed in its countless corporeal manifestations; its coldness if it has been poured from a container in the fridge, forming a skin if it is hot, leaving a white moustache on our upper lip, tasting sour because it has gone off, giving us succour as did our mother, making us sick if we are lactose-intolerant. Nor does its rendering process resemble the rendering of a physical casting, which is a tactile form of making. This is a point of difference that we will take up in relation to Whiteread's casting process. A resin casting is made by physical approximation. It is a tactile rendering. The model is

produced through physical contact without being touched by human hand. Tactile rendering is objective without thinking, exact without effort, replicates form but not appearance, and the result is never life-like enough (Terebess 1981).

The computer model formulates its own unseen parameters of sensibility. It allows us to see exactly (enough to distinguish between skimmed and whole milk) in adherence with its abstracted optical-physical reality. This is the “inherently false” reality of a formal arrangement that cannot be realized materially. We cannot experience the material nature of milk based on how we are given to understand the condensed milk model’s virtual essence.⁸ Jensen’s model is a distillation of translucent appearance, generated by the actions of an automated, virtual light. The paths of virtual photons are calculated to lose and change position between entering and leaving the simulated material, thereby reflecting its internal structure and material makeup. In the interstices, light ceases to be a radiant beam of energy and becomes something more akin to an animated, ambient light, indistinguishable from and informative of its digital-milk environment. The characteristic optical behavior occurs spontaneously but its nature is pre-scripted. The degree of luminous “density” and “bleeding” is calculated in advance by an invisible structuring medium that determines the image’s translucent appearance. In the modeling process, the invisible ordering of a materially given, luminous sensuality is recast as a sensuality of automated light flows.

Despite the model’s artificially imposed order, it is not disorienting to a viewer. Translucent appearance is instantaneously perceived new structure. The simulation amounts to a formal re-arrangement of the visible based on the indices of “subsurface scattering”. The model creates its own invisible parameters of complexity (complex variations in the translucent appearance of different materials). The ordered appearance of soft, smooth diaphanousness is spontaneously computer-generated, but is based on a quantifiable light-scattering function that must be calculated in advance. Altering the variables can produce unexpected effects, but once scripted the actual behaviors of virtual photons interacting with virtual surfaces is fixed. As well as being motivated to act in a formally specified way, the random scattering of light is a calculated variable.

In traditional character animation, individual character is inscribed in the created movements, gestures, voice and behavior of each animated figure. New methods of procedural animation allow character variables to be built-in “genetically” to all scripted elements that make up a scene. Character is determined by descriptions of the multiple individual procedures that define the animation. Any of the variables in the make-up of a character or an effect can be individually edited and transformed at random.

Appearance models describe the procedures for animating the variables that apply to the characteristic appearance of materials and media. The model for subsurface scattering can be applied to the simulation of optical effects such as flows of light through virtual architectural models, about which more will be said later. It is also used in 3D computer animation to photorealistically render the appearance of human skin, which is a multilayered translucent substance. A model for subsurface scattering can replicate the “optical character” of physical space, or the optically defined

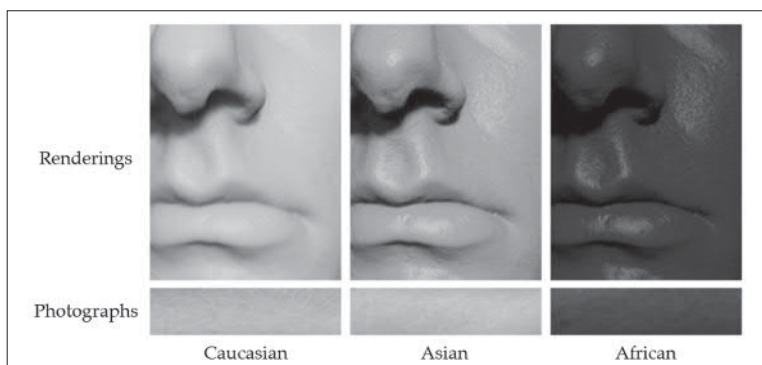
physical characteristics of “race”. Just as the milk model can render the appearance of different types of milk (whole and skim, etc.), the variables for modeling human skin can be altered to render shades of difference in its translucent appearance. The prototype for skin (skin1) was modeled as a facial close-up of a pale pink translucent cheek and fleshy, lipstick-reddened lips.⁹ The modeling of various skin-shades has followed (Caucasian, Asian and African) (Donner and Jensen 2006).

While the realistic appearance of simulated milk does not strike an uncanny note, the photorealism of a computer-generated actor can. The simulation of the “inner light” occluded in human facial-skin initially exposed something of the pre-formulated “soul” occluded in the polygonal faces of digital humans. These characters stumbled into what Masahiro Mori (1982) calls the “uncanny valley”. Here, instead of completely identifying with a figure that appears human, audiences are repelled by traces of its robotic nature. Rather than knowing what they are looking at, viewers are disturbed by a presence they are not quite able to identify. When synthesians first appeared in 3D animations, audiences scoured their faces of for traces of hidden engineering.¹⁰ Viewers were forced to adopt an interrogative mode of perceiving while watching the moving images. Their vision was moved by practical concerns, born of an uncertainty about how they should view the photorealistic approximations of cinema screen actors they were seeing.

The simulation of human facial-skin announced a particular way of seeing the world, and with it a novel way of being (in both an existential and social sense). It announced a new form of animated presence – simulated matter with its own built-in, manipulable, photorealistic character. It also announced new cultural image-forms – simulated photorealistic Caucasian, Asian, African characters that posed automated racial identities with their own optically defined order.

These optical forms of human being were no longer disclosed in the co-existence of film-making and the philosophy of consciousness, but in the industrialization of simulation and a predilection for obtaining “results” from methodically applied sets of language-tools and systems of thought that Lyotard dubbed the “systems theory” age (Lyotard 1984).

Models do not replicate the natural world. They demolish a naturalized metaphysical viewpoint and replace it with systems that generate a world-order composed of quantifiable, manipulable results. Disclosed according to this methodical replacement, translucency is the automatic outcome of a model with adjustable parameters that renders an empirical description as believable as the appearance of material objects in photographic images. Twentieth century avant-garde photography and cinema practices confounded indexical readings of their art by privileging material process. We have yet to discover procedures that render the fixed behavior of each shade of material character seen in photorealistic translucency in a questioning way. For this we need alternative ways of encountering simulated lighting effects, apart from blindly incorporating the perspective of the computer model.



15. Half tone of color image: different skin types simulated with model (top) compared to actual photographs of real skin samples (bottom). (Donner and Jensen 2006: Fig. 8)

A Physical Model of Translucency: Rachel Whiteread's *Water Tower*

Rachel Whiteread is an artist who works through the contradictions implied in translucent substances in her sculptural elaborations of inner space. This does not involve any empirical knowledge or abstraction of the optical properties of translucent substances. Whiteread's process is given over to the efficacy of material elaboration as opposed to participating in the illumination of the material world. Her technique involves making resin castings of the spaces in-between, under or inside objects and buildings. She abstracts these spaces in an unexpected way, by solidifying them. In *Water Tower* (1998), Whiteread rendered the inside-space of one of New York's ubiquitous roof top water towers as a translucent, hollow, sculptural object cast from resin. The sculpture was re-installed in place of the original water tower. Having served as the mold, the original was destroyed in the process, as occurs with many of Whiteread's castings.

Whiteread's work invites observations that the destruction of a space by filling it gives us back the object attached to it, formally and conceptually, but also altered (Bradley 1997: 12). This was also the case in *Water Tower*. The work resembled a water tower, which is an object that provides space for the collection of water, yet no water could occupy the space inside the sculpture. Many commentators also regard Whiteread's cast objects as having multiple equivalences with the mortifying effects of photography (when understood as a type of index).¹¹ However, *Water Tower* is not as easily attached to the strings of so-called indexical forms as many of Whiteread's other works are. Installed high up on a rooftop, it was well out of tactile reach. Nor was close visual scrutiny of its surface possible, although the ghostly imprint of the inside surface of the original water tank's interior became faintly visible when framed by a cloudy sky. Aloof yet unassuming, the public artwork was alternately a beacon unexpectedly gathering the changes in light, or a ghostly form in the gathering darkness. It did not register as a filled in absence so much as a solitary, uncanny object. Although the casting bore an obvious resemblance to the water tower it replaced, and those around it, observers noted that its strange translucency de-familiarized an

item so ubiquitous that it had become invisible on the New York skyline (Malvern 2003: 392–5). If Whiteread's innovative use of translucent materials in her work in general is not as remarkable or alarming as the reversals of the casting process (Morgan 1997: 26), the translucent effect of *Water Tower* stood out as an exception.

With *Water Tower* Whiteread made a conscious move away from solidity as an effect, in favor of an “inherent transparency” (Lee 2004: 130). Here transparency is meant in the sense of phenomenal transparency – a seeing both through and by implication, the internal as well as the external structure.¹² Rather than freezing space (or emptying it of displaced life), Whiteread was after an effect that she described as more like “the intake of breath” (Whiteread, in Neri 1999: 166). Whiteread's previous castings have been described as having an obstinate material opacity (Lee 2004: 130), but in this statement the artist's emphasis is not on perceiving an impenetrable, apparitional presence. It is on taking in a lost animating connection (the vitality of water to human life).

Viewed in the context of its physical surroundings, the translucency of *Water Tower* liquidated any conscious understanding of its appearance. It resonated with something other than an indexically produced form of frozen after-life. To begin with, *Water Tower* was a casting of inner space, but the cast object was not simply located in space. It inhabited and was inhabited by multiple spaces: the rooftop, other water towers, passing clouds, the upturned gaze of passers by, the sunlit or night sky. It materialized itself in and with all these, but was not contained either in itself or in them. Its shifting appearance resonated with the mobility of phenomenal transparency.

We can account for the mobility of *Water Tower's* appearance from a phenomenological perspective, but not how translucency gives the object the quality of being strangely altered (rather than simply novel). Unlike a characteristic translucency that can be notionally turned around to call the character of a glass of milk into question, for example when Alfred Hitchcock renders a glass of milk suspect in *Suspicion*,¹³ translucency was not a material characteristic of the wooden water tower whose interior the casting replicated. As well as the fact that a physical casting replicates form but not appearance, *Water Tower* was materially different from any other water tower, as could be gathered while encountering its novel optical behavior. The installation's apparent materialization and dematerialization was a function of intensities of light, an image fluctuating between density and airiness. The appearance of *Water Tower* struck a contradictory note in relation to the world seen through it, the water tower it replaced, and those that surrounded it. The installation's appearance called into question its relation to the surrounding water towers and a water tower re-drawn from memory.¹⁴ From the way *Water Tower* shifted about in its appearance, the interior of a water tower was no longer the forgotten space inside an equally forgotten wooden structure.

In filling space in, Whiteread transformed it into a life-giving airy space that had the capacity to fill with water. An invisible void became paradoxically, a dynamic substance. *It was* – no longer transparent, self-erasing, displaced, overlooked, when seen through Whiteread's working through of her making process. The lost vitality of the water tower turned, like the transparent liquid resin, into a solidified liquid translucence.

The translucent substance of *Water Tower* cast its own impression of being seen through. It was not a space that is unseen, or whose contents are taken for granted. Looking up at the figure and, through it, the sky, viewers experienced an image of a transfigured void. The transformed form didn't resemble a photograph (although it was much photographed). It became the setting of a philosophical conversation about post-auratic monumental art¹⁵ and the monumentally shaky grounds of Western philosophy. As Irigaray exhorts Heidegger and all modern philosophers to recall, metaphysical concepts are founded on the solid. They are not written in fluids, although they circulate freely, without thought of the source of their fluidity. That which enables the binding of beings and their worlds into a whole is unthinkable by philosophy (Irigaray 1999: 2–10). The connection is just solidly there – transparently – in unseen air.

Air remains the groundless ground of mediation for all that “is” in idealist philosophy. Paradoxically, it is also the material forgotten by modern materialist philosophy. The functioning of knowledge and perception are unaffected by the forgetting of air. There is no imperative for either to recognize its prior existence, or its existence as substance. The need for air is something that simply goes unnoticed in the acts of consciousness – until we can no longer breathe in the space we inhabit: until the air is no longer there.

Water Tower posed the unthinkable stuff of fluid movement. Here was a liquefaction of air that neither imposed itself, nor allowed its existence to be foreclosed. It became a substance with its own memory, interiority and a face full of moody character, inhaling and exhaling light shared with others, burning fiercely and thinning to a smudge in the night. Interiority was not a dimension that was implied in *Water Tower*. It was breath that could not be suppressed. Whiteread's casting was not quite as life-like as the wooden water tower it replicated, but it was way too life-like for a water tower.

Inner Light and Unseen Vitality

Algorithms for depicting photorealistic translucency are just one example of a multitude of computer graphic models that simulate the way light rays interact with material surfaces and with each other. As previously mentioned, these models can be used to study the flow of light in complex architectural models. Jensen demonstrates a technique for synthesizing the movement of light in an architectural model: the un-built *Courtyard House with Curved Elements* by Ludwig Mies van der Rohe (Jensen 2001). The scattering of light throughout an imaginary building, as visualized in this animation, is an exercise in calculating a moving image of illuminated space. The exercise does not support the degree of immersion in an animated universe that viewers experience with interactive 3D environments, but it shares something in common with the magic realism of two boys deluding their house in currents of light from a broken lamp bulb.

In “Light is Like Water” liquefaction is a literary device that overruns the limits of metaphoric depiction. It becomes an act of making – liquid. In the Jensen team's model of translucency, an algorithm controls the appearance of liquefaction. Jensen

is not working as an artist when he creates images of this lighting effect. His models are tools for automatically applying empirical descriptions of the physics of light to digital image-synthesis. Yet insofar as the computer scientist uses the models to characterize the behavior of light and matter interacting with each other, he is, like Totó and Joel, a “material-character” animator.

The fictional universe of a magic realist story is different from the visible universe of simulated photorealism. Totó’s and Joel’s liberation of liquid light creates a chaotic flood of movement. An uncontrollable bleeding together of real and magical worlds occurs in the story by fleshing out a metaphor. When Jensen uses light like water he activates the flow to approximate a particular visual effect. His technique of image synthesis contains the chaos of a physical universe within its complex mapping of light’s movement. By replicating the way light and the surfaces of physical objects inter-act, Jensen can realistically depict their material characteristics – the way light “bleeds” in translucency, for instance.¹⁶

The refinement of appearance modeling serves an essential role in overcoming undesirable uncanny effects in 3D computer animation, games and simulation industries. Furthermore, the value of appearance models is not limited to naturalistic depiction. Architectural design is increasingly applying digital lighting programs to choreograph spaces that address human perception and multi-sensory experience. Here experimentation with computer-generated visual effects aims to evoke “animate” rather than optically defined spatiality. However, in all these applications of lighting programs viewers acquire a new perspective by intuitively understanding the model’s virtual order. The lighting adheres to a fixed description that does not leave any space to work through the voided carnality of stunningly exact computer models.

Whitread’s creative art practice, as exemplified in *Water Tower*, discovers overlooked openings in familiar structures. In this respect her process suggests how it would be possible to recast the abstracted material passages of “inner light” in simulated translucency. Unlike a computer model, the translucent appearance of *Water Tower* has no fixed attributes. Viewers are captivated by the way seeing happens in a tactile rendering of spontaneous material transformation.

There is an inherent mobility in Whitread’s method of physical approximation that is lacking in Jensen’s precise depictions. Jensen’s model gives virtual objects a life-like appearance by calculating the behavior of light interacting with matter. The model for subsurface scattering results in a controlled appearance of inner light that is indicative of different types of material character. Whitread’s casting process animates physical forms by rendering their unseen interiority to indicate an absent vitality.

This unnoticed absence is enacted in another of Whitread’s works, *Place (Village)* (2006–2008), an installation made of over 200 ready-made empty dolls houses resting on their packing crates. These are all arranged into a temporary “village” in a darkened room. The interiors of the dolls houses are lit by tiny light bulbs, which reveal no trace of inhabitation. To envisage that a living soul might actually reside inside defies the imagination, despite the fact that the glowing windows of the assembled colony project an aura of interiority back at viewers. *Place (Village)* is a model space made of chunks of hollow illumination. Light leaves an impression of an inside space devoid

of character: “the light is what’s animating the work. All the light is doing is casting the interior of the piece so you get to see the interior” (Whiteread 2008). Whiteread works with light in the same way as she works with resin: to render the unseen, immemorial, vital connections of quotidian life to memories, emotions and a knowledge of physicality that invisibly shapes perceptual experience and gives it character.

Whiteread continues, in the vein of de-familiarization, as does magic realism, to elaborate the architectural uncanny. She announces unseen realities in the strange perspectives incorporated in her castings. Rather than working by either formulation or affection, Whiteread’s method of production allows for unexpected, disorienting material transformations. Her artworks do more than animate spaces attached to physical objects. They animate solidified forms of thinking and perceiving. The capacity for movement is given to both in the disjunction, creating an opening for thought and sensation to affect and stimulate each other, to become energized together, through each other.

I began by wanting to pose what is unseen and unthinkable in a simulation of translucency. This question concerns a computer scientist only to the extent that an empirical description of the optical properties of matter renders an image that does in fact appear to act like a physical model. It takes a different approach to the process of approximation to shake our faith in the “recovery” process of computer models. It requires bypassing perception to take into account that virtual light is fluid like water only after the program that calculates its movement is rendered. One approach would be to extend art practices that privilege material process to re-workings of photorealistic lighting. As Whiteread demonstrates, a materialist art practice can work by abstraction to produce a knowledge that eludes perception and thought. In the case of a simulation, this might include recasting the flow of virtual light to show that the material character of the world it models is returned to us strangely altered in its formal exactness.

Both processes considered here involve the isolation, objectification and minute depiction of the stuff of everyday reality, but each entails a different aesthetic ethos: naturalistic depiction and de-familiarization. Jensen’s simulation approximates photorealism by giving precedence to the naturalistic depiction of an empirical viewpoint. Appearance modeling experiments with translucency as a quantifiable effect. Whiteread’s technique of physical approximation gives precedence to an objective viewpoint created by a tactile rendering of a transformative physical connection. Her work experiments with translucency as a quality of life.

There will be many more refinements to modeling processes based on the physics of subsurface scattering and global illumination, as related to psychophysical descriptions of human visual perception. However, there is more than meets the eye to rendering life-like translucency. Whiteread’s process overruns the minute depiction of material interactions in favor of a vital breathing space. This is the same moment of inspiration that animates Totó and Joel’s world of chaotic light: an unthought knowledge of physicality that invisibly shapes perceptual experience.

INSTRUMENTAL VISION

Rose Woodcock

Introduction

Stereo-immersive “virtual reality” (VR) technology enables the perceiver to experience what it is like to be immersed in a simulated environment. While the simulation is powered by the “geometry engine” (Cutting 1997: 31) associated with high-end stereoscopic computer imaging technology, the visual experience itself is powered by ordinary human vision: the vision system’s innate capacity to see “in 3D”. By way of a specific critique of the visuality of stereo-immersive VR, this chapter suggests that we think about the “practice” of vision, and consider on what basis vision can have its own “materiality”. In an oblique approach to the notion of a material practice associated with the making of (pictorial) art, preference here is given to pictorial perception over picture making, which prompts the question: What comes first: pictures, or the capacity to see pictorially? A quick answer to this is that pictures are only pictures, because they are perceived pictorially (otherwise, they would be sundry objects whose visible attributes are only “pictorial” inasmuch as they are arbitrarily deemed so). Yet we only *perceive* pictorially in any meaningful, knowledge-bearing sense, when we are perceiving a picture *as* a picture: thus, the answer would seem to require that pictures and pictorial perception be understood as part of a peculiar dialectical ontology, in which neither one can be meaningfully grounded in experience without the other. To preference their perception over their material fact, then, is simultaneously a defence of the picture as an object. Pictures do not necessarily need defending (as a class of objects, they continue to proliferate), but they do warrant attention. It is not the “primary heuristic function” (Hagen 1986: 82) of pictures to educate the viewer about aspects of the ordinary visual world. However, as a “special” perceptible object amidst all the ordinary objects in the visible world, pictures – through the perceptual experiences they elicit – can teach us important things about the possibilities of human visual awareness.

Understanding why pictures matter (given the options in a cultural milieu rich in alternative visualities) necessitates that we consider what it is like to see a picture as a picture; but it is also helpful to ask what it is *not* like. What is the antithesis of a picture, and (therefore) of pictorial perception? For the purposes of the argument, a critique of the visuality of stereo-immersive VR reveals an unlikeness that is exem-

plary: namely, a difference between what it is like to perceive pictorially, and what it is like to perceive ordinarily. Moreover, where a dialectic (“heuristic”) can be found at work in pictorial perception – that is in response to the material artifact whereby one encounters a paradoxical “duality of both a scene and a surface” (Gibson 1978: 231) – in stereo-immersive VR there is no such material artifact; and for perceptual experience, no such dialectical effort required.

What are the consequences, if any, of stereo-immersive VR’s release (for vision) from the burden of a materiality of the perceptible? In this chapter, a particular perceptual materiality is associated with the flat, pictorial imagery we encounter in a painting. However, aside from the painting as a material artifact, of greater interest here is the relationship between the artifact and the pictorial mode of perception it elicits. For, insofar as this relationship has a definable ontology it is, ironically, of a more “virtual” kind than that encountered by a perceiver in a “virtual” environment. Certainly, stereo-immersive VR presupposes a relationship between perceiver and the object of perception. However, for all its sophistication as a virtual imaging technology, the possibilities for perception afforded in a “virtual” environment return the perceiver to the conditions of an ordinary visual experience. That is, objects and surfaces appear perceptually robust in their solid three-dimensionality. We should therefore study not VR’s purported “virtuality”, but the specifications of its visuality.

The Invisible Interface

Clearly, an environment, real or virtual, is not just like seeing a very large, all-encompassing picture, although ordinary environments do have aspects that are “pictorial”: looking at objects in the distance assumes a “flat” or monoscopic mode of perception. In the vision science literature, monoscopic information is termed “pictorial” (Palmer 1999: 248). Yet VR is an art too. It assumes a technique (and a studio) and must also assume a specific materiality – both for its makers and its audience: yet its strategies, technologies and affects are part of an apparatus that is designed specifically to erase all visible traces of the very conditions by which it would be possible for an “immersant” to report on *how* they perceive the contents of a virtual environment. The question of what it is like to perceive “environmentally”, requires a different approach to understanding visual perception than the question of what is it like to perceive pictorially.

This essay considers the significance of James J. Gibson’s (1904–1979) contribution to the fields of vision research and computer-generated stereo-immersive VR, acknowledging how Gibson’s innovative work in the mid-1940s (developing rudimentary flight simulators for fighter-pilot training) foreshadowed the work to be done for later stereo-immersive VR developers (Gibson 1947). Gibson’s chief contribution was that he developed an ecological approach for the study of visual perception, which theorized vision in terms of the reciprocal relationship between perceiver and environment. For Gibson, the formulation of a theory of perception should begin by understanding the nature of the visible, external environment in which the perceiver is immersed. Thus, for Gibson, the question “what it is like to perceive?” already assumes a percept, whose ecological reality (its validity as an

environment) is grounded in its being “ordinary” for the perceiver. Environments, according to Gibson, cannot be paradoxical. According to this logic, they cannot be dualistic or inherently ambiguous in the manner of the perceptual experiences they afford. Thus, certain key indices associated with perceiving “environmentally”, emerge when ordinary perceptual experience is compared with pictorial perception. While such a comparison does not obviate questions of ontology, it does suggest that the ontological basis of a “reality” sufficient for immersion in VR, is to be found in vision itself rather than in the virtuality associated with the idea of VR. This is important, because it allows for the experience of immersion in a so-called “virtual” environment to be defined as a *perceptual reality*, obviating the need for recourse to ill-defined (that is, metaphysical) notions of “the virtual”. The nature of the stereo-immersive experience can be defined, then, through a Gibsonian ecological approach to visual perception.

The question of reality (whether virtual or actual) can be addressed (as above) by emphasizing how the perceptual system encounters its worlds (reality), rather than by questioning whether or not these worlds or environments are “real”. This can be further elaborated, with the argument that, for a perceptual reality to operate like an environment, three distinctive visual-perceptual aspects must also be in play:

- the absence of a tangible interface (such as a surface or screen);
- the state of seeming immersed in a three-dimensional space amidst solid-looking objects (stereo-immersion);
- the opportunity for mobility of the observer-immersant in that virtual space.

The practicalities and problems involved in achieving these three aspects or “affordances” are technically demanding. Yet the lack of an apparent interface, the appearance of 3D stereo-scopic (“solid”) spaces, and the affordance of self-mobility in the virtual environment are not so much techniques and strategies devised by VR developers, as those very qualities of an external environment to which human visual perception has adapted. Insofar as our ordinary perceptual experience is of a reality of some kind, the defining quality of this reality, as far as we can truly *experience* it, is its “realness”. My critique of the viscosity of stereo-immersive VR, centers on what I regard as an implicit collapsing, or merging of the inherent affordances (possibilities) of visual perception, into the technical requirements of a simulated “realness”.

Ordinary stereoscopic binocular perception of three-dimensional depth, while strictly relative, does enable us to reach and grasp near objects and gauge distances between things with reasonable accuracy. Human stereo-acuity “specifies ratios” (Palmer 1999: 207) rather than simply designate that one object or surface is in front of or behind another. The reciprocal relationship between perceiver and environment posits binocular stereo-acuity as the fulcrum around which VR developers achieve their main objective, defined in the VR literature as the “perceptual illusion of non-mediation” (Lombard & Ditton 1997: n.p.). The danger of this particular and powerful reciprocity is twofold. First, vision is aligned to the production of per-

ceptually narrow, or ordinary ends. Second, it suggests a shift toward an instrumental approach to the possibilities of visual experience, which threatens to overwhelm what might otherwise be an interest (for the VR discourse) in vision's transformative and performative possibilities. As an instrument, vision – our inherent binocular visuality – risks becoming a means to an end, reduced to functioning as an extension of the “geometry engine”. This endows vision with great potential. But in addition to gaining instrumental power, vision loses (or is not required to submit to) a special potency that is also inherent to human visual perception: the capacity to see more than what is environmentally “valid”.

In “Art as Technique”, Viktor Shklovsky (1893–1984) develops a similar thesis through his concept of “estrangement” (de-familiarisation), stating that, “the process of perception is an end in itself and must be lengthened” (Shklovsky 1965). Most of us see the world with two eyes, and it is the two eyes’ lateral displacement that produces disparity between the two retinal images for a given scene. This disparity triggers *stereopsis*, or “3D” stereoscopic vision; evidence that the visual system has adapted to perceive (more or less correctly) the visual structure of a three-dimensional environment. Yet, as vision scientist David Marr (1945–1980) notes, “the eye apparently regards changes in depth as so important that they must be made explicit everywhere, including places where there is no direct evidence for them” (Marr 1982: 51). This implies that the visual system does not require specifically three-dimensional (stereoscopic) information to have perceptual experiences that are “spatial”, since the range of information that is spatial in structure includes both monoscopic (pictorial) and stereoscopic sources. Marr’s observation is suggestive: it implies that the visual system has an innate capacity – an “appetite” – for seeing something as something else, and that perception is preordained for being “lengthened” just as Shklovsky implores.

The inter-workings of the three affordances of VR outlined above define a difference between the *realness* of ordinary reality and the *realism* of conventional media such as paintings, drawings, photographs and films. The significance of the term “media” arises here. Related to Shklovsky’s notion of “estrangement” is the sense in which a picture “mediates” our perception of its contents, in ways that an ordinary environment cannot. Art historian Barbara Marie Stafford articulates how pictures do not yield their contents unproblematically, when she describes what a painting does *pictorially*. The painting (by Pehr Hillestrom c. 1800):

[S]hows how attaining the goal of a visual search can be knowingly deferred, prolonging the capture of the eye within material substance and deepening visual attention so that the cognitive system can examine additional properties. (Stafford 2007: 146)

Importantly, this understanding of what it is like to look at a picture, gives due attention to what a picture is as a *particular* class of mediatory objects in the world: that is, “a surface so treated [that it] always specifies something other than what it is” (Gibson 1978: 228–229).

Gibson's Contribution

Gibson was a perceptual psychologist with a special interest in the epistemology of vision (Reed & Jones 1982: ix). As Gibson worked to overturn the incumbent structuralism of Classical understandings of human vision, he also rejected the Kantian idealization of space. The *concept* of space, Gibson argued, “has nothing to do with perception” (Gibson 1979: 3). In the reciprocal relationship between perceiver and environment, a “perceiver” belongs to a perceptual *system* – involving eyes, head, and body in motion – evolved in response to an external environment. An environment is a mute, physical externality, but it is also fundamentally a repository of sensory information. The term “ecology” emphasizes how the illuminated layout of surfaces and textures in an environment, provides all the visual information needed by an active perceiver on the move. For Gibson, what is *not* perceptible to an observer is redundant, irrelevant to theorizing vision: an ecological approach overrides the need to theorize vision according to internal “mental processing”. Gibson’s rejection of the notion of mental images relates to his argument that, instead of such intangibles, we think about real surfaces and their layout within an illuminated environment. Rather than recourse to internal processes inside the perceiver’s head, so Gibson argues, let this external illuminated layout be the basis for the formulation of a theory of perception. Associated terms such as “optic flow” and “invariant structure” similarly relate the actions (self-motion) of an observer to Gibson’s notion that visual information is immanent, always available; embedded in the very structure of ambient light. Information about direction, distance, orientation, etc. is made perceptible through the active perceptual behavior of the perceiver. Thus the realness of the perceived environment is grounded in the reciprocity between perceiver and environment.

Reference to Gibson’s theory is important, because it suggests an epistemological link between the (Gibsonian) ecological approach to understanding visual perception, and the “instrumentalization of vision” within the discourse of stereo-immersive VR. Gibson’s contribution to the development of stereo-immersive VR has imbued that discourse with an emphasis on realness through an “instrumental” approach to vision. For example, in an autobiographical passage from 1966, Gibson explains his decision to make an instructional film for AAF Fighter Pilots, and compare it with the best possible instructional manual and illustrated lecture available at the time to see what was needed to improve the pedagogy. Gibson and others analyzed AAF Production Unit training films to “develop a theory of what a motion picture shot could do that nothing else could” (Reed & Jones 1982: 17). The results were that aviation cadets “learned significantly more” from the moving image display with its flow of information, than from the conventional instructional material:

The reasons were fairly clear. What had to be learned was a system of how to aim at a moving target (fighter plane) from a moving platform (bomber). As the situation changed, the action changed. The film *showed* how one thing varied with another; the book and the talk could graph it, represent stages of it, and describe it in several ways, but could not display the continuous covariation in time. Moreover, and this impressed me, the film could make use of

the ‘subjective camera,’ taking the point of view of the learner and displaying how the situation would look to *him*, not merely what things looked *like*. (Reed & Jones 1982: 17)

Gibson’s emphasis on how the animated display gives the observer a sense of “continuous covariation in time” and particularly, how it inserted the observer’s own view-point at the center of the flow of images, is more like a description of a virtual reality display than of conventional cinema. Gibson’s own account of the achievement, can be interpreted in terms of what *would* be an ideal visual experience for a trainee pilot: a seamless virtual environment which appears real, not just in terms of the content of the imagery, but in terms of how this imagery behaves in relation to the observer in motion, in real time.

While “virtual reality” (VR) means different things to different people, the *idea* of VR has lasting cultural significance (Ryan 2001: 1). For the purposes of the argument, I have privileged vision, and explicitly linked VR to stereoscopic 3D imaging technologies worn by an “immersant”: typically a stereoscopic head-mounted display (HMD) and motion tracking sensors, linked to a computer system, which provides real-time visual input. “Stereo-immersive” refers to the way 3D imagery surrounds a perceiver, immersing them in the imagery that thereby becomes a visual “world” for the duration of the experience. What the perceiver’s head is inside of in the case of stereo-immersive VR, is not just a piece of equipment, but a “perceptual ecology” in which the requisite perceptible information is generated by a high-end computer imaging system. The reciprocity, here, between perceiver and “virtual reality” is exemplary in its Gibsonian perceptual-ecologic. Essentially this is a relationship in which “[t]otal sensory immersion depends in the long run on our ability to enter into what our senses receive” (Heim 1998: 28). The visual sense need not discriminate between “actual” and “virtual” sources of imagery: the visual system – eyes, head, body in motion – is inherently adaptable, and is equanimous toward the perceptual affordances of virtual and actual ecologies alike.

Realism and Realness

A material practice of vision can be articulated by comparing the realism of painting with the realness of ecological perception. Stereo-immersive VR can have its reality in ways that a picture cannot: no discernible interface, three-dimensional “solid” space, and capacity for self-motion that reaffirms the first two conditions. What is missing from ordinary perception is a mode of vision peculiar to visual engagement with an object which itself emerged from the material practice in the studio. However, it is not that the materiality of the object simply instigates a response from vision that corresponds with the pictorial object’s material genealogy. Rather, in addition to this correspondence is the visual-perceptual effort which must be engaged to bring the mute visible object into being as a picture. According to this way of understanding, it might be said that *pictorial perception makes pictures*, not painters! This could be elaborated to argue that pictorial perception and pictures are themselves an

ecology, involving a reciprocity between perceiving and making pictures, from which the notion of a *poiesis* in and of vision emerges.

As a category of pictorial art, realism includes a broad range of styles, traditions, and artistic and cultural values and intentions. The kind of realism of interest here belongs to the broad category of photorealism (which helps in the comparison against computer graphic representation, which is photorealistic by default). Reference works would include those by such painters as Giotto and Lorenzetti from the post-Byzantine era, and even Cimabue before them, as well as Caravaggio and contemporary realist painters such as William Bailey. In this regard, the realism of interest is tacitly mimetic. For while their paintings are clearly not photorealistic as such, Cimabue, Giotto and other early European artists have contributed to the tradition of illusionism by establishing the break with Medieval and Byzantine systems of representation. For these artists from the thirteenth century onwards, the establishment of pictorial strategies for representing space was vital. In particular, “artificial” and (later) “synthetic” perspectives were highly significant conventions in systematizing, in pictorial terms, the shape and texture of external reality. John White writes in *The Birth and Rebirth of Pictorial Space*:

The new artificial perspective was a tool finely fashioned to perform the three tasks for which the greatest masters used it from the moment of its inception: the creation of pictorial space; the organisation of the composition; and the maintenance of a perfect harmony between this new reality and the surface upon which it has its being. (White 1967: 167)

This mimetic tradition related to the notion of “perfect harmony” finds a willing ally in stereo-immersive VR (and computer graphics generally), since VR is an approach to representation that not only references, but reconstitutes the way an illuminated environment, real or imagined, displays itself visually as a surface layout (note the Gibsonian terminology). Immersive stereoscopic VR is uniquely empowered to instantiate the lit environment, since it alone, as a system of visual representation, can array surfaces in three-dimensions; and it can illuminate itself. Virtual imaging technology is thus well fashioned as a fine tool for the creation, not of pictorial illusionistic space, but three-dimensional, corporeal “actual” space. However, in terms of the stereoscopic imaging system harmonizing its “new reality” with the “surface upon which it has its being”, the analogy falls apart: where or what exactly is the “surface” in stereo-immersive VR?

Of course, Caravaggio’s realism is not reducible to just its material means: the organization of those means in pictorial terms is particularly significant. It shows how the limited corporeality of paint across a flat surface, gives us volumetric forms in space that are present to us pictorially. In achieving his corporeality *pictorially*, Caravaggio’s painterly realism differs fundamentally from the “real” corporeality of virtual objects in a space shared by an observer-immersant: in the latter, the corporeality is volumetrically, phenomenally real. In stereo-immersive VR, this achievement



16. Caravaggio, *Supper at Emmaus*, 1601–1602, oil on canvas, 139 x 195 cm

is not a matter of pictorial *interpretation* but of perceptual realness. This use of the term “realism”, then, concerns the operations of pictorial conventions in efforts to represent reality pictorially; whereas *realness* pertains to the quality of someone’s experience of something that is real (their perception of that reality; their sense that what they are perceiving is real, irrespective of its ontological status as Reality).

Emphasis here is on the manner of the visual *structuring* of the virtual content in stereo-immersive VR, rather than on how the contents – objects, figures, scenes – are depicted (for example, whether through figurative, expressionistic, impressionistic, abstract, photorealistic, or other modes of representation). To help clarify this point, consider the historic trend toward photographic realism in Western art: that is, (arguably) “any stylistic change in the photographic direction would produce an impression of realism, to the extent that photographic fidelity provides the criterion of realism” (Hagen 1986: 81). Let us accept for the moment that photographic fidelity is an accepted criterion of a certain kind of realism in pictorial art. Further, that photorealism can be qualified to mean: the observable behavior of light (photons; hence *photo-*) hitting various textured and colored surfaces in an environment, as recorded by the artist, such that the picture’s contents resemble the visual structure of that environment from a fixed point of observation, thus capturing to the best of the artist’s abilities, what that environment is really like, visually, for an observer (*realism*), from that particular point of observation. In pictorial, static art, conventions such as artificial perspective, and greater finesse in rendering shading across volumetric form (with the advent of oil paint), were devices that brought Renaissance painting closer to the qualified definition of photorealism set out above. Pictorial realism in (say) a Caravaggio, attends indexically (via mimesis) to the realness of visual aspects of the actual world or its imagined equivalent. Caravaggio’s painting of *Supper at Emmaus* (Figure 16) subscribes to the criterion of photographic fidelity in this regard, through

his use of pictorial conventions of realistic representation. The way light falls across surfaces and casts shadows to reveal volumetric form (*chiaroscuro*); the way reflected light bounces the color from one surface to another, for instance on the skin of the hands and face near the red cloak (“radiosity”); the way the outstretched hand of the figure on the right appears larger than that of the central figure (foreshortening); the way near objects occlude far objects, and the way the table surface takes on a tapered rhomboid shape (artificial or synthetic perspective), are all visual aspects that can be readily observed in the actual world. Pictorial realism borrows from, but does not purport to recreate, this world in full as a “reality”. As Svetlana Alpers observes in Dutch painting of the Seventeenth Century, this style of representational painting is (among other things) an art of *describing* what is observable in the visible world.

Certainly, something is recreated in the painting – pedagogical, moral, aesthetic, perceptual and existential aspects or “values” that the observer can interpret from the content, formal configuration of elements, style, and cultural and historical contexts; from what the painting is “about”. Caravaggio surely did not expect to recreate the world of *Supper at Emmaus*, except according to the limited means of pictorial communication available to him at the time. As well as the various technologies of oil paint and brush, and the skills in applying them to the canvas, these limited means include the pictorial conventions of shading, foreshortening, perspective, etc. They are *limited* because they are tied to the conventions of pictorial art, which must operate across a flat surface of some designated size. They are conventions insofar as they arise from and address these limitations in specifically pictorial and painterly terms.

Realism does not adequately account for the experience of realness in immersive virtual environments, because the two are different orders of experience: and the difference is of an entirely other, additional dimension. This is not simply because it is the stated goal of developers of VR to construct virtual “realities”, and because public hype about VR leads expectations in that direction. What is significant to the perceptual experience of stereo-immersive VR, and what is not available or expected from the experience of looking at Caravaggio’s *Supper at Emmaus*, is that objects and spaces appear perceptually, not pictorially, real and solid. Gibson is helpful on this point, stating that when we “scan” a picture for information, we are “generally in search of meaning”, not its “structure”. Anything other than meaning from a picture is, for Gibson, “a sort of perceptual luxury” (Gibson 1966: 250). This difference in perceptual structure, between notional pictorial space and corporeal actual space, marks an important epistemological shift, brought about by virtual imaging technologies, from the representation to the *presentation* of perceptual spaces.

Instrumental Realities

In a critique aimed at the positivist discourse of Western science, Bruno Latour states that laboratories “are now powerful enough to define *reality*” (Latour 1987: 87). Science can deem something to be real, to really exist: it can be named, measured, synthesized, or even cloned. However, a curious problem in defining an environment as a reality built entirely from information arises, because a “virtual reality” will typically be a short term experience, and its world will only contain whatever

the designers and technicians have put into it. As an artificial, constructed reality, it has no immanent “history” (although its contents may be historical in character) and no “future” because it can be turned off or “crash” at any time. Thus, a virtual environment seems on this basis to be a limited and transitory case of a real-like world. However, this makes the Gibsonian task all the more vital. Gibson states that “insofar as ambient light has *structure* it specifies the environment, and that *differences* in the light at the point of observation structure that light as containing ‘information’” (Gibson 1979: 51). Thus, the reality conditions required for it to be like an environment are not determined by factors outside and beyond immediate, temporal perception – such as real time and space (which become abstractions from the point of view inside the VR apparatus). Rather, reality can be “specified” on demand, and rendered through technical solutions based on a strategy of temporality: a world that is *real enough* for the time being. Crucial to the realness this implies, is the role of self-motion for the perceiver.

Gibson developed the concept of a visible “invariant structure” to explain how perception picks up information about the structure of an environment. To elaborate the concept, Gibson argues that the *motion* of the observer helps reveal the persistent (“invariant”) layout, the actual structure of environmental features, such that buildings, other people, trees, rocks, etc. appear as solid objects. Invariant structure in relation to a moving observer indicates changes in perspective (or point of view) toward an object – that is, it confirms that the object itself does not physically distort. Thus, objects appear larger the closer we are to them (they do not normally grow in physical size); surfaces of objects similarly appear to change (the top of the cup does not physically change from a circle to an ellipse), and so on. In turn, this subtle translation in their appearance confirms that objects in the layout are solid, and occupy three-dimensional space in a manner continuous with the persistent stability of the surrounding environment. Hans Jonas explains the point succinctly:

The possession of a body in space, itself a part of the space to be apprehended, and that body capable of self-motion in counterplay with other bodies, is the precondition for a vision of the world. (Jonas 2001: 156)

That stereo-immersive VR enables an immersant to move – even just to lean in or out from a standing position, turn the upper body, or move the head – establishes a relationship between observer and environment that is perceptually real. The observer moves: yet only to the physical extent of the VR laboratory, which also defines the extent of the fictive world in which the observer is “free” to move physically. The immersant remains in the same physical location but experiences a shift of so called realities. This effective movement indicates the extent to which vision in stereo-immersive VR becomes an extension of the imaging technology. For the only thing that really happens – transforms – is that the visual imagery changes (again, leaving other sensory modes aside): initially, to replace whatever was in view before the head-mounted display is put on, and thereafter in accord with the tracked movements of the observer, to which the visual-perceptual faculty responds as though

those movements were real in relation to the virtual content. Any residual pictoriality of objects in a virtual environment is largely overwhelmed by the perception that the objects in it are volumetrically solid; that they corporeally occupy space at distances that co-vary with the observer's movements.

The Origins of VR

The notion of a timeless "Edenic" natural world is an alluring one, and Nature might be the conceptual foundation for notions of a pre-existing, external reality. Nature (or "the natural") provides a necessary existential counterpoint to the experience of Being. Leaving existentialism aside, it must be said that natural scenes are "the obvious proving ground for realism" (Blake 1990: 403), in the discourse of "solid" VR. The application of Gibson's research on ecological optics to the design of computer-generated "virtual" environments, is evidence of the fit between "nature" and "the real". In Western epistemology, progress assumes a modernist, scientific paradigm. However, the discourse of stereo-immersive VR entails a peculiar "philosophical regression" (Penny 1994: 200) in which nature is again idealized as the representational perceptual schema for "the real".

Yet, the implicit ideology of VR is simultaneously Modernist, for in presupposing that if Nature, that is to say, "reality" can be known and represented, then it can be re-made (re-presented) synthetically. The theme of Nature as something to be abstracted, controlled and manipulated in order that we understand it, underlies Western epistemology. Nature as a "resource" to be controlled and exploited, is associated with scientific practices that emerged at the beginnings of Modernity, which saw nature as the "'inexhaustible' source of 'resources' which must be obedient to man's demands" (Ihde 1991: 56). This is also Heidegger's concern in his essay "The Question Concerning Technology", an early, thorough, and still relevant critique of Modernity as a manifestation of an instrumentalising "technoscience". Heidegger regards modern science (technology) as a way of representing the world that "pursues" and entraps nature as a calculable coherence of forces. Modern physics is particularly suspect for Heidegger:

Because physics, indeed already as pure theory, sets nature up to exhibit itself as a coherence of forces calculable in advance, it orders its experiments precisely for the purposes of asking whether and how nature reports itself when set up this way. (Heidegger 1977: 303)

The problem of reality (whether it can be known) was especially the case in the past, when understanding of human physiology and the physics of light was limited. The neurophysiologist Margaret Livingstone notes that "the vertebrate eye is so complex that it was (and sometimes still is) used as an argument against evolution and in favor of divine intelligence" (Livingstone 2002: 24). Thus vision is empowered, here, as a theological instrument: it constitutes reality as an instrument of (a "proof" for) intelligent design. In these now relatively secular times, we might remain wary of a potential "confounding" from such interfering ideologies. Yet VR

developers have absorbed certain practices – which could be called ideologies – in relation to the way they represent the inherent capabilities of human vision. The heavy reliance on vision science, and in particular a Gibsonian ecological theory, is a particular instance where there has been a collapsing of the physiological with the epistemic in vision. This is a potent mix, for Gibson's ecological model is so successful, as a model of how a perceiver picks up ecologically valid information from the visible environment in order to disambiguate its rich complexity, that VR developers have not needed to look elsewhere for an understanding (a representation; a model) of how *else* we might encounter perceptual reality.

It is important, therefore, to consider how VR researchers and developers have acquired and developed their understanding of space as something habitable which affords exploratory motion within. Stereoscopic immersive VR did not emerge solely from experiments with computer imaging technology. Further back in VR's technical history, and inscribed into Western enlightenment ideas about the structure (and ontology) of reality is the spatial paradigm of Euclidean space. The Cartesian coordinate grid, as derived from Euclidean geometry, is "built into our culture and our perception" such that it "structures our thinking" (Penny 1994: 200), suggesting that to subvert this visual-spatial paradigm would be to render things incomprehensible.

The Euclidean coordinate system provides a method and representational framework for analytic geometry. As a system, it operates on the basis of axes. Two such axes, the horizontal (x) and vertical (y), account for two-dimensional planar space. With the addition of the " z " axis designating depth, a three-dimensional coordinate system can be represented. Theoretically, in such a system every point in space can be mapped and represented. Accordingly, two- and three-dimensional objects can be described symbolically using mathematical systems, since every point in space can be ascribed a "place number". Metaphysical reality is, in this sense, literally a matter of rationalizing space in the abstract. Indeed, Bruno Latour proposes that "reality" is precisely how laboratories define it, using the objectification of protein as an example: Latour envisages a time when protein was an undifferentiated "whitish stuff" (Latour 1987: 87) for biologists; not an object *per se*. "Reality" is also by Bolter and Gromala described as "80 million polygons per second", a statistic that refers to the underlying image processing required to produce fully realistic images on a computer screen):

This would be the ultimate victory for the strategy of transparency because it would no longer be possible for the viewer to see the medium (computer graphics animation) at all. ... What began in the Renaissance, or with Zeuxis in ancient Greece, would culminate in cyberspace. (Bolter & Gromala, 2003: 40)

In computer imaging, a polygon is the basic unit of surface structure of a more complex virtual object. Polygons are visual representations (usually triangular planes), displayed in pixels, of the mathematically defined solid geometry that make up the surfaces of cylinders, boxes, pyramids, and more organic forms of more complex

computer models. It is easy to see how Latour's "undifferentiated white stuff" is analogous to the basic unit of surface structure in virtual imaging systems. Unlike protein in its pre-rendered state, the "stuff" of stereo-immersive VR imagery had no primordial ontology. However, the analogy does hold: Latour's point is that the undifferentiated becomes "real" as an object, through some technique in the laboratory context, and is represented and promulgated through the discourse of technoscience. This latter point is the significant one in the analogy, since a VR "lab" is both the source of the basic "stuff" *and* the world in which the undifferentiated matter becomes a manifest object.

Conclusion

An exemplary virtual world exploits the "perception and action knowledge base the user already has from acting and interacting in the ordinary world" (Hagen 1991: 76). Exploiting vision's innate capacity, VR offers a substitute world through a transfer of visual-perceptual and motor skills; a transfer between "the real" and "the virtual" uncompromised by any apparent intervening interface. This transferability founds the link between seeing and knowing in stereo-immersive VR, in which "the epistemic importance of vision" (Dretske 1969: 78) becomes something of particular importance to technicians and designers of stereo-immersive "virtual" spaces. This raises a question concerning whether the success of stereo-immersive VR is simultaneously the source of its own perceptual redundancy, because to bring into being the perceptual basis of ordinary "real" reality, is to return the perceiver to where they were, perceptually, before entering into the VR world. This leaves vision with no further purpose other than to align itself with the ecological reciprocity by which reality can be substituted for itself.

Gibson puzzled over the perplexities of vision for fifty years, and while he has his critics, Gibson's contribution to understanding visual perception reaches well into recent studies of vision and cognition (Port & Van Gelder 1995: 373). The theory of ecological optics, for instance, has been applied to machine vision in the emerging field of "ecological robotics" – a transfer of knowledge and methodology that confirms Gibson's ideas about what vision is for, by demonstrating with success that it does not matter what kind of organism ("agent") is doing the perceiving. In a similar vein, Jerry Fodor and Zenon Pylyshyn discuss what is required by a perceiver in order to grasp visually the three dimensional shape of an object, noting that the organism need not make complex, knowledge dependent inferences:

At least in this case, it appears that specific properties of the ambient array are able, as Gibson would put it, to specify the three dimensional shape uniquely. Such results have helped kindle the current interest in Gibson's theory. (Fodor & Pylyshyn 2002: 212–213)

Fodor and Pylyshyn do not specify that the interest comes from computer visualization or VR developers. Yet as far as determining that the organism: (the immersant in the case) will see the virtual world as a three-dimensionally structured environment

through information provided directly from the ambient optic array, Gibson's contribution to the field of virtual imaging technology is evident in the "striking parallels" between (Gibsonian) ecological psychology and new developments in computer vision, particularly with regard "how agents interact with the [virtual] environment" (Duchon, Warren, & Kaelbling 1998: 474).

The perceptual mode specific to pictorial art – the "material practice" of vision as elicited in pictorial perception – presupposes a non-naïve viewer and is rarely if ever confused with the kind of visual attention an observer would direct toward objects and scenes when perceiving the real world. Moreover, as Margaret Hagen points out:

Pictures – good pictures, successful pictures – educate in terms of the *pictorial* possibilities of perception. Progressive pictures [according to a continuum of stylistic change over time] point out perceptual realities new to the *pictures* of a culture, not new to its people [or to their experience of perceiving the ordinary world] (Hagen 1986: 81).

Hagen thus provides an additional way to bracket the notion of *realism*, broadly understood as associated with a style of pictorial art, from the notion of *realness* proposed above, associated with the visual structure of stereo-immersive VR. Both photorealist painting and immersive VR draw reference from the visual structure of the ordinary world. But the difference between their respective referencing is definitive, epistemological: it concerns the possibilities of perceptual experience that each offer. Pictorial possibilities, as Hagen suggests, include stylistic, semantic, aesthetic, poetic, formal, physical and other aspects that vivify the pictorial art of various cultures. Importantly, these possibilities designate the capacity to see a picture as referring to something else (which seems to be a universal capacity. My use of the term "perceptual possibilities" builds on this notion of the capacity to see a picture as referring beyond itself, to include a more specific meaning: in representational painting, perhaps all painting, the painted world is given in addition to the real world – as a *supplement* to real world perception. In contrast, stereo-immersive VR gives the observer its world as a substitute, real-like world – as a temporary, notional replacement for the real one. For stereo-immersive virtual technology, this is no mean achievement. However, the kind of perceptual possibilities this realness invokes are arguably less engaging than the perceptual possibilities of pictures. Pictures give us supplementary, non-habitable, worlds-within-the-world. To perceive virtual objects as three-dimensional volumetric forms, is to perceive them as *already* real forms; to have them in their *realness*.

"Realism" as opposed to "realness" thus marks a definitive difference between pictorial and real-world perception respectively. This difference is epistemological, rather than a matter of degree (for example, of detail or resolution), and corresponds to the way assumptions about what vision is *for*, find expression and manifest differentially within the enterprises of pictorial representation and the design of virtual worlds.

IMMATERIAL LAND

Brian Martin

I would like to acknowledge the traditional owners of this land, the Wurundjeri people of the Kulin nation. I pay my respects to elders both past and present and convey thanks and gratitude to be able to create this writing on their ancestral lands. I also pay my respects to Bunjil, the great creator spirit of the Kulin nation.

The above acknowledgement is necessary as it identifies the vital importance of what country we are on. In Aboriginal Australian culture there is an extensive tradition to pay respects to the people and country that you are on or entering, as you are a visitor. The different language groups that exist in Australia would proceed to do this to identify who they are and to pay respect to another's country. It is also important to be welcomed onto someone's country out of mutual respect. This aspect of an ancient tradition is one that opens this chapter in order to situate myself into an ontological dynamic of Indigenous thinking and ideology. This act of acknowledgement introduces and demonstrates a way of thinking that is applied to culture in Indigenous ideology that is premised on "country". The focus "country" is central throughout this chapter as "country" invokes an embodied subjectivity. In Indigenous terms, one "belongs to country", not the reverse and there is a reciprocal relationship between people and "country". This also reveals a way of thinking in terms of a dynamic ontology that exists within Indigenous cultural ideology where this reciprocity is grounded in country. This will be examined within this chapter to demonstrate how the immaterial is materialized within an Indigenous way of thinking about ontology and cultural practices.

The relationship of Western thinking to ideology has created what I term "tranquilization", which means that Western culture is complacent and apathetic about its own surroundings. This trajectory is twofold; it is both an effect of amnesia and one that perpetuates amnesia. This is not only significant for Australia, but illustrates a Western conception of art and culture which is based on representation, something separated from culture and life. This is vital to understanding how art and culture in a Western sense become merely representational ideology. Art is vital to the formation of cultural identity as cultures rely on practices that articulate memories, which are embedded within a collective consciousness. In Western terms, art has become

focused on the action of transgressing itself and is performed within its own domain which is separate from everydayness. Although it has an ontological premise, it is also masked because it has been hijacked by the ontic and representationalism. Indigenous cultural practices are crucially different as cultural productions have inter-relatedness to the real everyday world and are not isolated from one another. Furthermore, the premise of producing works extends from the real world itself, “country”, as will be demonstrated in my discussion of the work of Indigenous artist Badger Bates. This examination therefore challenges an established value system that considers Western ideology as legitimate and any “other” to be of a lesser value. Indeed, it is this notion of value that contributes to the effect of amnesia.

This chapter investigates and recasts ideology in terms of material conditions. At the outset I analyze Louis Althusser’s definition of ideology in order to expose its own false consciousness, which, in his own framework, is founded and premised on imaginary relations to existence. This permits me to introduce the notion of amnesia, which, I demonstrate, is a result of such false consciousness and ideology. I take up Andreas Huyssen’s definition of amnesia in order to reveal how a Western framework of ideology not only has an imaginary relationship to existence but embeds an amnesiac conditioning within its own constructed framework. Martin Heidegger’s explanation of the relationship between art and ontology provides an opening to an understanding of an alternative framework. By extending this ontology, I employ Barbara Bolt’s account of a radical ontology that sees the image as having “real material effects”, a conception that has resonances with the radical material beliefs of Indigenous culture. By elaborating this framework, I will provide an alternative “ground” and understanding of ideology by presenting how ideology in Indigenous culture is not based on the imaginary conditions of existence but is grounded in the “real”. Through an examination of the relationship between art, ontology and ideology in Indigenous Australian culture, it is possible to arrive at an alternative understanding of “cultural ideology” and ontology. In doing so, we may recast the notion of ideology, materialism and ontology through their basis in “land”. The Western notion of “land” denotes an object to be owned. Drawing on Paul Carter’s ideas concerning metaphor and layered meaning as they relate to ideology, I provide a counter-argument and will demonstrate that Indigenous cultural ideology involves an ontological relationship, a cohesion between life, art, country (land), practice and memory rather than an imaginary relationship to existence. Aboriginal “cultural ideology” is the “real” world.

By examining the work of artist Badger Bates, a Paakantji man from Wilcannia, far west New South Wales, I will demonstrate this alternative framework and examine how Bates’ work operates through a material practice that produces ontological effects through its articulation of immateriality. An examination of Bates’ practice demonstrates this alternative framework of ideology since it clearly refuses the binary between the material and the metaphysical, but rather shows that ideology operates within both realms concurrently. His work illustrates an ontological framework and revolves around extracting images and revising techniques from the ancient tradi-

tions of his ancestors, to be created in contemporary form. Although Bates works in various media, I will focus on his lino prints. The implications of recasting ideology in this manner and through Indigenous artistic practice will be examined in light of how “country” grounds the metaphysical into the material. The spiritual or immaterial nature of Aboriginal culture is a continual practice created by the reality of country. The important concept in this alternative framework is “country”. Country is the basis of Indigenous ideology and it specifically constitutes and is constituted by the relationship between memory, life and culture, which are embedded in land (country). Indigenous art practices manifest this trinity. Within this framework, the immaterial is materially constituted by the real material conditions of existence, where the immaterial itself, becomes a reality. This “immateriality” in Indigenous cultural ideology is manifest in the real existence of country and ever continuing cultural practices.

Material Representations

Western philosophical discourse in relation to materialism and the arts, grounds its thinking in an ideology that has little relationship to the existence of everyday living. The fundamental problem here is that a binary is created where ideology and lived experience have an ephemeral relationship. Since the inception of Western philosophical discourse, there has been a pursuit to represent an ideology of existence that is empirically and scientifically grounded. It is this line of thinking that has established a Western framework. Such a framework is confined by the limitations of Western ideology and representation in so far as it does not accept other forms of metaphysical beliefs. How can such ideology be grounded when it remains an ideology that is merely representational? The representational is problematic as will be discussed in more detail later in this chapter. The critical issue at stake here is the notion of ideology itself. Althusser states: “Ideology represents the imaginary relationship of individuals to their real conditions of existence” (Althusser 1971: 163).

In his exploration of ideology and the state, Althusser examines the means of production as a reproduction of labor power in its submission to the ruling ideology. In doing so, Althusser presents a “new reality”: “Ideology” (Althusser 1971: 133). It is in this context that Althusser claims there is a ruling ideology that unifies the diversity of production. He works though the following premises:

- ideology has no history: it does not have a history of its own as its history is external to it;
- ideology is a pure dream.

Althusserian ideology is related to Freud’s notion of the unconscious and formulates the view that ideology in general, has no history; it is eternal like the unconscious. From this, Althusser claims that ideology itself does not correspond to reality.

However, while admitting that they do not correspond to reality, i.e. that they constitute an illusion, we admit that they do make allusion to reality, and that

they need only be ‘interpreted’ to discover the reality of the world behind their imaginary representation of that world (ideology = *illusion/allusion*). (Althusser 1971: 162)

Ideology and the Subject

The vital point that Althusser arrives at from the above illusion/allusion relationship is that humans transpose and represent reality to themselves in an imaginary form. Challenging this is vital to any attempt at redefining ideology, as in this instance ideology becomes a construction of imaginary representations. Althusser further argues that humans construct themselves through an alienated representation since the conditions of existence around them are alienating. The crucial part of this argument is founded on the idea that the relationship that humanity has to its own reality is like ideology itself. Humanity in Althusserian terms has no history. Within this dilemma, Althusser moves on to claim that “ideology has a material existence”; that ideology exists in its practice or practices and therefore, this type of existence is material (Althusser 1971: 165). Within this materiality, ideology itself is made possible by subjectivity as “Ideology interpellates individuals as subjects” (Althusser 1971: 170). This is vital to Althusser’s understanding of ideology in a material existence as having no history as it is manufactured by and manufactures the subjects themselves. However, we must acknowledge the separation the subject has in relation to ideology in this framework. This is the reason why the subject has to be interpellated by ideology in order to attempt to resolve this existing binary. However, this further adds to the separation due to the nature of ideology itself within this framework. Opposing this split or separation is the notion of Land, where there is no need for interpellation as there is no existing separation between the subject and ideology. In this instance, land can take on the subjective position, which is the fundamental basis of an Indigenous ideology.

For Althusser, the existence of ideology is premised by the subject and for the subject. This and the points made above are crucial to an understanding of ideology in Western terms. If ideology is an imaginary relation to reality and has no history, and this being an aspect of material existence itself, then humanity itself, in the light of the above, has no relationship to its own reality and history. Not only does ideology represent a false consciousness,¹ but the ideology of humanity does itself operate as false consciousness. On the basis of the Althusserian relationship between ideology, subjects and the real conditions of existence, the notion of an amnesiac condition presents itself.

Perpetual Amnesia

For Andreas Huyssen, amnesia is produced in the context of the fading of generational memory. Huyssen uses the creation of the museum as a precursor to our obsession with memory, where the museum becomes a larger metaphor for holding onto culture and experience in everyday life. This is demonstrated by our restoration of things around us and the recording of life by technology. Hence the metaphor

of the museum becomes a paradigm for contemporary culture. The issue here is that this obsession does not solve the onset of amnesia, because by categorizing cultural productions and memories within the metaphor of the museum, it promotes an amnesiac conditioning. Our obsession with the past through the safekeeping of memories in institutions give a false sense of capturing memory, because what is fundamentally locked away remains so: by safekeeping memories we forget that they are in fact there.

Huyssen looks at the issue of representation in relation to temporality and memory:

It does not require much theoretical sophistication to see that all representation – whether in language, narrative, image or recorded sound- is based on memory. *Re* – presentation always comes after, even though some media will try to provide us with the delusion of pure presence. But rather than leading us to some authentic origin or giving us verifiable access to the real, memory, even and especially in its belatedness, is itself based on representation. (Huyssen 1995: 2–3)

The key point in the above is that amnesia threatens to consume memory itself as it is based on representation. As with ideology, memory, in this sense, has little relationship to our conditions of existence. This dilemma is stated by Huyssen in the following: “It will have sealed the very forgetting of memory itself: nothing to remember, nothing to forget” (Huyssen 1995: 9). However, Huyssen does recognize the tenuous crevice between the past and the present as one that is alive. It is this dynamic space where culture and memory can be inter-related and through artistic practice can be re-integrated back into living memory. Furthermore, it is this dynamic space that Indigenous cultural production utilizes in bringing cultural memory into consciousness:

The past is not simply there in memory, but it must be articulated to become memory. The fissure that opens up between experiencing an event and remembering it in representation is unavoidable. Rather than lamenting or ignoring it, this split should be understood as a powerful stimulant for cultural and artistic creativity. (Huyssen 1995: 3)

However, “creating” an ideology can trap us into Western modes of thinking, such as the deferral of meaning as it is posited within a postmodern framework. It is in the postmodern space where art and ideology and their meanings are deferred. This implies not only a condition of amnesia, but further perpetuates amnesia. Postmodernism in art demonstrates another mode where ideology is based on the imaginary relationship to existence, as it is also predicated on representation. How can we conceive of a different mode of thinking that does not involve this separation or split?

Teresa de Lauretis’ notion of an “elsewhere space” presents another type of space where reality, ontology and culture co-exist. Although her “elsewhere space” is theorized in reference to gender, I would argue that the same can be applied to the

construction of race and identity. By looking for the blind spots in the margins of hegemonic discourses, de Lauretis wants to create spaces within the “power-knowledge apparatus.” She states this is why:

The critique of all discourses concerning gender, including those produced or promoted as feminist, continues to be as vital a part of feminism as is the ongoing effort to create new spaces of discourse, to rewrite cultural narratives, and to define the terms of another perspective – a view from ‘elsewhere?’ (de Lauretis 1987: 25)

De Lauretis is referring to the space that is not represented within existing representations. It is within this space that we can operate to create an alternative materialist cultural ideology not predicated on the fissure between discourse and representation. This space of lived and situated experience is not embedded in “false consciousness” and imaginary relations to real existence. The relationship of art, ontology and culture, in an Indigenous mode of living, acts as a means of resolving the trap of deferral that this “open discourse” threatens to create, by presenting a “real” alternative ideology. This “real” ideology is an Indigenous conception of what I will term “cultural ideology”, and has a connection to the real social world as real and ontological experience. “Cultural ideology” is ontological. This is opposed to the notion of the ontic as elaborated by Heidegger in his discussion of the work of art and truth in relation to art (Heidegger 1971: 40). The ontic is in the realm that Heidegger refers to as “arts business”. This is a place where the ontic relates to our “everydayness” and our experience, as it is mediated by ideology. The ontic as the realm of everydayness, removes the work of art from its origins. In this sense, the ontic moves us further away from “Being” and from the ontological or the real conditions of our existence. The ontological represents the world of reality that is masked by the veil of “everydayness”, the ontic. The ontological is the space where the work of art opens up for itself. This is the space of “cultural ideology”, which being ontological, refers to the cohesion between life, art, land, practice and memory. This is also opposed to the notion of ideology put forward by Althusser.

Empirical Ontology²

Western modes of thinking about art are preoccupied with relating ideology to art and vice versa. This obsession within the history of Western art is premised on a desire to accurately represent a stable ideological basis whilst moving between the binary of the material and the metaphysical. How can this be achieved if there is a predicament with ideology itself? In dealing with this dilemma within the discourse of art, Heidegger presents an alternative within his notion of ontology, by exploring art’s potential for revealing “truth”. For Heidegger art operates in the realm of Beings.

Heidegger views art ontologically rather than aesthetically. Art in the ontological sense is where art opens up a space for itself in order to operate its purpose of demonstrating essence and truth. This space is one where art does not have an imaginary relationship to existence. Opposed to this notion is the idea that art is established

on aesthetic material principles based on perception. This perception is predicated on the imaginary relationship that ideology has to real existence. Within his work, *The Origins of the Work of Art* (1971), he paradoxically articulates his notion of art within the space opened by Immanuel Kant. Heidegger provocatively asserts “Matter is the substrate and field for the artist’s formative action” (Heidegger 1971: 27). This is a strategic intervention intended to move us out of preconceptions about art and take us to the understanding of art as “revealing” by its own nature. Matter, understood in this sense, extends out of the comprehension of matter according to Aristotle. In contrast to Plato’s idea of art as *mimesis*, Heidegger confirms the notion of art practice as the movement of *potential* to the *actual* of *substances* as presented in Aristotle’s fourth book on *Metaphysics*.³ He utilizes this dynamic to demonstrate the revealing nature of art. This resonates with the dynamics out of which Indigenous cultural practices operate.

Art, for Heidegger, exists as a sanctuary that stands in, building over the void of nihilism, as it is the “setting-into-work-of-truth”. In his search for essence, Heidegger examines the “thingly” character of objects. He points out that if a thing is let be, it will come to the senses, as it is “self-revealing” (Heidegger 1971: 69). Because we don’t access the bare sensation of sound, what is close to us is the “thing” in its non-essential form. In such a relation it is not the thing in itself, but a representation of the “thing”. So we can say that the problem lies with perception and it is here that Heidegger proposes art’s ontological function. The aesthetic perception and function in this instance removes us further from the “truth” and can be paralleled to the notion of ideology been predicated on subjective temporality.

Heidegger proposes the concept of form and matter, and the removal of the characteristics of the usefulness of the thing, hence, defining the “thing-being” as consisting of what is left over. This examination also portrays art as a preserving tool of truth of “essence” or “being”. For Heidegger, art is seen as an origin or essence, a happening of “truth”. His notion of “standing-reserve” is one where things come forth from concealment (Heidegger 1971: 17). Meaning is concealed in “standing-reserve” and acts as a *potential* where art can create a pattern of movement in a visual order to reveal true meaning, becoming an *actual*. In this instance, art’s function is about revealing the true nature of things by its strategy of isolation. Isolation in this instance refers to the way that art isolates an essence of the “thing”, by focusing on the thing itself. Through its isolating properties, art preserves the true nature of things that are then accessible to be embedded into memory through practice (Heidegger 1971: 69–71).

The “self-revealing” nature of art proposed by Heidegger has important implications for my argument as Indigenous cultural practices do not operate from a formal academic perception of aesthetics and a restricted Western ideological basis. The dynamic force created by the movement of revealing as described by Heidegger is the premise for arguing an alternative mode of ideology as realized by Indigenous practices. In such practices the notion of standing-reserve could be viewed as a metaphor of the notion of “country”, where all things are extracted through memory and practice.

However, this movement/happening that Heidegger discusses is still confined to Western modes of ideology. Western ideology, as defined earlier, is a representation therefore art becomes a representation. This is elaborated by Barbara Bolt in her discussion of the performative potential of art:

For Heidegger, as I have argued, representation traps us in a mode of thought that insists on grasping reality through imposed conceptual structures. (Bolt 2004: 55)

These imposed structures are outside our real existence evidenced by the fact that they exist as constructed representations in an “ideological” sense. Interestingly the premise of Bolt’s investigation lies in the examination of the notion of performativity, where images are not only representational, but are also performative, as they extend themselves to real bodies. It is here that Bolt extends our understanding of the ontological dynamic of art. She observes:

The suggestion that the material practice of art has real material effects and there could possibly be a mutual exchange between the matter of bodies and the image of bodies, has limited currency in Western art history and theory. (Bolt 2004: 168)

Bolt recognizes that a new “materiality” in the sense of the metaphysical has little acceptance in Western discourses and by presenting the notion of art as being more than just representational, Bolt asks a vital question:

How does Western culture respond to the challenge that an image just might transcend its own structure as representation? (Bolt 2004: 163)

The answer is that dominant Western culture cannot, as its premise is based on constructed ideologies that are still bound by the material in a limited sense. In reference to art as demonstrated by Bolt, the issue is with representation itself and the ideology of representation can be understood by looking at the term “ideology” as discussed in relation to Althusser. The representational falls in the same category as the imaginary relationship to existence: images represent reality, therefore are imaginary. However, this type of “new materiality” proposed by Heidegger still operates within a constructed Western ideological discourse. Bolt recognizes the limitations within Heidegger:

Whilst Heidegger’s theorization of the revealing potential of the work of art goes beyond representation, it continues to operate within Enlightenment modes of thought. (Bolt 2004: 188–189)

Bolt also distinguishes the conception of light in Australia to the European conception of light as elaborated by Heidegger. She discusses the blinding glare of

Australian light and the other, a softer European light and concludes that ontological experiences are therefore situated. It is this situated experience of country that gives Indigenous people a more objective account of the world as a real ontological experience. In Indigenous culture there is no division between the real and the ideological or split between memory and temporality which perpetuates a subjective amnesia.

Within the context of Australia, the above mode of enlightenment in a Western sense is experienced not in the sense of “light” but in “Land”. This chapter proposes a fundamental turning point by positing an alternative way of looking at materialism through an introduction of the notion of Indigenous ideology predicated on Country (Land). My argument is in accord with Bolt’s criticism of Heidegger’s conception of light, but also suggests that the notion of “light” used by Heidegger and Bolt is still within the framework of a Western discourse because in this framework, knowledge is understood through a metaphor of light. Within an Australian and Indigenous context, conceptions of enlightenment and light are not experienced or understood in the same way as they are within a European context. The alternative to this notion of “light” is “land”. All cultural productions within Australian Indigenous culture are created by the basis of “country”. One belongs to country which provides all sustenance in relation to living and all cultural practices, including “art”. As Bolt observes: “In the dynamic productivity of material practice, reality can get into images. Imaging, in turn, can produce real material effects in the world” (Bolt 2004: 8).

Bolt’s theorization opens up an alternative way of looking at materiality in relation to art. The images in Indigenous cultural practices are embedded in a living and vibrant culture and performativity for the purpose of ritual and practice. They are real in the sense of the “real” world *and* in an ontological sense, since the real world (both material and immaterial or imaginary) in Indigenous culture, is ontological. What is crucial here is the notion of land. Land, and in an Indigenous view, “country” is the basis of this ideology. Interestingly, this lineage, which resonates with Bolt’s notion of the “image” transcending representation has not held much value throughout Western thought. This is because Western meta-narratives of light and enlightenment were established and have been used to devalue other frames of thought. The situated space proposed here formulated from and by an Indigenous cultural framework is one that builds over the constructed void of ideology and its inherent amnesia.

Bolt elaborates the concept of *methexis*⁴ in order to extend her argument of performativity. She draws on Paul Carter’s (1996) elaboration of *methexis* as a principle that can be used to understand the performative nature of Indigenous Australian cultural practices. *Methexis* is understood as a sense of action from a fundamental basis such as the Dionysian in Greek culture, which encompasses performance that is not individualized and has a direct relationship to the group and the commemorative act. Carter observes:

Methexis was the ‘non-representative’ principle behind Celtic, and Aranda, art, whose spirals and mazes reproduced by an act of concurrent actual production a pattern danced on the ground. (Carter 1996: 84)

Carter's *methexis* emphasizes a physical ground and it is through this ground that Indigenous practices resonate. Indigenous culture is based on an ancestral history where various aspects of culture are not isolated. For example, there is no distinction between art, culture and living in the mediated experience of human beings. It is through the concept of *methexis* that Bolt proposes a view of art as productive materiality, a relationship between bodies and objects and it is in this context that Bolt introduces a radical ontology. "Matter is transformed in the exchange between objects, bodies and images" (Bolt 2004: 150). Bolt builds her argument by stating that materialization is a dynamic of reciprocity rather than a one way dynamic. In answer to the question "How do we experience a work of art as both an act of concurrent actual production and a sign?" (Bolt 2004: 173). Bolt turns to Charles Sanders Peirce's notion of semiosis to argue for a relationship between signification and the material world.

The dynamic object operates as a pressure on, or pulse in, the see-able. The insistence of the dynamic object constitutes a key energy or force in the work of art. Thus, a picture is not just the coded, immediate object. A picture also bears the pressure of the dynamical object. In this way, the dynamic object prevents the picture from being reduced to just a sign. (Bolt 2004: 175)

This force that prevents the image from being reduced to a sign is vital to my elaboration of Indigenous imaging. Bolt's discussion of Peirce's categorization of signs as iconic, symbolic and indexical extends on the idea of this dynamic force. The iconic refers to how an image looks like a thing, insofar that it is a mirror image. The symbolic is such that it "symbolizes" the thing, it is a representation of something, and does not necessarily look exactly like the thing it represents. The indexical is where the image creates a dynamic relation between the things it represents in relation to the viewer. In this instance, the image becomes exactly the thing itself, materializing itself by its causal relationship to the referent. This is a crucial point in relation to *methexis* and the indexical in the analysis of Indigenous ideology and practices which are grounded in "country". However it is important to point out that the above theories still operate within Western ideological discourses that are predicated on notions of light and enlightenment. In this sense we are therefore pulled back into ideology as a Western construct. The Indigenous relationship to land provides an altogether different basis for conceiving of ideology and casts a new perspective on the notion of performativity as elaborated by Bolt. The positing of land as a foundation of all Indigenous cultural practices thus points to a different conception of art – a conception that is "grounded" because we belong to "country".

Grounded Ideology

"Belonging to country" is *methexis* or "grounded practice" and all cultural productions extending from this encompass the notion of the indexical. The premise of "country" and its definition in an Indigenous ideology is the framework that provides

an understanding of how the indexical and radical ontology operate. What is important in understanding this alternate framework is that country also demonstrates the limits of Western ideological discourses in terms of “grounding” its thinking:

What is the ‘ground’ in question which we inexplicably fail to touch? Philosophers may urge other philosophers to ground their thought more thoroughly, but this is merely a metaphor and assumes (once again) that the ground itself can be taken for granted. What is that fundamental surface that seems to inform every dimension of our oneiric, physiological and architectural fantasy but remains, hitherto, metaphorical, metaphysical-ungrounded? (Carter 1996: 3)

Carter refers to this elusive ground where philosophers establish their thinking and inquiry yet they fail to discover its location both in a material and metaphysical sense. What is this “ground”, this fundamental basis that philosophical inquiry extends from or lays its foundations upon? This ever-escaping basis that eludes us, but is ever present is one that exists within the binaries of Western philosophical discourses, moving constantly with apparent instability, and creating a sense of amnesia about our own reality. Within this instability, materialism defines itself, partially real and ephemeral at the same time, eluding our vision appearing sporadically in what Plato refers to as “fleeting appearance of Being”. Materialism is embedded within the construct of ideology itself which is specifically a Western one; therefore any reverberation of a “new materiality” operates from this basis. A question is posed: Can we look outside this construct in order to explore ways of redefining a different type of materiality that Western discourse is hoping to find?

Western discourses based on an ephemeral materiality and the construct of ideology attempt to strip away appearances in order to uncover an essence or truth. This action removes us further from the truth. Carter suggests that Western philosophy is no different to the bulldozers clearing the ground (Carter 1996: 3). In his discussion of the colonization of Australia he comments:

No one appears to worry about what was cleared away when the streets were laid out according to a two-dimensional plan, when the natural topography was neutralized and in its place artificial vistas were carefully mortgaged. At no point in the process of arrival, survey, settlement and residence does the ground make any claim upon our attention. (Carter 1996: 1–2)

This represents a form of amnesia that underpins the colonization of Australia and all other colonizations. This underlying amnesia has a direct impact on the cultural foundations of Australia that have negative implications upon a cohesive cultural identity. It is through such a colonization of terra nullius⁵ that cultural amnesia evolves and has an impact on the value given to Indigenous culture which is predicated on “country”. Apart from this amnesia about the ground of settlement, land is also seen as a type of “other”, land is something to be controlled and defined by

the use of power. Land is also seen as the “Antipodes”, romanticized and feared. The important point here is that Western thought works through a metaphors of light/sight as it relates to knowledge.

We build in order to stabilize the ground, to provide ourselves with a secure place where we can stand and watch. But this suggestion that the ground is treacherous, unstable, inclined to give away, is the consequence of our own cultural disposition to fly over the earth rather than walk with it. (Carter 1996: 2)

The type of value given to “land” in Western terms differs greatly to an Indigenous one. Crucial here is the distinction between an Indigenous and Western value of “land” and how a Western definition is one that determines the value of an Indigenous one. Opposing the notion of clearing the metaphysical ground, is an Indigenous conception of art and a worldview that provides an alternative to cultural amnesia and ideology based upon imaginary relationships to real existence. It does so by locating materiality in a whole cultural practice. The most vital aspect of Indigenous ideology is the fact that all aspects of culture do not stand alone. For example, art does not exist in isolation from life and culture as it plays a crucial role in cultural life, extending its relationship with country. This will be illustrated with reference to the works of Kathleen Petyarre and Rover Thomas and further elaborated through an examination of the works of Badger Bates as mentioned earlier in this chapter.

Indigenous Cultural Ideology and Arts Practices

Grand narratives and representation within Western discourses bring about the onset of “amnesia” as it is grounded in a subjective position related to ideology. This also applies in Western conceptualizations of “art” and its art practices. Indigenous cultural ideology is based on how our ancestors brought into existence the *methexical* relationship between country and people. From this ground, all cultural productions are created with a holistic inter-relatedness as they do not exist in isolation to one another. The vital part of this ideology is the movement and dynamic force of everything including the movement across “country” and all culture throughout the mediation of everyday life. Within this dynamic, there is no split between memory and temporarily or between the ontic and the ontological.

The argument that Indigenous cultural ideology is ontological recasts Heidegger’s notion of ontology. Heidegger’s notion of ontology still operates within Western modes of thinking. Indigenous Australian ontology refuses the binaries of Western thought. Furthermore it permits a reconceptualization of ontology that involves a shift from “light” to “land” as an underpinning metaphor for ideology. Whilst Heidegger’s thinking on ontology is underpinned by the metaphors of light and revelation, the Indigenous imaginary is grounded in materiality and the “real” – in ontology. The immaterial is materially constituted by the real material conditions of existence, where the immaterial and the metaphysical become a reality through the practice of creating cultural productions, through the imaginary or immater-

ial dimensions of art-making. This immateriality in Indigenous cultural ideology is manifest in the real existence of land. “Real immateriality” offers an alternative way of defining the concept of “ground” that Western philosophical discourse seeks and which underpins artistic practices and materiality. This has numerous and fundamental ramifications for cultural identity and the discourse of artistic production by highlighting how the relationship of the everyday world and cultural productions inform and give basis to Indigenous cultural identity. The ontological relationship that people have to country is vital in Indigenous cultural practices where the casual relationship between the referent and the sign is reciprocal:

Art provides a sacred charter to the land and producing art is one of the conditions of existence. It keeps the past alive and maintains its relevance to the present. (Morphy 1998: 5)

This relationship is demonstrated in the works of both Alyawarr and Eastern Anmatyerr artists Kathleen and Margaret Petyarre and Warmun artist Rover Thomas. Petyarre’s work focuses on her country as a foundation to her identity and she states:

That’s the dreaming. The Dreaming journeys were made for real people, for Aboriginal people. And now we really follow that Dreaming – now Aboriginal people keep following it. (Willstead 2007: 215)

Petyarre’s work focuses on the relationship of the Dreaming and the location of ancestors on her country. Her work also points out important information such as the location of food, animals and water. These things and the location of important sites and tracks are vital in the foundation of “belonging to country” and her work demonstrates a reality founded in real country which is the ontological premise of Indigenous cultural ideology. It is the reciprocal relationship that she has with “country” that informs her work, where country is not passive in the sense of been



17. Kathleen Petyarre and Margaret Petyarre, *Emu, Thorny Devil, Yam and Dogwood Seed Dreamings at Albalcker and Akweranty*, 1989, batik on silk, 592 x 224cm

taken for granted. For example, in the work *Emu, Thorny Devil, Yam and Dogwood Seed Dreamings at Alhalker and Akweranty*, batik on silk, 1989, Petyarre demonstrates her own and her families relationship to country. This specific work outlines a women's ceremony from Alhalker country.

The complexity of this work is the relationship of this ceremony to ancestors, animals, dreamings and country itself. This country is Petyarre's grandfathers' country, where various marks in the work signify emu tracks, tracks of dancing women, dingoes, yams, yam leaves, haistring skirts, dogwood seeds, women's body paint designs and the dreaming journey of the mountain devil lizards. All aspects of this work extend from country, not only the images, but the materials used in its production, for example, the ochres. It is this constant reference to all things associated with country that creates this work, and it is the artist's role to continue to bring this into existence into material reality. Country in this instance plays an active role in the materialization of the work.

This relationship can also be observed in the work of Rover Thomas which draws on recollections of the Kukaja/Wangkanjunga world of the Gibson Desert region. His work directly articulates country in the sense of a performative mapping. For example, *Landscapes at Kalumpiwarra, Yalmanta, and Ngulalintji*, natural pigments and gum on board, 1984. This work shows a person's spirit journey where they cannot decide which direction to go. This work is a direct mapping of country, it performs country as it is the artist's role to articulate, in this case, Thomas' country. The top band of yellow ochre demonstrates Blackfellow Creek, the large red ochre shape at the top is the rock at Ngarkarlin, the small white shape on the left is the grass at Yalmanta, the repeated shapes in orange show a wooded area of trees and the lower yellow ochre shape shows Ashburton Hill with a creek running through it. His work demonstrates a *methexis*, a performative rhythm that is part of the real surface of country and this performativity is real as it extends from the real world. This real world is immaterial and material concurrently and it is Thomas' causal relationship to this world that is evident in his work. The work *Landscapes at Kalumpiwarra, Yalmanta, and Ngulalintji* portrays this country in a material sense from the immaterial world. This work is a performative mapping, not a landscape. If anything, we could term it a countryscape. The importance of the artist to articulate the relationship to country is vital, as it is our role to bring the world into existence through our interaction with it. Art is the result of this reciprocal relationship. This demonstrates an Indigenous conception of ideology as being our "real" relationship to existence, not one based on the imaginary. This further demonstrates the purpose of Indigenous cultural practices of relaying the ontological premise of country into the material work. The social ramifications of this are vital to the ongoing evolution of Indigenous cultural practices as they are intrinsically linked to the real world and refuse to work within the binary of the material and the metaphysical as it both at the same time. This is vital to a reconfiguration to ideology in Indigenous terms. Country is the foundation of this ideology and it is the causal relationship that we have to country which is manifest in cultural productions that becomes the role of cultural practices. The causal relationship is not based on an imaginary relationship to real existence as in

this instance real existence is evident in real country. The aspect of country in relationship to cultural productions demonstrates the ontological premise of Indigenous cultural practices and it is the materialization of the metaphysical aspect of country in these practices that refuse the split between the ontic and the ontological, the material and the immaterial. Carter refers to this as *methexis* and it is the ongoing *methexical* relationship to country that is brought to the fore in these practices. This is evident in the works of Petyarre and Thomas and will be further analyzed in the work of Badger Bates.

This grounded materiality is evident in both the language and material practices of Indigenous Australians. However, it is important to recognize that the status of “art” in Indigenous culture varies from Western understandings. Among the various languages in Aboriginal Australia, the word “art” is not at all common. For example, in Southern Queensland, Murri country, the word “Mulka” translates as painting. Also, in Wergaia language of the Horsham district of Victoria, “Yuka” translates as “to paint”. These two examples illustrate how language relating to art denotes action, “to paint”. Art in the Western sense has become a stagnant term, whereas in Indigenous Ideology, cultural productions are active and inclusive for the whole of society. The visual arts in Indigenous culture have a language that is always associated, and parallel to, other languages, stories, dance, song, and ritual. In an Aboriginal view, the meaning of the visual is inscribed in one’s awareness only when it is absorbed through languages that affect both mind and body. Languages in this instance are not only the spoken oral tradition but the visual languages created in cultural productions.

The dynamic of ritual and practice within Indigenous culture is vital to overcome the Western experience of amnesia and an imaginary relationship to existence. Indigenous cultural ideology is understood as an “actual” and not a “potential” trying to build over the void of nihilism. Indigenous culture has its ideology grounded both materially and metaphysically in “country” itself, which overcomes the notion of the void experienced in Western discourses.

Immaterial Land, Memory and Practice

Indigenous art is a process that reveals memory and is a practice that preserves culture. This line of inquiry examines and formulates the potential of art as a practice that preserves culture. Indigenous “cultural ideology” links ideology, identity and art in a cohesive ontology that creates a different “value” upon our existing understanding of “art”.

The important point here, as shown in my reference to Petyarre and Thomas, is that an Indigenous framework is connected with the “real” world. Both Petyarre’s and Thomas’ works are grounded in the everyday world. The works *Emu*, *Thorny Devil*, *Yam and Dogwood Seed Dreamings at Althalker and Akweranty* and *Landscapes at Kalumpinvarra, Yalmanta, and Ngulalintji* extend out of the artists’ relationship to country. This relationship is based on something real existing in both the world and the artists’ world, as they are inseparable. Aboriginal “cultural ideology” is the “real” world. This Indigenous framework also proposes different layers of ontology, from the physical to the spiritual. This is demonstrated in the work of Petyarre and Thom-

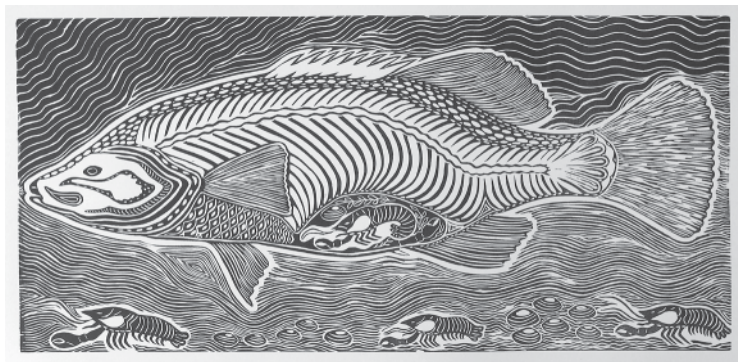
as and is the foundation to Indigenous cultural practices. However, Western thought confines this as mere “myth”. A Western conception of culture is one that has a dramatic impact on contemporary Australia. Edward Said states the following:

Now the trouble with this idea of culture is that it entails not only venerating one’s one culture but also thinking of it as somehow divorced from, because transcending, the everyday world. (Said 1993: xiii)

Much of contemporary art practice is divorced from the everyday or real world. For example, postmodern art may at times involve social critiques but it is primarily built on a negation of any grounded truths. I will now turn to Indigenous cultural productions to illustrate the ideas presented here and to further demonstrate my argument.

A Visit to Bates’ country and ancestral lands provides the opportunity to view ancient rock engravings that are around 20,000 years old. This type of etching into rock is what Bates has transformed into mark making with the process of lino printing. The materiality of Badger’s work has direct lineage to an ancient past and practice. The ontological premise of his work lies in its reality, as his work is directly derived from his “country”, the basis of Indigenous cultural ideology. The work *Parntu Thayilana Wiitbi – Cod Eating Yabbies*, lino print, 2004 is an example of cultural productions that encompass the metaphysical, the material and the ontological. This lino print illustrates a number of points raised. Bates has constructed this work from memory, a real experience. The cod is part of his real experience as he has opened this fish numerous times creating a continual ontological relationship. This is evidenced by the relationship that Paakantji have with the Darling River:

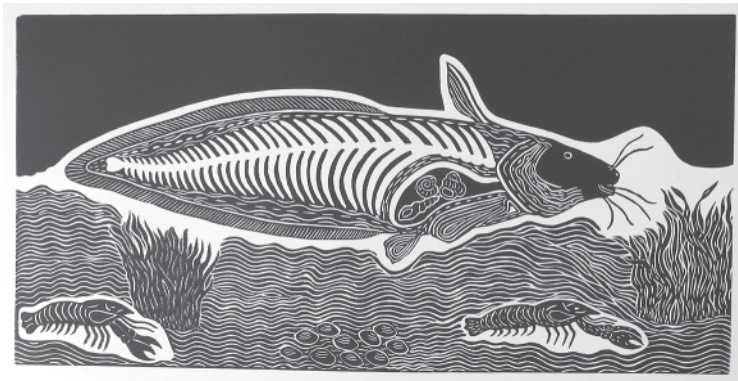
When you open the swim bladder of the Murray Cod, the shape of its skin shows the place where this fish was born, under the same shaped tree where it was born. (Bates: 2008)



18. Badger Bates, *Parntu Thayilana Wiitbi: Cod Eating Yabbies*, 2004, lino print, 37 x 79cm

It is also believed that people who experience this part of the cod are also born under the same tree that the shape forms. There are various stories such as this that are passed through the generations that illustrate different relationships between people creating skin groups. This relationship illustrates the connection between people and their country and further, the relationship between the real physical world and the metaphysical. These both have a causal effect on real land, real people and real country and it is this causal relationship that is demonstrated in the indexical images produced by Bates. This ontology is shown in Bates' work. In Bolt's terms, this image has real material affects, as we can see its construction, how the artist has performed this print and through its indexicality, the cod has real material effects and becomes the cod itself. This is the crucial part of a new materiality explained in the following. Bates has made this work from memory, and has presented the cod in an iconic and symbolic form. However, the work becomes the cod itself through having a real material effect on the viewer by its constant movement, created by linear patterns that have been created within the ancient rock etchings of his country. There is nothing hidden or ephemeral in this work. It does not claim what it is not, but rather this work simply relates to the real world conveying a story that is embedded in an ancient culture. This is the sophistication in his work: the ability to borrow from one tradition and to communicate through another, one through the work and one through experiences as real material experience. The viewer re-experiences this materiality even as one views it as representation.

This ability is also illustrated by the work *No More Catfish*, lino print, 2004. This work has a direct relationship to Badger's experience in watching catfish as a youth. By utilizing mark making techniques Badger brings this work to life and also returns us to question why there is no more catfish. In his country and in this work, Badger is illustrating a direct connection to his surroundings and further illustrates the performative power of the image. In discussing his work, Badger states: "I look at the fish, I respect it, I eat it and bring it back to life in art" (Bates: 2008).



19. Badger Bates, *No More Catfish*, 2004, lino print, 37 x 75cm

It is this performative action that brings to reality the indexical quality of his work. The statement above is an action; it is movement, which is also demonstrated pictorially within the work itself. It is this dynamic of experiencing this fish that enables Badger to bring it to life within the work. This is an ancient performative tradition. There are many parallels existing in his work. Firstly, he is relating memory, a vital factor in Indigenous culture, to present experience through cultural production and repetition. This memory is embedded in country and incites a transformation in the thought and awareness of viewers. The physical work also moves constantly by the placing of marks within the pictorial space. These layers also illustrate the Indigenous perception of all existence in congruent layers from metaphysical to physical dimensions. The lineage of cultural productions in an Indigenous view is also important as images have been renewed in present experience from ancient traditions which are premised in "country".

By engaging with and discussing his work, one forms a relationship of immediacy with Bates. This immediacy encourages listeners not only to memorize but also to visualize at each stage. This further illustrates the performative power of his images, relating ancient traditions in both a metaphysical and material sense in contemporary form. His images are extracted from the foundation of his country and are further demonstrated by his origins: Paakantji, meaning "River People".

His work demonstrates an Indigenous "cultural ideology", an ideology that is created through the real conditions of existence, and not by our imaginary relationship with them, as is the case in the Western construct of ideology and its relationship to a separate culture. This raises a number of questions: Is it possible to incorporate ideas from Indigenous culture within a system of Western thought in ways that would contribute to the formation of a more cohesive cultural identity? By understanding a new view of ideology within a Western framework and the importance of art in relation to culture, we view art in the instance of Indigenous cultural ideology as producing and forging a more coherent Australian ideology and identity. If the viewer of the work shares in the experience there is a re-living of the experience by the artist and a transfer between artist and viewer. This can extend to individuals beyond culture and this can be seen as the emergence of an empathetic relationship made through art. This aspect of Indigenous culture is vital to refiguring the function of art as there is no false consciousness within this framework and art has a different function within the whole broader spectrum of cultural practices. It is in this sense of ideology and culture and where there is no false consciousness that "reality" and cultural ideology can be transferred between cultures.

To further analyze Indigenous cultural productions, we have to look at the context in which they are produced. Ian McLean's review of Smith's *Transformations of Australian Art* (2002), suggests that:

The legacy of Aboriginal art might, ironically, be the transformation of a largely nationalist cultural agenda to a post national one that repositions the global and local ways we are yet to see. (McLean 1998: 31)

In order to achieve the above we have to reconfigure our understanding of art and Indigenous cultural productions. For example, Jean-Francois Lyotard, in *Driftworks*, discusses art as a deconstructing force and in relation to reconstituting figural forms, he states:

Primitive culture cannot be invented; it is *given* by definition, which means that, even in a revolutionized and revolutionary society, man's type of coexistence could not have primitive society as its model. The function of a revolutionary art is thus very precisely determined. (Lyotard 1984: 72)

Instead of assigning an ascribed value to a cultures practice, the Western world needs to acknowledge the importance of how practice plays a vital role in marking the relationship between people and expressing identity. In the context of Australia, contemporary contextualizing of Aboriginal art and culture has occurred in a confined terrain that has seen Aboriginal ideology and art practice devalued. Lyotard's notion of not being able to use so called "primitive" culture's ideology in revolution is created through the value given to oppressed cultures that their ideology is based on mythology. However maybe our collective identity lies before us. In looking at Indigenous art and ideology, we can reconceptualize an ontological relationship to identity, culture and the real. In reference to Aboriginal art, Maurice O'Riordan suggests that:

This movement signals a broad, ongoing political consciousness with radical implications for artistic practice and cultural sustainability that have yet to be fully appreciated. (O'Riordan 2003: 33)

Indigenous cultural productions not only demonstrate a practice that relies on the foundations of what Carter states as "ground" but they are founded in the real existence of country and not by an ephemeral relationship to an ideology needing to build over the top of nothingness, or in search of an elusive foundation. They demonstrate an alternate way of defining ideology by opening an understanding of art not as transgression, but as an articulation of the notion of a cohesive cultural identity. Through its foundation in "country" Indigenous cultural productions present a different basis of ideology which is predicated on linking the material and the immaterial. The foundations of ideology in a Western sense rely on modes of light and empirical experience. An Indigenous Australian conception of culture and life encompasses the material with the metaphysical and recasts this binary relationship into a complete whole where the immaterial is materially constituted.

I would like to acknowledge that this chapter was completed on the ancestral country of the Wiradjuri people. I pay my respects to elders of the Wiradjuri both past and present. I also extend my respects to Guunaary, goanna totem of the Wiradjuri peoples.

The acknowledgement demonstrates the living action of movement, a reciprocal dynamic between people and country. It is this act that demonstrates a culture that is far from static. This action produces a real effect and, like art and cultural practices, an acknowledgement demonstrates an action of our role, which is to constantly bring to the foreground the basis and premise of our culture: our “country”.

THE PRIMACY OF MOVEMENT:
 VARIATION, INTERMEDIALITY
 AND BIOPOLITICS IN
 TERO SAARINEN'S *HUNT*
 Milla Tiainen and Jussi Parikka

Notions of movement as qualitative variation in time have attained at least a double-edged urgency *vis-a-vis* the contemporary stage of capitalism and the info-technological networks of power. Key writers scrutinizing these developments (Hardt and Negri 2009; cf. Berardi 2009) have argued for and theorized the vital importance of the constant potential of living bodies to connect, perceive and act with a difference for political existence and resistance. In their broad contours, these insights could be traced back to a view of life as an unforeseeable delay between stimulus and response. Henri Bergson (1988) was among the famous earlier advocates of this conception. It seems to be exactly the said kind of delay that certain recent mappings of (bio)politics have put forward as the fundamental resource for the ways in which instances of (human) life may overcome their positioning in existing frameworks of subjection and power or control.

And yet, capacities for movement and difference, for making new connections that enhance new inventions (of ideas, experiential contents and associated avenues for value acquisition) are simultaneously celebrated as the cornerstones of the contemporary labor markets of late capitalism. Indeed, such abilities are consistently attached to the figure of the “creative individual” as it is promoted within these frameworks of current commodity and service economy. There thus exists an uncomfortable overlap between recent conceptual developments in radical cultural theory and in discourses surrounding contemporary forms of production. Specific strands of the first field strive to (re)introduce the body-as-movement and the related concept of *pure difference*, which we will address below, into accounts of culture, agency and politics. At the same time, the reconfigurations of productivity and work aligned with post-Fordism draw on apparently very similar terms to supposedly very different effects. It hence becomes critical to attend to the divergent settings and mutually conflicting political agendas within which the ideas of movement and variation may today figure.

This raises the question: How can we interrogate the theme of movement-as-continuous-difference while endorsing a dissonance, rather than a “love affair”, between cutting edge artistic and theoretical strivings and the timely tactics of capitalism?

This is a challenge that our chapter for its part wishes to take on. Expressed firstly in conceptual terms, the analysis to come taps into a problematic that is encapsulated by the assemblage of the following notions: temporality as differentiating continuity or *duration*; variation as the emergence of the new; and the biopolitical (techno-capitalist) interventions into the body as a temporal – and therefore necessarily varying – moving, feeling and thinking entity. Secondly, and more significantly in the context of this book, we explore these notions in how they both get summoned by and help give sense to an instance of contemporary performative arts: the dance piece *Hunt*, choreographed and performed by Finnish dancer Tero Saarinen and premiered in 2002 at the Venice Biennale. Our focus on Saarinen’s choreography-performance is motivated above all by the ways that it, through what we shall call its *movement techniques* and the recurrent connection of these to multimedia/lightning elements that were designed by feminist new media artist Marita Liulia along with Mikki Kunttu, thematizes temporality and variation in relation to corporeality, perception and experience with distinct acuteness. Any dance works of course involve the body – or bodies – in movement whilst accentuating, playing on and stylizing their temporal unfolding by a wide range of means. Yet this artistic example seems effectively to concentrate and turn on the, as we will argue, near imperceptible but ontologically crucial *moments of transition* that (all) moving, receptive, interactive bodies undergo.

At the same time, *Hunt* also stages striking tensions between this difference or self-excess movement puts at the heart of (sentient) bodies and attempts to contain or channel their variability. More precisely, the interplays of *Hunt*’s components from dance to image and lighting patterns gesture toward broad but still particular enough paradigms of bringing that containment about, whether this means pinning bodies to positions – in terms of their physical and social existence – or modulating (restraining, consolidating, altering) their experiential capabilities via their immersion in specifically designed and executed sensuous and temporal environments. Now in contrast to suppression or straightforward enclosure, these tropes of confinement feature subtler forms of scrutiny and corporeal/mental manoeuvring (with respect to the attunement of feelings, recollection, and a sense of capacity for example). Further, while their beginnings can certainly be located in earlier stages of capitalism and multi-sensory experience shaped by modern media technologies – as we will elaborate below – it is according to many thinkers the current realms of capitalist production and media culture that, in close conjunction with one another, increasingly pronouncedly and sophisticatedly rely on them (Deleuze 1992; Lazzarato 2006, 2008).¹ Tellingly, the tensions that *Hunt* creates between a body’s “transitional immediacy” (Massumi 2002a: 5) and its containment within positions or technologically saturated milieus for experience (productivity and consumption?) occur at the interface of Saarinen’s moving-perceiving body and the multimedia effects.

Our working assumption is, then, that this artistic project ingeniously resonates with the prevalence of variation and attempts at its management within much wider

contemporary practices of culture and economy. Most importantly, it resonates with variation's expressly ambivalent, or as argued above dissonant, relationship to those practices: dissonant in the sense that it provides constantly a potential source for alternative (socio-culturally transformative, non-profit driven) modes of production *as well as* for capitalist production processes proper. To use the term "resonate" is not to contend *Hunt* would just illustrate these themes as if standing in a re-presentational relation to preexisting, fully formed problems. We rather want to offer that there are connective and dialogic – or, as Gilles Deleuze and Félix Guattari would have it, rhythmic² – threads to be discerned and drawn between the heterogeneous and highly dynamic worlds of artistic, here particularly "dancerly" expressions, media technologies, cultural industries, capitalist logic, and ongoing academic debates around these issues. It is such threads, instead of any simple correspondences between disparate yet inter-relational registers of reality (cf. Massumi 2002b: xiv), that we embark on mapping in what follows.

To arrange these mapping endeavors, we have divided the chapter into three sections. Following an overture where key points about putting difference and not substance first in considerations of the body are explained, the sections, called respectively "Variation; or, Movement and Affect", "Intermediality", and "Biopolitics", each have their conceptual gravitational core and related lines of inquiry. Yet these lines constitute a "fragmentary" totality (Deleuze & Guattari 1994: 23): they pursue interdependent facets of the article's overall problematic that concerns the historically changing biopolitical tropes or environments involving bodies-in-becoming. Our approach could also be described as one of exemplification. What the term implies here is an attempt to approach *Hunt* as a singularity that belongs to itself in the sense of not being subsumed by supposedly general ideas (like "body as movement" or "affective economy") of which it would only serve as a particular instantiation. Eventually, however, the "exemplary" force of *Hunt*, as intimated above, lies in the ways its very singularity is connect-able to other components and registers spanning from media-cultural and socioeconomic engagements of the body to theoretical concepts. (Massumi 2002a: 17–18; Massumi 1993: 96–98).

Through notions of variation and affect, the first of the said three chapter sections investigates (the primacy of) movement in *Hunt*. Linked to this, we will present the claim that a new materialist inquiry into bodies and/in the arts is crucially characterized by an acknowledgment and meditation of the circuitous, mutually impacting relations between corporeality and incorporeality (or the actual and the virtual). The second section concentrates on the interfaces between the dancing body and multimedia/light elements in *Hunt* in connection with several modern media technologies of movement capture and perception. The final section draws out and summarizes the associations between this dance piece and historically transforming, albeit also overlapping, biopolitical agendas.

An Overture: Bodies, Ecosophy, and Pure Difference

Saarinen's choreography comprises a thirty-five minute long solo performance for his own dancing body. This performance is sonically accompanied by Igor Stravinsky's

emblematically modernist ballet music *The Rite of Spring* (1913). Entertaining over the course of the piece several types of relation with the body that dances, the multimedia images designed by Liulia and light effects provided by Kunttu add key visual and temporal dimensions to the composition of *Hunt*. The images basically inform the piece in two modes. They appear as repetitive, continuously and steadily layering series and as abruptly shifting, hence discontinuous, collages. Of the lighting effects the most relevant for our discussion are the successions of strobe flashes, occurring a couple of times in the late and final moments of the performance respectively, that startlingly manipulate its amenability to vision.

As will be shown, each of these modes of visibility-visibility significantly orients our inquiries into variation, intermediality and biopolitics in *Hunt*. In all, we aim to address the co-unfolding motions of dance and technical image(s) in the piece as ecosophical objects or processes (Guattari 1995: 120–135) which yield important insights into the field of questioning laid out in the preceding section. Félix Guattari's concept of *ecosophy* is indeed an apt interlocutor here insofar as the syntheses of the moving body, (media) technologies, and technologically imbued modifications of experience *Hunt* brings forth make up an ensemble that by condensed and peculiarly artistic means arguably points to the kinds of elements of which contemporary systems of interaction between environment – both physical and cultural – social relations and individual subjectivity/variation are comprised.

More concretely, the piece assembles its expressions out of movements, factors and encounters between them that imply the wider media-technological, associated scientific and capitalist contexts where bodies live, get managed, and vary in the modern age in general, but especially since the rise and intensification of post-Fordist tendencies as well as digital media and networks. *Hunt* does not miniaturize these developments of the labor markets and everyday life. It innovatively translates or mutates aspects of them into an artistic vision of how the variation of bodies – and thus the infinite in-process nature of their very materiality – occurs at the intersections of those bodies, particular techno-social milieus and configurations of relation. For us, the vision of *Hunt* concerns first and foremost the sorts of complex inter-relational systems that might be endemic to the present age and modes of existence (the study of which is precisely the impetus of Guattari's ecosophical approach). Nonetheless, we will also comment on a certain genealogical awareness that seems to underlie *Hunt*'s choreography and body – technical media interfaces. This awareness pertains to the historically developing modalities of biopolitical power (in brief, of the organization, employment and modulation of life) and their relationship to distinctive paradigms of institutional, economic and techno-scientific governance. It is the ecosophical perspective and, within it, the question about biopower or biocontrol, that embodies the political dimension of our new materialist cultural analysis of the arts predicated on the notion of the primacy of movement.

But this political emphasis does not come without an ontology. Let us therefore clarify the key premises of the ontology of movement – or of becoming and identity-defying difference – that our investigations of *Hunt* expand on. Our understandings in this area draw prominently from Deleuze and Guattari's philosophical-on-

tological project. This is an ontology within which movement does not designate a mere interim between two states of a thing that would in its essential respects remain self-identical. Movement is not a secondary, derivative attribute of identities. Rather than issuing from beings that would retain a basic self-sameness, it concerns “the very dynamism of change” (Stagoll 2005: 21) or “the being of becoming” (Deleuze 1994: 41). This according to Deleuze and Guattari constitutes the primary existential layer of entities and configurations, whether these are cultural or natural, individual or collective. A body in motion, then, does not so much make moves as exist by moving. It *is* all the variations – temporal, interactive and broadly existential – that its movements without definite starting and end points bring about. Inasmuch as one wishes to use that term, the identity of a body (and of an embodied mind) is composed of their continuity across these differentiations.

To be sure, this idea of *pure difference* – a difference that does not emanate from finalized entities but keeps on differentiating from itself (while being perceptible as the changing states and qualities of entities) – has by now been discussed with great insight by Deleuze and Guattari “infected” scholars. As far as ontological reconsiderations of the body are concerned, those of Brian Massumi (2002a, 2007) belong amongst the most pertinent for our inquiry. They consistently seek to show how cultural theory, insofar as it wishes to rethink bodies beyond the assumption that they are always already mediated by signification, should acknowledge that what is specific to bodies *qua* bodies partakes of two different, albeit interlocking, modes of reality. On the one hand, bodies do exist as relatively permanent matter. They consist of distinguishable parts that occupy space. Always accompanying this mode of spatial concreteness, however, is that of movement. Bodies (human and others) move without a stop, as movements are an inescapable trait of living bodies. These movements, moreover, comprehend so many reactions to and actions upon the material and psychic environment. It is hence the bodies’ manners of sensing, participating and being able (or unable) that co-constitutively vary with their mobility under way (Massumi 2002a: 4, 15). Most important, it is this second mode that stands for the *abstractness* of bodies in that all their potential re/actions can never be given in an individual here and now. The futurity of these capacities remains to be determined. They are in a ceaseless process of interactive metamorphosis: becoming. What Massumi therefore highlights, is the existence of bodies in two mutually disjunctive but still inseparable dimensions. To grasp them properly, we must grant that bodies figure both as already differentiated things (in terms of their so far unfolded states and materiality) and as differentiating becomings; self-differing instances of living open to a yet non-present otherwise.

In addition to the terms differentiation/differentiation, the above premises could also be couched for the purposes of our chapter with the concepts of actual/virtual and extensive/intensive. We will return to these below. Suffice it to say for now that the views on the endless “in transit” nature of bodies after Deleuze-Guattari and related ontological stances have recently inspired a series of investigations into dance. To mention some whose discoveries approximate our aims, José Gil argues in his essay “Paradoxical Body” (2006) for a specific *space of the body* in which the

dancing dancer evolves while that space itself is carved out by his or her movements. This space differs “from objective space” (Gil 2006: 21) in that it cannot be captured within the quantitative parameters of standardized systems of measurement. It rather unfolds along with the transformations of the energy flow of moving bodies, differentiating in correspondence with each of these transformations (Gil 2006: 27). The space of the body thereby collapses the distinction between a fixed, objectively calculable space and the body as a bounded interiority that moves in that space. It establishes a relation of reversibility between the exterior and the interior. The external space folds *into* the dancer’s body in terms of sensation and perception.³ At the same time, the energy shifts of her movements modify immanently the intensive, energetic-rhythmic, constitution of their exteriority. To paraphrase Gil (2006: 23–24), the space of the body involves a moving body and its surroundings as co-habitants in a joint becoming.

Gil furthermore claims that the dancing body multiplies (to itself and to the co-perceivers). Not only is each of its actual motions accompanied by virtual images of its other movements that either have passed or lie ahead. The registration of these motions is also inevitably multi-sensorial. Besides being perceived (and recollected, imagined) visually, they are listened, sensed, touched. (Gil 2006: 24.)

A number of Deleuze-Guattarian concepts have consequently been invoked for rethinking the nature, political potential, and study of dance.⁴ This might, however, raise the question why produce another mapping of dance with these conceptual tools? Might not the “sympathies” between the tools and dance as an art form *par excellence* of bodily transitions start to become too readily observable? After all, inasmuch as one wishes to explore the acuity of Deleuze-Guattari’s insight (1987: 281–282) that movement is all but imperceptible *per se* while it simultaneously cannot but feature as a “background perception” to the differentiations it engenders, then surely dance, and other time-based (media) arts, provide evident testing grounds for such an exploration.

However, we would argue that Saarinen’s *Hunt* contributes new aspects to the presently burgeoning cultural theoretical investigations of movement and pure difference. For one, it deliberately seems to play with the body’s twin-character. Indeed, it is as if many of Saarinen’s movements were urging the spectators to realize that although we may perceive body motion through the body’s consecutive spatial displacements, it is in the connective and vaguely but insistently felt *in-between* of those “results” where bodies as movement, and change, really take place – or time. Second, by manoeuvring the ways Saarinen’s body can be perceived the images and lights by Liulia and Kunttu foreground the ontological difference at the heart of the body. As we shall demonstrate later, they engage both the body’s coinciding with its temporal alterations and the ways it partakes of or is containable within the domain of spatial positionality, *extension*, for varied scientific, cultural and economic purposes.

I Variation or Movement and Affect

If techniques are understood in the manner proposed by Erin Manning after Gilbert Simondon, as technologies “of emergence [...] through which new complex systems

are composed” (Manning 2009: 71), Saarinen’s *Hunt* involves a set of “dancerly” techniques through which the body-as-variation emerges and displays impact. To put it more broadly, there are diverse ways in which *Hunt* “moves the relation” (Manning 2009: 34–35) – that between the dancer and his surroundings (including the stage space, the multimedia and lighting variables and the audience) as well as that of the dancing body to itself. We foreground two ways Saarinen employs to exemplify this understanding of technique.

To help readers engage with these techniques, let us however first outline what kind of overall space of gestures, velocities and relations – or space of the body – they partake in. In order to enhance appreciation of the techniques’ functioning and their powers to affect the space, audience perception as well as theoretical thought it is useful, in other words, to provide an overview of *Hunt*. For purposes of clarification, we can divide the piece into three sections whilst at the same time acknowledging the migration of the typical qualitative ingredients of its expressive style between these, which is exactly what happens with the movement techniques we will look into.⁵ Roughly speaking, *Hunt* commences with a section that spans the first fifteen minutes of its thirty-five-minute length. In terms of the dancing body, this section is mostly characterized by emphatically prolonged and slow movements (that actually provide the initial articulations of the first technique of our upcoming analysis). More agitated and spatio-temporally agile movements – leaps, twitches of the torso and hands, hesitating reachings out – do nevertheless enter the section around 12 minutes into the piece. During this section, the stage baths in dim, yet pervasively spread out golden yellow light. Naked from waist up, Saarinen is clad in a thin, white, loosely draped cloth reaching down to his legs. Particularly distinctive for this section is the frequently evident obliviousness or apparent lack of relation and sensitivity of Saarinen’s dancing to the temporal, dynamic and timbral events of the music. Whether accompanied by the subtle introductory melody played in legato by high bassoon or the shorter, more eruptive figures and rapid shifts between instrument groups (winds, strings) that begin to feature in the piece after its opening minutes, Saarinen’s body seems to be confined within a temporal evolution of its own.

The first more global expressive change within *Hunt* occurs by the time its opening fifteen minutes are completed, relatively soon after the transition of the music into a sharply noticeable, repetitive and dissonant, irregularly accentuated string motif/chord. A spotlight whose harsh glow focuses on Saarinen instantaneously replaces the hazy lighting. Saarinen’s body simultaneously freezes into a posture of tense expectancy: perhaps of the intuited, albeit for the moment still incipient and undefined, arrival of a transformative encounter. What then comes into view is a slowly appearing and moving configuration of projected image that gradually descends from above the stage toward Saarinen. Once the configuration reaches his body, it is possible to see that the images comprising it draw their for now circular routes and shapes on another piece of clothing; a wide and stiffly geometric skirt into which Saarinen, as if hypnotized, slips himself.

After a short anticipatory period the stage suddenly darkens and, interconnecting with increasingly fierce, dynamically charged musical gestures, the former movement

patterns of the multimedia material transform into a collage of disjointedly and violently relocating images that progressively take over Saarinen's entire body as their "screen", performing *their* dance upon its surface. While this synthesis comes to a provisional close after a few minutes and Saarinen performs his body's exiting of that relation with disoriented torso and whole body gestures, his dancing nonetheless continues to echo or re-mediate the distinct temporal/relational features of the encounter. This is evident through the increased figuring of discontinuous and convulsive movements within his performance although glimpses of a similar movement style do appear during the first section of *Hunt*, too. Lastly, in addition to the forceful returns of the interface between the sensing/feeling/thinking body and multimedia assemblages, it is the arrival and effects of the strobe flashes over the four closing minutes of the piece that mark the final section within it. Interjecting enigmatic instants of invisibility into Saarinen's at that stage large-scale movements and high-reaching leaps, it is these pulsating alternations between light and darkness that interlock with and affect *Hunt's* dancing body and performative space up until their very last variations.

To turn now, against the above background, to the two particular movement techniques we want to address, technique number one concerns, as tentatively indicated, sequences of wide and languish body motions that Saarinen creates especially before the multimedia components enter *Hunt*. While returning at later junctures, these motions dominate especially the first section of *Hunt*. Bending his torso backwards, moving his arms in curves and turning upon himself, Saarinen appears to be caught in a curious mode of slow-motion. This is all the more accentuated due to the drastically different temporal shapes of the music – its rapid and rhythmically diverse wind and brass instrument gestures – at many points of this phase. These ways of moving seem emblematic of Gil's space of the body. Namely, they keenly foreground how a moving body can modify its associated milieu beyond the latter's "objective", measurable properties; or, how the space of the body depends upon the time a movement takes in outlining it (Gil 2006: 27). From the vantage of energy transformations the viscous slowness of Saarinen's motions *dilates* the environing space. Gil in effect uses this very term when discussing Vaslav Nijinsky, the first dancer-choreographer of *The Rite of Spring*, and more specifically the latter's famous over-articulated movements that gave the impression he had "all the time in the world" for actualizing them. That extended temporality of movement moving created, Gil argues, a "dilation-effect" with regard to the space of the body born along with Nijinsky's dancing (Gil 2006: 27). However, it is not just because of this potential stylistic connection internal to the field of modern dance that Saarinen's slow motions warrant scrutiny here. Rather, they resonate with the core ideas of a new materialist theory of bodies we are advocating. This holds for the priority assigned to temporality as variation over variants, and for the notion that bodies in movement are as ineluctably incorporeal (*virtual*) as they are corporeal (*actual*), as much *intensive* as they are *extensive*.

Contrary to commonsense perception it is not variants – the successive displacements in space of a supposedly preformed body – that movement consist of. Instead, it is a continuum of variation. It is a passing of time, whereby difference differentiates



20. Still from *Hunt*, 2002. © Tero Saarinen company

itself and the body does not precede but immanently coincides with its transitions in space-time. It is exactly these characteristics of movement that the timings of space brought about by Saarinen's dance moves highlight. Put otherwise, the specific pace of the motions that compose the first technique of *Hunt* emphasizes two aspects pertaining to the primacy of movement. First, these unfoldings of energy lend support to the crucial claim of the ontology of becoming, stating that "real transformation and change seem to require that the distinction between movement (the process) and moving (the agent or patient) be abandoned" (Boundas 1996: 84). Indeed, when following Saarinen's motions taking shape, we become sentient of how the movements of this dancing body are not subsidiary to but constitutive of it. Saarinen's body exists *as* variation before our eyes. The slowness of its motions renders tangible how its continuity (or coinciding) with itself is a continuity with transformation. All the time this body is already becoming other than it was. The "other-than" in question is the incorporeal, virtual dimension of bodies: their potential to vary.

The second and inherently related aspect Saarinen's dilated motions flag is the nature of movement as a dynamic, *indivisible* unity (Massumi 2002a: 6). Similarly as instances of moving are not decomposable into an *a priori* body and the motions it then performs, they are not decomposable, either, into a series of fixed points in space that the body allegedly inhabits along its "kinesthetic journeys". Assuming the possibility of such decomposition would reduce movement to extension: to that dimension of reality where things are arrested in position and thus (momentarily) confined to sheer actuality. This assumption would overlook the very manner in which the moving body does *not* reside in a state(s) of halted actuality (how would it then jump from one of these states to the next?), but differ from itself in a con-

tinuous passage as its virtuality – potential for movement and relations – *actualizes* as long as the motion in question endures. It is this processuality which, when occurring, cannot be pinned down to a succession of points in space that comprises the intensive dimension of movement. Intensity is, moreover, akin here to duration. Temporality understood as duration signifies precisely time as continuous differentiation or multiplicity where discrete instants do not follow and replace one another but the present *becomes* the past whilst also flowing into the other direction: toward the undetermined future (for example, Boundas 1995: 92–93). Now the temporally “elongated” motions of Saarinen’s dancing render particularly evident this self-divergence of time, too. They make felt how it is due to duration that a moving body is always already outside of itself. Its present keeps on passing into the past (of the present it just was) while simultaneously opening up to what that body will become.

Hence the “new materialist” distinctions between actual and virtual and extension and intensity with which *Hunt* singularly resonates are not opposed to materialism – except to a Newtonian materialism of inert bodies. Rather, they propose a reconfigured materiality of duration and variation as wonderfully summed up by Eric Alliez (2004a) in the wake of Bergson: “pure duration is what differs from itself in coexistence with itself through a non-chronological time, since what differs from itself is *immediately* a coexistence of the past with the present, the *contemporaneous* unity of being with becoming, substance with object, in the *élan vital* which raises up difference to the absolute of a potential or a virtual, and forces thought *to begin with the materiality of difference* insofar as this materiality designates *the new in the making*” (2004a: 111).

The second movement type in *Hunt* differs sharply from what was described above. This mode is, at the early stages of *Hunt*, inserted several times in-between the movements that are marked by uninterrupted continuity. It is, furthermore, characterized in itself by the in-between. The motions that constitute this second “technique” are punctuated by series of cuts, momentary suspensions. Whereas eventually taking many forms within *Hunt* from movements of arms and bends of the torso and knees to interfaces with the multimedia elements and especially the strobe flashes, these instances of Saarinen’s dancing involve essentially *discontinuity*. It is as if his body would reside recurrently in a fleeting state of suspended or *incipient* action. Put another way, the actualization of Saarinen’s movements is repeatedly interrupted even if each time so briefly as to almost escape perception.

Why this technique of moving is of interest here owes to its connect-ability with the concept of affect. Conjoining insights from Bergson and Baruch Spinoza (via Deleuze), Massumi asserts that affect is “a state of suspense, potentially of disruption” and hence “a hole in time” insofar as time is conventionally conceived in terms of linear action–reaction sequences or narrativizable progressions of causes and effects (2002a: 26). Affect, in other words, amounts to a “third” condition that precedes the distinction between activity and passivity. Although implying suspense, it is not an instant of existential void. It rather signals an uncontainable *excess* of impulses – from sensory stimuli to a wealth of “pretenders” to expression and thought – that impinge upon a body and mind in the event of their specific encounter with their environment (Masumi 2002a 30, 32). A further way of comprehending affect

is to address it as a state of emergence that occupies the dimension of “*intension*”. Phrased differently, affect stands for the above-mentioned incipience of the actions or responses that a body and mind will perform in relation to the instance they are embedded in. It encompasses these re/activities in their budding stage, *before* they assume an extensive form as concrete, goal-oriented acts or become signified within discourse as distinct psychic and corporeal experiences. More aptly still, affect involves, for a blinking moment, a multitude of potential pathways for feeling and action that coexist in an incorporeal or virtual state – though only one of these can ultimately actualize at a time (Masumi 2002a 30). Affect, therefore, functions as another name for movement as pure difference. As Deleuze announces in his treatise on Spinoza, it is “purely transitive”; a “passage from one state to another” (1988: 49).

Accepting the above delineations of affect, the serial cuts into Saarinen’s movements could be perceived precisely as tiny holes in (linear) time that are of the order of intensive potential, occurring in-between an actual unfolding of a motion and the next. Indeed, these interruptions could be regarded as *jerk*s, discussed by Masumi as exemplary “bifurcation points” – points of indeterminacy – amidst a movement whose continuity they, just for a flicker, suspend and decompose. In this capacity, a jerk almost unnoticeably intercalates “a flash of virtuality into the actual movement under way”. This is because “at each jerk, at each cut into the movement, the potential is there for the movement to veer off in another direction, to become a different movement” (Masumi 2002a: 40–41). Hence, insofar as “the jerk” is considered an instant of virtuality whilst also taking into account the inherent connection between body movements and sensation, it does not designate only a body’s as-yet un-actualized potential for *kinetic* movement. It also engages the related potentials for *experience and thought* that pertain to the brain/mind of the one who moves and of the possible others who perceive that (suspended) movement. If understood along these lines the jerk could be employed as a synonym for affect. It would qualify as a specific manifestation of that flashing moment when a body (and mind) teem with a virtual multiplicity of re/actions prior to the selection of one actual response.

We thus propose that the above-portrayed motions comprising the second movement technique of *Hunt* – or rather, the near-imperceptible intervals between them – gave rise to series of affective bifurcation points. These of course involved Saarinen’s individual, performing “mind/body/brain” system (Powell 2005: 4). Still, their impact was also necessarily, albeit unpredictably, transposed on those of the audience: “the spectator will endure simultaneously a process of becoming-dancer” (Gil 2006: 25). This, however, does not yet say much of such biopolitical implications that could be mapped onto these movements. Let us hence, by way of concluding this section, make a couple of initial suggestions. Echoing the opening statements of the chapter, they should be conceived as two sides of the same, undecidedly spinning coin.

Firstly, by experimenting, beyond the readily available everyday habits, with how a body can move, self-relate and connect with coexisting entities (from other human bodies to imaging technologies), *Hunt* arguably conjures up space-times where likewise experimental – unexpected and “refreshing” – itineraries of feeling and thought may emerge. As an instance of artistic expression that implies no neatly definable

goal or straightforward use value, *Hunt*, in other words, may sensitize us to such aspects of experience that tend to get overshadowed within the daily contexts of task- and utility-orientated action. It encourages us to attend to the constant becoming of bodies; to the ways that our vision and other senses in fact register not just the actuality of entities but also their virtuality (pastness as well as further relational potential); and to the origin of any activity in passion, in that impingement of a virtual excess of potentials upon the mind and body which is stubbornly felt in the “background” even though it only ever actualizes highly selectively (Massumi 2007: 73–74, 76; Massumi 2002a: 32). In all these respects, the “jerky” movements of Saarinen and *Hunt* could be seen as illustrative. If, to quote Felicity Colman (2005: 11), “Affect is part of the Deleuzian project of trying-to-understand, and comprehend, and express all of the incredible, wondrous, tragic, painful and destructive configurations of things and bodies as temporally mediated, continuous events”, then this movement technique seems surely capable of intensifying the passage of affects – both within and between the participants involved. Indeed, based on the two modes of movement explored up until now, the biopolitics of *Hunt* might be described as an artistic *politics of more life* (Massumi 2007: 80). What the epithet refers to are the capacities of the arts, insofar as they experimentally assemble novel aesthetic (sensible) and spatiotemporal distributions, to engender experiential qualities, occurrences of thought, and related life potentials that may elude pre-established frameworks of symbolic categorization and utilitarian value.

Secondly, however, there are significant complications to the argument just made. For starters, artistic practices, especially those that reside in the more experimental end of the spectrum while not being, as a rule, objectified into permanently tangible outcomes (the latter feature of which of course characterizes all performative arts), may well lack use value and even exchange value. They might rather be said to have *event value* (see Massumi 2007: 79). This, nonetheless, does not distance these practices in any definitive manner from the currently prevalent, late capitalist modes of industrialized production and acquisition of surplus value. Instead, performative arts, and Saarinen’s internationally acclaimed *Hunt* along with them, allegedly both contribute to and provide models for what various theorists of the socio-economic have come to call an “affect economy” or “experience economy” (Thrift 2008: 71–74). These notions will be inquired further in the following sections. To put it bluntly for now, they signal two related, broad lines of development. The first concerns a purported shift of emphasis in economic production whereby the attainment of profit thrives increasingly on the ways that both the workers’ and the consumers’ affective capacities – interactive powers to feel, elicit or relay effects and vary – can be tapped into and modulated. The primary focus of advertising, for instance, might not in the wake of this shift reside in the delivery of such narratives and identity categories that can be exhaustively recognized by particular target groups. It might instead lay in the transmission of *a force of potential*: a sensation of an excess of possible meanings with respect to a given image/product which consequently invokes in the receivers a capacitating feeling that there is a virtual more to their every actual appropriation (Wissinger 2007: 244).



21. Still from *Hunt*, 2002. © Tero Saarinen company

Closely connected to the above ideas, a second notion of experience economy designates such contemporary modes of economic output that capitalize on the knowledge of “bare life”, or *zoe*, of the human sensory-motor apparatus. These range from the kinaesthetic element of adventure tourism to the carefully designed experiences of multi-sensorial immersion within the arts, entertainment industries and the whole service sector that draw on technologies old and new (Thrift 2008: 71–73). One of the functions of *Hunt* as a production is undeniably to act as a beacon for the flourishing export cultural industries of Finland, moving – or varying and affecting – as a product across national borders. To this extent, it could with notable reasons be linked to the biopolitical tactics that underlie these intensified tendencies of political economy of digital culture. In order to pursue the movement techniques and biopolitical associations of *Hunt* further with a specific view to modern *media* technologies (with their scientific underpinnings and artistic-affective potential), we turn now to the next section.

II Intermediality

In terms of the sonic, musical aspect, *Hunt* by turns aligns with and counterpoints the qualities of Stravinsky's composition. The famous delicate melody lines that open *The Rite of Spring* soon give room to intensive, explosive and asymmetrically accentuated rhythmic and orchestral elements that Saarinen at times taps into and at others departs from with his dancing body, thus keeping the dialogue (or synthesis) between the involved bodily and sonorous movements in vibrant and heterogeneous motion. What distinguishes *Hunt* from the numerous other *Rite of Spring* adaptations are the highly effective digital imaging and screening elements that Marita Liulia designed with her team for Saarinen's choreography. Liulia used a bug found in Director-multimedia software to make the images interact fluently with the music of Stravinsky – and consequently, to produce what Liulia referred to as a *Rite of Spring* for the media age. Liulia explains how she could use the software to “follow” the dancer on stage and then “paint” his body with the animation tools. What the digital design beamed on bodies had to tackle was the problem of how not to kill the dance with the moving image. Eating up the dancer was a key problem that moving digital images might entail, Liulia explains. By using the dancing body as the screen she and her team came up with the solution that actually intensifies the movement by doubling it up through a minimalistic use of the digital elements.⁶

Saarinen's body is at several junctures of the performance entangled with and literally taken over by the lighting effects and multimedia material provided by Liulia's team. When it comes to the latter, it comprises two kinds of moving image. The first involves footage on Saarinen himself, shot from above and showing his body in seemingly infinite, circular movement, turning time and again upon itself. Therefore, when this imagery is superimposed on Saarinen's torso and limbs, which remain motionless during those instances of visual “carving”, his body on stage becomes doubled by its own previous continuity in motion. In this manner, the images hint at the complex temporal layering of bodies. More precisely, they point to the ways in which a body's actuality is constantly and variably pierced and shaped by the virtuality of its past. The body is embedded not only in its actuality and the fleshiness of its phenomenological being in space and in time. It is continuously displaced in temporal terms; thrown out-of-sync.

Whereas this first kind of imagery can hence be seen to concern the moving body's self-relationality, the second kind of digital imaging material is composed of montage-like constellations of diverse images that disappear and return at abrupt speed. If the first image type implies how the body is, in terms of its habits and affective capacities, always inscribed by its pastness, then the second set of images gestures toward the ways its sensorial and perceptual powers are manipulated and overwhelmed by the flows of image and sound within contemporary media-technological environments. In a spirit of Nietzschean territorial inscription and recalling the “original” sacrificial subtext of the *Rite of Spring*, Saarinen's body, by way of acting as a receptive surface for the images, becomes itself sacrificed, as it were, for their disruptive rhythms. The same goes for the snapshot effects engendered by the strobe lights that seize Saarinen in various postures, breaking some of his bodily

movement into separate, observable units. This is the violence of the ephemeral lights that, despite their incorporeal feel, act on the materiality of the body. However, what the strobes, again and paradoxically, illuminate is that there is indeed a body that does not coincide with itself but with its variation, and that the variation is the affording element for the time-slots that the viewer perceives. In the biopolitical regimes that paid special attention to the modulation of attention together with the socially mobile body, a living environment defined by moving bodies needed particular modes of knowledge about such mobilities (see Crary 2001).

What is moreover interesting is that Saarinen engages in “jerky” movements right from the start of *Hunt* and thus prior to the appearance of the strobe light effects. The passages of intensity of his body thereby anticipate the affects and percepts those elements bring into the piece. Considering the inter-medial totality of *Hunt* it could be said that Saarinen enters into assemblages with, or *transduces*, a set of forces in the course of the performance (on transduction, see Mackenzie 2002). Not only does his dance affect and get affected by the sonic “body” of Stravinsky’s music, consisting of once revolutionary relations between the rates and shapes of sound waves in the form of dissonant harmonies and poly-rhythms. It also acts and reacts in an electrified fashion to the generation and manipulation of affect enabled by the previous and recent developments of modern technical media. The electrified flashes, the images whirling and taking over the body of the dancer and the intrusive imagery that possesses Saarinen’s body are, then, not *signifiers* or *metaphors* for the disturbing powers of media capitalism or such like. Rather, they amount to actual intertwinings with the moving, sensing and thinking body. What is being danced is *the actuality of that encounter of differing scales of bodies*: the body of the dancer, the intensive embodiment of the space, the ephemeral bodies of the images, the bodies of flickering as a temporal structuration of the space, and so forth.

The music, the space, the jerking body in relation to itself, seem to work not so much as back-groundings for the body in movement, but as its accomplices. The body is only in this framing that happens through the temporal unfolding of sound in a vibrating acoustic space. This resonates with an analysis of dance proposed by Gil (2006) where the intensity of the dance is a creative enterprise, not just registering the already existing movements. Furthermore, the jerk as an art of the interval is something that cuts the phenomenological continuity of the body and transposes it explicitly in a media technological context.

The body in movement in *Hunt* is transposed with images, but the body itself works through a specific kind of a notion of image in movement and more specifically image in time. The Deleuzian concept of the time-image has been continuously evoked in various analyses of the moving image, and indeed it gives tools to grasp the interval of between images. For Deleuze (2005: 252), cinema acted as psychomechanics where the materiality of the image has a direct relation with the body and the brain. Deleuze outlines the emergence of cinema as an art of industrialized movement (and hence intimately connected to mass production and capitalism). This in itself is a “shock to thought” (Deleuze 2005: 151) that constitutes a resonance and a vibration between the industrialized automated movement and the nervous and

cerebral systems. At this point Deleuze talks about the spiritual automaton in the human body as a reaction to that movement. With the time-image that emerges after the Second World War, the representation of time is no more indirect and the cinematic movement achieves a new level of constancy. This is the Kantian moment of cinema where “time is out of joint” and not anymore subordinated to space.⁷ It is time itself that becomes the direct focus and enabler of the image and movement (Deleuze 2005: 260). Already this would be interesting in terms of Hunt’s jerking movements. The subject of the time-image is the subject of situations where direct and clear responses are no longer possible. The optical and sound elements achieve a force or constancy over the logical continuity of space and time.⁸ In addition to these concepts of the image, Deleuze outlines the emergence of the electronic image, or the numerical image, that dismantles on a new scale the privileged positions of bodies, and space coordinates (horizontal and vertical). Instead, a variation becomes primary and new images can, to paraphrase Deleuze, emerge from any-point-whatever in the constellation. Here, information replaces nature and such concepts like the “brain-city” emerge according to Deleuze (2005: 254) as primary.

What we can pick up from this kind of understanding of the resonating milieus of media technologies and bodies is an understanding of the co-rhythmic elements of bodies, movements and technologies. Deleuze presents a strong case for a materialist ontology to understand the abstract movements of bodies as crucial for their media technological framing. For Deleuze, the notion of the brain emerges as a key concept to understand this ephemeral and ecological materiality, and indeed in the light of *Hunt* (as an ecosophical object), the brain or more widely the nerve can be seen to suggest a fruitful understanding of the event. The jerks, the pulsations, the body-in-variation all remind of not only the interval between the images but of the novel regime of the image as a fluid information surface that scans the body and uses it as a screen. The flicker effects that are familiar from experimental cinema act here as well to remind us of the embodiment of even visual perception in a distributed system of nerves. Recently, such ideas have been suggested in media theory occupied with the primacy of the body as a relay of information flow (Hansen 2004: 27). In addition, the flickering effects of strobe lights cutting through the body in movement in *Hunt* provide a screening (and hence an image, or a body) of a kind where the time-image and the duration of time work through the interstices of darkness. In other words, in the heightened ending of *Hunt*, the flashing stop poses are the image par excellence, if with image we understand the aesthetic and biopolitical modulations of bodies with technologies. So the flickering actually is a frame for the body emerging continuously in the midst of the strobe lights, and where the time-image emerges from the “between-images” as Mark Hansen (2004: 241) argues based on Deleuze. The image does not represent, and the time cannot be a function of the image, but only its limit and condition of existence. Whilst Hansen proceeds to specify the singularities of the video image, we want to point toward another direction suggested by the element of dance and movement in *Hunt*.

The pulsations on stage and the embodiment of variation provide a vocabulary relevant for a biopolitical evaluation of bodies, creativity and work in the age of

network capitalism. Indeed, this is where the aesthetics of movement and materiality intertwine with the regimes of tapping into difference, movement and the interval as the crucial concerns of capitalism.

III Biopolitics

We have already sought to hint at the arguable links between *Hunt* and the vibrant contemporary debates around biopolitics. To be sure, historically the techniques and institutions around dance, music, singing, and in general performance are an explicit part of the disciplinary regimes of molding bodies through spatial/izing practices.

Training the body whether it is the muscles of the dancer, or the throat of the singer are prime examples of how memory is inscribed in bodily habits and potentials. Yet, there are fresh examples that can be extracted from *Hunt* as an arrangement of forces of intensity and individuation, and which point more concretely toward its existence as part of “creative industries” and the understanding of vibrant materialism at the heart of the capitalist machines of capture. Indeed, the spatializing disciplinary practices have given way to a much more open-ended modulation of the intensive potentials of life that bodies in movement express. This resonates with Deleuze’s (1992) notion of the control society for sure, but also with Michel Foucault’s (2007) much discussed points about security as a specific mode of tapping into the potentialities of bodies. Nigel Thrift addresses more specifically dance and performance and their role in the constitution of post-Fordist engineering of movements and sensations into industries of profit. Intuitively Thrift points toward the opening up of the spatialized institutions in these regimes of bodies in movement since the 1960s, where “knowledge of performance [...] has moved out from the stage to fill all manner of venues – from corporate presentations to the street” (Thrift 2008: 71). This he connects to the ways in which capitalism and the creative industries tap into “the various knowledges of bare life they are producing” in order to engage in an “ethology of the senses” (Thrift 2008: 71).

What is important about dance and performance in this framework are their non-representational aspects. Indeed, the dancing and moving body can be seen in historical terms as a specific form of knowledge production with an increasing economic importance. Dance is the perfect interface for cultural theories of movement (bodies in variation) to understand the complexity of interaction, an ethology of forces/bodies and the object of cultural industries of affect and experiences. Thrift writes, “...dance can sensitize us to the bodily sensorium of a culture, to touch, force, tension, weight, shape, tempo, phrasing, intervalation, even coalescence, to the serial mimesis of not quite a copy through which we are reconstituted moment by moment” (Thrift 2008: 140). As said already in the beginning, the moving body has become a crucial mode of knowledge production for Deleuzian and radical empiricist theories of relationality that do provide a much more thorough picture of embodiment than representational theories. However, without claiming that it has been neglected, we want to underline the importance in seeing how such theories operate undeniably in contexts where the issue of capitalism cannot be forgotten. One would be tempted to write that this is where the neo-materialist theories of bodies meet up with the

“old materialist” Marxist viewpoints of cultural production, and where the intensity of moving bodies is a topic fruitful both for the philosopher as it is for the cultural industry.⁹ Yet, this is a new materialism not solely of logic of production seen in dialectical terms, but a political economy where the intensities of movement and, in that sense, “creative” embodiment are closely aligned together with immaterial labor, creative economy and innovation culture.

In his short commentary on *Hunt*, Saarinen raises the rhetorical questions “Who owns us? Who pulls our strings?” The piece proposes a connection to the post-Fordist regime of labor. Indeed, as suggested earlier the “jerky” movements and dancer-become-puppet that Saarinen showcases with an impact-full bodily touch imply as such the plausibility of considering *Hunt* in this framework. Furthermore, the questions could also be understood as pointing toward the “inorganic double” of the production or relaying of affects within the piece (Deleuze & Guattari 1994: 172). The dancerly forces of movement and temporality are not “owned” by Saarinen’s body. Instead, they display a spectral quality in the sense that affects and percepts have a duration of their own that surpasses (even though traversing) that of the material. The spectrality of the forces is continuously present, however, or at least visible, through the framing images that seem to arrest the dancing body. The body becomes an image frame, quirkily reminiscent of a TV set frame, for the images that are thrown back at Saarinen. He becomes a surface for the second-order phenomena of visual projection of himself. These images are what position the body in a grid – just like the strobe lights halted bodies in movement for the purpose of biopower measurements around the 1930s. In the act of dance, the body has to be stable for the images to appear.

In addition, the questions Saarinen poses/dances are what signal the biopolitical dimension of this dance piece and return us once more to the question of the apparatuses of capture at work in the age of network capitalism. The body is not “mine” in *Hunt* for two main reasons: first, the body is varying and never “mine” even in the phenomenological sense; secondly, the varying body is increasingly the key object for capitalism tuned toward containing/employing temporal movements and processes of difference. The plasticity, intensity, creativity and the ethological excavating of the thresholds of the body and its capacities are what characterize the body in movement in *Hunt*, but more so they are the tempting characteristics of any productive body in the age of post-Fordist production. This is the relay point where the body as a materiality of lived abstractions and abstract potentiality (Manning 2007: 136) meet with the capitalism of incorporeal events. Indeed, as Manning explains the Body-Without-Organs as the element of becoming or pure potentiality of bodies stems from an absolute difference that cannot be contained by biopolitics of capture (Manning 2007: 137). However, we have to note that difference and pure potentiality are the ideals that contemporary capitalism of incorporeality, events and movements aims toward. The BwO might remain an anathema to the body-politic, as Manning argues, but it is, simultaneously, an increasingly emphasized topos of capitalist capture as well.

This process of capturing the variation has been noted by a bunch of recent post-Fordist commentators, including, for example, Maurizio Lazzarato. For Lazzarato, “The paradigmatic body of our societies is no longer the mute body molded by discipline, but

rather it is the bodies and souls marked by the signs, words, and images (company logos) that are inscribed in us" (Lazzarato, quoted in Thrift 2008: 52). Lazzarato's Deleuze and Foucault inspired take has provided insightful analyses on the centrality of the event in control societies. Hence, it is not only the advertisement business that is characteristic of making the body speak and giving the business a soul, but furthermore the importance of the event, invention and creativity for the control society and its apparatuses of capture and regulation (Lazzarato 2006: 178). Here the temporal ephemerality and difference are the new objects of biopower, which places performing arts into an interesting light. Lazzarato explains how "play" and "work" cannot anymore be understood as dialectical opposites, but part of a continuum of production where modes and activities of bodies, labor and relationality are arranged (Lazzarato 2008: 27).

As Thrift (2008: 71) outlined, dance and performance become exemplary probe-heads for social forms of creativity and capture for the control society. Perhaps one could say that they provide vocabularies both for an understanding of the transitional intensity of bodies, but also the late capitalist logic of the event. The nature of the event, notions such as virtuosity important in Paolo Virno's (2004) writings, and the essential temporality that acts as the unifying modulation are in fact the crucial issues identified by radical empiricist-informed theories of control and performativity. Whereas art and the artist served as the paradigmatic models in modernity for freedom, heterogeneity, difference and deviance (Lazzarato 2008: 26), this is no longer certain. What then is their political status in the era of generalized creativity? As questioned by Lazzarato following a Guattarian line of thought relating to the ethico-politico-aesthetic production of subjectivity, this is a question where sensibilities and creativity are, by definition, at the core of the question of politics as well, as well as the redefinition of work. Furthermore, from this follows the "impossibility of separating political revolution from the revolution of the sensible [...]" (Lazzarato 2008: 29).

So what is the body of Saarinen performing? A working, artistic, creative body? The body is performing the rite for forces and sensations taking over, a whole ecology of forces reaching from the microperceptions of a body overwhelmed by vibrations and rhythms, but also the forces of creative industries, capitalism, and the relay where the body in variation becomes the motor of surplus. Capitalism functions through surplus, circulation and distribution: all relevant movements for the body of the performer as well which we can think here as a beacon of the export creative industries. Indeed, there is in a sense no opposition of work and play in the performative assemblage of *Hunt*, where the cultural industrial networking across the globe takes the theme of bodies in movement to another level. In an ecological perspective (that is also a non-representational one), what constitutes *Hunt* are the multiple linkages and relations that enable its existence as an aesthetico-political and technological event. While capitalism is not an all-subsuming apparatus of power, as our claims about variation have indicated from the start, we still need to urgently acknowledge that forces of aesthetic creation cannot be seen as contradictory to the powers of capitalism either, and that politics or the (bio)political economy of bodies centrally take place nowadays on the aesthetic level. It is on this level where *both* revolt *and* capture occur.

Let us hence summarize, by way of conclusion, our main points about the relations amidst which the dancing body of *Hunt* relays us. First of all and on a more historical note, the use of strobe lights and the entire repertory of jerky motions displayed in the course of this piece suggest a connection to the modern media technologies of inspecting, dissecting, and hence “hunting down” the complexities of corporeal movement. These socio-technological devices include, amongst others, the “acts of whirling” (*strobos* in Greek) that were used from the early twentieth century on in time based analysis of moving bodies to take apart their intensive continuity. Strobe lights, too, were harnessed in the 1930s for similar biopower-imbued purposes of division and measurement. To follow Friedrich Kittler, modern media are essentially defined by time axis manipulation: by their capacities to manipulate, reverse, sample, and play back time (see Krämer 2006). This is something that also Saarinen’s body’s by turns stretched and disrupted gestures seem to refer to.

Secondly, however, the halts and disruptions of a body in motion that *Hunt* involves can be seen as emblematic of affect, which within our approach signals the body’s inexhaustible potential to become: to relate, sense, perceive and think otherwise than what it is or does in any particular here and now. A jerky body is a changing and thus an un-decidable body. It is precisely this un-decidability that the media technologies and related creative industries of societies of control, indicated in Saarinen’s performance by its multimedia components, purportedly eagerly tap into. As we have argued, a key target of the operations of biopower in relation to contemporary informational and communicative technologies and capitalism is the management of sensations, perceptions and memories – that is, of the incorporeal and virtual of bodies.

This leads us to our final summative remark that concerns the persistently unresolved and productive tension *Hunt* appears to stage. For sure, it presents us with a body that is, to paraphrase Elizabeth Grosz (2006: 22), qualitatively more; more intensive, interactive, transformative, than what our habituated daily patterns of movement and perception would seem to allow. But then again, transformation, flexibility, and creativity are exactly the forces that the contemporary practices of advanced capitalism and post-Fordism favor and attempt to harness for various ends. Far from offering a solution to this dilemma, the very last moments of *Hunt* seem to dramatically toss it in the air as a pressing question, the complexities of which, require sustained and imaginative address. While the media technologies figuring as part of the piece over and again seize Saarinen’s moving and affective body, what remains after the last visible flash of his leap, is the continuity of that body’s differentiation we cannot anymore, actually, see: its becoming-imperceptible. Perhaps, then, the idea this open-ended conclusion of *Hunt* gestures toward, is that while the techno-scientific and capitalist procedures of capture do not cease to catch up with bodies’ capacities to vary, the next transition and difference, because of the insistent remainder of unpredictability accompanying its eventual outcomes, is always worth experiencing and launching into – simply as the eternally recurring chance for relations, creation and a life that might become otherwise.

NOTES

Chapter 1

- 1 The field data gathered between October 2004 and May 2005 at Susana Nevado's studio in Turku, Finland, is part of research material I have collected with three different artists between the years 2003–2005 for my PhD study that aims at elaborating a new materialist theory and methodology for Art history. For another case study essay base on this material, see Kontturi (2009). For more information about Nevado's art visit at: www.susananevado.com.
- 2 Put differently, it might be claimed that my effort of creating a cartography of an art process speaks for a situated, embodied knowing. For to get a better understanding of the concept of particle-sign, I am offering a detailed description of its evolvment/involvement in relation to a certain material art process.
- 3 By referring to a new materialist ontology rather than to a new materialist knowledge production I would like to highlight the importance of conceiving art as process and becoming that has its own singular dynamics irreducible to the human domain of knowledge production. Thus, if we want broaden our understanding of what kind of new (forms) knowledge art can provide us with, we need first to attend art's own dynamics, observe its own laws and ways and capabilities, its autopoiesis. For what "new" in new materialism stands for, is not only a novel understanding of materialism; but also, and crucially, it refers to the ontology of change, world *anew*ing itself continuously.
- 4 For worth-mentioning exceptions see Bolt's (2004) reference to her own artistic work, Elizabeth Grosz's (2008) discussion of Indigenous Australian art, Judy Purdum's (2000) essay on Nancy Spero, Brian Massumi's (2006) account of Bracha L. Ettinger's work, and Erin Manning's *Relationscapes* (2009). Also, Simon O'Sullivan (2006: 5) makes a corresponding, though gender-neutral point not to attend the artists considered by Deleuze himself.
- 5 When scrutinized closely, it is quite obvious that the two navels did not emerge in the same way; their processes of becoming differ. Whereas one is more or less clearly painted on the surface of the piece, and is part of the most recent layer that plays a major role in the emergence of the girl figure, another one is deeper, thus going through more than one layer. What makes the case even more interesting is that neither of the navels seems to lie in the middle of the girl's stomach, which obviously, and maybe also disturbingly, undermines the idea that one of them would be more "original" than the other.
- 6 Interestingly enough, these are images that feminists have accused of distributing submissive gender-role models, but which have also made a critical comeback as popular feminist accessories, as magnets and postcards with slogans that challenge those very

roles. A whole range of these critical products, “visual wise-cracks”, can be viewed at <http://www.ephemera-inc.com/> [9 January 2010]

- 7 For example, work of American artists such as Barbara Kruger illustrates well this deconstructive tradition, which critically recycles oppressive imagery. In early 1980's the deconstructive strategy was introduced as a counterforce for subjective and essentialist body art. See also for example, Barry and Flitterman (1987). For a contemporary account of feminist art that runs more parallel to Nevado's “material” practice see Rosemary Betterton's (2004) edited collection *Unframed. Practices and Politics of Women's Contemporary Painting*, which also includes Barbara Bolt's essay “Painting is not a representational practice”.
- 8 Powers of a-signifying semiotics, are however, more than anything else, affective. This is why a-signifying signs affect us, our bodies – for better and for worse – without involvement of conscious interpretation. This has made it very profitable for capitalism. For a range of examples of a-signifying signs varying from pincodes in credit and bank cards to contemporary cinema see Genosko (2009).
- 9 More than interestingly, Guattari (1995: 39) attributes the “double lack” to the Lacanian signifier: “[Which] is too abstract in that it makes heterogeneous, expressive materials translatable, it lacks ontological heterogenesis, it gratuitously uniformises and syntaxises divers regions of being and, at the same time, it is not abstract enough because it is incapable of taking into account the specificity of these machinic autopoietic nodes...”

Chapter 2

- 1 This image is in the public domain http://commons.wikimedia.org/wiki/File:The_Earth_seen_from_Apollo_17.jpg?uselang=fr
- 2 Bachelard (1942) quotes a certain Bousquet (maybe Jacques, 1923 –) : « Une image coûte autant de travail à l'humanité qu'un caractère nouveau à la plante. »

Chapter 3

- 1 See Clark, A and Chalmers, D., “Extended Mind” in *Analysis* 58: 10–23, 1998. “In these cases [where] the human organism is linked with an external entity in a two-way interaction, creating a coupled system that can be seen as a cognitive system in its own right. All the components in the system play an active causal role, and they jointly govern behavior in the same sort of way that cognition usually does. If we remove the external component the system's behavioral competence will drop, just as it would if we removed part of its brain. Our thesis is that this sort of coupled process counts equally well as a cognitive process, whether or not it is wholly in the head.” (1998 HREF)
- 2 See Ezekiel Di Paolo “Extended Life” (2009) and Jill Bennett (2010) *Vibrant Matter*.
- 3 A proposal based on the available information in a shared common world must recognize the importance of ecological psychologist, James J. Gibson notion of direct perception and affordances (1966) and the reformulations of mono-nature and multi-culture by Bruno Latour (2004).
- 4 William James' (1967) radical empiricist position asserts that experiences are themselves experientiable and demystifies the status of consciousness to the activities of thinking without going as far as Gibson (1966) and omitting or separating the content of thought from the experientiable or “directly perceivable”. James' descriptions in *Essays in Radical Empiricism* (1967) and in *The Meaning of Truth* (1978) have a strong resonance with Gibson definition of direct perception: “The directly apprehended universe needs, in short, no extraneous trans-empirical connective support, but possesses in it own right a concatenated or continuous structure” (James 1909/1978: 173). He observes that relations are “transitional experiences *which the world supplies*” (James 1912/1967: 14).

- 5 Di Paulo (2009), Thompson and Stapelton (2009) notions of extended life through the theory of enaction is to be understood in relation to and as a qualification of the extended mind theory of Clark and Chalmers (1998). The difference is that while extended mind expands the cognitive domain it still settles on the isolation of the components from the surrounding environment by virtue of the coupling activity. Extended life goes much further by dissolving the rigid distinction between the cognitive system and the vibrancy of the environment.
- 6 See Di Paulo (2009) "Extended life" in *Topoi* 28: 7–21 and Thompson and Stapelton (2009) "Making sense of sense making: reflections on enactive and extended mind theories", *Topoi* 28: 23–30.
- 7 Within this phrase, Marx anticipates a contemporary debate in cognitive science regarding the relation of attention and perception. Inattention blindness or change blindness is the term researchers use to describe the degree to which attention is determined by prior knowledge (Mack and Rock 1998). It turns out that change blindness is easy to illustrate with experimental examples. The number of change blindness videos on YouTube indicate the growing awareness of the power that attention, expectation and have upon the threshold of awareness and the continuity perception. When viewing a change blindness video and asked to "count the number of times a ball is passed between the 8 participants" a person will usually not perceive the person in a bear costume moon-walking through the middle of the scene.
- 8 The alternative translation of the passage reads "As soon as it emerges as a commodity, it changes into a thing, which transcends sensuousness. It not only stands with its feet on the ground, but, in relation to all other commodities, it stands on its head, and evolves out of its wooden brain grotesque ideas, far more wonderful than if it were to begin to dance of its own free will" (Marx 1992: 163). This version erases the reference to mysticism, clairvoyance and séances embedded in the "table-turning" phrase and replaces it with a more poetic and perhaps Nietzschean image of agency: dance.
- 9 De Landa outlines three types of thinking or styles of thinking made operative by a Deleuzian materialism: population thinking (in which evolution requires large heterogeneous populations to searches for opportunities); intensive thinking (in which thermodynamics provides qualitative measures such as speed, temperature and pressure phase transitions); and topological thinking (in which Gauss and Reinmann's non-Euclidian and non-metric use of differential calculus assigns intensive values to measure rates of change in surfaces) (De Landa 2004: video HREF).
- 10 There is no part of the biosphere that is not cleaving – constantly joining and separating – thus bioscleave (Gins and Arakawa 2006: 208).
- 11 Arakawa, a protégé of Duchamp and Arakawa and Gins collaborators and friends of Duchamp repositioned their work from the domain of art, which they see as having special consideration to the realm of built environment, which does not take recourse in a critical position outside collective human endeavour. In devising for a common world the art games are re-gifted as material tactics.
- 12 Thomas McEvelley (2005) maps the origins of anti-art in Marcel Duchamp's work, recounting the period after his rejection by the Salon in which Duchamp retreats into a part-time job as a librarian and discovers the work of Pyrrhon of Elis (2005: 20–21). What Duchamp found in Pyrrhon was an antidote to the law of the excluded middle in which things transform under pressure from new contexts and imported relationships to open up possibilities. Another approach might be to "cultivate a neglect of opinions reflected frequently in Duchamp's discourse" (McEvelley 2005: 22).

- 13 Gins and Arakawa (2002) *Architectural Body*, in the chapter entitled Daily Research (2002: 95–100) refer to the activities of the resident-researcher in one of their procedural architectural surrounds that lives a life of learning that enlivens all the ready and possible connections available through exploration of the organism–environment relationship.
- 14 It is blank that gives a diffuse blurring over to the state of consciousness in conjunction with the narrowly focused delineations of actions. Providing the sense of volume to any fiction of place, blank helps form that out of which any act arises (Arakawa and Gins 1987: 30). *Landing sites* are a body-wide form of attention and awareness. They are a neutral marker, a simple taking note of. They exist on many different scales of action at once. And they configure with other ways of landing as a site, which include perceptual landing sites, imaging landing sites and dimensional sing sites (Gins and Arakawa 2006: 210).
- 15 Both the sequence of panels in *The Mechanism of Meaning* (Arakawa and Gins 1997: 54) and the sequence of room on the *Bridge of Reversible Destiny* show a midway changeover from dismantling to reassembly (1997: 125).
- 16 The excluded middle, form the point of view of this essay, is the background environment that acts as a context. In the case of Arakawa and Gins they activate the architectural surround/tactically posed surround/tutelary abode to prompt the “organism that persons”. They state “In responding to the ubiquitous call that comes from nooks, crannies and non-nooks and crannies of an architectural surround – most observers feel that they ought to eventually get around to noting everything around them – a person assembles and takes on an architectural body” (Gins and Arakawa 2006: 205). *Atmospheric intricateness* is a mode of connection engineered to take full advantage of the ability to coordinate degrees of emphasis across the bioscleave.

Chapter 5

- 1 The director is usually listed as being Bill Morrison although as *Decasia* is fabricated from found footage many can, of course, be said to have had a hand in its production. It is a tissue of film quotations.
- 2 For a discussion of the breakdown in language experienced by concentration camp inmates see Edith Wyschogrod, (1985) *Spirit in Ashes: Hegel, Heidegger, and Man-made Mass Death*, New Haven: Yale University Press: 106–149.
- 3 Robert Jay Lifton (1967) *Death in Life: The Survivors of Hiroshima*, London: Weidenfeld & Nicolson: 172–181. *Hibakusha* is a coined word meaning “explosion-affected person” used in Japan to describe the survivors of Hiroshima and Nagasaki.
- 4 A photograph of this effect of the bombing can be found on page 96 in Donald M. Goldstein, Katherine V. Dillon, & J. Michael Wenger (1999) *Rain of Ruin: A Photographic History of Hiroshima and Nagasaki*, Dulles: Brassey’s.
- 5 Ibid. Our emphasis.
- 6 For a fuller description of the operations of transference and their interpretation see Julia Segal, Melanie Klein (1992) London: Sage 1992: 65–67.
- 7 Julia Kristeva (1981) “Giotto’s Joy” in *Desire in Language: A Semiotic Approach to Literature and Art*, Trans. Thomas Gora, Alice Jardine and Leon S. Roudiez, Oxford: Blackwell: 210–236.
- 8 Sigmund Freud (1981) The Penguin Freud Library *Volume 7: On Sexuality*, London: Penguin: 83.
- 9 Jameson, F. (1995) *Signatures of the Visible*, London: Routledge: 196, 98. For Jameson color is also “associated with phenomena of the body.”

Chapter 7

- 1 The spiral form of the installation mimics Trajan's column (113 CE) erected in celebration of the emperor's victory in the Dacian wars. Here, the talking head form embodies a critical function regarding the imperial practices that wage their might on European citizens. My experience of the installation is from Ataman's *Mesopotamian Dramaturgies* show at MAXXI, the National Museum for twenty-first century Arts in Rome over the summer of 2010.
- 2 The video belongs to an on-going installation project – *untitled* – in which Salloum deals with the interstitial spaces and subjectivities caused by war. The variant setup of the installation includes footage from the Federal Republic of Yugoslavia (FRY) as well as Lebanese prison camps. Interviews in and about Yugoslavia are projected in pairs with footage Salloum shot from a bus window while he was traversing the country at the time of the NATO bombings in 1999. Footage from Lebanese prison camps is paired with views of clouds. In the installation setup, *everything and nothing* stands out: it is a single-channel loop played on a TV monitor. See Salloum's production notes for a detailed setup of the installation, available from http://www.111101.net/Artworks/Jayce_Salloum/untitled/JS_untitled.pdf [accessed 31 January 2009].
- 3 The “moments of affection” in *everything and nothing* are analogous in function to the “musical moments” Amy Herzog locates in a selection of filmic works. She describes these moments as points of rupture in the image-sound hierarchy and argues that musical moments may invert and restructure the spatiotemporal coordinates of the narrative and thus change the impetus of a film (Herzog 2010: 7–8).
- 4 Connolly's project is to come to terms with the materiality of perception by placing Maurice Merleau-Ponty, Michel Foucault and Gilles Deleuze into conversation. He notes that the link between Merleau-Ponty and Foucault is in how they “attend to the preconscious, affective dimensions of discipline and experience” and how they both outline a prior disciplining of the senses in which a history of inter-involvement organizes perception (Connolly 2010: 187, 190). The connection to Deleuze is construed on this level with Deleuze's proposal of enhancing our existential attachment to the world in the face of “our universal schizophrenia”.

Chapter 8

- 1 “Orthonymity” is signing with one's legal name; in “anonymity” there is no signing at all; “pseudonymity” refers to signing with “a false name”; “antonymity” takes place when an author chooses a name opposite in meaning to one's legal name; “homonymity” occurs when there are at least two authors with the same name; “mononymity” happens when several persons agree to sign with one common name; “allonymity” refers to an author who impute her work to somebody else in order to receive more publicity.
- 2 “Materiality of authorship” acts as an oxymoron if insistence on separating “writer” from “author” is maintained: “author” referring to a construction formed by the critic through reading cannot be “material” due to its representational nature – author becomes an image which differs strategically from “the writer”, the empirical and historical person, who in this scheme of thinking seems to be doomed to the sphere of materiality, of life, with no access to the entrance leading elsewhere. In this paper the distinction between “the writer” and “the author” is not kept up; “materiality” persists in versatile ways. In *Kafka – Toward a Minor Literature* (1975/1986) with Félix Guattari and e.g. in *Dialogues* (1977/1987) with Claire Parnet, Deleuze reserves the concept of “writer” (*écrivain*) to representatives of “minor literature”; “author” (*auteur*) refers to makers of “major literature”. This paper does not follow this Deleuzian distinction although it welcomes the political nature of this division to separate

different ways of entering the writing-machine instead of emphasizing differences based on ontological essence. This paper wishes to de-territorialize also the concept of author and the ways it has been used.

- 3 The materiality of Algot Untola's names *per se* effectuates various connections for those who know Finnish language. His paternal last-name, Tietäväinen, literally means "the one who knows". Untola, the last name he chose for himself, alludes to both dreams (*uni* in Finnish) and to oblivion (*unohdus*). The author-name of Irmari Rantamala is a curious mixture of two common first names, that is the female name Irma and the male name Ilmari are intertwined in Irmari as well as two female names of Irma and Mari. Rantamala refers to "a place on the lakeshore". The female name Maiju Lassila presents itself as a very "ordinary" name for a Finnish woman, simultaneously it appears as totally opposed to ordinariness: *Maiju* is the Finnish version of "Maya", which means "Illusion". Indeed, in the totality of Untola's authorship the intensive forces of creating "illusion" are at work!
- 4 "The Death of the Author" by Roland Barthes (1968) has been said to constitute with Michel Foucault's "What Is an Author?" (1969) the founding statements of much subsequent theoretical and critical work on (literary) author (Bennett 2005: 11).
- 5 All translations from Finnish to English, if not otherwise stated, are mine. The citations of the letters refer to the collection of letters titled "Maiju Lassilan saamia kirjeitä" ("The Correspondence of Maiju Lassila"), Collection 137, Helsinki: Finnish Literature Society.
- 6 The correspondence of Algot Untola remains in the archives of Finnish Literature Society in Helsinki. This collection of letters is titled "The Correspondence of Maiju Lassila" although the letters are also signed by Irmari Rantamala, A. Rantala and Väinö Stenberg. The title is yet another expression of "faciality": the authorship(s) of Untola are reduced to one single proper name and one "face", the face of Maiju Lassila, the most canonized author-name. The immanent paradox, of course, is in the fact that Maiju Lassila does not really have a face – only a name.
- 7 Since writing is understood as "desire-production" (see Deleuze & Guattari 1996), the problematics of "intentionality" – whether "fallacious", "hypothetical" or "true" – do not enter the realms of authorship.
- 8 Traces of "conceptual persona" can be seen already for example in *Logic of Sense* (1969/1990) and *Difference and Repetition* (1968/1994).

Chapter 10

- 1 See Harvey, (2008: 49) and Tseelon, (1995: 13–14) for examples of this.
- 2 In the case of women, however, this has only been partially successful for, as Entwistle points out, the suit, when worn by women, is still seen as a more eroticized garment than when it is worn by a man because of the long-standing sexualization of women's bodies. (2001: 53)
- 3 The de-materialization of the body reached its apotheosis in the costumes Oskar Schlemmer created for his various dance works which were designed to re-configure the bodies of the dancers so that they resembled abstract, geometric forms. (Schlemmer, 1979 [c. 1927]: 97)
- 4 Similarly, while the act of acquiring a tattoo or piercing makes the wearer conscious of his body as a corporeal presence, it subsequently converts his body into an object of spectacle, as indicated by the fact that most tattoos acquired by males today are highly visible and are designed to attract the viewer's attention. The acquisition of the tattoo thus serves only as a temporary reminder of the visceral nature of the body. This might account for the

fact that it is not uncommon for tattooees to continually acquire new tattoos rather than being satisfied with one, as their experience of bodily alienation is ultimately unassuaged.

Chapter 11

- 1 Márquez anchors the transfiguration of light and water to the believability of everyday experience, as does the architect Luis Barragán's indoor pool at la Casa Gilardi, in Mexico City (de Vasquez 2002: 211).
- 2 For example the "see-through" effect took a luminous turn in art criticism that championed transparency in construction and critical reflection. Understood as the experiencing of things "being what they are" (Sontag 1966: 13–14), transparency was a liberating, luminous clarity. It applied to minimalist art that is open to, and addressed directly to, the transformative exercise of our senses, rather than thick with hidden crafting and meaningful content.
- 3 I have offered only the briefest sketch of transparency here, setting aside discussion of the "myth" of transparency that haunted Modernity (revealed in all its opacity by Breton, Benjamin and Duchamp), and its revival as a new principle of "good modernism", exemplified by making a monument disappear in relation to its context (as in the pyramid of the Louvre). These and other aspects of architectural transparency in particular are discussed by at length by Vidler (1992: 217–225).
- 4 Robert Slutzky's influence as a painter on the idea of phenomenal transparency in architecture is discussed at length by Anthony Vidler (2003: 6–7). Rowe and Slutzky acknowledge Gyorgy Kepes' initial description of phenomenal transparency, which they elaborated in their essay.
- 5 "Any seeing of an object by me is instantaneously reiterated among all those objects in the world which are apprehended as co-existent, because each of them is all that the others 'see' of it...the completed object is translucent, being shot through from all side by an infinite number of present scrutinies which intersect in its depths leaving nothing hidden." (Merleau-Ponty 1962: 68–9).
- 6 The Department of Computer Graphics at Cornell University is home of the Cornell Box, which is a physical model housed at Cornell University, and also the ubiquitous test scene for global illumination. Since 1984 it has been used in the development and testing of algorithms for rendering light naturalistically, through physically based lighting models and perceptually based rendering procedures. The aim of such rendering methods is to produce images that are visually and measurably indistinguishable from real world images. I have discussed the role of Cornell Box in relation to the phenomenon of "virtual light" elsewhere (Vasseleu 2003).
- 7 For a full account of the process of photon mapping see Jensen's book on the subject (2001).
- 8 I am referring here to the beguiling perspective the model conjures, as discussed for example by Robert Harbison (1997: 84–98). There are other grounds for incommensurability, such as that between the model's abstracted lighting and a situated, phenomenal "lighting" which invisibly constitutes the visible (Merleau-Ponty (1962: 311). The translucence of digital milk presents to us as a technically accessible, mechanically rendered light-likeness, which is produced by data that is unintelligible except in relation to human-machine systems.
- 9 The recovery of the optical properties of "skin1" is indicative of one of the fantasy pursuits that appearance modeling can serve – in this case a familiar formal aesthetic endeavor involving a desired love object (that doesn't resemble Wall-E's gleaming white girlfriend).

- 10 At least one reviewer was unconvinced that Jensen's formulae could ever master the subtle intricacy with which light radiates from inside the face, that is, the so-called "inner light" of consciousness (Weschler 2002: 120–122).
- 11 In this reading the castings are indicative of something that is missing, left behind in its material traces. Aligned with other indexically produced forms, including death masks (with which Whiteread consciously associates her work), the replica is resonant with the sense that it has been cast (physically or optically) from life. At the same time it has the sense of trying to preserve life by producing it in the form of a frozen pose (Krauss 1997: 76–77). Mortification, viewed as temporal arrest or a space "frozen" in the mold, is also echoed in Whiteread's use of translucent casting materials. The freezing of the moment, reminiscent of a shutter closing, functions as a thickening or blindspot that can be seen though. This is replicated in the closed off inner core of translucent substances. Realized in both form and substance, the eclipse is a temporary death or moment of detachment of the subject, partially suspended from the field of vision (Fer 2000: 77–78).
- 12 See Jennifer Gross (2004: 34–50) for a discussion of Whiteread's work as a revival of an early modernist endeavor to show or remove the interiority of sculptural form and the audience's anticipation of, or search into, form for meaning in the public monument. Here Marcel Duchamp's use of glass in *Large Glass* (1915–23) to confirm the constructive nature of perception inherent in visual experience is an apparent influence.
- 13 Deleuze singles out this famous scene in Alfred Hitchcock's *Suspicion* (1941) as an instance in which cinema creates a "mental image" by making relation itself the object of an image. The appearance of a thing may relate to the thing itself in a familiar, natural way, or the same appearance might leap out and strike a contradictory note (Deleuze 1986: 203). The image created by Whiteread's resin casting is different. It does not have the same appearance as a wooden water tower, yet its translucent appearance embodies an unseen relation between them.
- 14 There was the memory of the original water tower but Water Tower was itself something of a personal memorial, as Whiteread stated: "This project is dedicated to my father, Thomas Whiteread (1928–1988), whose interest in industrial archaeology enabled me to look up." (Neri 1999: inside front cover).
- 15 Whiteread's work in general has been discussed in these terms by Rosalind Krauss (1997: 81).
- 16 Jensen describes the behavior of light in a simulation of a translucent object (in this case a translucent box) in terms of fluidity: "Note how the light bleeds from the light source and bleeds down through the box" (Jensen and Buhler 2002: 576). The image is viewable online. See "Cornell Box Images" (Jensen 2005).

Chapter 13

- 1 The term "false consciousness" is used here in similar terms to an eastern conception where consciousness is determined by an individual's relationship to the real and in this case the real being the ontological. The argument here is that the subject's relationship to the real is based on representation. Generally ideology represents "false consciousness" as it is predicated as the ontic as opposed to the ontological.
- 2 Empirical Ontology refers to the "real" being based on lived experience as opposed to theorised ideology. This relationship between the real and lived experience is the basis of an Indigenous cultural ideology, an ontology grounded in reality and creates an alternate space to present a radical materialist ideology.
- 3 Franz Brentano. (1975) *On the Several Senses of Being According to Aristotle*, edited and trans. by George and Roderick Chisholm.

- 4 *Methexis*, a Greek term and hence a Western term that has been garnered to understand the performative power of Indigenous cultural practices.
- 5 A Latin term derived from Roman law meaning “land belonging to no one”. In the context of Australia, colonization was justified through Terra Nullius. It was believed that Aboriginal Australians did not exercise any sovereignty over land, although unwritten legal codes did exist amongst traditional owners groups.

Chapter 14

- 1 We will detail this argument further in later sections of the article, for example when discussing Gilles Deleuze’s notion of control societies.
- 2 Essentially, Deleuze and Guattari’s concept of rhythm needs to be distinguished from meter as a coded system of measure comprised of uniform units. In film and sound scholar Amy Herzog’s helpful formulation, “This rhythm is not a literal beating-out of metered time nor the repetition of identical elements.” It rather operates “in the spaces of transition between milieus, bringing different milieus into dialogue with one another.” Herzog (2010: 81). Rhythms, therefore, are movements of connection and coordination between *disparate* elements and contexts. They are processes by which such elements both come to collaborate – forming new assemblages with their particular affective capacities – and retain their mutual as well as internal differences. Rhythms could also be described as ways in which varied elements affect one another; not through appropriation under one homogeneous principle but by producing changes in each other’s state and relational capabilities. In sum, rhythm is an occasion of relation and translation between bodies, components and milieus. As Deleuze and Guattari state in their plateau “Of the Refrain” in *A Thousand Plateaus*, “There is rhythm whenever there is a transcoded passage from one milieu to another, a communication of milieus, a coordination between heterogeneous space-times” (Deleuze and Guattari 1987: 313).
- 3 Sensation designates here the embeddedness of a body/mind in an experiential continuity. Perception refers in its turn to an object-orientated mode of experience that “segments” things out of that continuity, being capable of arresting and signifying them. (Massumi 2002: 258–259 n11); see also, for example, (Conley 2005: 244).
- 4 Whereas Gil’s exposition amalgamates impulses from Deleuze-Guattari and Merleau-Ponty’s phenomenology, other scholars have lately explored dancing along more exclusively Deleuze-Guattarian lines. Valerie A. Briginshaw (2005) weds Deleuze’s key idea of repetition as differentiation and not identical replication of the same with her analysis of the relations between two men in the dance piece *Both Sitting Duet* that transgress the dominant Western paradigms of male subjectivity. Meanwhile, Stamatia Portanova (2005) has striven to rethink dancing (and dance-perceiving) bodies through the notion of viral rhythm, and by conceptualizing the encounters between dance and digital technologies in terms of the radically novel capacities for movement and thought they can yield rather than through phenomenological, functional or mimetic models. (See also Markula 2007.)
- 5 Of course, the sections also become connected and mutually referential in the experiences of the audience members, most prominently through the workings of recollection that retain “the before in the after.” See Deleuze and Guattari (1994: 212).
- 6 Jussi Parikka’s e-mail exchange with Marita Liulia, 10/07/2008. See also an interview with Tero Saarinen, online at <http://www.liikekieli.com/index.php?name=News&file=article&sid=489>
- 7 What Eric Alliez (2004b: 98) has named as Deleuze’s interest for the heterogenesis of Kant’s thought, provides the philosophical reference point for this media technological

turn. As Deleuze notes, and as explained by Alliez, time becomes out of joint and reversed “in relation to the cardinal movements of the world”.

- 8 Deleuze writes: “These are pure optical and sound situations, in which the character does not know how to respond, abandoned spaces in which he ceases to experience and to act so that he enters into flight, goes on a trip, comes and goes, vaguely indifferent to what happens to him, undecided as to what must be done” (Deleuze 2005: 261).
- 9 For example Balibar (2007) promotes an understanding of Marx’s ontology as a relational one.

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